STUDY MATERIAL FOR GP RATING ENTRANCE TEST for T S RAHAMAN AND NUSI (GOA)

BY-

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YOUTUBE CHANNEL- *MERCHANT NAVY CAREER*

T S RAHAMAN

GP RATING

ENTRANCE TEST PATTERN

TIME- 1hr 45 mins

ENGLISH	**= 50 × 1**	**= 50 MARKS**
GK	**= 20 × 1**	**= 20 MARKS**
MATHS	**= 15 × 2**	**= 30 MARKS**
TOTAL		**= 100 MARKS**

TIME- 1hr 45 mins

T S RAHAMAN

G.P Rating

Entrance Test

held on 02.10.2021

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ENGLISH (50 questions = 50 marks)

Question 1

_____ he playing Tennis now?

A) Did B) Was **C) Is** D) Does

Question 2

He has not yet _____the death of his dear ones.

A) tolerated **B) believed** C) mourned D) convinced

Question 3

She_____ half of her time in travelling.

A) **spends** B) is spending C) has spent D) will spent

Question 4

What _____ you like for dinner tonight?

A) **would** B) must C) can D) wood

Question 5

I haven't seen her _____ ages.

A) in B) of C) since **D) for**

Question 6

You look _____. Didn't you sleep well?

A) tiredsome B) tiredly **C) tired** D) none of the above

Question 7

It looks _____ good.

A) **pretty** B) prettier C) prettily D) prettiest

Question 8

It was _____ hot to drink.

A) **too much** B) too C) so D) such

Question 9

It is desirable to take _____ in any business.

A) profit **B) risk** C) success D) victory

Question 10

The greater the demand _____ the price.

A) higher **B) the higher** C) the high D) high

Question 11

He's _____ friend of mine.

A) isn't B) never **C) not** D) no

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Question 12

It is not _____ mine.

A) same like B) same as **C) the same as** D) the same like

Question 13

Children are more _____ than adults

A) adaptable B) conservative C) resourceful **D) susceptible**

Question 14

The radio was _____ from my car last week.

A) Stole **B) stolen** C) steal D) stealing

Question 15

John held the plate _____.

A) careful **B) carefully** C) cared D) none of the above

Question 16

Correct emergency procedures were not _____.

A) **followed** B) took C) achieved D) important

Question 17

Why did you confide your secrets _____ a stranger?

A) **to** B) on C) with D) at

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Question 18

She dropped her kids _____ at school.

A) **off** B) to C) up D) at

Question 19

_____ not the way to do it.

A) That **B) That's** C) These D) That

Question 20

She did it _____ her own.

A) by B) with C) in **D) on**

Question 21

If I went, I _____ speak to her.

A) should **B) would** C) will D) must

Question 22

I don't know where my book is. I have to _____ it.

A) look up B) look in C) look upto **D) look for**

Question 23

It's warm inside. _____ your coat.

A) Take of **B) Take off** C) Take out D) Take in

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Question 24

There is a bridge _____ the river

A) on B) in **C) across** D) to

Question 25

The person _____ interviewed me was very friendly.

A) that B) which **C) who** D) whom

Question 26

Every _____ is a new experience.

A) body **B) day** C) days D) one

Question 27

She _____ it easy to finish the test.

A) finding B) find **C) found** D) fond

Question 28

Where _____ the sun rise?

A) **does** B) did C) do D) has

Question 29

Julia is a _____ person.

A) **careful** B) carefully C) cared D) none of these

Question 30

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You must provide something _____ the evil days.

A) with **B) against** C) for D) into

Question 31

I won't be _____.

A) this **B) there** C) their D) the

Question 32

She lives _____.

A) her alone B) own C) her own **D) alone**

Question 33

It was _____ hot that we couldn't concentrate.

A) so **B) so much** C) such D) so many

Question 34

I didn't see _____ people there.

A) much of B) many of C) much **D) many**

Question 35

She's good _____ maths.

A) for B) on C) in **D) at**

Question 36

I was certain _____ she would be there.

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A) **that** B) the C) this D) these

Question 37

She spoke _____ him yesterday.

A) by **B) to** C) for D) at

Question 38

It is _____ to study daily.

A) necessity B) neccesary C) necessarily **D) necessary**

Question 39

Does your little brother _____ ghosts?

A) believe on B) believe to **C) believe in** D) believe for

Question 40

Our mom was angry. She spoke to us _____.

A) **angrily** B) happily C) politely D) sadly

Question 41

During my childhood, I liked _____ in the rain.

A) to playing **B) to play** C) play D) played

Question 42

If I'd been there, I _____ that.

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A) have done **B) would have done** C) would have D) would had done

Question 43

She did it _____ Tuesday.

A) to B) for **C) last** D) in

Question 44

I am saving my money _____ a CD player.

A) **to buy** B) buying C) buy D) buys

Question 45

The watchkeeper lacked clear _____ of the situation.

A) **understanding** B) following C) study D) acquisition

Question 46

Victory made a sound _____ before the collision.

A) **signal** B) blast C) study D) understanding

Question 47

Shakespeare is the _____ playwright in English.

A) great B) greater **C) greatest** D) none of the above

Question 48

Vegetables are good _____ you.

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A) in B) by **C) for** D) to

Question 49

I could not resist _____ him why he avoided me.

A) to ask **B) asking** C) to have asked D) asked

Question 50

_____ I saw her yesterday, she was on her way home.

A) What B) Whenever C) Who **D) When**

G.K (20 questions = 20 marks)

Question 1

_____ is known as the Missile Man of India.

A) Dr. U R Rao B) Dr. Homi Baba C) Kalpana Chawla **D) Dr. A P J Abdul Kalam**

Question 2

Ram Nath Kovind is the current _____ of India.

A) Defence Minister B) Chief Minister C) Prime Minister **D) President**

Question 3

_____ is the currency of Japan.

A) **Yen** B) Won C) Euro D) Shekel

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Question 4

_____ is the capital of West Bengal.

A) **Kolkata** B) Mumbai C) Delhi D) Chennai

Question 5

The Indian state with the smallest land area

A) Tripura B) Nagalan **C) Goa** D) Sikkim

Question 6

Cristiano Ronaldo is a famous footballer from _____.

A) France **B) Portugal** C) Argentina D) Italy

Question 7

At the _____ days and nights are of equal length.

A) longitudes **B) equator** C) poles D) latitudes

Question 8

Kumbh Mela is held every _____ years,

A) **12** B) 14 C) 10 D) 15

Question 9

The blue color of the sky is due to _____.

A) diffraction B) reflection C) dispersion D) refraction

Question 10

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What is the capital of Sri Lanka?

A) Kathmandu B) Colorado C) Columbia **D) Colombo**

Question 11

What is the capital of Tamilnadu?

A) Telangana **B) Chennai** C) Hyderabad D) Bhubaneshwar

Question 12

_____ is one of the COVID vaccines developed in India.

A) **Covishield** B) Covimask C) Covisafe D) Covitech

Question 13

The study of heavenly bodies is called _____.

A) Astronautics B) Astrophysics **C) Astronomy** D) Astrology

Question 14

Sambhar lake is situated in _____.

A) Maharashtra B) Gujarat C) Bihar **D) Rajasthan**

Question 15

Boat races are special features of _____ festival.

A) Bihu **B) Onam** C) Pongal D) Baisakhi

Question 16

The tropic of cancer passes through which of these Indian states?

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A) Bihar B) Orissa C) Goa **D) Gujarat**

Question 17

Wagah is situated on the boundary line between which two countries?

A) India and China B) India and Afghanistan **C) India and Pakistan** D) India and Nepal

Question 18

Paradip sea port is situated in _____ state of India.

A) **Odisha** B) Tamil Nadu C) Andhra Pradesh D) Kerala

Question 19

The ozone layer restricts _____.

A) x-rays B) infrared rays C) radio waves D) **ultraviolet radiation**

Question 20

Which is the original language of the Mahabharat?

A) Hindi **B) Sanskrit** C) Hindustani D) Arabic

MATHS (15 questions = 30 marks)

Question 1

The marks obtained by 17 students in a mathematics test (out of 100) are given below:

91, 82, 100, 100, 96, 65, 82, 76, 79, 90, 46, 64, 72, 68, 66, 48, 49. The range of the data is

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A) 46 B**) 54** C) 90 D) 100

Question 2

[(-10) * (+9)] + (-10) is equal to

A) 100 **B) -100** C) – 80 D) 80

Question 3

Area of a right angled triangle is 54 cm sq. If one of its legs is 12 cm long. Its perimeter is

A) 18 cm B) 27 cm **C) 36 cm** D) 54 cm

Question 4

In a cylinder if radius is halved and height is doubled, the volume will

A) same B) double **C) halved** D) four times

Question 5

If two interior angles on the same side of a transversal intersecting two parallel lines are in the ratio 2:3, then the greater of the two angles is

A) 54 **B) 108** C) 120 D) 136

Question 6

If x and y are directly proportional and when x=13, y = 39, which of the following is not a possible pair of corresponding values of x and y?

A) 1 and 3 B) 17 and 51 **C) 30 and 10** D) 6 and 18

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Question 7

Construction of a cumulative frequency table is useful in determining the

A) mean **B) median** C) mode D) all the above three measures

Question 8

The class-mark of the class 130-150 is

A) 130 B) 135 **C) 140** D) 145

Question 9

The linear equation 3x-y = x-1 has

A) A unique solution B) Two solutions **C) infinitely many solutions** D) No solution

Question 10

If x^2+kx+6=(x+2)(x+3) for all x, then the value of k is

A) 1 B) -1 **C) 5** D) 3

Question 11

The 10th term of the AP: 5, 8, 11, 14 …. is

A) **32** B) 35 C) 38 D) 185

Question 12

A card is selected at random from a well shuffled deck of 52 playing cards. The probability of its being a face card is

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A) **3/13** B) 4/13 C) 6/13 D) 9/13

Question 13

A bag contains 3 red balls, 5 white balls and 7 black balls. What is the probability that a ball drawn from the bag at random will be neither red nor black?

A) 1/5 **B) 1/3** C) 7/15 D) 8/15

Question 14

A shirt with marked price Rs 800 was sold at Rs 680. The rate of discount allowed on the shirt is

A) 10% **B) 15%** C) 20% D) 25%

Question 15

If angle between two radii of a circle is 130°, the angle between the tangents at the ends of the radii is

A) 90° **B) 50°** C) 70° D) 40°

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Entrance Test

held on 02.05.2020

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ENGLISH (50 questions = 50 marks)

Question 1

INTERESTING (Synonym)

A) refer B) reward C) relieve **D)fascinating**

Question 2

A person who presents a radio/ television programme.

A)partial B) unbiased **C) anchor** D) particular

Question 3

_____ I come in?

A)should **B) may** C) can D) do

Question 4

You _____ maintain a healthy body, if you keep exercising?

A)can B) was keep C) is keep D) keep

Question 5

Which is the pronoun in the sentence-

I like to walk in the rain for long time because no one could see my tears

A)long B) **I** C) no one D) could

Question 6

He_____ tennis every Sunday morning.

A)play **B) plays** C) played D) playing

Question 7

Successful people always ___two things, one determination and second, sincerity.

A)having B) doing C) did **D) have**

Question 8

Which is the pronoun in the sentence-

Unless they do all of their homework, they won't allow to play

A)they B) won't C) do D) unless

Question 9

LITTLE (Synonym)

A)tiny B) explain C) exclaim D) experience

Question 10

BETTER (opposite word)

A)discuss B) dispute **C) worse** D) distance

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Question 11

STRICT (opposite word)

A)leave B) loud C**) lenient** D) laughter

Question 12

Adam , who is my brother, _____ a famous cricketer.

A)are B) were C) was **D) is**

Question 13

I would have bought a new car for her but it ___ too expensive.

A)were B) are **C) was** D) can

Question 14

Which is the pronoun in the sentence-

A friend of mine is visiting me today.

A)today B) A C) of D) **mine**

Question 15

What type of word (part of speech) is the underlined word in the sentence-

Fifteen people were arrested when hundreds of police officers raided dozens of

homes.

A)verb B) conjunction C) noun D) pronoun

Question 16

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Don't make so much noise, she ____ to study for her test.

A)tries B) try **C) is trying** D) tried

Question 17

A journalist who ____ detained in a city for more than a year.

A)is be B) can be C) might been **D) has been**

Question 18

They _____ some saving to fall back on.

A)be B)been C) being **D) have**

Question 19

FALL (synonym)

A)down B) agreement **C) descend** D) agree

Question 20

LOOSE (opposite word)

A)partial B) unbiased **C) tight** D) particular

Question 21

What type of word (part of speech) is the underlined word in the sentence-

I could not recognize her voice on the phone.

A) pronoun B) preposition C) adverb D) noun

Question 22

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A tropical cyclone occurring in the western Pacific or Indian ocean.

A)turmoil B) tremors **C) typhoon** D) tsunami

Question 23

EXACT (synonym)

A)specific B) exist C) exclude D) excuse

Question 24

EDUCATED (opposite word)

A)learn B) agreement **C) ignorant** D) agree

Question 25

Complete shutdown _____ observed today against new law.

A)is being B) can be C) is been D) was been

Question 26

HARSH (opposite word)

A)refer B) reward C) relieve **D) mild**

Question 27

What type of word (part of speech) is the underlined word in the sentence-

When we were going up the hill, we have seen an old castle.

A)pronoun B) adjective C) adverb D) noun

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Question 28

What time _____.

A)the train leaves?

B) leaves the train?

C) does the train leave?

D) is the train leave?

Question 29

She is a mother , and a great teacher_____.

A)also B) and **C) too** D) to

Question 30

DECIDE

A)freed B) free C) freely **D) determine**

Question 31

HURRY

A)part **B) rush** C) partly D) party

Question 32

A piece of enclosed land planted with fruit trees.

A)hard work B) handiwork C) handicraft **D) orchard**

Question 33

Related to water.

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A)aquatic B) saline C) glass D) rain

Question 34

A person who sells and arranges cut flowers.

A)depositor B) curator C) pharmacist **D) florist**

Question 35

SMART

A)discuss B) dispute **C) clever** D) distance

Question 36

IDENTICAL (opposite word)

A)part **B) different** C) partly D) party

Question 37

JOY (opposite word)

A)freed B) free C) freely **D) grief**

Question 38

What type of word (part of speech) is the underlined word in the sentence-

Let’s go to the park and have fun with our friends

A)verb B) article C) noun D) pronoun

Question 39

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What type of word (part of speech) is the underlined word in the sentence-

Swimming is the best exercise to build stamina.

A)pronoun B) adjective C) verb D) noun

Question 40

Keep yourself away from bad company if you____to study well.

A)want B) wants C) wanted D) wanting

Question 41

What type of word (part of speech) is the underlined word in the sentence-

What did you think about them?

A) pronoun B) adjective C) adverb D) noun

Question 42

An arrangement of flowers that is usually given as a present.

A)leave B) loud **C) bouquet** D) laughter

Question 43

What type of word (part of speech) is the underlined word in the sentence-

When my mom did shopping, at that time, we spent our time in the cafeteria.

A) adverb B) adjective C) pronoun D) noun

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Question 44

A measure of the amount of salt dissolved in sea water.

A)partial B) beach C**) salinity** D) sand

Question 45

Some time I like to spend my time alone, it ____ my mind.

A)refreshes B) refresh C) refreshed D) is refreshes

Question 46

LANDLORD (opposite word)

A)tenant B) explain C) exclaim D) experience

Question 47

MOST (opposite word)

A)deposit B) dispose C) deport **D) least**

Question 48

A violent disturbance of the atmosphere with strong winds and usually rain.

A)manoeuvre B) berth C) bank **D) storm**

Question 49

HEROIC (opposite word)

A)cowardice B) exist C) exclude D) excuse

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Question 50

What type of word (part of speech) is the underlined word in the sentence-

I could not recognize her voice on the phone.

A) noun B) adjective C) pronoun D) adverb

G.K (20 questions = 20 marks)

Question 1

Light year is a unit of

A)time of light **B) distance** C) light D) intensity

Question 2

The tides in the sea are due to

A)atmospheric effect of the Earth.

B) gravitational effect of Venus on the earth
C) gravitational effect of Mars on the earth
D) gravitational effect of the Sun and the Moon on the earth

Question 3

Currency of Japan is

A)Dollor B) Won C) Ruble **D) Yen**

Question 4

Emperor Ashoka belonged to the

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A)Maurya Dynasty

B) Gupta Dynasty
C) Ghazni Dynasty
D) Vardhana Dynasty

Question 5

Mary Kom is a _____ champion.

A)boxing B) wrestling C) weight lifting D) martial arts

Question 6

One inch is equal to ___ cms.

A)1.5 B) 2 **C) 2.5** D) 3.5

Question 7

What does 'OS' on a computer refer to?

A)Operating System

B) Order of Significance
C) Open Software
D) Optical Sensor

Question 8

The Indian Council of Medical Research had constituted a

National Task Force _____ to fight against COVID-19.

A)Ativan

B) Lexapro

C) Hydroxychloroquine

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D) Ibuprofen

Question 9

The capital of Karnataka is

A)Belgaum **B) Bengaluru** C) Mangalore D) Hubli

Question 10

The water body on the east coast of India is-

A)Arabian Sea

B) Bay of Bengal

C) Arctic Ocean

D) Pacific Ocean

Question 11

Panambur port, in Karnataka is also known as New_____.

A)Gujarat Port

B) Mangalore Port

C) Kerala Port

D) Kolkata Port

Question 12

Who was known as the Iron Man of India.

A)Pandit Nehru

B) Mahatama Gandhi

C) Sardar Vallabhbhai Patel

D)Bhagat Singh

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Question 13

The major seaport in Odisha is

A)Kandla port **B) Paradip port** C) JNPT D) Haldia port

Question 14

Who is the father of our Constitution?

A)Mahatama Gandhi

B) Dr. B R Ambedkar

C) Rabindranath Tagore

D) Subhash Chandra Bose

Question 15

Who discovered Radium?

A)Isaac Newton

B) Albert Einstein

C) Benjamin Franklin

D) Marie Curie

Question 16

Finance Minister

A)Nitin Gadkari

B) Amit Shah

C) Nirmala Sitharaman

D) Mansukh Mandaviya

Question 17

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A body of water or air moving in a definite direction,

especially through a surrounding body of water or air

A) **current** B) currant C) bay D) river

Question 18

Visakhapatnam port is situated in

A)Odisha **B) Andhra Pradesh** C) Tamil Nadu D) Kerala

Question 19

Dirham is the official currency of

A)Iran **B) UAE** C) UK D) USA

Question 20

The award presented to the outstanding sports coach is

A)Arjuna Award

B) Filmfare Award

C) Vir Chakra

D) **Dronacharya Award**

MATHS (15 questions = 30 marks)

Question 1

Area of a right angled triangle is 30 cm sq. If its smallest side is 5cm, then its hypotenuse is

A)14cm **B) 13cm** C) 12cm D) 11cm

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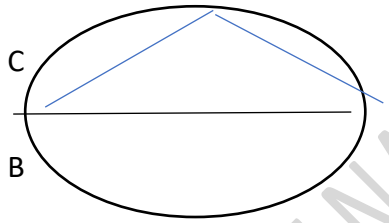
Question 2

The interest on Rs 5000 at the rate of 15% per annum for one month is

A) 750 B) 75 C) 625 **D) 62.5**

Question 3

In fig. if AOB is a diameter of the circle and AC=BC, then ∟CAB is equal to



A)30° B) 60° C) 90° **D) 45°**

Question 4

The angles between North and East and North and West are

A)complementary angles

B) supplementary angles

C) both acute angles

D) both obtuse angles

Question 5

Any point on y- axis is of the form

A)(x,0) B) (x,y) **C) (0,y)** D) (y,y)

Question 6

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Two parallelograms are on equal bases and between the same parallels. The ratio of their areas is

A)1:2 **B) 1:1** C) 2:1 D) 3:1

Question 7

The coefficient of xy in 3x2zy + 7xyz – 2z2x

A)3z B) -2 C) 7yz **D) 7z**

Question 8

In triangle PQR if angle R **>** angle Q, then

A)QR **>** PR **B) PQ > PR** C) PQ <PR D) QR< PR

Question 9

In a cylinder if radius is halved and height is doubled , the volume will be

A)same B) doubled **C) halved** D) four times

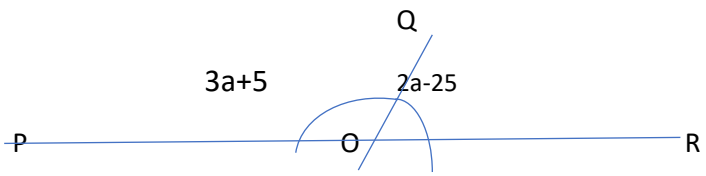
Question 10

The value of $\frac{-4}{3}$ - $\frac{-1}{3}$ is

A)-2 B) -3 C) 2 **D) -1**

Question 11

In fig, POR is a line. The value of a is



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A) **40°** B) 45° C) 55° D) 60°

Question 12

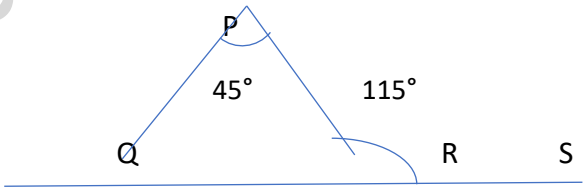
Axioms are assumed

A) universal truths in all branches of mathematics

B) universal truths specific to geometry
C) theorems
D) definitions

Question 13

In fig, side QR of a triangle PQR has been produced to the point S. If angle PRS = 115° and angle P= 45°, then angle Q is equal to



A)70° B) 105° C) 51° D) 80°

Question 14

The number of teeth and the age of a person vary

A)directly with each other

B) inversely with each other
C) neither directly nor inversely with each other
D) sometimes directly and sometimes inversely with each other

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IMPORTANT GK QUESTIONS

1	Name the tallest building in the world. **Burj Khalifa, UAE**
2	Which is the highest mountain peak in the world? **Mount Everest**
3	Which is the tallest animal in the world? **Giraffe**
4	Name the longest river in the world. **Nile**
5	Which is the biggest and deepest ocean in the world? **Pacific Ocean**
6	Which is the smallest bird in the world? **Humming Bird**
7	Which animal is the fastest runner in the world? **Cheetah**
8	Which is the smallest country in the world? **Vatican City**
9	Which is the world's biggest sea animal? **Blue Whale**
10	Which is the fastest bird in the world? **Peregrine Falcon**
11	Which is the smallest continent in the world? **Australia**
12	Which is the world's longest railway platform? **Gorakhpur (Uttar Pradesh, India)**
13	Which is the highest waterfall in the world? **Angel Falls (Venezuela, South America)**
15	Which is India's first National Park established in 1936 **Jim Corbett National Park**
16	Bandhavgarh National Park is located in which state? **Madhya Pradesh**
17	Where is Ranthambore National Park ? **Rajasthan**
18	Kaziranga National Park is in which state? **Assam**
19	Where is Bandipur National park ? **Karnataka**
20	The endangered Asiatic lions can be found in which National Park? **Gir National Park**

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21	Jim Corbett National Park was earlier known as? **Hailey National Park**
22	The National Park that was the first tiger reserve in India is: **Jim Corbett National Park**
23	The national park which is famous as the home of "Big Five" is **Kaziranga National Park**
24	Satpura National Park is located in_____. **Madhya Pradesh**
25	Pin Valley National Park, best known for the Siberian ibex, is located in **Himachal Pradesh**
26	Located in Ladakh,_____is globally famous for its Snow Leopards. **Hemis National Park**
27	We can find the Nanda Devi National Park in_____. **Uttarakhand**
28	Mahatma Gandhi Marine National Park is situated in_____. **Andaman and Nicobar Islands**
29	Similipal National Park is located in which state? **Odisha**
30	Where is Sundarban National Park located? **West Bengal**
31	_____is the largest National Park of India. **Hemis National Park, J & K**
32	The smallest National Park of India is_____. **South Button National Park, (Andaman And Nicobar Island), 2 square miles**
33	Being home to 19% of India's tiger population,_____is called the Tiger State of India. **Madhya Pradesh**
34	The Great Himalayan National Park is located in which state of India? **Himachal Pradesh**
35	_____is the state with the highest number of National Parks in India. **Madhya Pradesh**
36	The Blackbuck National Park is located in which state of India? **Gujarat**
37	Khangchendzonga National Park is located in which state of India? **Sikkim**
39	Which is the hottest planet in our solar system? **Venus**
40	Which planet of our Solar System does the largest moon Ganymede belong to? **Jupiter**
41	Which is the smallest planet in our Solar System? **Mercury**

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42	Which is the largest moon of Saturn and the second-largest moon in our Solar System? **Titan**
43	Which planet in our Solar System takes less than 10 hours to complete its rotation? **Jupiter**
44	Which planet has the most prominent ring system around it in our solar system ? **Saturn**
45	Which planet is known as the blue planet? **Earth**
46	Which planet is known as the red planet ? **Mars**
47	Which is the brightest planet in our solar system? **Venus**
48	Which is the biggest planet in our solar system ? **Jupiter**
49	Which is the closest star to Earth? **Sun**
50	The gas in most abundance on the Earth is **Nitrogen**
51	_____is composed of molten rocks that comes out during a volcanic eruption. **Lava**
52	Earth completes one rotation in approximately_____hours. **24**
53	_____of Earth results in day and night. **Rotation**
54	_____of Earth results in the change of seasons. **Revolution**
55	The Rotation of Earth happens when **The Earth spins on its own axis**
56	In the_____season, the Earth is closest to the Sun. **Summer**
57	Alteration in the length of the shadows is caused due to **Earth's Rotation**
58	Our Solar System is a part of which Galaxy? **Milky Way**
59	In our solar system the asteroid belt is found between_____and _____ **Mars and Jupiter**
60	Solar System is in_____galaxy **Milky Way**
61	How many planets does our solar system consist of? **Eight. (Mercury, Venus, Earth, Mars, Jupiter, Saturn, Uranus, and Neptune)**

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62	Which is the closest planet to earth? **Venus.**
63	Which is the farthest planet from earth? **Neptune**.
64	How many moons does the earth have? **One**
65	Which planet has the most moons? **Saturn**
66	How many moons does Saturn have? **82**
67	Which is the largest moon in our solar system? **Ganymede**
68	How many layers are there in the earth's atmosphere? **Five.**
69	What is the cause of the Sun's energy? **Fusion of Hydrogen and Helium.**
70	Which is the nearest planet to the Sun? **Mercury.**
71	How much time does the sunlight take to reach the Earth? **8 minutes 20 seconds.**
72	How much time does the moonlight take to reach the Earth? **1.26 seconds.**
73	14. What is the largest moon of Saturn? **Titan**
75	Who was the first President of India? Dr. Rajendra Prasad
76	Who was the first Prime Minister of India? Pandit Jawahar Lal Nehru
77	Who was the first Indian to travel to space? **Rakesh Sharma**
78	Who was the first Indian pilot? **J.R.D. Tata**
79	Who was the first Home Minister of India? **Sardar Vallabh Bhai Patel**
80	Who was the first Indian to win a Nobel Prize? **Rabindranath Tagore**
81	Who was the first Indian woman to win Miss World contest? **Reita Faria**
82	Who was the first Chief Justice of India of the apex court, Supreme Court? **Justice Harilal Jekisundas Kania**

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83	Who was the first woman to receive the Bharat Ratna? **Indira Gandhi**
84	Name the Indian who won the Miss Universe crown for the first time. **Sushmita Sen**
85	Who was the first Speaker of Lok Sabha? **Ganesh Vasudev Mavalankar**
86	Who was the first woman Chief Minister in India? **Sucheta Kripalani**
87	Who was the first Railway Minister of India post-independence? **John Mathai**
88	Name the first Indian who won the Grammy Award. **Pandit Ravi Shankar**
89	The First Foreigner To Receive Bharat Ratna? **Khan Abdul Gaffar Khan**
90	The First Vice President Of India? **Dr. Radhakrishnan**
91	The First Home Minister Of India? **Sardar Vallabh Bhai Patel**
92	Who is the first Indian to climb Mount Everest? **Avtar Singh Cheema.**
93	Who is the first Indian woman to climb Mount Everest? **Bachendri Pal.**
94	Who was the first President of India? **Dr.Rajendra Prasad**
95	Who was the first Prime minister of India? **Jawaharlal Nehru**
96	Who was the first woman to become President of India? **Pratibha Patil**
98	Who is the inventor of the Telephone? **Alexander Graham Bell**
99	Who invented the Light Bulb? **Thomas Edison**
100	The Steam engine was invented by_____. **Thomas Savery**
101	Who invented the Aeroplane? **The Wright brothers (Orville and Wilbur Wright)**
102	Who is the inventor of the Computer? **Charles Babbage**
103	_____invented the mobile phone. **Martin Cooper**

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104	Which famous scientist who discovered gravity? **Sir Issac Newton**
105	The famous scientist from Sweden, known as the inventor of dynamite. **Alfred Nobel**
106	William Cullen made the first _____ **Refrigerator**
107	Who invented the radio? **Guglielmo Marconi**
108	In which country did chess originate? **India**
109	Who discovered Penicillin? **Alexander Fleming**
111	Which river is famous as the Sorrow of Bengal? **Damodar River**
112	Which river is also known as Dakshin Ganga? **Godavari**
113	_____is known as the Sorrow of Bihar. **Koshi River**
114	Which river in India crosses the Tropic of Cancer twice? **Mahi River**
115	Which river has the largest basin in India? **Ganga**
116	Which river in India is called the salt river? **Luni**
117	Which is the longest river in India? **Ganga**
118	What are the two headstreams of Ganga? **Bhagirathi and Alaknanda**
119	Which of the following rivers flows through the rift valley in India? **Narmada**
120	_____is the place where Alaknanda and Bhagirathi join and take the name Ganga. **Devprayag**
121	Which of the following is a habitat for freshwater dolphins? **Ganga**
122	The_____river originates from Amarkantak, Madhya Pradesh. **Narmada**
123	Which two major rivers form the biggest delta system in the world? **Ganga and Brahmaputra**
124	Which is the longest river system in India? **Ganga**

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125	Delhi is situated on the bank of which river? **Yamuna**
126	In Which sea Ganga river falls? **Bay of Bengal.**
127	Mohenjo-Daro Site of Indus Valley Civilization was associated with which river? **Indus**
128	What is the origin of the Tapi river? **Satpura Ranges**
130	The national currency of India is_____. **Rupee**
131	The national currency of the United States of America (USA) is_____. **Dollar**
132	_____is the national currency of Russia. **Ruble**
133	What is the national currency of the United Kingdom? **Pound Sterling**
134	The national currency of Australia is_____. **Dollar**
135	What is the national currency of South Africa? **Rand**
136	_____is the national currency of New Zealand. **Dollar**
137	_____is the national currency of Afghanistan. **Afghani**
138	The national currency of Bangladesh is_____. **Taka**
139	What is the national currency of Nepal? **Rupee**
140	What is the national currency of China? **Yuan**
141	The national currency of Pakistan is_____. **Rupee**
142	What is the national currency of Saudi Arabia? **Riyal**
143	_____is the national currency of Bhutan. **Ngultrum**
144	The national currency of Brazil is_____. **Real**
145	_____is the national currency of Indonesia. **Rupiah**

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146	The national currency of Japan is_____. **Yen**
147	The national currency of Malaysia is_____. **Ringgit**
148	_____is the national currency of Maldives. **Rufiyaa**
149	_____is the national currency of Myanmar. **Kyat**
150	The national currency of Belgium is_____. **Euro**
151	_____is the national currency of France. **Franc**
152	What is the national currency of Germany? **Euro**
153	The national currency of Sri Lanka is_____. **Rupee**
154	What is the national currency of South Korea? **Won**
155	What is the currency of Iraq? **Iraqi Dinar**
156	which is the highest crowned currency in the world? **Kuwaiti Dinnar**
158	Who won the first T20 World Cup? **India**
159	Which country has won the Cricket World Cup three times in a row? **Australia**
160	How many times has India won the ICC World Cup? **2**
161	Who is known as the Flying Sikh of India? **Milkha Singh**
162	How many times has India won the Hockey World Cup (Men's)? **1**
163	Name this only woman boxer to have won a medal in each one of the seven world championships :- **Mary Kom**
164	Name this woman who is the first Indian to win a medal in Badminton at the Olympics **Saina Nehwal**
165	Identify this famous personality who is known as the God of Cricket _____ **Sachin Tendulkar**
166	At the Beijing Olympic Games in 2008,_____became the first Indian to win an individual gold medal at the Olympic Games. :-**Abhinav Bindra**

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167	This sports personality has won medals in various badminton tournaments including a gold at the 2019 World Championships :- **P.V.Sindhu**
168	Sania Mirza is famous for playing which sport? **Tennis**
169	PV Sindhu is associated with which sport? **Badminton**
170	What is the capital of Argentina from where the famous football player Lionel Messi belongs?
172	_____is the capital of Russia. **Moscow**
173	What is the capital of Canada? **Ottawa**
174	Canberra is the capital city of which island continent? **Australia**
175	What is the capital city of Italy which is also referred to as the "Eternal City" of the world? **Rome**
176	Thimphu is the capital of which South Asian country? **Bhutan**
177	Wellington is the capital of which country? **New Zealand**
178	Lima is the capital of_____. **Peru**
179	What is the capital of Iraq? **Baghdad**
180	_____is the capital of Malaysia. **Kuala Lumpur**
181	Athens is the capital of which southeastern European country? **Greece**
182	Vienna is the capital city of_____. **Austria**
183	Nairobi is the capital city of which country? **Kenya**
184	_____is the capital city of India. **New Delhi**
185	Amsterdam is the capital city of which country? **Netherlands**
186	What is the capital city of Oman? **Muscat**
187	_____is the capital city of South Korea. **Seoul**

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188	_____is the capital city of the United States of America (USA). **Washington, D.C.**
189	Abu Dhabi is the capital city of which country? **United Arab Emirates (UAE)**
190	What is the capital city of Indonesia? **Jakarta**
191	Dublin is the capital city of_____. **Ireland**
192	What is the capital of Spain? **Madrid**
193	_____is the capital city of Bangladesh. **Dhaka**
194	_____is the capital city of France. **Paris**
195	What is the capital city of China? **Beijing**
196	Cairo is the capital city of which country? **Egypt**
197	_____is the capital city of Mauritius. **Port Louis**
198	What is the capital city of the Philippines? **Manila**
199	What is the capital of Uruguay? **Montevideo**
200	What is the capital of Thailand? **Bangkok**
201	What is the capital city of Germany? **Berlin**
203	How many Union Territories are in India? **8**
204	Leh is the capital of _____ **Ladakh**
205	Hirakud dam is the longest dam in India. It is located in _____ **Orissa**
206	_____is known as the Iron man of India. **Sardar Patel**
207	Ellora Caves is a UNESCO world heritage site located in _____ **Maharashtra**
208	Number of lions in Sarnath pillar is _____ **4**

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209	Anamudi is a mountain located in the Indian state of _____ **Kerala**
210	Kathak is a classical dance form which originated in the Indian state of_____. **Uttar Pradesh**
211	Victoria Memorial is located in _____ **Kolkata**
212	Largest Port in India is located in _____ **Mumbai**
213	Gir National Park is situated in _____ **Gujarat**
214	_____is the second largest state in India by area **Madhya Pradesh**
215	Quit India Movement started in _____ **1942**
216	Who was the last Governor-General of India, when India became a Republic in 1C **Rajagopalachari**
217	Which is the national fruit of India? **Mango**
218	Which is the national game of India? **Hockey**
219	How many spokes are there in the wheel of an Indian Flag? **24**
220	What is the length to width ratio of the Indian Flag? **3:2**
221	Who is India's first citizen? **President of India.**
222	Who is India's second citizen? **Vice-President of India.**
223	Who is India's third citizen? **Prime Minister of India.**
224	Which is the National Tree of India? **Banyan Tree**
225	Which is the holy river of India? **Ganga**
226	India’s largest freshwater lake is? **Wular Lake, J & K**
227	Which state has the highest literacy rate in India? **Kerala**
228	How much is Indian Standard Time (IST) is ahead of Greenwich mean time (GMT)? **5 Hr 30 Min**

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229	In which Indian state the Sun temple is located? **Odisha**
230	Which is the national bird of India? **Peacock**
231	Name the state which is largest in India according to the area. **Rajasthan.**
232	Which city of India is called the City of lakes? **Udaipur.**
233	Where is Ranthambore National Park situated? **Rajasthan**
234	What is the name of India's longest Highway having a length of 4,112 km? **NH44**
235	Which public sector bank is the largest in India? **State Bank of India**
236	Where does the Bhil tribe find? **Rajasthan**
237	The oldest stock exchange of Asia is situated in India? Which is it? **Bombay Stock Exchange**
238	Where is the Reserve Bank of India located? **Mumbai**
240	Christopher Columbus discovered the Americas. He was from _____ **Italy**
241	Dalai Lama is the spiritual and political leader of _____ **Tibet**
242	_____was known as Vishwa Kavi **Rabindranath Tagore**
243	_____is famously known as Flying Sikh **Milkha Singh**
244	Who was the first Indian woman in Space? **Kalpana Chawla**
245	Who was the first Indian in space? **Rakesh Sharma**
246	Who built the Jama Masjid? **Shah Jahan**
247	Who wrote the Indian National Anthem? **Rabindranath Tagore**
248	Who was the first Indian Scientist to win a Nobel Prize? **CV Raman**
249	Who is the first Indian to win a Nobel Prize? **Rabindranath Tagore**

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250	Who was the first Indian woman to win the Miss World Title? **Reita Faria**
251	Who was the first President of India? **Dr. Rajendra Prasad**
252	Who was the first Indian to win the Booker Prize? **Arundhati Roy**
253	Who was the second Prime Minister of India? **Lal Bahadur Shastri**
254	Who is known as the Father of India? **Mohandas Karamchand Gandhi**
255	Who is the chief architect of the Indian Constitution? **Bhimrao Ambedkar**
256	The first woman Prime Minister of India,_____is also called the Mother of India. **Indira Gandh**i
257	Who designed the national flag of India? **Pingali Venkaya**
258	Who is the author of the famous book My country My Life? **L.K Advani**
259	Who built the Taj Mahal? **Shah Jahan**
260	Who is known as Human-Computer? **Shakuntala Devi**
261	Who is the founder of Facebook? **Mark Zuckerberg**
262	Who is known as the Father of Geometry? **Euclid**
263	Who is known as the Nightingale of India? **Sarojini Naidu**
264	Who is popularly known as Frontier Gandhi? **Khan Abdul Gaffar Khan.**
265	Lala Lajpat Rai is popularly known as: **Punjab Kesari.**
266	Who was the first president of the USA. **George Washington**
268	Which is the largest country in the world? **Russia**
269	Which country has the largest population in the world? **China**
270	What are the names of the five oceans? **Pacific, Atlantic, Indian, Arctic and Southern**

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271	How many states of India share its border with Bhutan? **4 (Arunachal Pradesh, Assam, West Bengal & Sikkim)**
272	What is the capital city of India? **New Delhi**
273	In which country would you find the Leaning Tower of Pisa? **Italy**
274	Which are the 7 continents in the world from largest in area to smallest? **Asia, Africa, North America, South America, Antarctica, Europe, and Australia**
275	Which is colder: The North Pole or the South Pole? **The South Pole**
276	Which planet is nearest to the Earth? **Venus**
277	What is the name of the biggest ocean on Earth? **The Pacific Ocean**
278	The United States consists of how many states? **50**
279	Which is the biggest desert in the world? **Sahara Desert**
280	Which is the hottest continent on Earth? **Africa**
281	Which is the highest mountain in the world? **Mount Everest**
282	Which is the coldest place on Earth? **Antarctica**
283	What do you call a chain of mountains? Answer: A range
284	What do you call land with water on 3 sides? **Peninsula**
285	What is Belgium's nickname? **Cockpit of Europe**
286	From whom did the USA purchase Alaska from? **Russia**
287	Which countries have pyramids older than egypt? **Sudan**
288	Which country is also known as the Netherlands? **Holland**
289	What are horizontal and vertical imaginary lines around the earth called? **Latitudes and Longitudes**
290	Which is the largest US state in terms of population? **California**

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291	What is the official language of Brazil? **Portuguese**
292	At which height does a hill become a mountain? **600 m**
293	What do you call smaller rivers that flow into another river? **Tributaries (It doesn't flow into the sea)**
294	Which is the smallest country in the world? **Vatican City**
295	Which country is called the Land of Rising Sun? **Japan**
296	Which country faces the threat of drowning due to global warming? **Maldives**
297	Where is K2 mountain located? **Between China and Pakistan border.**
298	Which is the world's biggest port? **Port of Shanghai**
299	What is the official language of Australia? **English language**
300	Which city is also known as 'The Eternal City'? **Rome**
301	In which country do you find the Yellow River, also known as Huang He? **China**
302	Which ocean lies between Europe and America? **Atlantic Ocean**
303	Which country is the second biggest in the world? **Canada**
304	What is the currency used in Japan? **Japanese Yen**
305	Which is the longest river in the world? **Nile**
306	Which is the only country through which both the equator and the Tropic of Capricorn pass?
307	In which continent is the world's longest river, the Nile? **Africa**
308	Which is the tallest building in the world? **Burj Khalifa in Dubai since 2008, 828-metre (2,717 ft)**
309	Mount Everest lies in which mountain range? **The Himalayas**
310	What is the biggest latitude called? **Equator**

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311	Where can we find the longest border? **Between Canada and US**
312	Yamuna is the tributary of? **Ganges river**
313	What is the full form of LoC **Line of control**
314	Where is the Blue Nile river located? **Ethiopi**a
315	Which device measures the speed of wind? **Anemometer**
316	Which is the country with the most number of islands? **Sweden (221,800 islands)**
317	Which is the second most populated country in the world? **India**
318	In which country is the holy cities of Mecca and Medina located? **Saudi Arabia**
319	Which is the country with the most number of borders? **China (14 countries)**
320	Which is the world's longest country from north to south? **Chile (2,647 miles long, spreads through 38 degrees of latitude)**
321	Which are the countries that share open borders? **New Zealand and Australia**
322	In which country does the River Nile meet the sea? **Egypt**
323	Which are the only two countries to have a land border with the US? **Canada and Mexico**
324	What is Earth's approximate water vs. land coverage ratio? **71 to 29**
325	Which day is celebrated as World Earth Day? **April 22**
326	Which rainforest makes up around half of the world's remaining rainforests? **The Amazon Rainforest**
327	What do you call a study of weather? **Meteorology**
328	Which country has the longest coastline in the world? **Canada**
329	Europe and Africa are separated by which sea? **The Mediterranean**
330	Which countries share the longest border in the world? **Canada – United states**

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331	When did Bangladesh get independence? **1971**
332	In which city Char Minar is located? **Hyderabad**
333	How many states are there in India? **28**
334	How many Union Territories are there in India? **Eight.**
335	In which city is Hawa Mahal located? **Jaipur**
336	How many oceans are there in the world? **Four (Atlantic, Pacific, Indian, and the Arctic).**
337	Which is the highest mountain peak in the world? **Mount Everest**
338	Which is the highest mountain peak in India? **Kangchenjunga**
339	Which is the biggest continent in the world? **Asia**
340	Which is the smallest continent in the world? **Australia**
341	The largest country in the world is? **Russia**
342	Which is the largest reef in the world? **The Great Barrier Reef**
343	Greenland is part of which European nation? **Denmark**
344	Which is the largest hot desert in the World? **Sahara Desert**
345	Name the largest ocean in the world. **Pacific.**
346	Which Channel divides the Indian Islands Andaman and Nicobar in the Bay of Bengal? **Ten Degree Channel**
347	Which strait lies between the Andaman Sea and the South China Sea? **The Malacca Strait**
349	Who is considered as father of Indian Space Program **Vikram Sarabhai**
350	Longest bone in human body is _____ **Femur**
351	_____is the most common element found in our universe. **Hydrogen**

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352	_____is the second most abundant element found on earth's crust. **Silicon**
353	_____is used in balloons as it is lighter than air **Helium**
354	Speed of light in vacuum is _____ **3,00,000 km/sec**
355	Which is the shortest bone in the human body? **Stapes(in ear)**
356	How many bones are there in the adult human body? **206**
357	How many teeth are there in adult humans? **32**
358	Which is the hardest part of the human body? **Enamel**.
359	Which is the biggest part of the brain? **Cerebrum**
360	What is the metal filament in a light bulb made of? **Tungsten**
361	Which is the human body's largest organ? **Skin**
362	What is used in making pencils? **Graphite**
363	An electric bulb is filled with which gas? **Nitrogen**
365	In which year did Gandhiji start the Quit India Movement? **1942**
366	Name the first metal used by Harappan Civilization. **Copper.**
367	Samudragupta is popularly known as:- **Napoleon of India.**
368	Which Mughal emperor is called Zinda Pir? **Aurangzeb.**
369	Which Mughal ruler is associated with Bibi ka Maqbara? **Aurangzeb**
370	When did World War I happen? **28 July 1914 to 11 November 1918**
371	Who was the founder of the Khilji Dynasty? **Jalaluddin Khalji**
372	**BORDERS**

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373	Niagara Falls is an international border between which two countries? **Canada and the USA**
374	Wagah Border is ? **Punjab (India and Pakistan)**
375	The 17th Parallel **It is the Line Between North & South Vietnam**
376	The 38th parallel **It is the Line Between North & South Korea (Before the Korean war)**
377	The 49th Parallel (also The Medicine Line) **It is the Boundary Between Canada & the USA**
378	The Siegfried Line **It is the Boundary Between France and Germany**
379	The Maginot Line **It is France's Defensive Line**
380	The Oder–Neisse line **It is the Boundary Between Germany & Poland**
381	The Radcliffe Line **It is the Boundary Between India & Pakistan**
382	The McMahon Line **It is the Boundary Between India & China**
383	The Line of Control (LoC) **It is the military control line between India and Pakistan**
384	The Line of Actual Control (LAC) **It is the Effective Boundary between India and China**
385	**MISCELLANEOUS**
386	Which country has no words but has only music in their National Anthem? **Spain**
387	Who is the author of the famous Harry Potter book series? **J K Rowling**
388	Which is the biggest animal on earth? **Blue Whale**
389	Name the folk dance of Assam? **Bihu**
390	Which city is called Pearl City? **Hyderabad.**
391	Victoria Memorial is located in which city? **Kolkata.**
392	Name one famous crop of Assam? **Tea.**

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393	Samsung is the company of which country? **South Korea.**
394	The scientific study of an earthquake is called: **Seismology.**
395	What is the rank of India in the world according to the area? **7th.**
396	When is National Youth Day celebrated? **12th Jan.**
397	How many members were there in the Constituent Assembly in the starting? **389.**
398	Patliputra was the ancient name of_____city. **Patna.**
399	ROM stands for: **Read-Only Memory.**
400	How many scheduled languages are there as per the Constitution of India? **22**
401	Who has the authority to issue currency in India? **Reserve Bank of India (RBI)**
402	What does 4G mean? **Fourth Generation**
403	Where is the World Health Organisation situated? **Geneva, Switzerland**
404	How many MBs are there in 1 GB? **1024 MBs**
405	What does COVID-19 mean? **Coronavirus Disease 2019**
406	What does "DC" mean in Washington DC? **District of Columbia**
407	Full form of ISRO. **Indian Space Research Organisation**
408	Which is the national sport of Japan? **Sumo Wrestling**

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PORTS


WHAT ARE THE MAJOR PORTS IN INDIA?
N
TAJIKISTAN
AFGHANISTAN
JAMMU & KASHMIR
HIMACHAL PRADESH
CHINA (Tibet)
PUNJAB
CHANDIGARH
UTTARAKHAND
PAKISTAN
HARYANA
DELHI
NEPAL
SIKKIM
BHUTAN
ARUNACHAL PRADESH
ASSAM
NAGALAND
RAJASTHAN
UTTAR PRADESH
BIHAR
MEGHALAYA
MANIPUR
BANGLADESH
JHARKHAND
TRIPURA
MIZORAM
Kandla
GUJARAT
MADHYA PRADESH
CHHATTISGARH
WEST BENGAL
MYANMAR (BURMA)
Kolkata
DAMAN & DIU
DADRA & NAGAR HAVELI
MAHARASHTRA
ODISHA
Paradip
Bay of Bengal
Mumbai
JNPT
TELANGANA
Visakhapatnam
Arabian Sea
Mormugao Port Trust
KARNATAKA
ANDHRA PRADESH
Ennore (Kamarajar Port)
Mangaluru
Chennai
PUDUCHERRY
LAKSHADWEEP (INDIA)
KERALA
TAMIL NADU
ANDAMAN & NICOBAR ISLANDS (INDIA)
Kochi
Tuticorin (V.O.Chidambaranar Port)
SRI LANKA
MALDIVES
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INDONESIA

STATES AND CAPITALS



S.NO	STATES AND CAPITALS	
1	Andhra Pradesh	Amaravati
2	Arunachal Pradesh	Itanagar
3	Assam	Dispur
4	Bihar	Patna
5	Chhattisgarh	Raipur
6	Goa	Panaji
7	Gujarat	Gandhinagar
8	Haryana	Chandigarh
9	Himachal Pradesh	Shimla
10	Jharkhand	Ranchi
11	Karnataka	Bengaluru
12	Kerala	Thiruvananthapuram
13	Madhya Pradesh	Bhopal
14	Maharashtra	Mumbai
15	Manipur	Imphal
16	Meghalaya	Shillong
17	Mizoram	Aizawl
18	Nagaland	Kohima
19	Odisha	Bhubaneswar

YT channel- MERCHANT NAVY CAREER

20	**Punjab**	**Chandigarh**
21	**Rajasthan**	**Jaipur**
22	**Sikkim**	**Gangtok**
23	**Tamil Nadu**	**Chennai**
24	**Telangana**	**Hyderabad**
25	**Tripura**	**Agartala**
26	**Uttar Pradesh**	**Lucknow**
27	**Uttarakhand**	**Dehradun**
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RATIONAL NUMBERS

1. An integer can be:

A. Only Positive B. Only Negative C. Both positive and negative D. None of the above

Answer: **C**

Explanation: An integer can be both positive and negative as well as zero.

i.e. …-3, -2, -1, 0, 1, 2, 3,….

2. A rational number can be represented in the form of:

A. p/q B. pq C. p+q D. p-q

Answer: **A**

Explanation: A rational number can be represented in the form p/q where p and q are integers and q is not equal to zero.

3. The value of ½ x ⅗ is equal to:

A. ½ B. 3/10 C. ⅗ D. ⅖

Answer: **B**

Explanation: ½ x ⅗ = (1 x 3)/(2 x 5) = 3/10

4. The value of (½) ÷ (⅗) is equal to:

A. 3/10 B. ⅗ C. 6/5 D. ⅚

Answer: **D**

Explanation: (½) ÷ (⅗) = (½) x (5/3) = (1 x 5)/(2 x 3) = ⅚

5. The value of ½ + ¼ is equal to:

A. ¾ B. 3/2 C. ⅔ D. 1

Answer: **A**

Explanation: ½ + ¼

Making the denominator equal:

2/4 + ¼ = (2 + 1)/4 = ¾

6. The value of (5/4) – (8/3) is:

A. 17/12 B. -17/12 C. 12/17 D. -12/17

Answer: **B**

Explanation: 5/4 – 8/3

Making the denominator equal:

[(5/4) x (3/3)] – [(8/3) x (4/4)]
= (15/12) – (32/12)

= (15 – 32)/12 = -17/12

7. The associative property is applicable to:

A. Addition and subtraction

B. Multiplication and division

C. Addition and Multiplication

D. Subtraction and Division

Answer: **C**

Explanation: As per associative property:

A + (B + C) = (A + B) + C

A × (B × C) = (A × B) × C

8. The value of (-10/3) x (-15/2) x (17/19) x 0 is:

A. 0 B. 22.66 C. 20 D. 35

Answer: **A**

Explanation: Any number multiplied by zero is equal to zero.

9. The additive identity of rational numbers is:

A. 0 B. 1 C. 2 D. -1

Answer: **A**

Explanation: Any number added to zero is equal to the number itself.

Ex: 5+0 = 5

Therefore, 0 is the additive identity of rational numbers.

10. The multiplicative identity of rational numbers is:

A. 0 B. 1 C. 2 D. -1

Answer: **B**

Explanation: Any number multiplied by 1 is equal to the number itself.

Ex: 5 x 1 = 5

Therefore, 1 is the multiplicative identity of rational numbers.

11. What is the sum of ⅔ and 4/9?

A. 6/3 B. 6/9 C. 10/9 D. 10/3

Answer: **C.10/9**

Solution: ⅔ + 4/9

⇒ ⅔ x (3/3) + 4/9

⇒ 6/9 + 4/9

⇒ 10/9

12. What is the product of 2/9 and ¾?

A. ⅙ B. ⅔ C. 1/9 D. ¼

Answer: **A. ⅙**

Solution: The product of 2/9 and ¾:

⇒ 2/9 x ¾

⇒ (2 x 3)/(9 x 4)

⇒ (2 x 3)/(3 x 3 x 2 x 2)

By cancelling the common terms from numerator and denominator, we get;

⇒ 1/(3×2)

⇒ ⅙

13. What is the reciprocal of 1/9?

A. 9 B. 0 C. 1 D. None of the above

Answer: **A. 9**

1/9 x 9 = 1

14. What is the value of 100 divided by 0?

A. 0 B. 100 C. 1 D. Undefined

Answer: **D. Undefined**

100/0 = undefined

15. Which of the following is commutative for rational numbers?

A. Addition and subtraction

B. Addition and multiplication

C. Multiplication and division

D. Subtraction and division

Answer: **B. Addition and multiplication**

16. Division of rational numbers is associative.

A. True B. False

Answer: **B. False.**

Explanation: Division of rational numbers is not associative.

For example,

2/5 ÷ (1/2 ÷ 1/4) = 0.2

(2/5 ÷ 1/2) ÷ 1/4 = 3.2

Hence, 2/5 ÷ (1/2 ÷ 1/4) ≠ (2/5 ÷ 1/2) ÷ 1/4

17. What is the value of ¾ + ⅚ + 2/7?

A. 10/84 B. 134/84 C. 157/84 D. 167/84

Answer: **C. 157/84**

Solution: Given,

¾+⅚+2/7

LCM of 4, 6 and 7 = 84

Now, making the denominators the same and adding the given rational numbers, we get;

⇒ 63/84 + 70/84 + 24/84

⇒ 157/84

18. Find the additive inverse of 11/7?

A. 7/11 B. -7/11 C. 11/7 D.-11/7

Answer: **D. -11/7**

Explanation: When a number is added to its additive inverse, then the result is zero,

11/7 – 11/7 = 0

19. How many rational numbers are there in between ¾ and 1?

A. 0 B. 1 C. 2 D. Countless

Answer: **D. Countless**

Explanation: We can write ¾ as 30/40 and 1 as 40/40.

Hence the rational numbers between them are:

31/40, 32/40, 33/40, 34/40,35/40,36/40, 37/40, 38/40, 39/40.

Note: There are countless rational numbers between any two rational numbers.

20. What should be subtracted from -⅔ to get -1?

A. ⅓ B. -⅓ C. ⅔ D. -⅔

Answer: **A. ⅓**

Solution: Let x is subtracted from -⅔.

-⅔ – x = -1, -x = -1 + ⅔, -x = -⅓, x = ⅓

LINEAR EQUATIONS IN ONE VARIABLE

1. Which of the following is not a linear equation in one variable?

A. 33z+5 = 0 B. 33(x+y) = 0 C. 33x+5 = 0 D. 33y+5 = 0

Answer: **B**

Explanation: In 33(x+y) = 0, x and y are two variables.

2. The solution of 2x-3=7 is:

A. 5 B. 7 C. 12 D. 11

Answer: **A**

Explanation: 2x-3=7

2x=7+3=10

x=10/2 = 5

3. The solution of 2y + 9 = 4 is:

A. 9/2 B. 4/9 C. -⅖ D. -5/2

Answer: **D**

Explanation: 2y+9 = 4

2y = 4-9 = -5

y=-5/2

4. The solution of y/5 = 10 is:

A. 15

B. 10

C. 50

D. 5

Answer: **C**

Explanation: y/5 = 10

y = 5×10 = 50

5. What should be added to -7/3 to get 3/7?

A. 21/58

B. 58/21

C. 47/21

D. 50/21

Answer: **B**

Explanation: Let the number be x

-7/3+x = 3/7

x=3/7+7/3 = (9+49)/21 = 58/21

6. The perimeter of the rectangle is 20cm. If the length of the rectangle is 6cm, then its breadth will be:

A. 4 cm

B. 6 cm

C. 10 cm

D. 14 cm

Answer: **A**

Explanation: Perimeter of rectangle = 2(Length+Breadth)

20 = 2(6+x)

6+x = 20/2

6+x = 10

x = 10-6

x=4 cm

7. The age of the father is three times the age of the son. If the age of the son is 15 years old, then the age of the father is:

A. 50 years

B. 55 years

C. 40 years

D. 45 years

Answer: **D**

Explanation: Let the age of the father is x

Given: x = 3 × (age of son) = 3 × (15) = 45 years

8. The difference between two whole numbers is 66. The ratio of the two numbers is 2: 5. The two numbers are:

A. 60 and 6

B. 100 and 33

C. 110 and 44

D. 99 and 33

Answer: **C**

Explanation: Let the two numbers be 2x and 5x since they are in the ratio of 2:5.

The difference between 5x and 2x = 66

5x – 2x = 66

3x = 66

x = 22

Hence, 2x = 2(22) = 44 and 5x = 5(22) = 110.

9. Three consecutive integers add up to 51. The integers are:

A. 16,17,18

B. 15,16,17

C. 17,18,19

D. 18,19,20

Answer: **A**

Explanation: Let the three consecutive integers be x, x+1, x+2

x+(x+1)+(x+2) = 51

3x+3 = 51

3x = 51 – 3

x = 48/3 = 16

x+1 = 16+1=17

x+2 = x+2 = 18

10. The solution for 3m = 5m – (8/5) is:

A. 8/5

B. ⅘

C. 5/4

D. 4/3

Answer: **B**

Explanation: 3m = 5m – (8/5)

8/5 = 5m – 3m

2m = 8/5

m = 8/10 = 4/5

11. The degree of $x^2 – 5x + 2 = x^3$ is:

A. 3
B. 2
C. 1
D. 0

Answer: A. 3

Explanation: Degree is the highest power of the variable in an equation. Therefore, in the given equation, the highest degree is 3.

12. The degree of equation $x^2 – 9 = 2x^2$ is:

A. 0
B. 1
C. 2
D. 3

Answer: C. 2
Explanation: Degree is the highest power of the variable in an equation. Therefore, in the given equation, the highest degree is 2.

13. What is the value of x if x + 9 = 12?

A. 2
B. 3
C. 8
D. 6

Answer: B. 3
Explanation: x + 9 = 12
X = 12 – 9
X = 3

14. If a number is divided by 8 it gives 6 as the value. Find the number.

A. 36
B. 42
C. 48
D. 56

Answer: C. 48
Explanation: Let X be the number
X/8 = 6
X = 8 x 6 = 48

15. Solve 2x + 9 = 4.

A. X = 6
B. X = -5/2
C. X = -3/2
D. X = -9/2

Answer: B. X = -5/2

2x + 9 = 4
2x = 4 – 9
2x = -5
x = -5/2

16. Find the value of x if 2x + 10 = 76.

A. 33
B. 7.6
C. 66
D. 32

Answer: A. 33
Explanation: 2x + 10 = 76
2x = 76 – 10
2x = 66

x = 66/2
x = 33

17. The perimeter of a rectangle is 40 cm. If its width is 10 cm, then find the length.

A. 10
B. 20
C. 30
D. 40

Answer: A. 10

Explanation: Perimeter of a rectangle = 40 cm
Width = 10 cm
Let the length be x.
Perimeter of rectangle = 2(length + width)
40 = 2 (x + 10)
40/2 = x + 10
20 = x + 10
x = 20 – 10 = 10
Thus, the length is also 10 cm.
Hence, we can say, that the given rectangle is basically a square, with all its sides equal.

18. If x is an even number, then the next even number is:

A. x+1
B. x+2
C. x+3
D. x+4

Answer: B. x+2
Explanation: If x = 2, then x + 2 = 2 + 2 = 4

19. The difference between the two numbers is 30. If the bigger number is x, then what is the smaller number?

A. x – 30
B. 30 – x
C. 30x
D. None of these

Answer: A. x – 30
Explanation: x – small number = 30
Small number = x – 30

20. When a number is added to itself, it becomes 24. What is the number?

A. 2
B. 4
C. 12
D. 21

Answer: C. 12
Explanation: Let the number be x.
x + x = 24
2x = 24
x = 24/2
x = 12

LINEAR EQUATIONS IN TWO VARIABLES

1) The linear equation 3x-11y=10 has:

a. Unique solution b. Two solutions
c. Infinitely many solutions d.No solutions

Answer: **c**

Explanation: 3x-11y=10 ,y=(3x-10)/11

Now for infinite values of x, y will also have infinite solutions.

2) 3x+10 = 0 will has:

a. Unique solution b. Two solutions c. Infinitely many solutions d.No solutions

Answer: **a**

Explanation: 3x+10 = 0

x = -10/3.

Hence, only one solution is possible.

3) The solution of equation x-2y = 4 is:

a. (0,2) b. (2,0) c. (4,0) d. (1,1)

Answer: **c**

Explanation: Putting x=4 and y = 0, on the L.H.S. of the given equation, we get;

4-2(0) = 4 – 0 = 4

Which is equal to R.H.S.

4) Find the value of k, if x = 1, y = 2 is a solution of the equation 2x + 3y = k.

a. 5 b. 6 c. 7 d. 8

Answer: **d**

Explanation: 2x + 3y = k

k=2(1)+3(2) = 2+6 = 8

5) Point (3, 4) lies on the graph of the equation 3y = kx + 7. The value of k is:

a. 4/3 b. 5/3 c. 3 d. 7/3

Answer: **b**

Explanation: 3y = kx + 7

Here, x = 3 and y = 4

Hence, (3×4) = (kx3) + 7

12 = 3k+7 , 3k = 12–7 , 3k = 5 , k = 5/3

6) The graph of linear equation x+2y = 2, cuts the y-axis at:

a. (2,0) b. (0,2) c. (0,1) d. (1,1)

Answer: **c**

Explanation: x+2y = 2 , y = (2-x)/2

If x=0, then; , y=(2-0)/2 = 2/2 = 1

Hence, x+2y=2 cuts the y-axis at (0,1).

7) Any point on line x = y is of the form:

a. (k, -k) b. (0, k) c. (k, 0) d. (k, k)

Answer: **d**

8) The graph of x = 3 is a line:

a. Parallel to the x-axis at a distance of 3 units from the origin

b. Parallel to the y-axis at a distance of 3 units from the origin

c. Makes an intercept 3 on the x-axis

d. Makes an intercept 3 on the y-axis

Answer: **b**

9) In equation, y = mx+c, m is:

a. Intercept b. Slope c. Solution of the equation d. None of the above

Answer: **b**

10) If x and y are both positive solutions of equation ax+by+c=0, always lie in the:

a. First quadrant b. Second quadrant c. Third quadrant d. Fourth quadrant

Answer: **a**

11) A linear equation in two variables is of the form ax + by + c = 0, where

(a) a = 0, c = 0 (b) a ≠ 0, b = 0 (c) a = 0, b ≠ 0 (d) a ≠ 0, b ≠ 0

Answer: **d**

Explanation: A linear equation in two variables is of the form ax + by + c = 0, where a ≠ 0, b ≠ 0.

If the values of “a” and “b” are equal to 0, the equation becomes c =0.

Hence, the values of a and b should not be equal to 0.

12) Any point on the x-axis is of the form

(a) (x, y) (b) (0, y) (c) (x, 0) (d) (x, x)

Answer: **c**

Explanation: Any point on the x-axis is of the form (x, 0).

On the x-axis, x can take any values, whereas y should be equal to 0.

13) Any point on the y-axis is of the form

(a) (y, y) (b) (0, y) (c) (x, y) (d) (x, 0)

Answer: **b**

Explanation: Any point on the y-axis is of the form (0, y).

On the y-axis, y can take any values and x should be equal to 0.

14) The linear equation 2x – 5y = 7 has

(a) No solution

(b) unique solution

(c) Two solutions

(d) Infinitely many solutions

Answer: **d**

Explanation: The linear equation 2x-5y has infinitely many solutions.

Because, the equation 2x-5y = 7 is a single equation, that involves two variables.

Hence, for different values of x, we will get different values of y and vice-versa.

15) The linear equation 3x – y = x – 1 has

(a) No solution

(b) unique solution

(c) Two solutions

(d) Infinitely many solutions

Answer: **d**

Explanation: The linear equation 3x – y = x – 1 has infinitely many solutions.

On simplification, the given equation becomes 2x-y= -1, which is a single equation with two variables.

Thus, 3x – y = x – 1 has infinitely many solutions.

16) The graph of the linear equation 2x + 3y = 6 cuts the y-axis at the point

(a) (2, 0) (b) (0, 2) (c) (3, 0) (d) (0, 3)

Answer: **b**

Explanation:

Given that the graph of the linear equation 2x + 3y = 6 cuts the y-axis at the point.

Let the point be “P”. Hence, the x -coordinate of point P is 0.

Now, substitute x= 0 in the given equation,

2(0) + 3y = 6

3y = 6

y=2

Hence, the cooridnate point is (0, 2).

17) The equation 2x + 5y = 7 has a unique solution, if x, y are:

(a) Rational numbers (b) Real numbers

(c) Natural numbers (d) Positive real numbers

Answer: **c**

Explanation: The equation 2x + 5y = 7 has a unique solution, if x, y are natural numbers.

In natural numbers, there exists only one pair (1, 1) which satisfies the given equation.

But for rational numbers, real numbers, positive real numbers, there exist many solution pairs to satisfy the equation.

18) The point of the form (a, a) always lies on:

(a) On the line x + y = 0

(b) On the line y = x

(c) x-axis

(d) y-axis

Answer: **b**

Explanation: The point of the form (a, a) always lies on the line y = x.

If the point having the same x and y values, it should lie on the same line.

19) If we multiply or divide both sides of a linear equation with the same non-zero number, then the solution of the linear equation:

(a) Remains the same

(b) Changes

(c) Changes in case of multiplication only

(d) Changes in case of division only

Answer: **a**

Explanation: If we multiply or divide both sides of a linear equation with the same non-zero number, then the solution of the linear equation remains the same.

20) If (2, 0) is a solution of the linear equation 2x + 3y = k, then the value of k is

(a) 2 (b) 4 (c) 5 (d) 6

Answer: **b**

Explanation:

Substitute x=2 and y=0 in the given equation, we get

2(2) + 3(0) = k

k = 4+0

k = 4.

Hence, the value of k is 4.

PAIR OF LINEAR EQUATIONS IN TWO VARIABLES

1. The pairs of equations x+2y-5 = 0 and -4x-8y+20=0 have:

(a) Unique solution

(b) Exactly two solutions

(c) Infinitely many solutions

(d) No solution

Answer: **(c) Infinitely many solutions**

Explanation:

$a_1/a_2 = 1/-4$

$b_1/b_2 = 2/-8 = 1/-4$

$c_1/c_2 = -5/20 = -¼$

This shows:

$a_1/a_2 = b_1/b_2 = c_1/c_2$

Therefore, the pair of equations has infinitely many solutions.

2. The pairs of equations 9x + 3y + 12 = 0 and 18x + 6y + 26 = 0 have

(a) Unique solution

(b) Exactly two solutions

(c) Infinitely many solutions

(d) No solution

Answer: **(d) No solution**

Explanation: Given, 9x + 3y + 12 = 0 and 18x + 6y + 26 = 0

$a_1/a_2 = 9/18 = 1/2$

$b_1/b_2 = 3/6 = 1/2$

$c_1/c_2 = 12/26 = 6/13$

Since, $a_1/a_2 = b_1/b_2 \neq c_1/c_2$

So, the pairs of equations are parallel and the lines never intersect each other at any point, therefore there is no possible solution.

3. If the lines 3x+2ky – 2 = 0 and 2x+5y+1 = 0 are parallel, then what is the value of k?

(a) 4/15 (b) 15/4 (c) ⅘ (d) 5/4

Answer: **(b) 15/4**

Explanation: The condition for parallel lines is:

$a_1/a_2 = b_1/b_2 \neq c_1/c_2$

Hence, 3/2 = 2k/5 , k=15/4

4.The solution of the equations x-y=2 and x+y=4 is:

(a) 3 and 1 (b) 4 and 3 (c) 5 and 1 (d) -1 and -3

Answer: **(a) 3 and 1**

Explanation: x-y =2 , x=2+y

Substituting the value of x in the second equation we get;

2+y+y=4 , 2+2y=4, ,2y = 2 , y=1

Now putting the value of y, we get; , x=2+1 = 3

Hence, the solutions are x=3 and y=1.

5. Ritu can row downstream 20 km in 2 hours, and upstream 4 km in 2 hours. Her speed of rowing in still water and the speed of the current is:

(a) 6km/hr and 3km/hr (b) 7km/hr and 4km/hr (c) 6km/hr and 4km/hr (d) 10km/hr and 6km/hr

Answer: **(c) 6km/hr and 4km/hr**

Explanation: Downstream = x + y km/h

Upstream = x – y km/h

As per the question given, 2(x+y) = 20

And, 2(x-y) = 4

so x = 6 and y = 4

Therefore, Speed of Ritu is still water = 6 km/hr
Speed of Stream = 4 km/hr

5. The father's age is six times his son's age. Four years hence, the age of the father will be four times his son's age. The present ages, in years, of the son and the father are, respectively

(a) 4 and 24 (b) 5 and 30

(c) 6 and 36 (d) 3 and 24

Answer: **(c) 6 and 36**

Explanation: (x + 4) = 4(y + 4)….(i)

Also, x = 6y….(ii)

From (i) and (ii), y = 6, x=36.

UNDERSTANDING QUADRILATERALS

1. Which of the following is not a quadrilateral?

A. Square B. Rectangle C. Triangle D. Parallelogram

Answer: **C**

Explanation: A quadrilateral is a four-sided polygon but triangle is a three-sided polygon.

2. Which of the following quadrilaterals has two pairs of adjacent sides equal and its diagonals intersect at 90 degrees?

A. Square B. Kite C. Rhombus D. Rectangle

Answer: **B**

3. Which one of the following is a regular quadrilateral?

A. Square B. Trapezium C. Kite D. Rectangle

Answer: **A**

Explanation: A square has all its sides equal and angles equal to 90 degrees.

4. If AB and CD are two parallel sides of a parallelogram, then:

A. AB>CD B. AB<CD C. AB=CD

D. None of the above

Answer: **C**

5. The perimeter of a parallelogram whose parallel sides have lengths equal to 12 cm and 7 cm is:

A. 21 cm B. 42 cm C. 19 cm D. 38 cm

Answer: **D**

Explanation: Perimeter of parallelogram = 2 (Sum of Parallel sides)

P = 2 (12 + 7)

P = 2 (19)

P = 38 cm

6. If ∠A and ∠C are two opposite angles of a parallelogram, then:

A. ∠A > ∠C B. ∠A = ∠C C. ∠A < ∠C D. None of the above

Answer: **B**

Explanation: Opposite angles of a parallelogram are always equal.

7. If ∠A and ∠B are two adjacent angles of a parallelogram. If ∠A = 70°, then ∠B = ?

A. 70° B. 90° C. 110° D. 180°

Answer: **C**

Explanation: The adjacent angles of parallelogram are supplementary.

∠A + ∠B = 180°

70° + ∠B = 180°

∠B = 180 – 70° = 110°

8. ABCD is a rectangle and AC & BD are its diagonals. If AC = 10 cm, then BD is:

A. 10 cm B. 5 cmC. 15 cm D. 20 cm

Answer: **A**

Explanation: The diagonals of a rectangle are always equal.

9. Each of the angles of a square is:

A. Acute angle B. Right angle C. Obtuse angle D. 180 degrees

Answer: **B**

Explanation: All the angles of square is at right angle.

10. The quadrilateral whose diagonals are perpendicular to each other is:

A. Parallelogram B. Rectangle C. Trapezium D. Rhombus

Answer: **D**

11. Which of the following is not a regular polygon?

A. Square B. Equilateral triangle C. Rectangle D. Regular hexagon

Answer: C. Rectangle
Explanation: A regular polygon is both equiangular and equilateral.
But all four sides of a rectangle are not equal, thus it is not a regular polygon.

12. If the two angles of a triangle are 80° and 50°, respectively. Find the measure of the third angle.
A. 50° B. 60° C. 70° D. 80°
Answer: A. 50°

Explanation: By the angle sum property of triangle, we know that;
Sum of all the angles of a triangle = 180°
Let the unknown angle be x
80° + 50° + x = 180°
x = 180° – 130°
x = 50°

13. In a parallelogram ABCD, angle A and angle B are in the ratio 1:2. Find the angle A.
A. 30° B. 45° C. 60° D. 90°

Answer: C.60°

Explanation: As we know, the sum of adjacent angles of a parallelogram is equal to 180° and opposite angles are equal to each other.
Thus, in parallelogram ABCD angle A and angle B are adjacent to each other
Let angle A = x and angle B = 2x.
So, x + 2x = 180°
3x = 180°
x = 60°

14. The angles of a quadrilateral are in ratio 1:2:3:4. Which angle has the largest measure?
A. 120° B. 144° C. 98°
D. 36°

Answer: B.144°

Explanation: Suppose, ABCD is a quadrilateral.
Let angle A is x
Then,
x + 2x + 3x + 4x = 360° [Angle sum property of quadrilateral] 10x = 360°
x = 36°
Hence, the greatest angle is 4x = 4 x 36 = 144°

15. The length and breadth of a rectangle is 4 cm and 2 cm respectively. Find the perimeter of the rectangle.
A. 12 cm B. 6 cm C. 8 cm
D. 16 cm

Answer: A. 12 cm
Explanation: Given, length of rectangle is 4 cm
Breadth of rectangle = 2cm
By the formula of perimeter of rectangle, we know that;
Perimeter = 2 (Length + Breadth)
P = 2(4+2)
P = 2 x 6
P = 12 cm

16. The diagonals of a rectangle are 2x + 1 and 3x – 1, respectively. Find the value of x.
A. 1 B. 2 C. 3 D. 4

Answer: B.2

Explanation: The diagonals of a rectangle are equal in length.
2x + 1 = 3x -1
1 + 1 = 3x – 2x
2 = x
Thus, the value of x is 2.

17. The diagonals of a kite:
A. Bisects each other B. Are perpendicular to each other C. Does not bisect each other D. None of the above

Answer: B. Are perpendicular to each other

Explanation: The diagonals of a kite are perpendicular to each other.

They intersect at 90 degrees but does not bisect.

18. A rhombus has a side length equal to 5 cm. Find its perimeter.
A. 25 B. 10 C. 20 D. 30

Answer: C. 20

Explanation: A rhombus is a parallelogram that has all its four sides equal. Thus, the perimeter of rhombus,
P = 4 x side-length
P = 4 x 5
P = 20 cm

19. ABCD is a parallelogram. If angle A is equal to 45°, then find the measure of its adjacent angle.
A. 135° B. 120° C. 115° D. 180°

Answer: A.135°

Explanation: The adjacent angles of a parallelogram sums up to 180°.
Thus,
45° + x = 180°
x = 180° – 45°
x = 135°

20. The kite has exactly two distinct consecutive pairs of sides of equal length.
A. True B. False

Answer: A. True

Explanation: A kite is a quadrilateral that has exactly two distinct consecutive pairs of sides of equal length.

DATA HANDLING

1. Pictorial representation of data using symbols is known as:

A. Bar graph B. Pictograph C. Pie chart D. None of these

Answer: **B**

2. Double bar graphs display _____ sets of data simultaneously.

A. Four B. Three C. Two D. No

Answer: **C**

3. _____ gives the number of times a particular entry occurs.

A. Organisation of data B. Collection of Data

C. Representation of Data D. Frequency distribution table

Answer: **D**

4. In the grouped data, each of the group is called:

A. Class interval B. Collection of data

C. Frequency distribution D. Grouped frequency

Answer: **A**

5. If 20-30 is the class interval of grouped data, then the lower class limit is:

A. 50 B. 30 C. 20 D. 10

Answer: **C**

6. The difference between the upper-class limit and lower class limit of a class interval is called:

A. Frequency B. Width C. Grouped data D. Ungrouped data

Answer: **B**

7. The width or size of the class interval 30-40 is:

A. 10 B. 30 C. 40 D. 70

Answer: **A**

Explanation: Width of class interval = upper class limit-lower class limit

= 40 – 30 = 10

8. If a coin is flipped in the air, what is the probability of getting a tail?

A. 0 B. ½ C. 1 D. 2

Answer: **B**

Explanation: A coin has two sides, Head and tails.

Probability = Number of possible outcomes/Total number of outcomes

We know that the tail is present on one side of a coin and in total there could be two outcomes H & T.

Hence, the probability of getting a tail = ½

9. A bag has 4 red balls and 4 green balls, what is the probability of getting a red ball randomly?

A. ¼ B. 1/8 C. ½ D. 0

Answer: **C**

Explanation: Total number of balls in bag = 4 + 4 = 8

Total number of red balls = 4

Probability of getting a red ball = 4/8 = ½

10. If a die is thrown in the air, the probability of getting even numbers is:

A. ½ B. ⅙ C. ⅓ D. ⅔

Answer: **A**

Explanation: A die has six sides each consisting numbers from 1 to 6, i.e. {1, 2, 3, 4, 5, 6}.

Total possible outcomes = 6

Number of even numbers on a die = 3

i.e. {2, 4, 6}

Probability = 3/6 = 1/2

Check the given table, and give the correct answers to the following questions.

Range of Marks Obtained	Number of students
250 – 300	10
300 – 350	12
350 – 400	14
400 – 450	13
450 – 500	11

11. What is the total number of students for all ranges?

A. 12 B. 14 C. 50 D. 60

Answer: D. 60

Explanation: Total number of students = 10 + 14 + 12 + 13 + 11 = 60

12. How many students secured marks between 250 – 300?

A. 10 B. 11 C. 13 D. 14

Answer: A

13. What is the upper limit in the given frequency distribution table?

A. 350 B. 450 C. 500 D. 400

Answer: A. 500

Explanation: The last class has the upper limit of 500, which is the upper limit for the whole table.

14. How many students come under the marks range of 300 – 400?

A. 12 B. 26 C. 14 D. 25

Answer: B.24

Explanation: Number of students from 300 – 350 = 12
Number of students from 350 – 400 = 14
Thus, the total number of students from 300 – 400 = 12 + 14 = 26

15. What is the size of the class interval?

A. 30 B. 40 C. 50 D. 55

Answer: C. 50

Explanation: For any class, we can determine the class interval, by subtracting the lower limit from upper limit.
500 – 450 = 50 or 450 – 400 = 50

16. A die is thrown. What is the probability of getting even prime number?

A. ⅙ B. ½ C. ⅓ D. ⅕

Answer: A. 1/6

Explanation: When a die is thrown,
Number of total possible outcomes = 6
Number of even prime numbers = 1 (2 is the only even prime number)
Thus, the probability of getting a even prime number, P = 1/6

17. When a coin is tossed, what are the possible outcomes?

A. 1 B. 2 C. 3 D. 0

Answer: B

Explanation: The possible outcomes are Head and Tail.

18. When two coins are tossed, then what is the probability of getting Tails for both the coins?

A. ½ B. ¼ C. ⅓ D. None of these

Answer: B.¼

Explanation: When two coins are tossed, then the possible outcomes will be:
HH, HT, TH, TT
Thus, probability of getting tails for both the coins = ¼

19. The number of times that a particular entry occurs is called:

A. Frequency B. Class-interval
C. Raw data D. Range

Answer: A. Frequency

20. What is the number of possible outcomes, when two dies are thrown, simultaneously?

A. 6 B. 12 C. 36 D. None of these

Answer: C. 36
Explanation: Number of possible outcomes of each dies = 6
For two dies, number of possible outcomes = 6 x 6 = 36

SQUARE AND SQUARE ROOTS

1. Which of the following is a perfect square?

A. 1057 B. 625 C. 7928 D. 64000

Answer: **B**

Explanation: The natural numbers which end with 0, 2, 3, 7 and 8 are not perfect squares.

Hence, 625 is a perfect square.

25 x 25 = 625

2. Which of the following will have 6 at unit place?

A. 19^2 B. 11^2 C. 24^2 D. 13^2

Answer: **C**

Explanation: 24^2 = 24 x 24 = 576

3. If 5278 is squared, then what will be at unit place?

A. 8 B. 7 C. 6 D. 4

Answer: **D**

Explanation: If we square the unit digit of the number 5278 we get:

8^2 = 64, Hence, at the unit place of 5278^2 the number is 4.

4. What will be the number of zeros in square of 400?

A. 2 B. 3 C. 4 D. 6

Answer: **C**

Explanation: 400^2 = 400 x 400 = 160000

5. How many natural numbers lie between 9^2 and 10^2?

A. 17 B. 18 C. 19 D. 20

Answer: **B**

Explanation: The number of natural numbers between n^2 and $(n+1)^2$ is equal to 2n.

Here, n = 9

Therefore, 2n = 2 x 9 = 18

6. What is the sum of the first four odd natural numbers?

A. 16 B. 17 C. 18 D. 20

Answer: **A**

Explanation: The sum of the first odd natural numbers = 1+3+5+7 = 16

7. The value of $9^2 - 1$ is equal to:

A. 81 B. 80 C. 79 D. None of the above

Answer: **B**

Explanation: 9^2-1 = 81 – 1 = 80

8. The sum of 1 + 3 + 5 + 7 + 9 + 11 + 13 + 15 + 17 +19 is:

A. 121 B. 120 C. 100 D. 110

Answer: **C**

Explanation: Sum of first n odd numbers =n^2 = 10^2 =100

9. The Pythagorean triples whose smallest number is 8:

A. 8, 16 17 B. 8, 17, 18 C. 8, 15, 17 D. 8, 15, 16

Answer: **C**

Explanation: The general form of Pythagorean triplets is 2m, m^2-1, m^2+1

Given, 2m = 8

So, m = 4

m^2 – 1= 4^2-1 = 16-1 = 15

m^2+1=4^2+1= 16+1 = 17

10. Which of the following is a perfect square? 32, 66, 81, 101.

A. 32 B. 66 C. 81 D. 101

Answer: C.81

Explanation: 81 = 9 x 9 = 9^2

11. A perfect square that lies between 40 and 50 is:

A. 42 B. 45 C. 46 D. 49

Answer: D. 49

Explanation: 49 is a perfect square.

49 = 7 x 7 = 7^2

12. Which of the following is not a square number?

A. 4 B. 9 C. 24 D. 100

Answer: C. 24

Explanation: 24 is not a square number.

It cannot be written as n2, where n is any natural number.

13. Find the greatest four-digit number that is a perfect square.

A. 9990 B. 9801 C. 9999 D. None of these

Answer: B. 9801

Explanation: 9801 is a perfect square.

$9801 = 99 \times 99 = 99^2$

14. The square of which of these will have 6 at the unit place?

A. 17^2 B. 16^2 C. 15^2 D. 19^2

Answer: B.16^2

Explanation: The square of 16,

$16^2 = 16 \times 16 = 256$

Hence, the unit place of the number has 6.

15. The square of which of the following number will be an even number?

A. 31 B. 69 C. 67 D. 38

Answer: D. 38

Explanation: If the square of unit place digit is an even number, then the square of the number will also be an even number.

Square of $8 = 8^2 = 64$

Since 4 is an even number, thus, square of 38 is also an even number.

16. How many zeros are there for the square of 200?

A. 2 B. 3 C. 4 D. 6

Answer: C. 4

Explanation: Square of 200 is:

$200^2 = 200 \times 200 = 2 \times 100 \times 2 \times 100 = 4 \times 10000 = 40000$

Hence, there are four zeros.

17. What will be the unit digit of square of 35789?

A. 1 B. 9 C. 3 D. 8

Answer: A. 1

Explanation: The square of 35789 will be having 1 at its unit place

Square of unit place digit of 35789 => 9 x 9 = 81

Since the square of the unit place digit has 1 at its unit place, thus the number will also have the same digit.

18. 49 can be expressed as the sum of how many same odd numbers?

A. 5 B. 7 C. 9 D. 11

Answer: B. 7

Explanation: 7 + 7 + 7+7+7+7 +7 = 49

This proves 7 times of 7 is equal to 49

Or $7^2 = 49$

19. What is the length of side of a square, if the area of square is 441 cm^2?

A. 21 cm B. 29 cm C. 31 cm D. 39 cm

Answer: A. 21

Explanation: Area of a square = 441 cm^2

By the formula, we know that;

Area of a square = $side^2$

$441 = side^2$

Side = $\sqrt{441} = 21$

Therefore, length of side of square is 21 cm.

CUBE AND CUBE ROOTS

1. The one's digit of the cube of 53 is:

A. 9 B. 3 C. 7 D. 1

Answer: **C**

Explanation: $53^3 = 53 \times 53 \times 53$

$3^3 = 3x3x3 = 27$ Hence, at the unit place, we will get 7.

2. The prime factorisation of 64 is:

A. 2 x 2 x 2 B. 4 x 4 x 4 C. 8 x 8 x 8 D. None of the above

Answer: D

Explanation: $64 = 2 \times 2 \times 2 \times 2 \times 2 \times 2 = 2^6$

3. Which of the following is not a perfect cube?

A. 216 B. 1000 C. 243 D. 1331

Answer: **C**

Explanation: $243 = 3 \times 3 \times 3 \times 3 \times 3 = 3^3 \times 3^2$

4. By what number should 81 be divided to get a perfect cube?

A. 3 B. 6 C. 7 D. 9

Answer: **A**

Explanation: The prime factorisation of 81 will be:

$81 = 3 \times 3 \times 3 \times 3$, $81 = 3^3 \times 3$

Hence, we need to divide 81 by 3 to get: $81/3 = 27 = 3^3$

5. By what number should we divide 135 to get a perfect cube?

A. 3 B. 5 C. 7 D. 9

Answer: **B**

Explanation: $135 = 3 \times 3 \times 3 \times 5$

We can see, 5 is the extra number which cannot be paired in a group of 3.

Hence, $135/5 = 27$

6. Cube root of 512 is:

A. 4 B. 6 C. 8 D. 12

Answer: **C**

Explanation: By prime factorisation, we get:

$512 = 2 \times 2 \times 2 \times 2 \times 2 \times 2 \times 2 \times 2 \times 2$

$\sqrt[3]{512} = 2 \times 2 \times 2 = 8$

7. What should be divided by 53240 to make it a perfect cube?

A. 5 B. 10 C. 15 D. 20

Answer: **A**

Explanation: The prime factorisation of 53240

$53240 = 2 \times 2 \times 2 \times 11 \times 11 \times 11 \times 5 = 2^3 \times 11^3 \times 5$

Hence, we need to divide 53240 by 5

$53240/5 = 10648$ is a perfect cube.

8. The value of $\sqrt[3]{1000}$ is:

A. 400 B. 40 C. 10 D. 4

Answer: **B**

Explanation: $\sqrt[3]{1000} = \sqrt[3]{(10 \times 10 \times 10)} = 4 \times 10 = 40$

9. A perfect cube of a number having 0 at its unit place, ends with _____ zeros.

A. 1 B. 2 C. 3 D. 4

Answer: **C**

Examples: $10^3 = 1000$, $20^3 = 8000$, $30^3 = 27000$

10. Cube root of 15625 is:

A. 5 B. 15 C. 25 D. 35

Answer: **C**

Explanation: 15625 = 5 x 5 x 5 x 5 x 5 x 5

$\sqrt[3]{15625} = 5 \times 5 = 25$

11. Which of the following is a perfect cube?

A.125 B. 36 C. 75 D. 100

Answer: A. 125

Explanation: $125 = 5 \times 5 \times 5 = 5^3$

12. What is the cube of 2a?

A. $2a^3$ B. $4a^3$ C. $8a^3$ D. $16a^3$

Answer: C. $8a^3$

Explanation:

$(2a)^3 = 2a \times 2a \times 2a = (2 \times 2 \times 2) \times (a \times a \times a)$

$= 8a^3$

13. 1. The cube root of 13824 is _____.

A. 36 B. 25 C. 24 D. 16

Answer: C. 24

Explanation: Prime factorisation of 13824 is:

$13824 = 2 \times 2 \times 2 \times 2 \times 2 \times 2 \times 2 \times 2 \times 2 \times 3 \times 3 \times 3$

$13824 = (2 \times 2 \times 2) \times (2 \times 2 \times 2) \times (2 \times 2 \times 2) \times (3 \times 3 \times 3)$

Taking cube root both the sides, we get;

(13824)⅓ = 2 x 2 x 2 x 3 = 24

14. Which of these numbers is not a cube number?

A. 10000 B. 343 C. 64 D. 729

Answer: A. 10000

Explanation: By prime factorisation method.

10000 = 2 × 2 × 2 × 2 × 5 × 5 × 5 × 5.

= $2^4 \times 5^4 = 2^3 \times 2 \times 5^3 \times 5$

Thus, the factors cannot be cubed.

15. When the square of a number is subtracted from the cube of the same number, it becomes 100. Find the number.

A.1 B. 2 C. 4 D. 5

Answer: D. 5

Explanation: Square of 5 = 5 x 5 = 25

Cube of 5 = 5 x5x5 = 125

125 – 25 = 100

16. Which is the smallest natural number by which 243 must be multiplied to make the product a perfect cube?

A. 3 B. 9 C. 8 D. 7

Answer: A. 3

Explanation: 243 x 3 = 729

Hence, by prime factorisation of 729 we get;

729 = 3 x 3 x 3 x 3 x 3 x 3

729 = 3^3 x 3^3

Taking cube root on both the sides, we get;

(729)⅓ = 3 x 3 = 9

17. The one's digit of the cube of the number 111 is:

A. 1 B. 2 C. 3 D. 9

Answer: A. 1

Explanation: The unit place of 111 has 1.

Cube of 1 = 1^3 = 1 x 1 x 1 = 1

18. The cube of 23 is:

A. 2304 B. 12167 C. 529 D. 23

Answer: B. 12167

Explanation: Cube of 23 = 23 x 23 x 23 = 12167

19. The one's digit of the cube of the number 242 is

A. 2 B. 4 C. 6 D. 8

Answer: D. 8

Explanation: The unit digit of 242 is 2.

Cube of 2 = 2 x 2 x 2 = 8

20. Find the smallest number by which the number 200 must be multiplied to obtain a perfect cube.

A. 2 B. 10 C. 5 D. 100

Answer: C. 5

Explanation: The prime factorisation of 200 gives:

200 = 2 × 2 × 2 × 5 × 5 = $2^3 \times 5 \times 5$

Now multiply by 5 on both sides.

200 x 5 = $2^3 \times 53$

(1000)1/3 = 2 × 5 = 10

COMPARING QUANTITIES

1. The ratio of speed of cycle 12 km per hour to the speed of scooter 36 km per hour is

A. 1:2 B. 1:3 C. 1:4 D. None of the above

Answer: **B**

Explanation: Speed of cycle/Speed of scooter = 12/36 = ⅓

2. The ratio of 10m to 10 km is:

A. 1/10 B. 1/100 C. 1/1000 D. 1000

Answer: **C**

Explanation: 10m/10km = 10m/10000m = 1/1000

3. The percentage of 1:4 is:

A. 75% B. 50% C. 25% D. 100%

Answer: C

Explanation: 1:4 = 1/4

(1×100)/(4×100) = 1/4 x 100% = 0.25 x 100% = 25%

4. The percentage of 2:5

A. 20% B. 50% C. 60% D. 40%

Answer: **D**

Explanation: 2:5 = ⅖ = ⅖ x 100% = 0.4 x 100% = 40%

5. If 50% of students are good at science out of 20 students. Then the number of students good at science is:

A. 10 B. 15 C. 5 D. 11

Answer: **A**

Explanation: 50% of students out of 20 students = 50% of 20

= (50/100) x 20= ½ x 20 = 10

6. The price of a motorcycle was Rs. 34,000 last year. It has increased by 20% this year. The price of motorcycle now is:

A. Rs. 36,000 B. Rs. 38,800 C. Rs. 40,800 D. Rs. 32,000

Answer: **C**

Explanation: 20% of Rs.34,000 = 20/100 x 34,000 = Rs.6800

New price = Rs. 34,000 + Rs.6800 = Rs. 40,800

7. An item marked at Rs. 840 is sold for Rs. 714. The discount % is:

A. 10% B. 15% C. 20% D. 25%

Answer: **B**

Explanation: Discount = Marked Price – Sale Price

= 840 – 714 = Rs. 126

Discount % = (126/840) x 100% = 15%

8. A person got a 10% increase in his salary. If his salary was Rs. 50000, then the new salary is:

A. Rs. 55000 B. Rs. 60000 C. Rs. 45000 D. Rs. 65000

Answer: **A**

Explanation: Previous salary = Rs. 50000

10% of Rs.50000 = (10/100) x 50000 = Rs. 5000

New salary = Rs. 50000 + Rs. 5000 = Rs. 55000/-

9. The cost of the article was Rs. 15500 and Rs. 500 was spent on its repairing. If it is sold for a profit of 15%. The selling price of the article is:

A. Rs.16400 B. Rs.17400 C. Rs.18400 D. Rs.19400

Answer: **C**

Explanation: Total cost = 15500+500 = 16000

Profit % = (Profit/Cost price) x 100

Profit = (Profit% x Cost price)/100

P = (15×16000)/100 = 2400

Selling Price = Profit + cost price = 2400 + 16000 = Rs.18400

10. Waheeda bought an air cooler for Rs. 3300 including a tax of 10%. The price of the air cooler before VAT was added is:

A. Rs. 2000 B. Rs. 3000 C. Rs. 2500 D. Rs. 2800

Answer: **B**

Explanation:10% VAT on Rs.100 will make it Rs.110

So, for price including VAT Rs.110, the original price is Rs.100

Then, Price including VAT Rs. 3300, the original price = Rs. (100/110) x 3300 = Rs. 3000.

11. The ratio of 50 paise to Rs. 1 is:

A. 1 : 2 B. 2 : 1 C. 1 : 1 D. 1 : 5

Answer: A

Rs. 1 = 100 paise

50 paise : 100 paise

50 : 100

1 : 2

12. The Ratio of 10m to 1 Km is Equal To:

A. 1/10 B. 1/100 C. 1/1000 D. 1000

Answer: B

Explanation: 10m/1km = 10m/1000m = 1/100

13. The ratio of 1m and 100 cm is:

A. 1:100 B. 1:10 C. 10:1 D. 1:1

Answer: D. 1:1

Explanation: One meter is equal to hundred centimeters.

1m = 100 cm

Thus, their ratio will be:

⇒ 1m:100cm

⇒ 100cm:100cm

⇒ 1:1

14. The percentage of ratio 2:3 is:

A. 44.44% B. 55.55% C. 66.66% D. None of the above

Answer:

Explanation: 2:3 = ⅔

Converting into percentage, we have to multiply the ration by 100.

⅔ x 100% = 200/3 = 66.66..%

15. The ratio 4 : 25 converted to percentage is:

A. 8% B. 4% C. 16% D. 25%

Answer: C. 16%

4 : 25 =4/25 x 100% = 16%

16. Convert the fraction ⅛ into a percentage.

A. 12.5 % B. 25% C. 8% D. 16%

Answer: A. 12.5%

Explanation: Multiply the given fraction by 100 to get the percentage value.

⅛ x 100 = 12.5%

17. The ratio of speed of a motorbike 50 km per hour and speed of cycle 20 km per hour is:

A. 2:5 B. 5:1 C. 5:2 D. 1:2

Answer: C. 5:2

Explanation:

Ratio of their speeds are = 50/20 = 5/2 = 5:2

18. If 40% of students passed in the Mathematics test, out of 40 students in the class. What is the number of students who have passed?

A. 10 B. 16 C. 5 D. 11

Answer: B

Explanation: 40% of students out of 40 students will be:

= (40/100) x 40= 2/5 x 40 = 16

19. 25% of 60 students of a class likes to play football. How many students does not like to play football?

A. 45 B. 15 C. 25 D. 50

Answer: A. 45

Explanation: Number of students = 60

25% of 60 students = 25/100 x 60 = 15

Number of students who does not play football = 60 – 15 = 45

20. Out of 40 students, 25% passed in a class. How many students did actually pass?

A. 10 B. 20 C. 30 D. 40

Answer: 10

Explanation: 25% of 40 = 25/100 x 40 = 10

ALGEBRIC EXPRESSION AND QUANTITIES

1. An algebraic expression that contains only one term is called:

A. Monomial B. Binomial C. Trinomial D. None of the above

Answer: **A** Example: 2x is a monomial

2. 5x+6y is a:

A. Monomial B. Binomial C. Trinomial D. None of the above

Answer: **B**

Explanation: The expression containing two terms is called binomial.

3. The algebraic expression 3x+2y+6 is a:

A. Monomial B. Binomial C. Trinomial D. None of the above

Answer: **C**

Explanation: The algebraic expression containing three terms is called a trinomial. Here, 3x, 2y and 6 are three terms.

4. A polynomial contains _____ number of terms:

A. One B. Two C. Three D. Any

Answer: **D**

Explanation: A polynomial can contain any number of terms, i.e. one or more than one.

5. In which of the following, the two expressions are like terms?

A. 7x and 7y B. 7x and 9x C. 7x and $7x^2$ D. 7x and 7xy

Answer: **B**

6. If we add, 7xy + 5yz – 3zx, 4yz + 9zx – 4y and – 3xz + 5x – 2xy, then the answer is:

A. 5xy + 9yz +3zx + 5x – 4y B. 5xy – 9yz +3zx – 5x – 4y

C. 5xy + 10yz +3zx + 15x – 4y D. 5xy + 10yz +3zx + 5x – 6y

Answer: **A**

Explanation: Given, 7xy + 5yz – 3zx, 4yz + 9zx – 4y and –3xz + 5x – 2xy.

If we add the three expressions, then we need to combine the like terms together.

(7xy – 2xy) + (5yz + 4yz) – 3zx + 9zx – 3xz – 4y + 5x

= 5xy + 9yz + 3zx + 5x – 4y

7. If we subtract 4a – 7ab + 3b + 12 from 12a – 9ab + 5b – 3, then the answer is:

A. 8a+2ab+2b+15 B. 8a+2ab+2b-15

C. 8a-2ab+2b-15 D. 8a-2ab-2b-15

Answer: **C**

Explanation: (12a – 9ab + 5b – 3) – (4a – 7ab + 3b + 12)

= 12a – 9ab + 5b – 3 – 4a + 7ab -3b-12

= (12 – 4)a – (9 – 7)ab + (5 – 3)b – 3 – 12

= 8a – 2ab + 2b – 15

8. If we multiply 5x and (– 4xyz), then we get:

A. $20x^2yz$ B. $-20x^2yz$ C. x^2yz D. -2xyz

Answer: **B**

Explanation: (5x) x (-4xyz)

= 5 × x × (-4) × x × y × z

= $-20x^{1+1}yz$

= $-20x^2yz$

9. The product of 4x and 0 is:

A. 4x B. 4 C. 0 D. None of the above

Answer: **C**

Explanation: Any value multiplied by zero is zero.

10. The volume of a cuboid with length, breadth and height as 5x, $3x^2$ and $7x^4$ respectively is:

A. $105x^7$ B. $105x^2$ C. $105x^4$ D. 105x

Answer: **A**

Explanation: Volume of cuboid = Length × breadth × height

V = 5x × $3x^2$ × $7x^4$

V = 105 x^{1+2+4}

V = $105x^7$ cubic units

11. The product of 5x and 3y is:

A. xy

B. 2xy

C. 5xy

D. 15xy

Answer: (D) 15xy

(5x)(3y) = 15xy

12. The product of 6x and -11x is:

A. $66x^2$ B. $-66x^2$ C. x^2 D. $-x^2$

Answer: B. $-66x^2$

$(6x)(-11x) = -66x^2$

13. The area of a rectangle whose length and breadth are 3y and $9y^2$ respectively is:

A. $12y^3$ B. $21y^3$ C. $27y^3$ D. y^3

Answer: C. $27y^3$

Explanation:

Area of rectangle = length x breadth = $3y \times 9y^2 = 27y^3$.

14. The area of a rectangle that has length = $2a^2b$ and breadth = $3ab^2$ is:

A. $6a^3b^3$ B. a^3b^3 C. $2a^3b^3$ D. $4a^3b^3$

Answer: A. $6a^3b^3$

Explanation:

Area of rectangle = $(2a^2b)(3ab^2) = 6a^3b^3$

15. The side of a cube is 2a. Find the volume of the cube.

A. $4a^2$ B. $2a$ C. $8a^3$ D. 8

Answer: C. $8a^3$

Explanation:

Volume of the cube = $2a \times 2a \times 2a = 8a^3$

16. Multiplication of monomials x^2, $(-x)^3$, $(-x)^4$ is equal to:

A. x^9 B. x^5 C. x^7 D. x^6

Answer: A. x^9

Explanation:
$(x^2).(-x^3).(-x)^4 = x^9$

17. The value of $(x - y)(x + y) + (y - z)(y + z) + (z - x)(z + x)$ is:

A. $x + y + z$ B. $x^2 + y^2 + z^2$ C. $xy + yz + zx$ D. 0

Answer: D. 0

Explanation:

$(x - y)(x + y) + (y - z)(y + z) + (z - x)(z + x)$

$= x^2 - y^2 + y^2 - z^2 + z^2 - x^2$ [By algebraic identity: $a^2 - b^2 = (a+b)(a-b)$]

$= 0$

18. $(a - b)^2$ is equal to:

A. $a^2 + b^2 - 2ab$ B. $a^2 + b^2 + 2ab$ C. $a^2 + b^2$ D. $2ab$

Answer: A. $a^2 + b^2 - 2ab$

Explanation: By algebraic identity,

$(a + b)^2 = a^2 + b^2 - 2ab$

19. The product of $3xy^2z$ and $4x$ is:

A. $12xyz$ B. $12xy^2$ C. $12x^2y^2z$ D. $12x^2yz$

Answer: C. $12x^2y^2z$

Explanation: The product of $3xy^2z$ and $4x$ is:

$\Rightarrow (3xy^2z)(4x)$

$\Rightarrow 3.x.y^2.z.4.x$

$\Rightarrow 12x^2y^2z$

20. Which of the following is a like term as 8xy?

A. 8 B. 8x C. 8y D. xy

Answer: D. xy

VISUALISING SOLID SHAPES

1. A cube has _____ faces.

A. 4 B. 5 C. 6 D. 8

Answer: **C**

Explanation: A cube is a three-dimensional shape having six faces.

2. Number of vertices a cone has:

A.One B. Two C. Three D. Four

Answer: **A**

3. A sphere has how many faces?

A. One B. Two C. Four D. None

Answer: **D**

Explanation: A sphere is a round-shaped object having no faces and edges.

4. An example of cone is:

A. Tube B. A tent C. A ball D. A box

Answer: **B**

5. The top-view of a cone looks like:

A. A Circle B. A Square C. A Rectangle D. A Triangle

Answer: **A**

Explanation: The base of a right-cone is always circular.

When we see it from the top, it looks like a circle with its vertex as the center of the circle.

6. The top-view of a cube looks like a:

A. Circle B. Square C. Rectangle D. Triangle

Answer: **B**

Explanation: Since all the faces of a cube are of equal length.

7. The top-view of a cuboid looks like a:

A. Circle B. Square C. Rectangle D. Triangle

Answer: **C**

8. The side-view of a cone appears as:

A. Circle B. Square C. Rectangle D. Triangle

Answer: **D**

9. The base of a prism is:

A. Circle B. Triangle C. Square D. Any shape

Answer: **D**

Explanation: A pyramid is a polyhedron whose base is a polygon (of any number of sides) and whose lateral faces are triangles with a common vertex.

10. If a polyhedron has 6 vertices and 12 edges. What is the number of faces it has?

A. 6 B. 8 C. 12 D. 18

Answer: **B**

Explanation: By Euler's formula we know that:

For any polyhedron,

F + V – E = 2

F = 2-V+E

F = 2 – 6+12 = 8

11. The example of cylindrical shape is:

A. Pipes B. Gas cylinder C. Cold drink cans D. All of the above

Answer: D. All of the above

12. A polyhedron is said to be regular if its faces are made up of regular polygons and the same number of faces meet at each vertex.

A. True B. False

Answer: A. True

13. Prisms and Pyramids are:

A. Polygons B. Cubes C. Polyhedrons D. Spherical shapes

Answer: C. Polyhedrons

14. A square is a 3D shape.

A. True B. False

Answer: B. False

A square is a 2D Shape.

15. A cube is a 3D shape.

A. True B. False

Answer: A. True

16. In the map, symbols are used to depict the different objects/places.

A. True B. False

Answer: A. True

17. 3D objects have the same views from different positions.

A. True B. False

Answer: B. False

Explanation: 3D objects have different views from different positions

MENSURATION

1. Area of a triangle with base (b) and height (h) is:

A. b.h B. ½ b.h C. 2b.h D. None of these

Answer: B. ½ b.h

2. Area of the square with side-length 'a' is:

A. 2a B. 4a C. a/2 D. a^2

Answer: D. a^2

3. The area of a parallelogram with length (l) and breadth (b) is:

A. ½ lb B. 2lb C. Lb D. $(lb)^2$

Answer: C. lb

4. Area of a circle with radius 'r' is:

A. πr^2 B. ½ πr^2 C. $2\pi r^2$ D. $4\pi r^2$

Answer: A. πr^2

5. 1 cm is equal to how many millimeters?

A. 10 B. 100 C. 1/10 D. 1/100

Answer: A. 10

1 cm = 10 mm

6. Surface area of cube of edge 'a' is:

A. $4a^2$ B. $6a^2$ C. $3a^2$ D. a^2

Answer: B. $6a^2$

7. Volume of a cuboid of length (l), width (w) and height (h) is:

A. lbh B. lb + bh + hl C. 2 (lb + bh + hl) D. 2 (l + b) h

Answer: A. lbh

8. Volume of a cylinder with base radius = r and height = h, is:

A. $2\pi rh$ B. πr^2h C. $2\pi r$ (r + h) D. 1/3 πr^2h

Answer: B. πr^2h

9. 1 litre is equal to how many cubic centimeters?

A. 10 cu.cm B. 100 cu.cm C. 1000 cu.cm D. 10000 cu.cm

Answer: C. 1000 cu.cm

10. The perimeter of a rectangle with length (l) and width (w) is:

A. l+w B. 2(l+w) C. $(l+w)^2$ D. lw

Answer: B. 2(l+w)

11. If the length and breadth of a rectangle are 15cm and 10cm, respectively, then its area is:

A. 100 sq.cm B. 150 sq.cm C. 115 sq.cm D. 200 sq.cm

Answer: **B**

Explanation: Length = 15 cm And breadth = 10 cm

Area of rectangle = Length x breadth

= 15 x 10 = 150 cm^2

12. The area of a rhombus whose diagonals are of lengths 10 cm and 8.2 cm is:

A. 41 cm^2 B. 82 cm^2 C. 410 cm^2 D. 820 cm^2

Answer: **A**

Explanation: Area of rhombus = ½ d_1 d_2

A = ½ x 10 x 8.2 A = 41 cm^2

13. The area of a trapezium is 480 cm², the distance between two parallel sides is 15 cm and one of the parallel side is 20 cm. The other parallel side is:

A. 20 cm B. 34 cm C. 44 cm D. 50 cm

Answer: **C**

Explanation: Area of trapezium = ½ h (a+b)

a=20cm, h = 15cm, Area = 480 sq.cm

480 = ½ (15) (20+b)

20+b = (480 x 2)/15

20+b=64 b=44cm

14. The area of a rhombus is 240 cm² and one of the diagonals is 16 cm. Find the other diagonal.

A. 16 cm B. 20 cm C. 30 cm D. 36 cm

Answer: **C**

Explanation: Area = 240 cm²

d_1 = 16 cm

Area of rhombus = ½ d_1 x d_2

240 = ½ x 16 x d_2

d_2 = 480/16 = 30 cm

15. A cuboid has _____ pairs of identical faces.

A. 2 B. 3 C. 4 D. 5

Answer: **B**

Explanation: All six faces are rectangular, and opposite faces are identical.

So there are three pairs of identical faces.

16. All six faces of a cube are:

A. Identical B. Different C. Circular D. Rectangular

Answer: **A**

Explanation: All six faces are squares and identical

17. A cylindrical box has ____ curved surface and ____ circular faces, which are identical.

A. One, One B. One, two C. two, one D. two, two

Answer: **B**

Explanation: A cylindrical box having circular bases have identical top.

One curved surface and two circular faces which are identical.

18. If a cuboidal box has height, length and width as 20 cm, 15 cm and 10 cm respectively. Then its total surface area is:

A. 1100 cm² B. 1200 cm² C. 1300 cm² D. 1400 cm²

Answer: **C**

Explanation: Total surface area = 2 (20 × 15 + 20 × 10 + 10 × 15)

TSA = 2 (300 + 200 + 150) = 1300 cm²

19. The height of a cylinder whose radius is 7 cm and the total surface area is 968 cm² is:

A. 15 cm B. 17 cm C. 19 cm D. 21 cm

Answer: **A**

Explanation: Total surface area = 2πr (h + r)

968 = 2 x 22/7 x 7 (7+h)

h = 15 cm

20. The height of a cuboid whose volume is 275 cm³ and base area is 25 cm² is:

A. 10 cm B. 11 cm C. 12 cm D. 13 cm

Answer: **B**

Explanation: Volume of a cuboid = Base area × Height

Height = Volume / Base area

H = 275/25 = 11 cm

EXPONENTS AND POWERS

1. The value of 2^{-2} is:

A. 4 B. ¼ C. 2 D. ½

Answer: **B**

Explanation: $2^{-2} = 1/2^2 = ¼$

2. The multiplicative inverse of 7^{-2} is:

A. 7^2 B. 7 C. $1/7^2$ D. 1/7

Answer: **A**

Explanation: The multiplicative inverse of any value is the one which when multiplied by the original value gives a value equal to 1.

$7^{-2} = 1/7^2$, Hence, $7^2 \times 1/7^2 = 1$

3. $2^2 \times 2^3 \times 2^4$ is equal to:

A. 2^{24} B. 2^{-5} C. 2^9 D. 2^{-9}

Answer: **C**

Explanation: By laws of exponents:

$a^m \times a^n = a^{m+n}$

$2^2 \times 2^3 \times 2^4 = 2^{2+3+4} = 2^9$

4. $3^{-2} \times 3^{-5}$ is equal to:

A. 3^{-7} B. 3^{-3} C. 3^{-10} D. 3^7

Answer: **A**

Explanation: $3^{-2} \times 3^{-5} = (1/3^2) \times (1/3^5)$

$= (1/3^{2+5})$ $= (1/3^7)$ $= 3^{-7}$

5. $5^4/5^2$ is equal to:

A. 5^6 B. 5^{-6} C. 5^{-2} D. 5^2

Answer: **D**

Explanation: By exponent law:

$a^m/a^n = a^{m-n}$

$5^4/5^2 = 5^{4-2} = 5^2$

6. The value of $(3^4)^3$ is:

A. 3 B. 3^{12} C. 3^7 D. None of the above

Answer: **B**

Explanation: By law of exponent:

$(a^m)^n = a^{mn}$

$(3^4)^3 = 3^{4\times3} = 3^{12}$

7. $3^2 \times 4^2$ is equal to:

A. 121 B. 49 C. 144 D. 156

Answer: **C**

Explanation: By exponent law;

$a^m \times b^m = (ab)^m$

$3^2 \times 4^2 = (3\times4)^2 = 12^2 = 144$

8. $5^7/6^7$ will give the value:

A. $(5/6)^7$ B. $(5/6)^0$ C. $(5/6)^{-7}$ D. $(6/5)^{-7}$

Answer: **A**

Explanation: By exponent law:

$a^m/b^m = (a/b)^m$

$5^7/6^7 = (5/6)^7$

9. $100^0+20^0+5^0$ is equal to

A. 125 B. 25 C. 1/125 D. 3

Answer: **D**

Explanation: By exponent law we know:

$a^0 = 1$

$100^0 + 20^0 + 5^0 = 1+1+1 = 3$

10. If $(-3)^{m+1} \times (-3)^5 = (-3)^7$, then the value of m is:

A. 5 B. 7 C. 1 D. 3

Answer: **C**

Explanation: $(-3)^{m+1} \times (-3)^5 = (-3)^7$

$(-3)^{m+1+5} = (-3)^7$

$(-3)^{m+6} = (-3)^7$

Since, base are equal on both the sides, hence if we compare the powers,

$m+6 = 7$

$m=7-6 = 1$

11. Find the multiplicative inverse of 5^{-3}.

A. 5^3 B. 1/5 C. 5^2 D. 5^{-2}

Answer: A

Explanation: The multiplicative inverse of 5-3 is 53.

$5^{-3} \times 5^3 = 1$

12. The multiplicative inverse of $1/3^2$ is:

A. 3^{-2} B. 3^2 C. 3 D. 1/3

Answer: B. 3^2

Explanation: The multiplicative inverse of $1/3^2$ is 3^2.

$1/3^2 \times 3^2 = 1$

13. What is the value of $4^2 \times 4^{-2}$?

A. 16 B. 8 C. 1 D. 0

Answer: C.1

Explanation: $4^2 \times 4^{-2}$

By law of exponents:

$= 4^{2+(-2)} = 4^{2-2} = 4^0 = 1$

14. $(⅓)^2$ is equal to:

A. 9 B. -9 C. -1/9 D. 1/9

Answer: D. 1/9

Explanation: $(⅓)^2 = (⅓ \times ⅓) = 1/9$

15. What is the value of $(2^2 + 3^2 + 4^2)^0$?

A. 9 B. 0 C. 1 D. 14

Answer: C.1

Explanation: The value of $(2^2 + 3^2 + 4^2)^0$ is 1.

By exponent law, any value raised to the power 0 is equal to 1.

16. $(-1)^{20}$ is equal to:

A. -1 B. 1 C. 0 D. 20

Answer: B.1

Explanation: $(-1)^{20}$ is equal to 1

because if a negative number is raised to the power of an even number,

then the resulting value will be positive.

17. 700000000 is equal to:

A. 7×10^8 B. 7×10^7 C. 7×10^6 D. 7×10^9

Answer: A. 7×10^8

18. 865000 is equal to:

A. 8.65×10^5 B. 8.65×10^3 C. 8.65×10^6 D. 8.65×10^4

Answer: A. 8.65×10^5

19. 1.8×10^{11} is equal to:

A. 180000000000 B. 18000000000

C. 1800000000 D. 1800000000000

Answer: A. 180000000000

20. 0.09×10^{10} is equal to:

A. 900000000 B. 9000000 C. 9000 D. 9

Answer: A. 900000000

DIRECT AND INVERSE PROPORTIONS

1. If x = 20 and y = 40, then x and y are:

A. Directly proportional B. Inversely proportional

C. Neither directly nor inversely proportional D. Cannot be determined

Answer: **A**

Explanation: x = 20 and y = 40 Clearly, 40 = 2 x 20 y=2x

$y \propto x$, where 2 is the proportionality constant.

2. If x = 15 and y = 1/30, then x and y are:

A. Directly proportional B. Inversely proportional

C. Neither directly nor inversely proportional D. Cannot be determined

Answer: **B**

Explanation: x = 15, y = 1/30 , $1/30 = 1/(2 \times 15)$

y = 1/2x Hence, $y \propto 1/x$, where 1/2 is the proportionality constant.

3. If x and y are directly proportional, then which of the following is correct?

A. x+y=constant B. x-y=constant C. xy=constant D. x/y=constant

Answer: **D**

Explanation: $x \propto y$, x=ky , k=x/y, where k is a constant.

4. If x and y are inversely proportional, then:

A. x+y=constant B. x-y=constant C. xy=constant D. x/y=constant

Answer: **C**

Explanation: $x \propto 1/y$, $x=k/y$, $xy=k$,Where k is a constant.

5. If $x \propto y$ and $x_1=5, y_1=210$ and $x_2 = 2$, then find y_2?

A. 200 B. 84 C. 99 D. 70

Answer: **B**

Explanation: $x \propto y$,

$5 \times 42 = 210$,

$2 \times 42 = 84$

6. If the weight of 12 sheets of thick paper is 40 grams, how many sheets of the same paper would weigh 2500 grams?

A. 750 B. 800 C. 850 D. 950

Answer: **A**

Explanation: For 12 sheets, weight of paper is 40 grams

Let number of sheets for 2500 is x.

Using direct proportion concept:

$12/40 = x/2500$,$x=(12\times2500)/40$,$x=750$

7. The scale of a map is given as 1:300. Two cities are 4 km apart on the map. The actual distance between them is:

A. 1000 km B. 1100 km C. 1200 km D. 1300 km

Answer: **C**

Explanation: 1:300 = x : y

1/300 = 4/y

y = 4 x 300 = 1200

Hence, the distance between the two cities is 1200 km.

8. If x and y are inversely proportional, then which one is true?

A. $x_1/y_1 = x_2/y_2$ B. $x_1/x_2 = y_1/y_2$ C. $x_1/x_2 = y_2/y_1$ D. $x_1.x_2 = y_1.y_2$

Answer: **C**

Explanation: $x \propto 1/y$

$x=k(1/y)$

$xy=k$

$x_1y_1=x_2y_2$

$x_1/x_2=y_2/y_1$

9. 6 pipes are required to fill a tank in 1 hour 20 minutes. If we use 5 such types of pipes, how much time it will take to fill the tank?

A. 120 minutes B. 96 minutes C. 80 minutes D. 85 minutes

Answer: **B**

Explanation: For 6 pipes, it takes 1 hour 20 minutes

1 hour 20 minutes = 60+20 = 80 minutes

For 5 pipes, let the time taken be x.

This is inverse proportion case:

$80 \times 6 = x \times 5$

$x = 480/5 = 96$

10. If 12 workers can build a wall in 50 hours, how many workers will be required to do the same work in 40 hours?

A. 10 B. 13 C. 14 D. 15

Answer: **D**

Explanation: $12 \times 50 = x \times 40$, $x = (12\times50)/40 = 15$

11. A man walks 20 km in 5 hours. How much time it will take for him to walk 32 km?

A. 3 hours B. 4 hours C. 6 hours D. 8 hours

Answer: D

Solution: Man walks 20 km in → 5 hours

That means it will take more time to walk 32 km. This is the case of direct proportion.

$20/5 = 32/x$, $x = 32/4 = 8$

12. If 300 Kg of coal cost 6000₹, then find the cost of 120 kg of coal?

A. 1200₹ B. 2400₹ C. 3200₹ D. 4200₹

Answer: B

Solution: 300/6000 = 120/x , x = 2400

13. Manvi types 200 words in 30 minutes. How many words she will type in 12 minutes?

A. 100 B. 80 C. 60 D. 50

Answer: B

Solution: 30/200 = 12/x

x = 12 x 200/30

x = 80

14. If it takes 40 days for 120 men to complete a work, how long will it take for 80 men to complete the same work?

A. 50 days B. 60 days C. 80 days D. 100 days

Answer: B. 60 days

40 × 120 = 80 × x

⇒ x = 60

15. 6 pipes are required to fill a tank in 80 minutes. How long will it take if only 5 pipes of the same type are used?

A. 96 minutes B. 102 minutes C. 106 minutes D. 108 minutes

Answer: A. 96 minutes

80 × 6 = x × 5

x = 96

16. If 12 labours can construct a road in 50 hours, how many labours will be required to construct the same road in 40 hours?

A. 10 B. 12 C. 14 D. 15

Answer: D. 15

12 × 50 = x × 40

x = (12 × 50)/40 = 15

17. If 100 students took 20 days to clean a garden. How many days it will take to clean the garden if 25 more students are added?

A. 10 B. 12 C. 16 D. 18

Answer: C. 16

100 × 20 = 125 × x

x = 16

18. If x = 10 and y = 20, then x and y are:

A. Directly proportional B. Inversely proportional

C. Neither directly nor inversely proportional D. Cannot be determined

Answer: A. Directly proportional

x = 10 and y = 20

y = 2 x 10 = 2x

Y is directly proportional to x, where 2 is the proportionality constant.

19. A car takes 18 hours to ride 720 kilometres. Time taken by the car to travel 360 kilometres is:

A. 10 hours B. 11 hours C. 9 hours D. 16 hours

Answer: C. 9 hours

720/18 = 360/x

20. The perimeter of a square and its side is in:

A. Direct Proportion B. Indirect Proportion

C. Neither direct nor indirect D. Cannot be determined

Answer: A. Direct Proportion

FACTORISATION

1. The factorisation of $12a^2b+15ab^2$ gives:

A. 3ab(4ab+5) B. 3ab(4a+5b) C. 3a(4a+5b) D. 3b(4a + 5b)

Answer: **B**

Explanation: $12a^2b+15ab^2$

$12a^2b$ = 3 x 4 x a x a x b

$15ab^2$ = 3 x 5 x a x b x b

The common factors are 3ab.

$12a^2b+15ab^2$ = 3ab(4a+5b)

2. The factorisation of 12x+36 is

A. 12(x+3) B. 12(3x) C. 12(3x+1) D. x(12+36x)

Answer: **A**

Explanation: 12x + 36

12 x + 12 . 3

=12(x+3)

3. On factorising 14pq + 35pqr, we get:

A. pq(14+35r) B. p(14q+35qr) C. q(14p+35pr) D. 7pq(2+5r)

Answer: **D**

Explanation: 14pq + 35pqr

= 2.7.p.q + 5.7.p.q.r

= 7pq(2+5r)

4. The factors of 6xy – 4y + 6 – 9x are:

A. (3x + 2) (2y + 3) B. (3x – 2) (2y – 3) C. (3x – 2) (2y + 3) D. (3x –+2) (2y – 3)

Answer: **B**

Explanation: 6xy – 4y + 6 – 9x

= 6xy – 4y – 9x + 6

= 2y (3x – 2) – 3 (3x – 2)

= (3x – 2) (2y – 3)

5. The factors of x^2+xy+8x+8y are:

A. (x+y) (x+8) B. (2x+y) (x+8) C. (x+2y) (x+8) D. (x+y) (2x+8)

Answer: **A**

Explanation: x^2+xy+8x+8y

= x(x+y)+8(x+y)

=(x+y)(x+8)

6. The factors of $4y^2$– 12y + 9 is:

A. $(2y+3)^2$ B. $(2y-3)^2$ C. (2y-3)(2y+3) D. None of the above

Answer: **B**

Explanation: $4y^2$– 12y + 9

$4y^2 = (2y)^2$ & $9 = 3^2$ & 12y = 2.3.2y

$4y^2$– 12y + 9 = $(2y)^2$ – 2 × 3 × (2y) + $(3)^2$

= $(2y – 3)^2$ [By algebraic identities: $(a-b)^2 = a^2+b^2-2ab$

7. The factors of $49p^2$ – 36 are:

A. $(7p+6)^2$ B. $(7p-6)^2$ C. (7p – 6) (7p + 6) D. None of the above

Answer: **C**

Explanation: $49p^2$ – 36 = $(7p)^2$ – (6)^2 = (7p – 6) (7p + 6)

8. The factors of m^2 – 256 are:

A. $(m + 4)^2$ B. $(m – 4)^2$ C. (m – 4) (m+4) D. None of the above

Answer: **D**

Explanation: $m^2 = (m)^2$ and $256 = (16)^2$

By using the algebraic identity, a^2-b^2 = (a+b).(a-b), we get (m+16).(m-16).

9. When we factorise x^2+5x+6, then we get:

A. (x +2) (x + 3) B. (x – 2) (x – 3) C. (x × 2) + (x × 3) D. (x × 2) – (x × 3)

Answer: **A**

Explanation: The factors of a form:

(x + a) (x + b) = x^2 + (a + b) x + ab

x^2+5x+6

a+b = 5 and ab = 6

x^2+5x+6 = (x +2) (x + 3)

10. The factors of $3m^2$ + 9m + 6 are:

A. (m + 1) (m + 2) B. 3(m + 1) (m + 2) C. 6(m + 1) (m + 2) D. 9(m + 1) (m + 2)

Answer: **B**

Explanation: $3m^2$ + 9m + 6 = 3(m^2 + 3m + 2)

= 3 [m^2 + m + 2m + 2]

= 3 [m(m + 1)+ 2(m + 1)]

= 3 [(m + 1) (m + 2)]

11. The common factor of a^3b^3 and ab^2 is:

A. a^2b^2 B. ab^2 C. a^2b D. ab

Answer: B. ab^2

a^3b^3 = a x a x a x b x b x b

ab^2 = a x b x b

The common factors are: $a \times b \times b = ab^2$

12. The common factor of a^3b^2 and a^4b is:

A. a^4b^2 B. a^4b C. a^3b^2 D. a^3b

Answer: D. a^3b

Explanation: Same as question no.11

13. The common factor 12a and 30 is:

A.6 B. 12 C. 30 D. 6a

Answer: A. 6

12a = 2 x 2 x 3 x a

30 = 2 x 3 x 5

14. The common factors of 10a, 20b and 30c are:

A. ab B. Ac C. 10abc D. 10

Answer: 10

10a = 2 x 5 x a

20b = 2 x 2 x 5 x b

30c = 2 x 3 x 5 x c

15. The common factor of $6x^3y^4z^2$, $21x^2y$ and $15x^3$ is:

A. $3x^2$ B. $3x^3$ C. $6x^3$ D. $6x^2$

Answer: A. $3x^2$

Explanation: Same as question number 14.

16. The common factor of $24a^3b^4$, $36a^4c^4$ and $48a^3b^2c$ is:

A. $12a^3$ B. $24a^3$ C. $36a^3$ D. $48a^3$

Answer: A. $12a^3$

Explanation:

$24a^3b^4 = 2 \times 2 \times 2 \times 3 \times a \times a \times a \times b \times b \times b \times b$

$36a^4c^4 = 2 \times 2 \times 3 \times 3 \times a \times a \times a \times a \times c \times c \times c \times c$

$48a^3b^2c = 2 \times 2 \times 2 \times 2 \times 3 \times a \times a \times a \times b \times b \times c$

17. The factorisation of $12x^2y + 15xy^2$ is:

A. $3xy^2$ (4x + 5y) B. $3x^2y$ (4x + 5y) C. 3xy (4x + 5y) D. $3x^2y^2$ (4x + 5x)

Answer: C. 3xy (4x + 5y)

$12x^2y + 15xy^2 = 3xy\ (4x + 5y)$

18. The factorisation of 5x – 20 is:

A. 5(x-5) B. 5(x-4) C. 5(x-3) D. 5(x-20)

Answer: B. 5(x-4)

Explanation: 5x – 20 = 5 (x – 4)

19. The factorisation of 8x + 4y is:

A. 8(x+4y) B. 4(2x+4y) C. 8(x+y) D. 4(2x+y)

Answer: D. 4(2x + y)

20. The factors of xyz are:

A. x B. Y C. Z D. All of the above

Answer: D.

x, y and z are all the factors of xyz.

INTRODUCTION TO GRAPHS

1. A _____ is a bar graph that shows data in intervals.

A. Bar-graph B. Pie-chart C. Histograph D. Line Graph

Answer: **C**

2. A graph that displays data that changes continuously over periods of time is called:

A. Bar-graph B. Pie-chart C. Histograph D. Line Graph

Answer: **D**

3. A line graph which is a whole unbroken line is called a:

A. Linear graph B. Pie-chart C. Histograph D. Bar-graph

Answer: **A**

4. Which point lies only on y-axis?

A. (-2,0) B. (2,0) C. (0,-2) D. (2,-2)

Answer: **C**

5. If we join (-3,2), (-3,-3) and (-3,4), then we obtain:

A. A triangle B. Straight-line without passing through origin

C. Straight-line passing through origin D. None of the above

Answer: **B**

6. The point (4,0) lies on which of the following?

A. x-axis B. y-axis C. Origin D. None of the above

Answer: **A**

7. The point (-2,-2) is:

A. near to x-axis B. near to y-axis C. near to origin D. Equidistant from x-axis and y-axis.

Answer: **D**

8. The point (-2, 5) is nearer to:

A. x-axis B. y-axis C. Origin D. None of the above

Answer: **B**

9. The point (-5, 2) is nearer to:

A. x-axis B. y-axis C. Origin D. None of the above

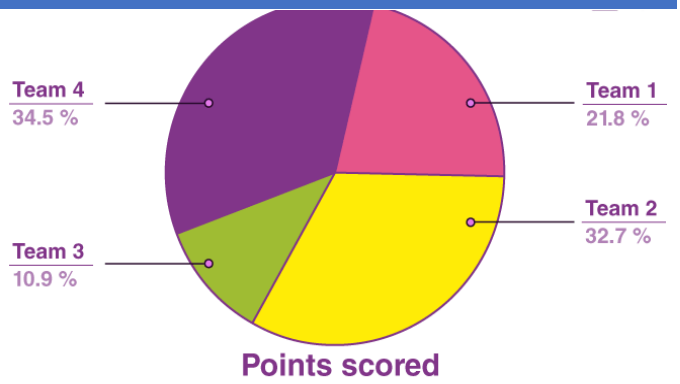
Answer: **A**

10. The point (0,0) lies at:

A. x-axis B. y-axis C. Origin D. None of the above

Answer: **C**

Observe the diagram, given below and find the correct answer to the following MCQs.



11. Which team has the highest score?

A. Team 1 B. Team 2 C. Team 3 D. Team 4

Answer: D Team 4

Explanation: Team 4 has the highest score with 34.5%.

12. Which team has the lowest score?

A. Team 1 B. Team 2 C. Team 3 D. Team 4

Answer: C. Team 3

Explanation: Team 3 has the lowest score with 10.9%.

13. Which team is coloured green?

A. Team 1 B. Team 2 C. Team 3 D. Team 4

Answer: C. Team 3

14. What is the average score of all the teams?

A. 22% B. 25% C. 27% D. 29%

Answer: B. 25%

Explanation: Average score = (sum of all scores)/(total number of scores)

15. Which team has the second-highest score?

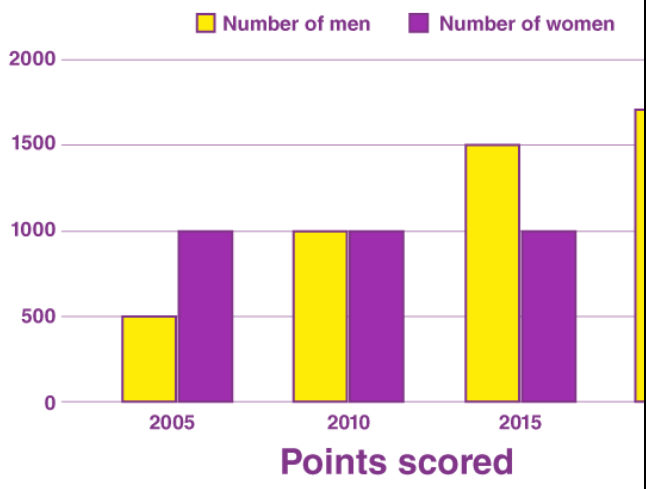
A. Team 1 B. Team 2 C. Team 3 D. Team 4

Answer: A. Team 2

Explanation: Team 2 has the second-highest score, equal to 32.7%.

Below is the data of the number of men and women in a village for different years.

Now based on this data answer the following MCQs with the correct option.



16. How many men were there in the village in 2010?

A. 500 B. 1000 C. 1500 D. 2000

Answer: B. 1000

Explanation: There were 1000 men in the year 2010 in the village.

17. The population of men and women in 2020 is the same?

A. True B. False

Answer: A. True

18. In which year the population of women is the highest?

A. 2005 B. 2010 C. 2015 D. 2020

Answer: D. 2020

19. When is the population of men, the minimum?

A. 2005 B. 2010 C. 2015 D. 2020

Answer: A. 2005

Explanation: In 2005, there were only 500 men in the village.

20. In which year population of men is more than women?

A. 2005 B. 2010 C. 2015 D. 2020

Answer: C. 2015

PLAYING WITH NUMBERS

1. The general form of abc is:

A. 100a + 10b + c B. 100b + 10c + a C. 100c + 10a + b D. None of the above

Answer: **A**

2. The generalised form of 129 is:

A. 100+90+2 B. 100+20+9 C. 100+2+9 D. None of the above

Answer: **B**

3. The usual form of 100 × 7 + 10 × 1 + 8 is:

A. 108 B. 708 C. 718 D. 170

Answer: **C**

Explanation: 100 × 7 + 10 × 1 + 8

= 700 + 10 + 8 = 718

4. Which of the following numbers are not divisible by 5?

A. 20 B. 125 C. 122 D. 200

Answer: **C**

Explanation: To be completely divisible by 5, the number should have 0 or 5 at it's one's digit place.

5. Which of the following numbers are divisible by 10?

A. 99 B. 45 C. 110 D. 75

Answer: **C**

Explanation: If any number has 0 at its one's digit place, then it is divisible by 10.

Hence, 110/10 = 11

6. Which of the following are divisible by 2?

A. 98 B. 99 C. 101 D. 121

Answer: **A**

Explanation: If one's digit of any number is even number, then the number is completely divisible by 2.

Hence, 98/2 = 49

7. If a number is divisible 9, then it is divisible by:

A. 6 B. 7 C. 3 D. 11

Answer: **C**

Explanation: If 99 is divisible by 9 then it is also divisible by 3.

99/9 = 11 and 99/3 = 33

8. If the three digit number 24x is divisible by 9, the value of x is:

A. 3 B. 7 C. 1 D. None of the above

Answer: **A**

Explanation: If 24x is divisible by 9, then:

2+4+x = 6+x is also divisible by 9.

It is possible only when, 6+x = 9, 18, …

Since x is single digit, therefore, 6+3 = 9 , x=3

9. The number 2146587 is divisible by:

A. 7 B. 3 C. 11 D. None of the above

Answer: **B**

Explanation: 2146587 = 2 + 1 + 4 + 6 + 5 + 8 + 7 = 33

Since, 33 is divisible by 3, hence 2146587 is divisible by 3.

10. The number 15287 is divisible by:

A. 3 B. 7 C. 9 D. None of the above

Answer: **D**

11. The general form of 1809 is:

A. 1 x 1000 + 8 x 100 + 9 B. 1800 + 9
C. 1000 + 809 D. None of the above

Answer: A

1809 = 1 x 1000 + 8 x 100 + 0 x 10 + 9

12. Which of the following numbers is divisible by 2?

A. 29 B. 19 C. 22 D. 47

Answer: C. 22

Explanation:
22/2 = 11

13. Find the value of Q.

```
  Q 1
+ 2 3
-----
4 4
-----
```

A. 1 B. 2 C. 3 D. 4

Answer: B. 2

```
2 1
+2 3
-----
4 4
-----
```

14. Suppose A is a digit. Find the value of A if 31A + 1A3 = 501.

A. 1 B. 2 C. 3 D. 4

Answer: A. 1

Explanation:
318 + 183 = 501

15. Which value is divisible by 5?

A. 36 B. 90 C. 26 D. 81

Answer: B.90
Explanation: At the unit place of 90 we have 0. Hence, as per the divisibility rule for 5, 90 is divisible by 5.

16. Which of the following is divisible by 9?

A. 108 B. 107 C. 109 D. 105

Answer: A. 108

Explanation: Sum of all digits of 108 = 1 + 0 + 8 = 9
9 is divisible by 9, thus, 108 is also divisible by 9.

17. Which of the following is divisible by 3?

A. 160 B. 170 C. 180 D. 190

Answer: C. 18

Explanation: Sum of digits of 180 = 1 + 8 + 0 = 9
Thus, 9 is divisible by 3.

18. If the three-digit number 80x is divisible by 9, what is the value of x?

A. 1 B. 9 C. 7 D. 4

Answer: A. 1
Explanation:
Given, 80x is divisible by 9.
Thus, sum of digits of 80x will also be divisible by 9.
8 + 0 + x is divisible by 9 only if, x = 1
8 + 0 + 1 = 9.

19. If the three-digit number 6 x 8 is divisible by 9, then what is the value of x?
A. 1 B. 2 C. 3 D. 4
Answer: D. 4

Explanation: 6 x 8 → 6 + x + 8 = 14 + x
14 + x is divisible by 9
14 + 4 = 18 is divisible by 9

20. Which number is divisible by 10?
A. 89 B. 70 C. 75 D. 15

Answer: B. 70
Explanation: 70/10 = 7

NUMBER SYSTEMS

1.) Can we write 0 in the form of p/q?

a. Yes b. No c. Cannot be explainedd. None of the above

Answer: a

Explanation: 0 is a rational number and hence it can be written in the form of p/q.

Example: 0/4 = 0

2.) The three rational numbers between 3 and 4 are:

a. 5/2, 6/2, 7/2 b. 13/4, 14/4, 15/4 c. 12/7, 13/7, 14/7 d.11/4, 12/4, 13/4

Answer: b

Explanation: There are many rational numbers between 3 and 4

To find 3 rational numbers, we need to multiply and divide both the numbers by 3+1 = 4

Hence, 3 x (4/4) = 12/4 and 4 x (4/4) = 16/4

Thus, three rational numbers between 12/4 and 16/4 are 13/4, 14/4 and 15/4.

3.) In between any two numbers, there are:

a. Only one rational number b. Two rational numbers c. Infinite rational numbers d. No rational number

Answer: c

Explanation: Take the reference from question number 2 explained above.

4.) Every rational number is:

a. Whole number b. Natural number c. Integer d. Real number

Answer: d

5.) √9 is _____ number.

a. A rational b. An irrational c. Neither rational nor irrational d. None of the above

Answer: a

Explanation: √9 = 3

Hence, √9 is a rational number.

6.) Which of the following is an irrational number?

a. √16 b. √(12/3) c. √12 d. √100

Answer: c

Explanation: √12 cannot be simplified to a rational number.

7.) 3√6 + 4√6 is equal to:

a. 6√6 b. 7√6 c. 4√12 d. 7√12

Answer: b

Explanation: 3√6 + 4√6 = (3 + 4)√6 = 7√6

8.) √6 x √27 is equal to:

a. 9√2 b. 3√3 c. 2√2 d. 9√3

Answer: a

Explanation: √6 x √27 = √(6 x 27) = √(2 x 3 x 3 x 3 x 3) = (3 x 3)√2 = 9√2

9.) Which of the following is equal to x^3?

a. $x^6 - x^3$ b. $x^6.x^3$ c. x^6/x^3 d. $(x^6)^3$

Answer: c

Explanation: $x^6/x^3 = x^{6-3} = x^3$

10.) Which of the following is an irrational number?

a. √23 b. √225 c. 0.3796 d. 7.478478

Answer: a

Explanation: √23 = 4.79583152331…

Since the decimal expansion of the number is non-terminating non-recurring.

Hence, it is an irrational number.

12.) 2√3+√3 =

a. 6 b. 2√6 c. 3√3 d. 4√6

Answer: c

Explanation: 2√3+√3 = (2+1)√3= 3√3.

.

16.) The irrational number between 2 and 2.5 is

a. √11 b. √5 c. √22.5 d. √12.5

Answer: b

Explanation: The irrational number between 2 and 2.5 is √5 because the approximate value of √5 is 2. 23606…

17.) The value of √10 times √15 is equal to

a. 5√6 b. √25 c. 10√5 d. √5

Answer: a

Explanation: √10×√15 =(√2.√5)×(√3. √5) = 5√6.

18.) The decimal representation of the rational number is

a. Always terminating b. Either terminating or repeating

c. Either terminating or non-repeating d. Neither terminating nor repeating

Answer: b

Explanation: As per the definition of rational number, its decimal representation is either terminating or repeating.

19.) Which of the following is a rational number?

a. 0 b. 2√3 c. 2+√3 d. π

Answer: a

Explanation: 0 is a rational number, and it can be written as 0/1 or 0/2 or 0/3 etc. Whereas 2√3, 2+√3, and π are irrational numbers.

20.) Which of the following is an irrational number?

a. √(4/9) b. √12/√3 c. √7 d. √81

Answer: c

Explanation: √7 is an irrational number. Because other options given are simplified into a rational number.

POLYNOMIALS

1) x^2-2x+1 is a polynomial in:

a. One Variable b. Two Variables c. Three variable d. None of the above

Answer: a

Explanation: x^2-2x+1 can be written as $x^2-2x^1+1x^0$. Hence, we can see that x is the only variable having powers as whole numbers: 2,1 and 0.

2) The coefficient of x^2 in $3x^3+2x^2-x+1$ is:

a. 1 b. 2 c. 3 d. -1

Answer: b

Explanation: The coefficient of x^2 in equation $3x^3+2x^2-x+1$ is the multiple of x^2.

3) A binomial of degree 20 in the following is:

a. 20x + 1 b. x/20 + 1 c. x^{20} +1 d. x^2+20

Answer: c

Explanation: A polynomial having two terms and the highest degree 20 is called a binomial of degree 20.

4) The degree of $4x^3-12x^2+3x+9$ is

a. 0 b. 1 c. 2 d. 3

Answer: d

Explanation: The degree is the highest power of a variable in an equation.

5) $x^2 - x$ is _____ polynomial.

a. Linear b. Quadratic c. Cubic d. None of the above

Answer: b

Explanation: A polynomial of degree two is known as a quadratic polynomial.

6) $x - x^3$ is a _____ polynomial.

a. Linear b. Quadratic c. Cubic d. None of the above

Answer: c

Explanation: A polynomial of degree three is known as a cubic polynomial.

7) 1+3x is a _____ polynomial.

a. Linear b. Quadratic c. Cubic d. None of the above

Answer: a

Explanation: A polynomial of degree one is known as a linear polynomial.

8) The value of $f(x) = 5x-4x^2+3$ when x = -1, is:

a. 3 b. -12 c. -6 d. 6

Answer: c

Explanation: When x= -1

$f(x)=5x-4x^2+3$

$f(-1)=5(-1)-4(-1)^2+3$

$=-5-4+3$

$=-6$

9) The value of $p(t) = 2+t+2t^2-t^3$ when t=0 is

a. 2 b. 1 c. 4 d. 0

Answer: a

Explanation: $p(0)=2+0+2(0)^2-(0)^3=2$

10) The zero of the polynomial f(x) = 2x+7 is

a. 2/7 b. -2/7 c. 7/2 d. -7/2

Answer: d

Explanation: f(x)=2x+7

$\Rightarrow 2x+7=0 \quad \Rightarrow 2x=-7 \quad \Rightarrow x=-7/2$

$\therefore x = -7/2$ is a zero polynomial of the polynomial f(x).

11) What is the degree of the polynomial √3?

a. 0 b. 1 c. ½ d. 2

Answer: c

Explanation: The polynomial √3 can also be written as $3^{1/2}$.

Hence, the degree of the polynomial √3 is 1/2.

12) The degree of the constant polynomial is

a. 0 b. 1 c. 2 d. 3

Answer: 0

Explanation: The degree of the constant polynomial is 0. For example, 3 is a constant polynomial that is equal to $3x^0$, and its degree is 0.

13) One of the linear factors of $3x^2+8x+5$ is

a. (x+1) b. (x-2) c. (x+2) d. (x-4)

Answer: a

Explanation: $3x^2+8x+5 = 3x^2+ 3x+5x+5$

$3x^2+8x+5 = 3x(x+1)+5(x+1)$

$3x^2+8x+5 = (3x+5)(x+1)$

Therefore, (x+1) is one of the factors of $3x^2+8x+5$.

14) The coefficient of x in $7x^2+6x-2$ is

a. 2 b. 6 c. -2 d. 7

Answer: b

Explanation: The coefficient of x in $7x^2+6x-2$ is 6. Because the number multiplied by x is 6.

15) Which of the following is an example of the quadratic polynomial?

a. 7x+3 b. $2x^2+x-1$ c. $x+3x^3-9$ d. None of the above

Answer: b

Explanation: $2x^2+x-1$ is a quadratic polynomial because the highest degree of the polynomial is 2.

16) Find the value of 7^2-5^2.

a. 22 b. 23 c. 24 d. 25

Answer: c

Explanation: 7^2-5^2 = 49 – 25 = 24.

17) If x^2+kx+6 = (x+2)(x+3) for all k, find the value of k.

a. -1 b. 1 c. 3 d. 5

Answer: d

Explanation: x^2+kx+6 = (x+2)(x+3)

x^2+kx+6 = x^2+3x+2x+6

x^2+kx+6 = x^2+5x+6

Hence, the value of k is 5.

18) What is the zero of the polynomial p(x)=cx+d?

a. –c b. –d c. -d/c d. d/c

Answer: c

Explanation: The zero of the polynomial p(x)= cx+d is -d/c.

cx+d = 0 ,cx = -d ,x = -d/c.

19) The zero of the polynomial p(x) = -5x+5 is

a. 0 b. -5 c. -1 d. 1

Answer: d

Explanation: p(x) = -5x+5

-5x+5 = 0 , -5x = -5, x = -5/-5 =1

20) which of the following is a constant polynomial?

a. 4x+1 b. 3 c. $2x^2$ d. 6x+3

Answer: b

Explanation: 3 is a constant polynomial, as 3 = $3x^0$.

Whereas 4x+1 and 6x+3 are linear polynomial and $2x^2$ is a quadratic polynomial.

COORDINATE GEOMETRY

1) The name of the horizontal line in the cartesian plane which determines the position of a point is called:

a. Origin b. X-axis c. Y-axis d. Quadrants

Answer: **b**

2) The name of the vertical line in the cartesian plane which determines the position of a point is called:

a. Origin b. X-axis c. Y-axis d. Quadrants

Answer: **c**

3) The section formed by horizontal and vertical lines determining the position of the point in a cartesian plane is called:

a. Origin b. X-axis c. Y-axis d. Quadrants

Answer: **d**

4) The point of intersection of horizontal and vertical lines determining the position of a point in a cartesian plane is called:

a. Origin b. X-axis c. Y-axis d. Quadrants

Answer: **a**

5) If the coordinates of a point are (0, -4), then it lies in:

a. X-axis b. Y-axis c. At origin d. Between x-axis and y-axis

Answer: **b**

Explanation: Since, x=0 and y=-4. Hence, the point will lie in the negative y-axis 4 units far from the origin.

6) If the coordinates of a point are (3, 0), then it lies in:

a. X-axis b. Y-axis c. At origin d. Between x-axis and y-axis

Answer: **a**

Explanation: Since, x = 3 and y = 0, therefore, the point will lie at the positive x-axis 3 units far from the origin.

7) If the coordinates of a point are (-3, 4), then it lies in:

a. First quadrant b. Second quadrant c. Third quadrant d. Fourth quadrant

Answer: **b**

Explanation: Since x = -3 and y = 4, then if we plot the point in a plane, it lies in the second quadrant.

8) If the coordinates of a point are (-3, -4), then it lies in:

a. First quadrant b. Second quadrant c. Third quadrant d. Fourth quadrant

Answer: **c**

Explanation: Since, x = -3 and y = -4, then if we plot the point in a plane, it lies in the third quadrant.

9) Points (1, 2), (-2, -3), (2, -3);

a. First quadrant b. Do not lie in the same quadrant c. Third quadrant d. Fourth quadrant

Answer: **b**

10) If x coordinate of a point is zero, then the point lies on:

a. First quadrant b. Second quadrant c. X-axis d. Y-axis

Answer: **d**

11) Signs of the abscissa and ordinate of a point in the second quadrant are respectively

a. +, + b. +, – c. -, + d. -, –

Answer: **c**

Explanation: The signs of abscissa (x-value) and ordinate(y-value) in the second quadrant are – and + respectively.

12) The point (-10, 0) lies in

a. Third quadrant b. Fourth quadrant c. On the negative direction of the x-axis d. On the negative direction of the y-axis

Answer: **c**

Explanation: The lies (-10, 0) lies in the negative direction of the x-axis.

13) A quadrant in which both x and y values are negative is

a. First quadrant b. Second quadrant c. Third quadrant d. Fourth quadrant

Answer: **c**

Explanation: In the third quadrant, both the abscissa and ordinate values are negative.

Example (-2, -2), which lies in the third quadrant.

14) Abscissa of all the points on the x-axis is

a. 0 b. 1 c. 2 d. Any number

Answer: **d**

Explanation: Abscissa of all the points on the x-axis can be any number.

The coordinates of any point on the x-axis is (x, 0), where x can take any value.

15) Ordinate of all points on the x-axis is

a. -1 b. 0 c. 1 d. Any number

Answer: **b**

Explanation: The ordinate of all points on the x-axis is 0. We know that the coordinates of any point on the x-axis is (x, 0).

Here, the abscissa can take any value and the ordinate is zero.

16) Abscissa of a point is positive in

a. I quadrant b. I and II quadrants c. II quadrant only d. I and IV quadrants

Answer: **d**

Explanation: In a coordinate plane, x can take positive values in the first and fourth quadrants.

For example, (2, 2) and (2, -4) lies on the first and fourth quadrants respectively.

17) Points (1, -1), (2, -2), (4, -5), (-3, -4)

a. lie in II quadrant b. lie in III quadrant c. lie in IV quadrant d. Does not lie in the same quadrant

Answer: **d**

Explanation: The point (1, -1), (2, -2) and (4, -5) lies in the fourth quadrant, where (-3, -4) lies in the third quadrant.

18) Abscissa of all the points on the y-axis is

a. 0 b. 1 c. -1 d. Any number

Answer: **a**

Explanation: The abscissa of all the points on the y-axis is 0. We know that the coordinates of any point on the y-axis is (0, y). Here, the ordinate can take any value and the abscissa is zero.

19) Ordinate of all the points on the y-axis is

a. 0 b. 1 c. -1 d. Any number

Answer: **d**

Explanation: The ordinate of all the points on the y-axis can be any number.

The coordinates of any point on the y-axis is (0, y).

Here, abscissa can take only the value of 0 and the ordinate can take any value.

20) The point which lies on the y-axis at a distance of 5 units in the negative direction of the y-axis is

a. (5, 0) b. (0, 5) c. (-5, 0) d. (0, -5)

Answer: **d**

Explanation: The coordinates of any point on the y-axis is (0, y).

Hence, abscissa should be 0.

Given that, the point lies in the negative direction of the y-axis. Hence, the value of y should be negative.

Therefore, the point which lies on the y-axis at a distance of 5 units in the negative direction is (0, -5).

COORDINATE GEOMETRY-2

1. If the distance between the points A(2, -2) and B(-1, x) is equal to 5, then the value of x is:

(a) 2 (b) -2 (c) 1 (d) -1

Answer: **(a) 2**

Explanation: $(-1-2)^2+(x+2)^2 = (5)^2$

$9+(x+2)^2=25$, $(2+x)^2=16$

Take square root on both the sides,

$2+x=4$, $x=2$

2. The midpoints of a line segment joining two points A(2, 4) and B(-2, -4)

(a) (-2,4) (b) (2,-4) (c) (0, 0) (d) (-2,-4)

Answer: **(c) (0, 0)**

Explanation: As per midpoint formula, we know;

$x=[2+(-2)]/2 = 0/2 = 0$

$y=[4+(-4)]/2=0/2=0$

3. The distance of point A(2, 4) from x-axis is

(a) 2 units (b) 4 units (c) -2 units (d) -4 units

Answer: **(b) 4 units**

Explanation: Distance of a point from x-axis is equal to the ordinate of the point.

4. The distance between the points P(0, 2) and Q(6, 0) is

(a) $4\sqrt{10}$ (b) $2\sqrt{10}$ (c) $\sqrt{10}$ (d) 20

Answer: **(b) $2\sqrt{10}$**

Explanation: $PQ = \sqrt{[(6-0)^2+(0-2)^2]}$

$PQ = \sqrt{(6^2+2^2)}$

$PQ=\sqrt{(36+4)}$, $PQ=\sqrt{40}$ $=2\sqrt{10}$

5. If O(p/3, 4) is the midpoint of the line segment joining the points P(-6, 5) and Q(-2, 3). The value of p is:

(a) 7/2 (b) -12 (c) 4 (d) -4

Answer: **(b) -12**

Explanation: Since, (p/3, 4) is the midpoint of line segment PQ, thus;

$p/3 = (-6-2)/2$, $p/3 = -8/2$, $p/3 = -4$, $p= -12$

6.The ratio in which the line segment joining the points P(-3, 10) and Q(6, – 8) is divided by O(-1, 6) is:

(a) 1:3 (b) 3:4 (c) 2:7 (d) 2:5

Answer: **(c) 2:7**

Explanation: Let the ratio in which the line segment joining P(-3, 10) and Q(6, -8) is divided by point O(-1, 6) be k :1.

So, -1 = (6k-3)/(k+1) , –k – 1 = 6k -3 ,
7k = 2 , k = 2/7

7. The coordinates of a point P, where PQ is the diameter of a circle whose centre is (2, – 3) and Q is (1, 4) is:

(a) (3, -10) (b) (2, -10) (c) (-3, 10) (d) (-2, 10)

Answer: **(a) (3, -10)**

Explanation: By midpoint formula, we know;

[(x+1)/2,(y+4)/2] = (2,-3) {since, O is the midpoint of PQ}
By equating the corresponding coordinates,

(x+1)/2 = 2 , x+1=4 , x=3

And ,(y+4)/2 = -3 , y+4=-6 , y=-10

So, the coordinates of point P is (3, -10).

8. The distance of the point P(–6, 8) from the origin is

(a) 8 units (b) $2\sqrt{7}$ units (c) 10 units (d) 6 units

Answer: **(c) 10 units**

Explanation: The distance of the point P(-6, 8) from the origin = $\sqrt{[(-6)^2 + (8)^2]}$

= $\sqrt{(36 + 64)}$ = $\sqrt{100}$ = 10 units

LINES AND ANGLES

1) A line joining two endpoints is called:

a. Line segment b. A ray c. Parallel lines d. Intersecting lines

Answer: **a**

2) An acute angle is:

a. More than 90 degrees b. Less than 90 degrees c. Equal to 90 degrees d. Equal to 180 degrees

Answer: **b**

3) A reflex angle is:

a. More than 90 degrees b. Equal to 90 degrees c. More than 180 degrees d. Equal to 180 degrees

Answer: **c**

4) A straight angle is equal to:

a. 0° b. 90° c. 180° d. 360°

Answer: **c**

5) Two angles whose sum is equal to 180° are called:

a. Vertically opposite angles b. Complementary angles c. Adjacent angles d. Supplementary angles

Answer: **d**

6) Intersecting lines cut each other at:

a. One point b. Two points c. Three points d. Null

Answer: **a**

Explanation: Two lines always intersect each other at one point.

7) Two parallel lines intersect at:

a. One point b. Two points c. Three points d. Null

Answer: **d**

Explanation: If two lines are parallel to each other, they don't intersect each other.

8) If two lines intersect each other, then the vertically opposite angles are:

a. Equal b. Unequal c. Cannot be determined d. None of the above

Answer: **a**

Explanation: If two lines intersect each other, then the angles formed at the point of intersection are vertically opposite angles and are equal.

9) An exterior angle of a triangle is 105° and its two interior opposite angles are equal. Each of these equal angles is

(a) 37 ½° (b) 72 ½° (c) 75° (d) 52 ½°

Answer: **d**

Explanation:

The exterior angle of triangle = 105°

Let the interior angles be "x".

By using, exterior angle theorem, Exterior angle = Sum of interior opposite angles

Therefore, 105° = x+x

2x = 105°

x = 52 ½°

10) If one of the angles of a triangle is 130°, then the angle between the bisectors of the other two angles can be

(a) 50° (b) 65° (c) 145° (d) 155°

Answer: **d**

11) If two interior angles on the same side of a transversal intersecting two parallel lines are in the ratio 2 : 3, then the greater of the two angles is

(a) 54°

(b) 108°

(c) 120°

(d) 136°

Answer: **b**

12) If one angle of a triangle is equal to the sum of the other two angles, then the triangle is

(a) a right triangle (b) an isosceles triangle (c) an equilateral triangle (d) an obtuse triangle

Answer: **a**

Explanation: If one angle of a triangle is equal to the sum of the other two angles, then the triangle is a right triangle. We know that the sum of interior angles of a triangle is equal to 180°. In the right triangle, one angle should be equal to 90°, and the remaining two angles are acute angles, and their sum is equal to 90°.

13) The angles of a triangle are in the ratio 5 : 3: 7. The triangle is

(a) a right triangle (b) an acute-angled triangle (c) an obtuse-angled triangle (d) an isosceles triangle

Answer: **b**

Explanation: If the angles are in the ratio of 5:3:7, then a triangle is an acute angle triangle.

We know that the sum of the interior angles of a triangle is 180°

Therefore, 5x+3x+7x = 180° , 15x = 180° , x = 180°/15 = 12°

Thus, 5x = 5(12°) =60° , 3x = 3(12°) =36° , 7x = 7(12°) =84°

Since all the angles are less than 90°, the triangle is an acute angle triangle.

14) Angles of a triangle are in the ratio 2: 4 : 3. The smallest angle of the triangle is

(a) 20° (b) 40° (c) 60° (d) 80°

Answer: **b**

Explanation:

We know that the sum of the interior angles of a triangle is 180°

Given that, the angles of a triangle are in the ratio of 2:4:3

Hence, 2x+4x+3x = 180°

9x = 180° , x= 20°

Therefore, 2x = 2(20) = 40°, 4x = 4(20) = 80°, 3x = 3(20) = 60°

Hence, the angles are 40°, 80° and 60°.

Therefore, the smallest angle of a triangle is 40°.

15) An obtuse angle is

(a) Less than 90° (b) Greater than 90° (c) Equal to 90° (d) Equal to 180°

Answer: **b**

Explanation: An obtuse angle is an angle that is greater than 90°

TRIANGLES

1) In triangle ABC, if AB=BC and ∠B = 70°, ∠A will be:

a. 70° b. 110° c. 55° d. 130°

Answer: **c**

2) For two triangles, if two angles and the included side of one triangle are equal to two angles and the included side of another triangle. Then the congruency rule is:

a. SSS b. ASA c. SAS d. None of the above

Answer: **b**

3) A triangle in which two sides are equal is called:

a. Scalene triangle b. Equilateral triangle c. Isosceles triangle d. None of the above

Answer: **c**

4) The angles opposite to equal sides of a triangle are:

a. Equal b. Unequal c. supplementary angles d. Complementary angles

Answer: **a**

5) If E and F are the midpoints of equal sides AB and AC of a triangle ABC. Then:

a. BF=AC b. BF=AF c. CE=AB d. BF = CE

Answer: **d**

Explanation: AB and AC are equal sides.

AB = AC (Given)

∠A = ∠A (Common angle)

AE = AF (Halves of equal sides)

Δ ABF ≅ Δ ACE (By SAS rule)

Hence, BF = CE (CPCT)

6) If ABC is an equilateral triangle, then each angle equals to:

a. 90° B.180° c. 120° d. 60°

Answer: **d**

7) If AD is an altitude of an isosceles triangle ABC in which AB = AC. Then:

a. BD=CD b. BD>CD c. BD<CD d. None of the above

Answer: **a**

8) In a right triangle, the longest side is:

a. Perpendicular b. Hypotenuse c. Base d. None of the above

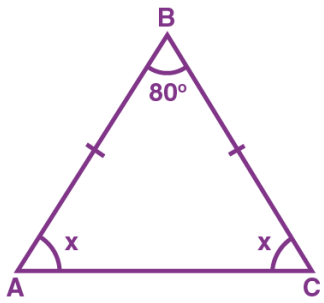
Answer: **b**

9) In Δ ABC, BC = AB and ∠B = 80°. Then ∠A is equal to

(a) 40° (b) 50° (c) 80° (d) 100°

Answer: **b**

Explanation: In a triangle, ABC, BC = AB and ∠B = 80°.



10) Two sides of a triangle are of lengths 5 cm and 1.5 cm. The length of the third side of the triangle cannot be

(a) 3.4 cm (b) 3.6 cm (c) 3.8 cm (d) 4.1 cm

Answer: **a**

Explanation: If two sides of a triangle are of lengths 5 cm and 1.5 cm, then the length of the third side of the triangle cannot be 3.4 cm.

Because the difference between the two sides of a triangle should be less than the third side.

11) In Δ ABC, AB = AC and ∠B = 50°. Then ∠C is equal to

(a) 40° (b) 50° (c) 80° (d) 130°

Answer: **b**

Explanation:

Given that, in a triangle ABC, AB = AC and ∠B = 50°.

Since the given triangle is an isosceles triangle, the angles opposite to the equal sides are also equal. Hence, ∠C = 50°.

QUADRILATERALS

1) The quadrilateral whose all its sides are equal and angles are equal to 90 degrees, it is called:

a. Rectangle b. Square c. Kite d. Parallelogram

Answer: **b**

2) The sum of all the angles of a quadrilateral is equal to:

a. 180° b. 270° c. 360° d. 90°

Answer: **c**

3) A trapezium has:

a. One pair of opposite sides parallel
b. Two pairs of opposite sides parallel to each other

c. All its sides are equal
d. All angles are equal

Answer: **a**

Explanation: A trapezium has only one pair of opposite sides parallel to each other, and the other two sides are non-parallel.

4) A rhombus can be a:

a. Parallelogram b. Trapezium c. Kite d. Square

Answer: **d**

5) A diagonal of a parallelogram divides it into two congruent:

a. Square b. Parallelogram c. Triangles d. Rectangle

Answer: **c**

6) In a parallelogram, opposite angles are:

a. Equal b. Unequal c. Cannot be determined d. None of the above

Answer: **a**

7) The diagonals of a parallelogram:

a. Equal b. Unequal c. Bisect each other d. Have no relation

Answer: **c**

8) Each angle of the rectangle is:

a. More than 90° b. Less than 90° c. Equal to 90° d. Equal to 45°

Answer: **c**

9) The angles of a quadrilateral are in the ratio 4: 5: 10: 11. The angles are:

a. 36°, 60°, 108°, 156° b. 48°, 60°, 120°, 132°
c. 52°, 60°, 122°, 126° d. 60°, 60°, 120°, 120°

Answer: **b**

10) If ABCD is a trapezium in which AB || CD and AD = BC, then:

a. ∠A = ∠B b. ∠A > ∠B c. ∠A < ∠B d. None of the above

Answer: **a**

11) Which of the following is not true for a parallelogram?

(a) Opposite sides are equal
(b) Opposite angles are equal

(c) Opposite angles are bisected by the diagonals
(d) Diagonals bisect each other.

Answer: **c**

12) Three angles of a quadrilateral are 75º, 90º and 75º. The fourth angle is

(a) 90º (b) 95º (c) 105º (d) 120º

Answer: **d**

Explanation: We know that the sum of angles of a quadrilateral is 360º.

Let the unknown angle be x.

Therefore, 75º+90º+75º+x = 360º

x = 360º – 240º = 120º.

13) ABCD is a rhombus such that ∠ACB = 40º. Then ∠ADB is

(a) 40º (b) 45º (c) 50º (d) 60º

Answer: **c**

Explanation: We know that the diagonals of the rhombus bisect each other perpendicularly.

14) The quadrilateral formed by joining the mid-points of the sides of a quadrilateral PQRS, taken in order, is a rhombus, if

(a) PQRS is a rhombus
(b) PQRS is a parallelogram

(c) Diagonals of PQRS are perpendicular
d) Diagonals of PQRS are equal.

Answer: **d**

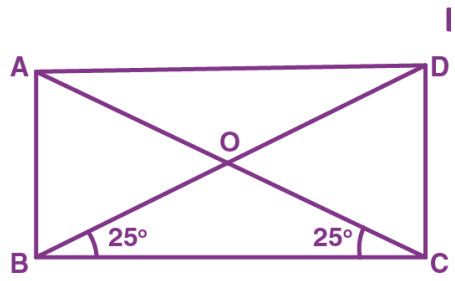
Explanation: The quadrilateral formed by joining the mid-points of the sides of a quadrilateral PQRS, taken in order, is a rhombus if the diagonals of PQRS are equal.

15) A diagonal of a rectangle is inclined to one side of the rectangle at 25º. The acute angle between the diagonals is

(a) 25º (b) 40º (c) 50º (d) 55º

Answer: **c**

Explanation: Consider the rectangle ABCD



16) If angles A, B, C and D of the quadrilateral ABCD, taken in order, are in the ratio 3:7:6:4, then ABCD is a

(a) Kite (b) Rhombus (c) Parallelogram (d) Trapezium

Answer: **d**

17) Which of the following is not a quadrilateral?

(a) Kite (b) Square (c) Triangle
(d) Rhombus

Answer: **c**

Explanation: square, kite and rhombus are quadrilaterals as it has four sides.

Whereas a triangle is not a quadrilateral as it has only three sides.

AREAS OF PARALLELOGRAMS AND TRIANGLES

1) If ABCD is a parallelogram, AE ⊥ DC and CF ⊥ AD. If AB = 10 cm, AE = 6 cm and CF = 5 cm, then AD is equal to:

a. 10cm B.6cm c. 12cm d. 15cm

Answer: **c**

2) If E, F, G and H are the mid-points of the sides of a parallelogram ABCD, respectively, then ar (EFGH) is equal to:

a. 1/2 ar(ABCD) b. ¼ ar(ABCD) c. 2 ar(ABCD) d. ar(ABCD)

Answer: **a**

3) If P and Q are any two points lying on the sides DC and AD respectively of a parallelogram ABCD, then:

a. ar(BQC) b. ar(APB) < ar(BQC)
c. ar(APB) = ar(BQC) d. None of the above

Answer: **c**

4) If ABCD and EFGH are two parallelograms between same parallel lines and on the same base, then:

a. ar (ABCD) > ar (EFGH) b. ar (ABCD) < ar (EFGH) c. ar (ABCD) = ar (EFGH) d. None of the above

Answer: **c**

5) A median of a triangle divides it into two

a. Congruent triangles b. Isosceles triangles c. Right triangles d. Equal area triangles

Answer: **d**

6) In a triangle ABC, E is the mid-point of median AD. Then:

a. ar(BED) = 1/4 ar(ABC) b. ar(BED) = ar(ABC) c. ar(BED) = 1/2 ar(ABC) d. ar(BED) = 2 ar(ABC)

Answer: **a**

7) If D and E are points on sides AB and AC respectively of ΔABC such that ar(DBC) = ar(EBC). Then:

a. DE is equal to BC b. DE is parallel to BC c. DE is not equal to BC d. DE is perpendicular to BC

Answer: **b**

Explanation: ΔDBC and ΔEBC are on the same base BC and also having equal areas.

∴ They will lie between the same parallel lines.

∴ DE || BC.

8) If Diagonals AC and BD of a trapezium ABCD with AB || DC intersect each other at O. Then,

a. ar (AOD) = ar (BOC) b. ar (AOD) > ar (BOC) c. ar (AOD) < ar (BOC) d. None of the above

Answer: **a**

9) If Diagonals AC and BD of a quadrilateral ABCD intersect at O in such a way that ar(△AOD) = ar(△BOC). Then ABCD is a:

a. Parallelogram b. Rectangle c. Square d. Trapezium

Answer: **d**

10) If a triangle and a parallelogram are on the same base and between same parallels, then the ratio of the area of the triangle to the area of the parallelogram will be:

a. 1:2 b. 3:2 c. 1:4 d. 1:3

Answer: **a**

11) The area of the figure formed by joining the mid-points of the adjacent sides of a rhombus with diagonals 12 cm and 16 cm is

(a) 48 cm^2 (b) 96 cm^2 (c) 64 cm^2 (d) 192 cm^2

Answer: **a**

12) Parallelogram is a

(a) Pentagon (b) Quadrilateral c) Heptagon (d) Octagon

Answer: **b**

Explanation: Parallelogram is a quadrilateral, as it has four sides.

13) The figure obtained by joining the mid-points of the adjacent sides of a rectangle of sides 8 cm and 6 cm is

(a) a square of area 25 cm^2 (b) a rectangle of area 24 cm^2 (c) a rhombus of area 24 cm^2 (d) a trapezium of area 24 cm^2

Answer: **c**

14) Two parallelograms are on equal bases and between the same parallels. The ratio of their areas is

(a) 1: 1 (b) 1: 2 (c) 2: 1 (d) 3: 1

Answer: **a**

Explanation: We know that the parallelogram on the equal bases and between the same parallels are equal in area. Hence, the ratio of their areas is 1 :1.

15) The mid-point of the sides of a triangle along with any of the vertices as the fourth point make a parallelogram of area equal to

(a) ar (ABC) (b) (1/2) ar(ABC) (c) (1/3) ar(ABC) (d) (1/4) ar(ABC)

Answer: **b**

16) The area of a parallelogram with base "b" and height "h" is

(a) b×h square units (b) b^2 square units (c) h^2 square units (d) b+h square units

Answer: **a**

Explanation: The area of a parallelogram with base "b" and height "h" is b×h.(i.e) A = base × height square units.

17) ABCD is a quadrilateral whose diagonal AC divides it into two parts, equal in area, then ABCD

(a) is always a rhombus (b) is a rectangle (c) is a parallelogram (d) need not be any of (A), (B) or (C)

Answer: **d**

Explanation: A quadrilateral ABCD need not be any of rhombus, parallelogram or rectangle.

If ABCD is a square, its diagonal AC divides it into two parts and are equal in area.

18) Triangles on the same base and between the same parallels are

(a) Greater in area (b) Equal in area (c) smaller in area (d) None of the above

Answer: **b**

Explanation: Triangles on the same base and between the same parallels are equal in area.

CIRCLES

1) The center of the circle lies in_____ of the circle.

a. Interior b. Exterior c. Circumference d. None of the above

Answer: **a**

2) The longest chord of the circle is:

a. Radius b. Arc c. Diameter d. Segment

Answer: **c**

3) Equal _____ of the congruent circles subtend equal angles at the centers.

a. Segments b. Radii c. Arcs d. Chords

Answer: **d**

4) If chords AB and CD of congruent circles subtend equal angles at their centres, then:

a. AB = CD b. AB > CD c. AB < AD d . None of the above

Answer: **a**

5) If there are two separate circles drawn apart from each other, then the maximum number of common points they have:

a. 0 b. 1 c. 2 d. 3

Answer: **a**

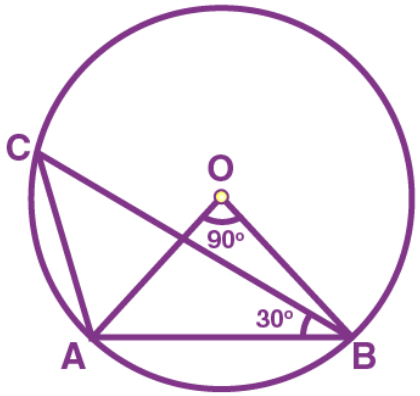
6) The angle subtended by the diameter of a semi-circle is:

a. 90 b. 45 c. 180 d. 60

Answer: **c**

Explanation: The semicircle is half of the circle, hence the diameter of the semicircle will be a straight line subtending 180 degrees.

11) In the given figure, ∠AOB = 90º and ∠ABC = 30º, then ∠CAO is equal to:



(a) 30º (b) 45º (c) 60º (d) 90º

Answer: **c**

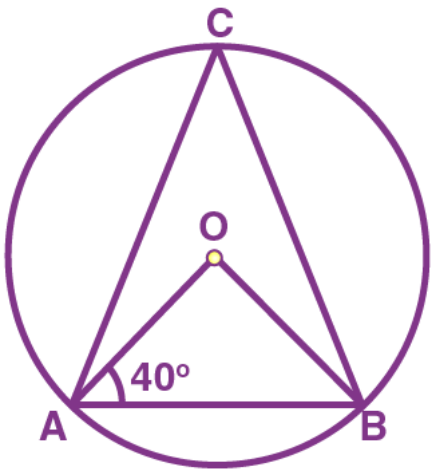
12) ABCD is a cyclic quadrilateral such that AB is a diameter of the circle circumscribing it and ∠ADC = 140º, then ∠BAC is equal to:

(a) 30º (b) 40º (c) 50º (d) 80º

Answer: **c**

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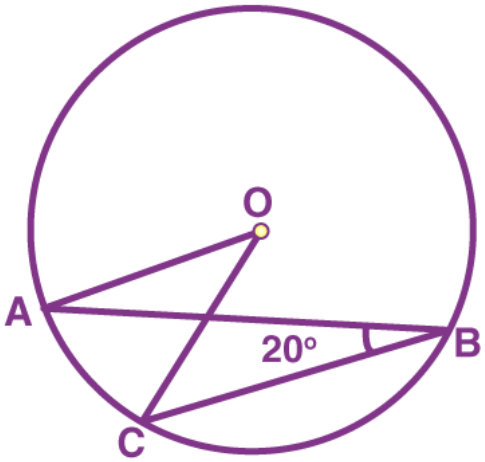
13) In the given figure, if ∠OAB = 40º, then ∠ACB is equal to



(a) 40º (b) 50º (c) 60º (d) 70º

Answer: **b**

14) In the given figure, if ∠ABC = 20º, then ∠AOC is equal to:



(a) 10º (b) 20º (c) 40º (d) 60º

Answer: **c**

Explanation:

Given that, ∠ABC = 20º.

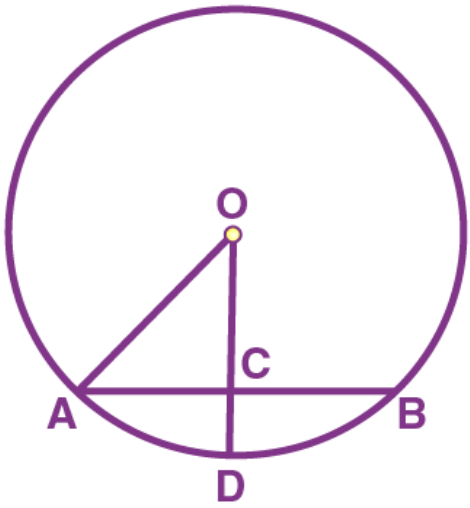
∠AOC = 2∠ABC (since the angle subtended by an arc at the centre of the circle is double the angle subtended at the remaining part.)

Now, substitute the values, we get

∠AOC = 2 × 20°

Therefore, ∠AOC = 40°.

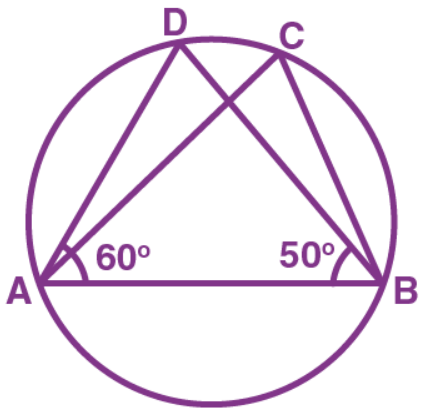
15) In the given figure, if OA = 5 cm, AB = 8 cm and OD is perpendicular to AB, then CD is equal to:



(a) 2 cm (b) 3 cm (c) 4 cm (d) 5 cm

Answer: **a**

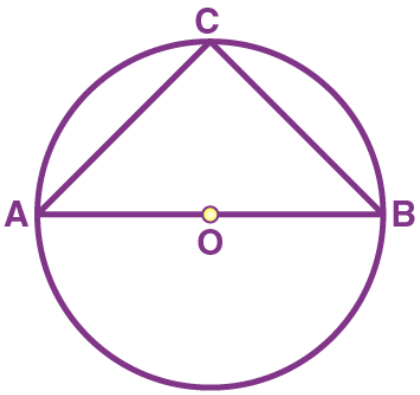
17) In the given figure, if ∠DAB = 60º, ∠ABD = 50º, then ∠ACB is equal to:



(a) 50º (b) 60º (c) 70º (d) 80º

Answer: **c**

18) In the given figure, if AOB is a diameter of the circle and AC = BC, then ∠CAB is equal to:



(a) 30º (b) 45º (c) 60º (d) 90º

19) If AB = 12 cm, BC = 16 cm and AB is perpendicular to BC, then the radius of the circle passing through the points A, B and C is:

(a) 6 cm (b) 8 cm (c) 10 cm (d) 12 cm

Answer: **c**

20) AD is the diameter of a circle and AB is a chord. If AD = 34 cm, AB = 30 cm, the distance of AB from the centre of the circle is

(a) 4 cm (b) 8 cm (c) 15 cm (d) 17 cm

Answer: **b**

CIRCLES-2

1. A circle has a number of tangents equal to

(a) 0 (b) 1 (c) 2 (d) Infinite

Answer: **(d) Infinite**

Explanation: A circle has infinitely many tangents, touching the circle at infinite points on its circumference.

2. A tangent intersects the circle at:

(a) One point (b) Two distinct point (c) At the circle (d) None of the above

Answer: **(a) One point**

Explanation: A tangent touches the circle only on its boundary and do not cross through it.

3. A circle can have _____parallel tangents at a single time.

(a) One (b) Two (c) Three (d) Four

Answer: (b) Two

Explanation: A circle can have two parallel tangents at the most.

4. If the angle between two radii of a circle is 110º, then the angle between the tangents at the ends of the radii is:

(a) 90º (b) 50º (c) 70º (c) 40º

Answer: **(c) 70º**

Explanation: If the angle between two radii of a circle is 110º, then the angle between tangents is 180º − 110º = 70º.

5. The length of the tangent from an external point A on a circle with centre O is

(a) always greater than OA (b) equal to OA (c) always less than OA (d) cannot be estimated

Answer: **(c) always less than OA**

Explanation: Since the tangent is perpendicular to the radius of the circle, then the angle between them is 90º.

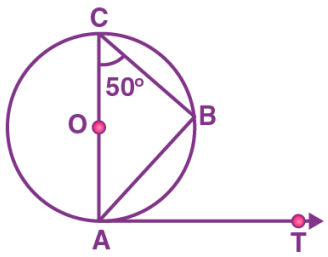
Thus, OA is the hypotenuse for the right triangle OAB, which is right-angled at B.

As we know, for any right triangle, the hypotenuse is the longest side. Therefore the length of the tangent from an external point is always less than the OA.

6. AB is a chord of the circle and AOC is its diameter such that angle ACB = 50°. If AT is the tangent to the circle at the point A, then BAT is equal to

(a) 65° (b) 60° (c) 50° (d) 40°

Answer: **(c) 50°**



$\angle A = 180° - (90° + 50°)$

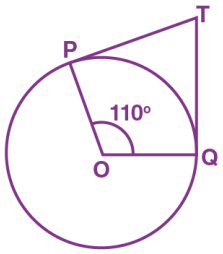
$\angle A = 40°$

Or $\angle OAB = 40°$, Therefore, $\angle BAT = 90° - 40° = 50°$

7. If TP and TQ are the two tangents to a circle with centre O so that $\angle POQ = 110°$, then $\angle PTQ$ is equal to

(a) 60° (b) 70° (c) 80° (d) 90°

Answer: **(b) 70°**



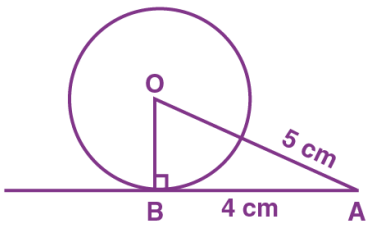
$\Rightarrow \angle PTQ + 90° + 110° + 90° = 360°$

$\Rightarrow \angle PTQ = 70°$

8. The length of a tangent from a point A at a distance 5 cm from the centre of the circle is 4 cm. The radius of the circle is:

(a) 3 cm (b) 5 cm (c) 7 cm (d) 10 cm

Answer: **(a) 3 cm**



Now, In $\triangle ABO$,

$OA^2 = AB^2 + BO^2$ (Using Pythagoras theorem)

$\Rightarrow 5^2 = 4^2 + BO^2$, $\Rightarrow BO^2 = 25 - 16$,$\Rightarrow BO^2 = 9$

$\Rightarrow BO = 3$

9. If a parallelogram circumscribes a circle, then it is a:

(a) Square (b) Rectangle (c) Rhombus (d) None of the above

Answer: **(c) Rhombus**

10. If angle between two radii of a circle is 130°, the angle between the tangents at the ends of the radii is

(a) 90° (b) 50° (c) 70° (d) 40°

Answer: **(b) 50°**

Explanation: We know that the sum of the angle between two radii of a circle and the angle between the tangents at the ends of the radii is 180°.

Therefore, the angle between the tangents at the ends of the radii = 180° – 130° = 50°

11. A line intersecting a circle in two points is called a _____.

(a) Secant (b) Chord (c) Diameter (d) Tangent

Answer: **(a) Secant**

A line intersecting a circle in two points is called a Secant.

12. The tangent to a circle is _____ to the radius through the point of contact.

(a) parallel (b) perpendicular (c) perpendicular bisector (d) bisector

Answer: **(b) perpendicular**

The tangent to a circle is perpendicular to the radius through the point of contact.

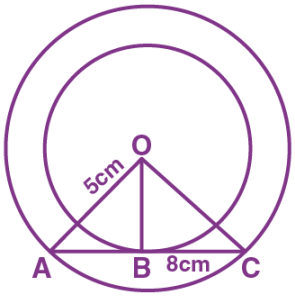
13. Out of the two concentric circles, the radius of the outer circle is 5 cm and the chord AC of length 8 cm is a tangent to the inner circle. The radius of the inner circle will be

(a) 3 cm (b) 4 cm (c) 2.5 cm (d) 2 cm

Answer: **(a) 3 cm**

OA = 5 cm and AC = 8 cm

AC is a chord which touches the inner circle at point B.



Thus, in right angled ∆AOB,

$OA^2 = AB^2 + BO^2$

$BO^2 = 5^2 - 4^2 = 25 - 16 = 9$,BO = 3 cm

14. The distance between two parallel tangents of a circle is 18 cm, then the radius of the circle is

(a) 8 cm (b) 10 cm (c) 9 cm (d) 7.5 cm

Answer: (c) 9 cm

Given,

Distance between two parallel tangents = 18 cm

That means, diameter = 18 cm

Therefore, radius of the circle = 18/2 = 9 cm

AREAS RELATED TO CIRCLES

1. The perimeter of a circle having radius 5cm is equal to:

(a) 30 cm (b) 3.14 cm (c) 31.4 cm (d) 40 cm

Answer: **(c) 31.4 cm**

Explanation: The perimeter of the circle is equal to the circumference of the circle.

Circumference = $2\pi r$ = 2 x 3.14 x 5 = 31.4 cm

2. Area of the circle with radius 5cm is equal to:

(a) 60 sq.cm (b) 75.5 sq.cm (c) 78.5 sq.cm (d) 10.5 sq.cm

Answer: **(c) 78.5 sq.cm**

Explanation: Radius = 5cm

Area = πr^2 = 3.14 x 5 x 5 = 78.5 sq.cm

3. The largest triangle inscribed in a semi-circle of radius r, then the area of that triangle is;

(a) r^2 (b) $1/2r^2$ (c) $2r^2$ (d) $\sqrt{2}r^2$

Answer: **(a) r^2**

Explanation: The height of the largest triangle inscribed will be equal to the radius of the semi-circle and base will be equal to the diameter of the semi-circle.

Area of triangle = ½ x base x height

= ½ x 2r x r = r^2

4. If the perimeter of the circle and square are equal, then the ratio of their areas will be equal to:

(a) 14:11 (b) 22:7 (c) 7:22 (c) 11:14

Answer: **(a) 14:11**

Explanation: Given,

The perimeter of circle = perimeter of the square

$2\pi r = 4a$, $a = \pi r/2$

Area of square = $a^2 = (\pi r/2)^2$, $A_{circle}/A_{square} = \pi r^2/(\pi r/2)^2$, = 14/11

5. The area of the circle that can be inscribed in a square of side 8 cm is

(a) 36 π cm^2 (b) 16 π cm^2 (c) 12 π cm^2 (d) 9 π cm^2

Answer: **(b) 16 π cm^2**

Explanation: Diameter of a circle = side of square = 8 cm

Therefore, Radius of circle = 4 cm

Area of circle = $\pi(4)^2 = 16\pi$ cm^2

6. In a circle of radius 21 cm, an arc subtends an angle of 60° at the centre. The length of the arc is;

(a) 20cm (b) 21cm (c) 22cm (d) 25cm

Answer: **(c) 22cm**

Explanation: Length of an arc = (θ/360°) × (2πr)

∴ Length of an arc AB = (60°/360°) × 2 × 22/7 × 21

= (1/6) × 2 × (22/7) × 21 Or Arc AB Length = 22cm

7. If the area of a circle is 154 cm^2, then its perimeter is

(a) 11 cm (b) 22 cm (c) 44 cm (d) 55 cm

Answer: **(c) 44 cm**

Explanation: Area of a circle = 154 cm^2 , $\pi r^2 = 154$

$(22/7) \times r^2 = 154$, $r^2 = (154 \times 7)/22$, $r^2 = 7 \times 7$, $r = 7$ cm

Perimeter of circle = $2\pi r = 2 \times (22/7) \times 7 = 44$ cm

8. Find the area of a sector of circle of radius 21 cm and central angle 120°.

(a) 441 cm^2 (b) 462 cm^2 (c) 386 cm^2 (d) 512 cm^2

Answer: **(b) 462 cm^2**

Explanation:radius (r) = 21 cm

Central angle = θ = 120

Area of sector = $(\pi r^2\theta)/360$ = $(22/7) \times (21 \times 21) \times (120/360)$

= 22 × 21 = 462 cm^2

9. If the perimeter and the area of a circle are numerically equal, then the radius of the circle is

(a) 2 units (b) π units (c) 4 units (d) 7 units

Answer: **(a) 2 units**

Explanation: Perimeter of circle = Area of circle

$2\pi r = \pi r^2$ ⇒ r = 2

Therefore, radius = 2 units

HERONS FORMULA

1) Area of the triangle is equal to:

a. Base x Height b. 2(Base x Height) c. ½(Base x Height) d. ½ (Base + Height)

Answer: **c**

2) If the perimeter of an equilateral triangle is 180 cm. Then its area will be:

a. 900 cm^2 b. 900√3 cm^2 c. 300√3 cm^2 d. 600√3 cm^2

Answer: **b**

Explanation: Given, Perimeter = 180 cm

3a = 180 (Equilateral triangle)

a = 60 cm

$A=(\sqrt{3}/4)\,a^2$

A = 900√3

3) The sides of a triangle are 122 m, 22 m and 120 m respectively. The area of the triangle is:

a. 1320 sq.m b. 1300 sq.m c. 1400 sq.m d. 1420 sq.m

Answer: **a**

Explanation: Given,
a = 122 m
b = 22 m
c = 120 m
Semi-perimeter, s = (122+22+120)/2 = 132 m
Using heron's formula:

$A=\sqrt{s(s-a)(s-b)(s-c)}$ $A=s(s-a)(s-b)(s-c)$

Put the values of s, a, b and c, to get the answer equal to 1320 sq.m.

4) The equal sides of the isosceles triangle are 12 cm, and the perimeter is 30 cm. The area of this triangle is:

a. 9√15 sq.cm b. 6√15 sq.cm c. 3√15 sq.cm d. √15, sq.cm.

Answer: **a**

Explanation: Given,
Perimeter = 30cm
Semiperimeter, s = 30/2 = 15cm
a = b = 12cm
c=?
a+b+c = 30
12 +12+c=30
c=30-24 = 6cm
By putting the values of s, a, b and c in Heron's formula, we can get the value of area.

5) The sides of a parallelogram are 100 m each and length of the longest diagonal is 160m. The area of a parallelogram is:

a. 9600 sq.m b. 9000 sq.m c. 9200 sq.m d. 8800 sq.m

Answer: **a**

Explanation: The diagonal divides the parallelogram into two equivalent triangles.

Hence, its area will be equal to the sum of the area of the two triangles.

Hence, we can determine the area of the two triangles using Heron's formula.

6) The base of a right triangle is 8 cm and the hypotenuse is 10 cm. Its area will be

(a) 24 cm^2 (b) 40 cm^2 (c) 48 cm^2 (d) 80 cm^2

Answer: **a**

Explanation:

Given: Base = 8 cm and Hypotenuse = 10 cm

Hence, height = $\sqrt{[(10^2 - 8^2)} = \sqrt{36} = 6$ cm

Therefore, area = $(½)\times b\times h = (½)\times 8\times 6 = 24 cm^2$.

7) The perimeter of an equilateral triangle is 60 m. The area is

(a) $10\sqrt{3}$ m^2 (b) $15\sqrt{3}$ m^2 (c) $20\sqrt{3}$ m^2 (d) $100\sqrt{3}$ m^2

Answer: **d**

Explanation:

Given: Perimeter of an equilateral triangle = 60m

3a = 60 m (As the perimeter of an equilateral triangle is 3a units)

a = 20 cm.

We know that area of equilateral triangle = $(\sqrt{3}/4)a^2$ square units

A = $(\sqrt{3}/4)20^2$, A = $(\sqrt{3}/4)(400) = 100\sqrt{3}$ cm^2.

8) If the area of an equilateral triangle is $16\sqrt{3}$ cm^2, then the perimeter of the triangle is

(a) 48 cm (b) 24 cm (c) 12 cm (d) 36 cm

Answer: **b**

Explanation:

Given: Area of equilateral triangle = $16\sqrt{3}$ cm^2

$(\sqrt{3}/4)a^2 = 16\sqrt{3}$

$a^2 = [(16\sqrt{3})(4)]/\sqrt{3}$, $a^2 = 64$, a = 8cm
Therefore, perimeter = 3(8) = 24 cm.

SURFACE AREAS AND VOLUMES

1) The formula to find the surface area of a cuboid of length (l), breadth (b) and height (h) is:

a. lb+bh+hl b. 2(lb+bh+hl) c. 2(lbh) d. lbh/2

Answer: **b**

2) The surface area of a cube whose edge equals to 3cm is:

a. 62 sq.cm b. 30 sq.cm c. 54 sq.cm d. 90 sq.cm

Answer: **c**

Explanation: Given, a = 3 cm

Surface area of cube = $6a^2$

SA = 6 x 3 x 3 = 54 sq.cm

3) The surface area of cuboid-shaped box having length=80 cm, breadth=40cm and height=20cm is:

a. 11200 sq.cm b. 13000 sq.cm c. 13400 sq.cm d. 12000 sq.cm

Answer: **a**

Explanation: surface area of the box = 2(lb + bh + hl)

S.A. = 2[(80 × 40) + (40 × 20) + (20 × 80)]

= 2[3200 + 800 + 1600]

= 2 × 5600 = 11200 sq.cm.

4) The volume of a hemisphere whose radius is r is:

a. 4/3 πr^3 b. $4\pi r^3$ c. $2\pi r^3$ d. ⅔ πr^3

Answer: **d**

5) If the radius of a cylinder is 4cm and height is 10cm, then the total surface area of a cylinder is:

a. 440 sq.cm b. 352 sq.cm.
c. 400 sq.cm d. 412 sq.cm

Answer: **b**

Explanation: Total Surface Area of a Cylinder = $2\pi r(r + h)$

TSA = 2 x 22/7 x 4(4 + 10)

= (2x22x4x14)/7

= (2x22x4x2)

= 352 sq.cm

6) The curved surface area of a right circular cylinder of height 14 cm is 88 cm^2. The diameter of the base is:

a. 2 cm b. 3cm c. 4cm d. 6cm

Answer: **a**

Explanation: Curved surface area of cylinder = 88 sq.cm

Height = 14 cm

$2\pi rh = 88$

$r = 88/2\pi h$. r=1 cm , Diameter = 2r = 2cm

7) The Curved surface area of a right circular cylinder is 4.4 sq.cm. The radius of the base is 0.7 cm. The height of the cylinder will be:

a. 2 cm b. 3 cm c. 1 cm d. 1.5 cm

Answer: **c**

Explanation: Curved surface area of cylinder = $2\pi rh$

$2\pi rh = 4.4$

$h = 4.4/(2\pi \times 0.7)$

h = 1 cm

8) The diameter of the base of a cone is 10.5 cm, and its slant height is 10 cm. The curved surface area is:

a. 150 sq.cm b. 165 sq.cm c. 177 sq.cm d. 180 sq.cm

Answer: **b**

Explanation: Diameter = 10.5, Radius = 10.5/2

Slant height, l = 10cm

Curved surface area of cone = $\pi rl = \pi(5.25)(10)$

CSA = 165 sq.cm

9) If slant height of the cone is 21cm and the diameter of the base is 24 cm. The total surface area of a cone is:

a. 1200.77 sq.cm b. 1177 sq.cm
c. 1222.77 sq.cm d. 1244.57 sq.cm

Answer: **d**

Explanation: Total surface area = $\pi r(l + r)$

r = 24/2 = 12 cm

l = 21 cm

TSA = $\pi(12)(21 + 12)$ = 1244.57 sq.cm

10) The surface area of a sphere of radius 14 cm is:

a. 1386 sq.cm b. 1400 sq.cm c. 2464 sq.cm d. 2000 sq.cm

Answer: **c**

Explanation: Radius of sphere, r = 14 cm

Surface area = $4\pi r^2$

= 4 x 22/7 x $(14)^2$ = 2464 sq.cm.

11) The radius of a sphere is 2r, then its volume will be

(a) $(4/3)\pi r^3$ (b) $4\pi r^3$ (c) $(8/3)\pi r^3$
(d) $(32/3)\pi r^3$

Answer: **d**

Explanation:

Given : r=2r

The volume of sphere = $(4/3)\pi r^3 = (4/3)\pi(2r)^3$

$V = (4/3)\pi(8r^3) = (32/3)\pi r^3$.

12) The radius of a hemispherical balloon increases from 6 cm to 12 cm as air is being pumped into it. The ratio of the surface areas of the balloon in the two cases is

(a) 1:4 (b) 1:3 (c) 2:3 (d) 2:1

Answer: **a**

Explanation:

We know that the total surface area of the hemisphere = $3\pi r^2$ square units.

If r= 6cm, then TSA = $3\pi(6)^2 = 108\pi$

If r = 12 cm, then TSA = $3\pi(12)^2 = 432\pi$

Then the ratio = $(108\pi)/(432\pi)$

Ratio = ¼, which is equal to 1:4.

13) The total surface area of a cube is 96 cm^2. The volume of the cube is:

(a) 8 cm^3 (b) 512 cm^3 (c) 64 cm^3 (d) 27 cm^3

Answer: **c**

Explanation:

We know that the TSA of the cone = $6a^2$.

$6a^2 = 96\ cm^2$, $a^2 = 96/6 = 16$, a =4 cm

The volume of cone = a^3 cubic units , $V = 4^3 = 64cm^3$.

14) The length of the longest pole that can be put in a room of dimensions (10 m × 10 m × 5m) is

(a) 15m (b) 16m (c) 10m (d) 12m

Answer: **a**

Explanation:

Given: l=10m, b= 10m, h= 5m

The length of the longest pole = $\sqrt{[10^2+10^2+5^2]}$

= $\sqrt{(100+100+25)} = \sqrt{225} = 15$ m.

15) In a cylinder, the radius is doubled and height is halved, the curved surface area will be

(a) Halved (b) Doubled (c) Same (d) Four times

Answer: c

Explanation:

We know that the curved surface area of a cylinder is $2\pi rh$

Given that, r = 2R, h= H/2

Hence, the CSA of new cylinder = $2\pi(2R)(H/2) = 2\pi RH$

Therefore, the answer is “Same”.

16) The lateral surface area of a cube is 256 m^2. The volume of the cube is

(a) 512 m^3 (b) 64 m^3 (c) 216 m^3 (d) 256 m^3

Answer: **a**

Explanation:

The lateral surface area of cube = $4a^2$

$4a^2 = 256$, $a^2 = 256/4 = 64$, a = 8 m

Hence, the volume of cube = a^3 cube units , $V = 8^3 = 512\ m^3$.

17) The radii of two cylinders are in the ratio of 2:3 and their heights are in the ratio of 5:3. The ratio of their volumes is:

(a) 10: 1 (b) 20: 27 (c) 17: 27 (d) 20: 37

Answer: **b**

Explanation:

Given that, the radii of two cylinders are in the ratio of 2:3

Hence, $r_1 = 2r$, $r_2 = 3r$

Also, given that, the height of two cylinders are in the ratio 5:3.

Hence, $h_1 = 5h$, $h_2 = 3h$

The ratio of the volume of two cylinders = V_1/V_2

= $\pi r_1^2 h_1/\pi r_2^2 h_2$

= $[(2r)^2(5h)]/[(3r)^2(3h)]$

Ratio of their volumes =$(20r^2h)/(27r^2h) = 20/27 = 20: 27$.

SURFACE AREAS AND VOLUMES-2

1. The shape of an ice-cream cone is a combination of:

(a) Sphere + cylinder (b) Sphere + cone (c) Hemisphere + cylinder (d) Hemisphere + cone

Answer: **(d) Hemisphere + cone**

2. If a cone is cut parallel to the base of it by a plane in two parts, then the shape of the top of the cone will be a:

(a) Sphere (b) Cube (c) Cone itself (d) Cylinder

Answer: **(c) Cone itself**

3. If we cut a cone in two parts by a plane parallel to the base, then the bottom part left over is the:

(a) Cone (b) Frustum of cone (c) Sphere (d) Cylinder

Answer: **(b) Frustum of cone**

4. If r is the radius of the sphere, then the surface area of the sphere is given by;

(a) $4 \pi r^2$ (b) $2 \pi r^2$ (c) πr^2 (d) $4/3 \pi r^2$

Answer: (a) $4 \pi r^2$

5. If we change the shape of an object from a sphere to a cylinder, then the volume of cylinder will

(a) Increase (b) Decrease (c) Remains unchanged (d) Doubles

Answer: **(c) Remains unchanged**

Explanation: If we change the shape of a three-dimensional object, the volume of the new shape will be same.

6. A tank is made of the shape of a cylinder with a hemispherical depression at one end. The height of the cylinder is 1.45 m and radius is 30 cm. The total surface area of the tank is:

(a) 30 m (b) 3.3 m (c) 30.3 m (d) 3300 m

Answer: **(b) 3.3 m**

Explanation: Total surface area of tank = CSA of cylinder + CSA of hemisphere

$= 2\pi rh + 2\pi r^2 = 2\pi r(h + r)$ $= 2 \times 22/7 \times 30(145 + 30)$ cm^2 $=33000$ cm^2 $= 3.3$ m^2

7. Two identical solid cubes of side a are joined end to end. Then the total surface area of the resulting cuboid is

(a) $12a^2$ (b) $10a^2$ (c) $8a^2$ (d) $11a^2$

Answer: **(b) $10a^2$**

Explanation: The total surface area of a cube having side a = $6a^2$

If two identical faces of side a are joined together, then the total surface area of the cuboid so formed is $10a^2$.

STATISTICS

1) The ratio of the sum of observations and the total number of observations is called:

a. Mean b. Median c. Mode d. Central tendency

Answer: **a**

2) The mean of x+2, x+3, x+4 and x-2 is:

a. (x+7)/4 b. (2x+7)/4 c. (3x+7)/4 d. (4x+7)/4

Answer: **d**

Explanation: Mean = (x+2+x+3+x+4+x-2)/4 = (4x+7)/4

3) The median of the data: 4, 6, 8, 9, 11 is

a. 6 b. 8 c. 9 d. 11

Answer: **b**

4) The median of the data: 155 160 145 149 150 147 152 144 148 is

a. 149 b. 150 c. 147 d. 144

Answer: **a**

Explanation: First arrange the data in ascending order.

144 145 147 148 149 150 152 155 160

Since, the number of observations here is odd, therefore,

Median = (n+1)/2 th = (9+1)/2 = 10/2 = 5th number = 149

5) The median of the data: 17, 2, 7, 27, 15, 5, 14, 8, 10, 24, 48, 10, 8, 7, 18, 28 is:

a. 10 b. 24 c. 12 d. 8

Answer: **c**

Explanation: Arrange the given data in ascending order:

2, 5, 7, 7, 8, 8, 10, 10, 14, 15, 17, 18, 24, 27, 28, 48

Since, the number of observations givere here is even, hence,

Median will be average of two middle terms.

n/2th = 16/2 = 8th term

(n/2 +1)th = (16/2 + 1)th = 9th term

Therefore,

Median = (10+14)/2 = 12

6) The mode of the given data: 4, 6, 5, 9, 3, 2, 7, 7, 6, 5, 4, 9, 10, 10, 3, 4, 7, 6, 9, 9 is;

a. 7 b. 9 c. 10 d. 6

Answer: **b**

Explanation: First arrange the data in order:

2, 3, 3, 4, 4, 4, 5, 5, 6, 6, 6, 7, 7, 7, 9, 9, 9, 9, 10, 10

Hence, mode = 9

7) The value which appears very frequently in a data is called:

a. Mean b. Median c. Mode d. Central tendency

Answer: **c**

8) The collection of information, collected for a purpose is called:

a. Mean b. Median c. Mode d. Data

Answer: **d**

9) The mean of the data 2, 3, 4, 5, 0, 1, 3, 3, 4, 3 is

a. 2 b. 2.2 c. 2.4 d. 2.8

Answer: **d**

Explanation: Mean = (2+3+4+5+0+1+3+3+4+3)/10 = 28/10 = 2.8

10) Which of the following is not a measure of central tendency?

a. Standard deviation b. Mean c. Median d. Mode

Answer: **a**

11) Find the range of the following data: 25, 18, 20, 22, 16, 6, 17, 15, 12, 30, 32, 10, 19, 8, 11, 20.

a. 10 b. 15 c. 18 d. 26

Answer: d

Explanation: Range = Maximum value – Minimum value

Range = 32-6 = 26.

12) What is the class mark of the class interval 90-120?

a. 90 b. 105 c. 115 d. 120

Answer: b

Explanation: Class mark = (upper limit + lower limit)/2

Class mark = (120+90)/2

Class mark = 105

13) In the class intervals 10-20, 20-30, 20 is included in which interval?

a. 10-20 b. 20-30 c. Both the intervals d. None of the intervals

Answer: b

Explanation: In the class intervals 10-20, 20-30, 20 is included in the interval 20-30, because the number is always included in the lower limit of the class interval.

14) Find the class width for the grouped frequency distribution of the class intervals 1-20, 21-40, 41-60, ..

a. 10 b. 15 c. 17 d. 20

Answer: d

Explanation: Class width is the same as the class size. The class size of the given intervals 1-20, 21-40, 41-60,.. is 20.

15) The arithmetic mean of the first 5 natural numbers is

a. 3 b. 4 c. 5 d. 6

Answer: a

Explanation: Arithmetic mean = (1+2+3+4+5)/5

Arithmetic mean = 15/5 = 3

16) Find the value of x, if the arithmetic mean of 4, 5, 6, 7, 8 and x is 7.

a. 4 b. 6 c. 8 d. 12

Answer: d

Explanation: (4+5+6+7+8+x)/6 = 7

4+5+6+7+8+x = 7(6)

4+5+6+7+8+x = 42

30+x = 42

x = 42-30 = 12

17) Find the mode of the following data: 15, 14, 19, 20, 14, 15, 16, 14, 15, 18, 14, 19, 15, 17, 15.

a. 14 b. 15 c. 16 d. 17

Answer: b

Explanation: The mode of the data 15, 14, 19, 20, 14, 15, 16, 14, 15, 18, 14, 19, 15, 17, 15 is 15, because the number 15 is repeated 5 times.

18) If each data in the observation is increased by 5, then the mean

a. Remains the same b. Increased by 5
c. Decreased by 5 d. None of the above

Answer: b

Explanation: If each data in the observation is increased by 5, then the mean is also increased by 5 because the mean is the average of the given values.

19) The difference between the maximum and minimum values of the given observation is called

a. Class b. Class interval
c. Classmark d. Range

Answer: d

Explanation: The difference between the maximum and minimum values of the given observation is called range.

20) Find the maximum value if the range is 38 and the minimum value is 82.

a. 60 b. 76 c. 120 d. 82

Answer: c

Explanation: We know that Range = Maximum value – Minimum value.

Let the unknown value, (i.e) Maximum value be x.

Now, substitute the values,

38 = x – 82

x = 38+82

x = 120.

Therefore, the maximum value is 120.

STATISTICS-2

1. The mode and mean is given by 7 and 8, respectively. Then the median is:

(a) 1/13 (b) 13/3 (c) 23/3 (d) 33

Answer: **(c) 23/3**

Explanation: Using Empirical formula,

Mode = 3Median – 2 Mean

3Median = Mode + 2Mean

Median = (Mode + 2Mean)/3

Median = [7 + 2(8)]/3 = (7 + 16)/3 = 23/3

2. The mean of the data: 4, 10, 5, 9, 12 is;

(a) 8 (b) 10 (c) 9 (d) 15

Answer: **(a) 8**

Explanation: mean = (4 + 10 + 5 + 9 + 12)/5 = 40/5 = 8

3. The median of the data 13, 15, 16, 17, 19, 20 is:

(a) 30/2 (b) 31/2 (c) 33/2 (d) 35/2

Answer: **(c) 33/2**

Explanation: For the given data, there are two middle terms, 16 and 17.

Hence, median = (16 + 17)/2 = 33/2

4. If AM of a, a+3, a+6, a+9 and a+12 is 10, then a is equal to;

(a) 1 (b) 2 (c) 3 (d) 4

Answer: **(d) 4**

Explanation: Mean of AM = 10

(a + a + 3 + a + 6 + a + 9 + a + 12)/5 = 10

5a + 30 = 50 ,5a = 20 ,a = 4

5. The class interval of a given observation is 10 to 15, then the class mark for this interval will be:

(a) 11.5 (b) 12.5 (c) 12 (d) 14

Answer: **(b) 12.5**

Explanation: Class mark = (Upper limit + Lower limit)/2

= (15 + 10)/2 = 25/2 = 12.5

6. Construction of a cumulative frequency table is useful in determining the

(a) mean (b) median (c) mode (d) all the above three measures

Answer: **(b) median**

7. The abscissa of the point of intersection of the less than type and of the more than type cumulative frequency curves of a grouped data gives its

(a) mean (b) median (c) mode (d) all the three above

Answer: (b) median

The abscissa of the point of intersection of the less than type and of the more than

type cumulative frequency curves of a grouped data gives its median.

PROBABILITY

1) The probability of each event, when a coin is tossed for 1000 times with frequencies: Head:455 & Tail: 545 is:

a. 0.455 & 0.545 b. 0.5 & 0.5
c. 0.45 & 0.55 d. 455 & 545

Answer: **a**

Explanation: Probability of Occurrence of Head P(E) = No. of heads/total number of trials

P(E) = 455/1000 = 0.455

Similarly,P(F) = No. of tails/total number of trials

P(F) = 545/1000 = 0.545

2) The sum of all probabilities equal to:

a. 4 b. 1 c. 3 d. 2

Answer: **b**

3) The probability of each event lies between:

a. 1 & 2 b. 1 & 10 c. 0 & 1d. 0 & 5

Answer: **c**

4) If P(E) = 0.44, then P(not E) will be:

a. 0.44 b. 0.55 c. 0.50 d. 0.56

Answer: **d**

Explanation: We know;

P(E) + P(not E) = 1

0.44 + P(not E) = 1

P(not E) = 1 – 0.44 = 0.56

5) If P(E) = 0.38, then probability of event E, not occurring is:

a. 0.62 b. 0.38 c. 0.48 d. 1

Answer: **a**

Explanation: P(not E) = 1 – P(E) = 1-0.38 = 0.62

6) The probability of drawing an ace card from a deck of cards is:

a. 1/52 b. 1/26 c. 4/13 d. 1/13

Answer: **d**

Explanation: There are 4 aces in a deck of card.

Hence, the probability of taking one ace out of 52 cards = 4/52 = 1/13

7) If the probability of an event to happen is 0.3 and the probability of the event not happening is:

a. 0.7 b. 0.6 c. 0.5 d. None of the above

Answer: **a**

Explanation: Probability of an event not happening = 1 – P(E)

P(not E) = 1 – 0.3 = 0.7

8) A dice is thrown. The probability of getting 1 and 5 is:

a. ⅙ b. ⅔ c. ⅓ d. ½

Answer: **c**

Explanation: The probability of getting 1 and 5 = 2/6 = ⅓

9) A batsman hits boundaries for 6 times out of 30 balls. Find the probability that he did not hit the boundaries.

a. ⅕ b. ⅖ c. ⅗ d. ⅘

Answer: **d**

Explanation: No. of boundaries = 6

No. of balls = 30

No. of balls without boundaries = 30 – 6 =24

Probability of no boundary = 24/30 = ⅘

10. Three coins were tossed 200 times. The number of times 2 heads came up is 72. Then the probability of 2 heads coming up is:

a. 1/25 b. 2/25 c. 7/25 d. 9/25

Answer: **d**

Explanation: Probability = 72/200 = 9/25

11) What is the probability of getting an odd number less than 4, if a die is thrown?

a. 1/6 b. ½ c. 1/3 d. 0

Answer: **c**

Explanation: Sample space, S = {1, 2, 3, 4, 5, 6}

Favourable outcomes = {1, 3}

Therefore, the probability of getting an odd number less than 4 = 2/6 = ⅓.

12) What is the probability of impossible events?

a. 1 b. 0 c. More than 1 d. Less than 1

Answer: **b**

Explanation: The probability of an impossible event is always 0.

13) Performing an event once is called

a. Sample b. Trial c. Error d. None of the above

Answer: **b**

Explanation: Performing an event once is called a trial.

14) A card is drawn from a well-shuffled deck of 52 cards. What is the probability of getting a king of the red suits?

a. 3/36 b. 1/26 c. 3/26 d. 1/16

Answer: **b**

Explanation: In a pack of 52 cards, there are a total of 4 king cards, out of which 2 are red and 2 are black.

Therefore, in a red suit, there are 2 king cards.

Hence, the probability of getting a king of red suits = 2/52 = 1/26.

15) Find the probability of a selected number is a multiple of 4 from the numbers 1, 2, 3, 4, 5, …15.

a. 1/5 b. 1/3 c. 4/12 d. 2/15

Answer: **a**

Explanation: S = {1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15}

Multiples of 4 from the sample space = {4, 8, 12}

Therefore, the probability of the selected number is a multiple of 5 is 3/15 = 1/5.

16) What is the probability of drawing a queen from the deck of 52 cards?

a. 1/26 b. 1/52 c. 1/13 d. 3/52

Answer: **c**

Explanation: Total cards = 52

Number of queens in a pack of 52 cards = 4

Hence, the probability of drawing a queen from a deck of 52 cards = 4/52 = 1/13

17) Which of the following cannot be the probability of an event?

a. 1 b. 0 c. 0.75 d. 1.3

Answer: **d**

Explanation: The probability of an event always lies between 0 and 1.

18) There are 4 green and 2 red balls in a basket. What is the probability of getting the red balls?

a. ½ b. 1/3 c. 1/5 d. 1/6

Answer: **b**

Explanation: Total balls = 4 green + 2 red = 6 balls

No. of red balls = 2.

Hence, the probability of getting the red balls = 2/6 = 1/3

19) If two coins are tossed simultaneously, then what is the probability of getting exactly two tails?

a. ¼ b. ½ c. 1/3 d. None of the above

Answer: **a**

Explanation: If two coins are tossed, then the sample space, S = {HH, HT, TH, TT}

Favourable outcome (Getting exactly two tails) = {TT}

Therefore, the probability of getting exactly two heads = ¼

PROBABILITY-2

1. The probability of event equal to zero is called;

(a) Unsure event (b) Sure Event (c) Impossible event (d) Independent event

Answer: **(c) Impossible event**

Explanation: The probability of an event that cannot be happened or which is impossible is equal to zero.

2. The probability that cannot exist among the following:

(a) ⅔ (b) -1.5 (c) 15% (d) 0.7

Answer: **(b) -1.5**

Explanation: The probability lies between 0 and 1. Hence it cannot be negative.

3. If P(E) = 0.07, then what is the probability of 'not E'?

(a) 0.93 (b) 0.95 (c) 0.89 (d) 0.90

Answer: **(a) 0.93**

Explanation: P(E) + P(not E) = 1

Or, P(not E) = 1 – 0.07

∴ P(not E) = 0.93

4. A bag has 3 red balls and 5 green balls. If we take a ball from the bag, then what is the probability of getting red balls only?

(a) 3 (b) 8 (c) ⅜ (d) 8/3

Answer: **(c) ⅜**

Explanation: Total balls in bag = 3+5 = 8

Probability of getting red balls = number of red balls/total number of balls = ⅜

5. A bag has 5 white marbles, 8 red marbles and 4 purple marbles. If we take a marble randomly, then what is the probability of not getting purple marble?

(a) 0.5 (b) 0.66 (c) 0.08 (d) 0.77

Answer: **(d) 0.77**

Explanation: Total number of purple marbles = 4

Total number of marbles in bag = 5 + 8 + 4 = 17

Probability of getting purple marbles = 4/17

Hence, the probability of not getting purple marbles = 1-4/17 = 0.77

6. A dice is thrown in the air. The probability of getting odd numbers is

(a) ½ (b) 3/2 (c) 3 (d) 4

Answer: **(a) ½**

Explanation: A dice has six faces having values 1, 2, 3, 4, 5 and 6.

There are three odd numbers and three even numbers.

Therefore, the probability of getting only odd numbers is = 3/6 = ½

7. A card is drawn from the set of 52 cards. Find the probability of getting a queen card.

(a) 1/26 (b) 1/13 (c) 4/53 (d) 4/13

Answer: **(b) 1/13**

Explanation: Total number of cards = 52

Number of queen cards= 4

The probability of getting queen card = 4/52 = 1/13

8. The sum of the probabilities of all the elementary events of an experiment is

(a) 0.5 (b) 1 (c) 2 (d) 1.5

Answer: **(b) 1**

The sum of the probabilities of all the elementary events of an experiment is equal to 1.

9. A card is selected at random from a well shuffled deck of 52 playing cards. The probability of its being a face card is

(a) 3/13 (b) 4/13 (c) 6/13 (d) 9/13

Answer: **(a) 3/13**

Explanation: Total number of outcomes = 52

Number of face cards = 12

The probability of its being a face card = 12/52 = 3/13

10. If an event cannot occur, then its probability is

(a) 1 (b) 3/4 (c) ½ (d) 0

Answer: **(d) 0**

If an event cannot occur, then its probability is 0.

11. The probability that a non leap year selected at random will contain 53 Sundays is

(a) 1/7 (b) 2/7 (c) 3/7 (d) 5/7

Answer: **(a) 1/7**

Explanation:

Non-leap year = 365 days

365 days = 52 weeks + 1 day

For 52 weeks, number of Sundays = 52

1 remaining day can be Sunday, Monday, Tuesday, Wednesday, Thursday, Friday, Saturday, Sunday.

Total possible outcomes = 7

The number of favourable outcomes = 1

Thus, the probability of getting 53 Sundays = 1/7

12. The probability of getting a bad egg in a lot of 400 is 0.035. The number of bad eggs in the lot is

(a) 7 (b) 14 (c) 21 (d) 28

Answer: **(b) 14**

Explanation: Total number of eggs = 400

Probability of getting a bad egg = Number of bad eggs/Total number of eggs

0.035 = Number of bad eggs/400

Number of bad eggs = 0.035 × 400 = 14

REAL NUMBERS

1. For some integer n, the odd integer is represented in the form of:

(a) n (b) n + 1 (c) 2n + 1 (d) 2n

Answer: **(c) 2n + 1**

2. HCF of 26 and 91 is:

(a) 15 (b) 13 (c) 19 (d) 11

Answer: **(b) 13**

Explanation: 26 = 2 x 13

91 = 7 x 13

Hence, HCF (26, 91) = 13

3. The addition of a rational number and an irrational number is equal to:

(a) rational number (b) Irrational number (c) Both (d) None of the above

Answer: **(b) Irrational number**

4. The multiplication of two irrational numbers is:

(a) irrational number (b) rational number (c) Maybe rational or irrational (d) None

Answer: **(c) Maybe rational or irrational**

5. If set A = {1, 2, 3, 4, 5,…} is given, then it represents:

(a) Whole numbers (b) Rational Numbers (c) Natural numbers (d) Complex numbers

Answer: **(c) Natural numbers**

6. If p and q are integers and is represented in the form of p/q, then it is a:

(a) Whole number (b) Rational number (c) Natural number (d) Even number

Answer: **(b) Rational number**

7. The largest number that divides 70 and 125, which leaves the remainders 5 and 8, is:

(a) 65 (b) 15 (c) 13 (d) 25

Answer: **(c) 13**

Explanation: 70 – 5 = 65 and 125 – 8 = 117

HCF (65, 117) is the largest number that divides 70 and 125 and leaves remainder 5 and 8.

HCF (65, 117) = 13

8. The least number that is divisible by all the numbers from 1 to 5 is:

(a) 70 (b) 60 (c) 80 (d) 90

Answer: **(b) 60**

Explanation: The least number will be LCM of 1, 2, 3, 4, 5.

Hence, LCM (1, 2, 3, 4, 5) = 2 x 2 x 3 x 5 = 60

9. The sum or difference of two irrational numbers is always

(a) rational (b) irrational (c) rational or irrational (d) not determined

Answer: (b) irrational

10. For some integer m, every even integer is of the form

(a) m (b) m + 1 (c) 2m (d) 2m + 1

Answer: **(c) 2m**

QUADRATIC EQUATIONS

1. The sum of two numbers is 27 and product is 182. The numbers are:

(a) 12 and 13 (b) 13 and 14 (c) 12 and 15 (d) 13 and 24

Answer: **(b) 13 and 14**

2. The roots of quadratic equation $2x^2 + x + 4 = 0$ are:

(a) Positive and negative (b) Both Positive (c) Both Negative (d) No real roots

Answer: **(d) No real roots**

3. The sum of the reciprocals of Rehman's ages 3 years ago and 5 years from now is 1/3. The present age of Rehman is:

(a) 7 (b) 10 (c) 5 (d) 6

Answer: **(a) 7**

4. A train travels 360 km at a uniform speed. If the speed had been 5 km/h more, it would have taken 1 hour less for the same journey. Find the speed of the train.

(a) 30 km/hr (b) 40 km/hr (c) 50 km/hr (d) 60 km/hr

Answer: **(b) 40 km/hr**

5. If one root of equation $4x^2-2x+k-4=0$ is reciprocal of the other. The value of k is:

(a) -8 (b) 8 (c) -4 (d) 4

Answer: **(b) 8**

Explanation: If one root is reciprocal of others, then the product of roots will be:

$\alpha \times 1/\alpha = (k-4)/4$, $k-4=4$, $k=8$

6. A quadratic equation $ax^2 + bx + c = 0$ has no real roots, if

(a) $b^2 - 4ac < 0$ (b) $b^2 - 4ac = 0$ (c) $b^2 - 4ac < 0$ (d) $b^2 - ac < 0$

Answer: **(c) $b^2 - 4ac < 0$**

A quadratic equation $ax^2 + bx + c = 0$ has no real roots, if $b^2 - 4ac < 0$.

That means, the quadratic equation contains imaginary roots.

7. The product of two consecutive positive integers is 306. To find the integers, this can be represented in the form of quadratic equation as

(a) $x^2 + x + 360 = 0$ (b) $x^2 + x - 360 = 0$ (c) $2x^2 + x - 360$ (d) $x^2 - 2x - 360 = 0$

Answer: **(b) $x^2 + x - 360 = 0$**

Explanation: Let x and (x + 1) be the two consecutive integers.

$x(x + 1) = 360$. $x^2 + x = 360$, $x^2 + x - 360$

8. The quadratic formula to find the roots of a quadratic equation $ax^2 + bx + c = 0$ is given by

(a) $[-b \pm \sqrt{(b^2-ac)}]/2a$ (b) $[-b \pm \sqrt{(b^2-2ac)}]/a$ (c) $[-b \pm \sqrt{(b^2-4ac)}]/4a$ (d) $[-b \pm \sqrt{(b^2-4ac)}]/2a$

Answer: **(d) $[-b \pm \sqrt{(b^2-4ac)}]/2a$**

9. The quadratic equation $x^2 + 7x - 60$ has

(a) two equal roots (b) two real and unequal roots (c) no real roots (d) two equal complex roots

Answer: **(b) two real and unequal roots**

Explanation: $x^2 + 7x - 60 = 0$

Comparing with the standard form,

$a = 1, b = 7, c = -60$

$b^2 - 4ac = (7)^2 - 4(1)(-60) = 49 + 240 = 289 > 0$

Therefore, the given quadratic equation has two real and unequal roots.

10. The maximum number of roots for a quadratic equation is equal to

(a) 1 (b) 2 (c) 3 (d) 4

Answer: **(b) 2**

ARITHMETIC PROGRESSIONS

1. In an Arithmetic Progression, if a = 28, d = -4, n = 7, then a_n is:

(a) 4 (b) 5 (c) 3 (d) 7

Answer: **(a) 4**

Explanation: For an AP,

$a_n = a+(n-1)d$ $= 28+(7-1)(-4)$ $= 28+6(-4)$ $= 28-24$, $a_n=4$

2. If a = 10 and d = 10, then first four terms will be:

(a) 10, 30, 50, 60 (b) 10, 20, 30, 40 (c) 10, 15, 20, 25 (d) 10, 18, 20, 30

Answer: **(b) 10, 20, 30, 40**

3. The first term and common difference for the A.P. 3, 1, -1, -3 is:

(a) 1 and 3 (b) -1 and 3 (c) 3 and -2 (d) 2 and 3

Answer: **(c) 3 and -2**

Explanation: First term, a = 3

Common difference, d = Second term – First term

⇒ 1 – 3 = -2

⇒ d = -2

4. 30th term of the A.P: 10, 7, 4, …, is

(a) 97 (b) 77 (c) -77 (d) -87

Answer: **(c) -77**

Explanation: Given,

A.P. = 10, 7, 4, …

First term, a = 10

Common difference, $d = a_2 - a_1 = 7-10 = -3$

As we know, for an A.P.,

$a_n = a+(n-1)d$. Putting the values;

$a_{30} = 10+(30-1)(-3)$, $a_{30} = 10+(29)(-3)$, $a_{30} = 10-87 = -77$

5. 11th term of the A.P. -3, -1/2, 2 …. Is

(a) 28 (b) 22 (c) -38 (d) -48

Answer: **(b) 22**

Explanation: A.P. = -3, -1/2, 2 …

First term a = – 3

Common difference, $d = a_2 - a_1 = (-1/2) -(-3)$

⇒(-1/2) + 3 = 5/2

Nth term;

$a_n = a+(n-1)d$,$a_{11} = 3+(11-1)(5/2)$,$a_{11} = 3+(10)(5/2)$,$a_{11} = -3+25$,$a_{11} = 22$

6. Which term of the A.P. 3, 8, 13, 18, … is 78?

(a) 12th (b) 13th (c) 15th (d) 16th

Answer: **(d) (d) 16th**

Explanation: Given, 3, 8, 13, 18, … is the AP.

First term, a = 3 Common difference, $d = a_2 - a_1 = 8 - 3 = 5$

Let the nth term of given A.P. be 78. Now as we know,

$a_n = a+(n-1)d$

Therefore, ,78 = 3+(n −1)5 ,75 = (n−1)5

(n−1) = 15 ,n = 16

7. The sum of the first five multiples of 3 is:

(a) 45 (b) 55 (c) 65 (d) 75

Answer: **(a) 45**

Explanation: The first five multiples of 3 is 3, 6, 9, 12 and 15

a=3 and d=3 n=5

Sum, $S_n = n/2[2a+(n-1)d]$

$S_5 = 5/2[2(3)+(5-1)3]$

=5/2[6+12] ,=5/2[18] ,=5 x 9 ,= 45

8. If the 2nd term of an AP is 13 and the 5th term is 25, then its 7th term is

(a) 30 (b) 33 (c) 37 (d) 38

Answer: **(b) 33**

Explanation:

Given,

$a_2 = 13$

a + d = 13….(i)

a + 4d = 25….(ii)

From (i) and (ii),

3d = 12, d = 4

So, a = 13 – 4 = 9 , $a_7 = a + 6d = 9 + 6(4) = 9 + 24 = 33$

9. What is the common difference of an AP in which $a_{18} - a_{14} = 32$?

(a) 8 (b) -8 (c) -4 (d) 4

Answer: **(a) 8**

Explanation:

Given,

$a_{18} - a_{14} = 32$

We know that, $a_n = a + (n - 1)d$

So, $a + 17d - (a + 13d) = 32$

$17d - 13d = 32$, $4d = 32$, $d = 8$

TRIANGLES

1. Which of the following triangles have the same side lengths?

(a) Scalene (b) Isosceles (c) Equilateral (d) None of these

Answer: **(c) Equilateral**

2. Area of an equilateral triangle with side length a is equal to:

(a) $\sqrt{3}/2a$ (b) $\sqrt{3}/2a^2$ (c) $\sqrt{3}/4\ a^2$ (d) $\sqrt{3}/4\ a$

Answer: **(c) $\sqrt{3}/4\ a^2$**

3. D and E are the midpoints of side AB and AC of a triangle ABC, respectively and BC = 6 cm. If DE || BC, then the length (in cm) of DE is:

(a) 2.5 (b) 3 (c) 5 (d) 6

Answer: **(b) 3**

Explanation: By midpoint theorem, ,DE=½ BC

DE = ½ of 6 ,DE=3 cm

4. The diagonals of a rhombus are 16 cm and 12 cm, in length. The side of rhombus in length is:

(a) 20 cm (b) 8 cm (c) 10 cm (d) 9 cm

Answer: **(c) 10 cm**

Explanation: $8^2+6^2=side^2$

$64+36=side^2$, side=10 cm

5. Corresponding sides of two similar triangles are in the ratio of 2:3. If the area of small triangle is 48 sq.cm, then the area of large triangle is:

(a) 230 sq.cm. (b) 106 sq.cm (c) 107 sq.cm. (d) 108 sq.cm

Answer: **(d) 108 sq.cm**

Solution: $A_2/48=(3/2)^2$

A_2=108 sq.cm.

6. Sides of two similar triangles are in the ratio 4: 9. Areas of these triangles are in the ratio

(a) 2: 3 (b) 4: 9 (c) 81: 16 (d) 16: 81

Answer: **(d) 16: 81**

Explanation: ∴ Area(ΔABC)/Area(ΔDEF) = $(4/9)^2$ = 16/81 = 16: 81

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YT channel- MERCHANT NAVY CLASSES

INTRODUCTION TO TRIGONOMETRY

1. In ∆ ABC, right-angled at B, AB = 24 cm, BC = 7 cm. The value of tan C is:

(a) 12/7 (b) 24/7 (c) 20/7 (d) 7/24

Answer: **(b) 24/7**

Explanation: AB = 24 cm and BC = 7 cm

tan C = Opposite side/Adjacent side

tan C = 24/7

2. (Sin 30°+cos 60°)-(sin 60° + cos 30°) is equal to:

(a) 0 (b) 1+2√3 (c) 1-√3 (d) 1+√3

Answer: **(c) 1-√3**

Explanation: sin 30° = ½, sin 60° = √3/2, cos 30° = √3/2 and cos 60° = ½

3. The value of tan 60°/cot 30° is equal to:

(a) 0 (b) 1 (c) 2 (d) 3

Answer: **(b) 1**

Explanation: tan 60° = √3 and cot 30° = √3

Hence, tan 60°/cot 30° = √3/√3 = 1

4. $1 - \cos^2 A$ is equal to:

(a) $\sin^2 A$ (b) $\tan^2 A$ (c) $1 - \sin^2 A$ (d) $\sec^2 A$

Answer: **(a) $\sin^2 A$**

Explanation: $\sin^2 A + \cos^2 A = 1$

$1 - \cos^2 A = \sin^2 A$

5. If cos X = ⅔ then tan X is equal to:

(a) 5/2 (b) √(5/2) (c) √5/2 (d) 2/√5

Answer: **(c) √5/2**

Explanation: cosx = 2/3 means base=3. hypotenuse=3.

Therefore perpendicular = V5

6. If cos X = a/b, then sin X is equal to:

(a) $(b^2-a^2)/b$ (b) (b-a)/b (c) $\sqrt{(b^2-a^2)}/b$ (d) $\sqrt{(b-a)}/b$

Answer: **(c) $\sqrt{(b^2-a^2)}/b$**

7. The value of sin 60° cos 30° + sin 30° cos 60° is:

(a) 0 (b) 1 (c) 2 (d) 4

Answer: **(b) 1**

Explanation: sin 60° = √3/2, sin 30° = ½, cos 60° = ½ and cos 30° = √3/2

8. The value of (sin 45° + cos 45°) is

(a) 1/√2 (b) √2 (c) √3/2 (d) 1

Answer: **(b) √2**

9. If sin A = 1/2 , then the value of cot A is

(a) √3 (b) 1/√3 (c) √3/2 (d) 1

Answer: **(a) √3**

10. The value of (tan 1° tan 2° tan 3° … tan 89°) is

(a) 0 (b) 1 (c) 2 (d) 1/2

Answer: **(b) 1**

Explanation: tan 1° tan 2° tan 3°…tan 89°

= [tan 1° tan 2°…tan 44°] tan 45° [tan (90° – 44°) tan (90° – 43°)…tan (90° – 1°)]

= [tan 1° tan 2°…tan 44°] [cot 44° cot 43°…cot 1°] × [tan 45°]

= [(tan 1°× cot 1°) (tan 2°× cot 2°)…(tan 44°× cot 44°)] × [tan 45°]

= 1 × 1 × 1 × 1 × …× 1 = 1

SOME APPLICATIONS OF TRIGONOMETRY

1. If the length of the shadow of a tree is decreasing then the angle of elevation is:

(a) Increasing (b) Decreasing (c) Remains the same (d) None of the above

Answer: **(a) Increasing**

2. The angle of elevation of the top of a building from a point on the ground, which is 30 m away from the foot of the building, is 30°. The height of the building is:

(a) 10 m (b) 30/√3 m (c) √3/10 m (d) 30 m

Answer: **(b) 30/√3 m**

3. If a tower 6m high casts a shadow of 2√3 m long on the ground, then the sun's elevation is:

(a) 60° (b) 45° (c) 30° (d) 90°

Answer: **(a) 60°**

:

4. The angle formed by the line of sight with the horizontal when the point is below the horizontal level is called:

(a) Angle of elevation (b) Angle of depression (c) No such angle is formed (d) None of the above

Answer: **(b) Angle of depression**

5. The angle formed by the line of sight with the horizontal when the point being viewed is above the horizontal level is called:

(a) Angle of elevation (b) Angle of depression (c) No such angle is formed (d) None of the above

Answer: **(a) Angle of elevation**

6. From a point on the ground, which is 15 m away from the foot of the tower, the angle of elevation of the top of the tower is found to be 60°. The height of the tower (in m) standing straight is:

(a) 15√3 (b) 10√3 (c) 12√3 (d) 20√3

Answer: **(a) 15√3**

7. A ladder makes an angle of 60° with the ground, when placed along a wall. If the foot of ladder is 8 m away from the wall, the length of ladder is

(a) 4 m (b) 8 m (c) 8√3 m (d) 16 m

Answer: **(d) 16 m**

8. If the angles of elevation of the top of a tower from two points at the distance of a m and b m from the base of tower and in the same straight line with it are complementary, then the height of the tower (in m) is

(a) √(a/b) (b) √ab (c) √(a + b) (d) √(a – b)

Answer: **(b) √ab**

SOME IMPORTANT FORMULAS

SIMPLE INTEREST

1. **Principal:**
   The money borrowed or lent out for a certain period is called the **principal** or the **sum**.
2. **Interest:**
   Extra money paid for using other's money is called **interest**.
3. Simple Interest (S.I.):
   If the interest on a sum borrowed for certain period is reckoned uniformly, then it is called **simple interest**.

   Let Principal = P, Rate = R% per annum (p.a.) and Time = T years. Then

   (i). Simple Intereest = $\left(\frac{P \times R \times T}{100}\right)$

   (ii). $P = \left(\frac{100 \times S.I.}{R \times T}\right)$; $R = \left(\frac{100 \times S.I.}{P \times T}\right)$ and $T = \left(\frac{100 \times S.I.}{P \times R}\right)$.

SPEED TIME AND DISTANCE

1. **Speed, Time and Distance:**

   $\text{Speed} = \left(\frac{\text{Distance}}{\text{Time}}\right)$, $\text{Time} = \left(\frac{\text{Distance}}{\text{Speed}}\right)$, Distance = (Speed x Time).

2. **km/hr to m/sec conversion:**

   x km/hr $= \left(x \times \frac{5}{18}\right)$ m/sec.

3. **m/sec to km/hr conversion:**

   x m/sec $= \left(x \times \frac{18}{5}\right)$ km/hr.

4. If the ratio of the speeds of A and B is $a : b$, then the ratio of the times taken by them to cover the same distance is $\frac{1}{a} : \frac{1}{b}$ or $b : a$.

5. Suppose a man covers a certain distance at x km/hr and an equal distance at y km/hr. Then, the average speed during the whole journey is $\left(\frac{2xy}{}\right)$ km/hr.

PROFIT AND LOSS

Cost Price:
The price, at which an article is purchased, is called its **cost price**, abbreviated as **C.P.**

Selling Price:
The price, at which an article is sold, is called its **selling prices**, abbreviated as **S.P.**

Profit or Gain:
If S.P. is greater than C.P., the seller is said to have a **profit** or **gain**.

Loss:
If S.P. is less than C.P., the seller is said to have incurred a **loss**.

IMPORTANT FORMULAE

1. Gain = (S.P.) – (C.P.)
2. Loss = (C.P.) – (S.P.)
3. Loss or gain is always reckoned on C.P.
4. Gain Percentage: (Gain %)

   $$\text{Gain \%} = \left(\frac{\text{Gain} \times 100}{\text{C.P.}}\right)$$

5. Loss Percentage: (Loss %)

   $$\text{Loss \%} = \left(\frac{\text{Loss} \times 100}{\text{C.P.}}\right)$$

6. Selling Price: (S.P.)

   $$\text{SP} = \left[\frac{(100 + \text{Gain \%})}{100} \times \text{C.P}\right]$$

7. Selling Price: (S.P.)

   $$\text{SP} = \left[\frac{(100 - \text{Loss \%})}{100} \times \text{C.P.}\right]$$

8. Cost Price: (C.P.)

   $$\text{C.P.} = \left[\frac{100}{(100 + \text{Gain \%})} \times \text{S.P.}\right]$$

9. Cost Price: (C.P.)

   $$\text{C.P.} = \left[\frac{100}{(100 - \text{Loss \%})} \times \text{S.P.}\right]$$

10. If an article is sold at a gain of say 35%, then S.P. = 135% of C.P.
11. If an article is sold at a loss of say, 35% then S.P. = 65% of C.P.
12. When a person sells two similar items, one at a gain of say x%, and the other at a loss of x%, then the seller always incurs a loss given by:

    $$\text{Loss \%} = \left(\frac{\text{Common Loss and Gain \%}}{10}\right)^2 = \left(\frac{x}{10}\right)^2.$$

13. If a trader professes to sell his goods at cost price, but uses false weights, then

$$\text{Gain \%} = \left[\frac{\text{Error}}{(\text{True Value}) - (\text{Error})} \times 100\right]\%.$$

TIME AND WORK

1. **Work from Days:**

   If A can do a piece of work in n days, then A's 1 day's work = $\frac{1}{n}$.

2. **Days from Work:**

   If A's 1 day's work = $\frac{1}{n}$, then A can finish the work in n days.

3. **Ratio:**

   If A is thrice as good a workman as B, then:

   Ratio of work done by A and B = 3 : 1.

   Ratio of times taken by A and B to finish a work = 1 : 3

EXERCISE

1. 1.14 expressed as a percent of 1.9 is:
   (A) 6% (B) 10%
   (C) 60% (D) 90%
2. 2 is what percent of 50?
   (A) 2% (B) 2.5%
   (C) 4% (D) 5%
3. 25% of 120+40% of 380=? of 637
   (A) $\frac{2}{7}$ (B) $\frac{1}{7}$ (C) $\frac{4}{7}$ (D) $\frac{3}{7}$
4. What is 27% of 36% of $\frac{5}{9}$ of 4500?
   (A) 239 (B) 241
   (C) 243 (D) 245
5. What is 20%, of 25% of 300?
   (A) 150 (B) 60 (C) 45 (D) 15
6. If x is 80% of y, what percent of x is y?
   (A) 75% (B) 80%
   (C) 100% (D) 125%
7. What percent of 1 day is 36 minutes?
   (A) 25% (B) 2.5%
   (C) 3.6% (D) 0.25%
8. If 40% of $(A + B)$ = 60% of $(A - B)$ then $\frac{2A-3B}{A+B}$ is
   (A) $\frac{7}{6}$ (B) $\frac{6}{7}$ (C) $\frac{5}{6}$ (D) $\frac{6}{5}$
9. If 120% of a is equal to 80% of b, then $\frac{b+a}{b-a}$ is equal to:
   (A) 5 (B) 6 (C) 7 (D) 8
10. If A exceeds B by 40%, B is less than C by 20%, then A:C is:

(A) 28 : 25 (B) 26 : 25
(C) 14 : 27 (D) 27 : 14

11. A rent collector receives $1\frac{1}{2}$%% for collecting rent. What will he receive for collecting Rs.5000?
(A) 25 (B) 50
(C) 75 (D) 125

12. 498 is 17% less than a number then the number is:
(A) 610 (B) 580
(C) 600 (D) 620

13. A number increased by $22\frac{1}{2}$% gives 98 as result. The number is:
(A) 45 (B) 18 (C) 80 (D) 81

14. In a village 30% of the population is literate. If the total population of the village is 6600 then the number of literate is:
(A) 1980 (B) 4620
(C) 2200 (D) 3280

15. First and second numbers are less than a third number by 20% and 40 respectively. The second number is less than the first by:
(A) 7% (B) 4%
(C) 3% (D) 25%

16. If the salary of Manoj is 40% less than that of Subhash, then by how much percentage is the salary of Subhash more than that of Manoj?
(A) 60% (B) $66\frac{1}{4}$%
(C) $66\frac{2}{3}$% (D) 65%

17. If the length of a rectangle is increased by 20% and breadth is decreased by 20%, then the area of the rectangle will be decreased by decreased by what percentage is
(A) 12% (B) 4% (C) 7% (D) 6%

18. The percentage change of a number when it is first decreased by 10% and then increased by 10% is
(A) .01 increase
(B) 1% decrease
(C) 0.1% decrease
(D) No change

19. If each side of a square is increased by 30% then by how much parentage is area of the square will be increased?
(A) 75% (B) 69%
(C) 72% (D) 39%

20. In an examination, 34% failed in Mathematics and 42% failed in English. If 20% failed in both the subjects, the percentage of students who passed in both subjects was:
(A) 54% (B) 50%
(C) 44% (D) 56%

21. In a group of students 70% can speak English and 65% can speak Hindi. If 27% of the students can speak none of the two languages, then what percent of the group can speak both the languages?
(A) 38% (B) 62%
(C) 28% (D) 23%

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22. In an examination 35% of total students failed in Hindi, 45% failed in English and 20% failed in both. Find the percentage of those who passed in both the subjects?
(A) 45% (B) 35%
(C) 20% (D) 40%

23. In an examination, 65% of the students passed in mathematics, 48% passed in Physics and 30% passed in both. How much percent of students failed in both the subjects?
(A) 17% (B) 43%
(C) 13% (D) 47%

24. A man spends 10% of his monthly salary on house rent and 20%, of the rest on the education of his children and 25% of the rest on other expenditures. Thus he saves Rs.3645. Find his total monthly salary.
(A) 7000 (B) 7500
(C) 6750 (D) 8000

25. A village lost 12% of its goats in a flood and 5% of remainder died from diseases. If the number before the flood?
(A) 1000 (B) 10000
(C) 100000 (D) 8360

26. A army lost 10% of its men in war, 10% of the remaining died due to disease and 10% of the rest were declared disabled. The s the strength of the army was reduced to 729000 active men. The original strength of the army was.
(A) 1500000 (B) 1000000
(C) 1200000 (D) 1100000

27. A student has to obtain 33% of total marks to pass. He got 25% of total marks and failed by 40% marks. The number of total marks is:
(A) 800 (B) 300
(C) 500 (D) 600

28. In a test a student got 30% marks and failed by 25 marks. In the same test another student got 40% marks and secured 25% marks more than the essential minimum pass marks. The maximum pass marks for the test were?
(A) 400 (B) 480
(C) 500 (D) 580

29. The population of a village has increased annually at the rate of 25%. If at the end of 3 years it is 10000, the population in the beginning of the first year was:
(A) 5120 (B) 5000
(C) 4900 (D) 4500

30. The population of a town 2 years ago was 62500. Due to migration to big cities, it decreases every year at the rate of 4%. The present population of the town is:
(A) 57600 (B) 56700
(C) 76000 (D) 75000

31. The value of equipment depreciates by 20% each year. How much less will the value of the equipment be after 3 years?
(A) 48.8% (B) 51.2%
(C) 54% (D) 60%

32. If the annual decrease in the population of a town is 5% and the present number of people is 40000 what will the population be in 2 years?

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(A) 31600 (B) 36100
(C) 36150 (D) 13600

33. The population of a town is 10000. It increases by 10% during the first year. During the second year, it decreases by 20% and increased by 30% during the third year. What is the population after 3 years?
(A) 441110 (B) 11440
(C) 11444 (D) 41410

34. If the annual decrease in the population of a town be 20% and the present population be 512000, what was it three years ago?
(A) 1000000 (B) 2621404
(C) 2621440 (D) 208000

35. The ratio of C.P. and S.P. of an article is 20:21. What is the gain percent?
(A) 5% (B) 5.5%
(C) 6% (D) 6.25%

36. On selling an article for Rs.651, there is a loss of 7%. The cost price of that article is:
(A) Rs.744 (B) Rs.751
(C) Rs.793 (D) Rs.700

37. If an article is sold for Rs. 550, the profit is 10%. At what price should the article be sold to make it a 10% loss.
(A) 400 (B) 400
(C) 450 (D) 300

38. If an article is sold at 5% profit instead of 10% loss then he receives Rs. 30 more than the first price. Find out the CP of the article.
(A) 150 (B) 200
(C) 250 (D) 300

39. If an article is sold at 5% gain instead Rs.5 more loss, the man gains Rs.5 more. Find the cost price of that article.
(A) Rs.100 (B) Rs.105
(C) Rs.50 (D) Rs.110

40. The cost price of 36 books is equal to the selling price of 30 books. The gain percent is:
(A) 20% (B) $16\frac{4}{6}\%$
(C) 18% (D) $82\frac{2}{6}\%$

41. If the cost price of 15 tables is equal to the selling price of 20 tables. The loss percent is:
(A) 20% (B) 30%
(C) 25% (D) 37.5%

42. If the selling price of 10 oranges is the cost price 12 oranges. Then the profit percent is:
(A) 20% (B) 22.5%
(C) 25% (D) 30%

43. A sells a cycle to B at a profit of 10%, B sells to C at a profit of 20%. If C pays Rs.264 for it, how much did A pay for it.
(A) Rs.200 (B) Rs.220
(C) Rs.225 (D) Rs.234

44. If would have purchased 11 articles for Rs.10 and sold all the article at the rate of 10 for Rs.11 the profit percent would have been.

Merchant Navy Classes

(A) 10% (B) 11%
(C) 21% (D) 100%

45. A TV and a freeze were sold for Rs.12000 each. If the TV was sold at a loss of 20% of the cost and the freeze at a gain of 20% of the cost, the entire transaction resulted in.
(A) No loss or gain/न ही हानि और न ही लाभ
(B) Loss of Rs.1000/1000 रूपये की हानि
(C) Gain of Rs.1000/1000 रूपये का लाभ
(D) Loss of Rs.2000/2000 रूपये की हानि

46. An article is sold at a loss of 10%. Had it been sold for Rs.9 more there would have been a gain for $12\frac{1}{2}$% on it. The cost price of the article is.
(A) 40 (B) 45 (C) 50 (D) 35

47. A man sells his goods at 20% profit. Had he purchase it for Rs.600 less and sold it for Rs.400 then he would gain 10% more profit. Find the initial cost price.
(A) 3900 (B) 3800
(C) 4200 (D) 4000

48. A man sells the rice at 10% gain but weights 20% less than the original weight. Find the profit percentage.
(A) 37% gain (B) 37.5% gain
(C) 37% loss (D) 12.5% gain

49. A dishonest dealer professes to sell his goods at 10% loss. But he uses a false weight of 800 gm instead of kilogram weight. Thus his profit percentage.
(A) 12% (B) $11\frac{1}{2}$%
(C) $8\frac{1}{2}$% (D) $12\frac{1}{2}$%

50. 1/3 of the total of an article is sold at a profit of 15%, ¼ is at a profit of 20% and the remained part is at a profit of 24%. If the total profit is Rs.6240 find the cost price of the article.
(C) 32000 (D) 31200

51. An article which is marked at Rs.975 is sold for Rs.897. The % discount is?
(A) 12% (B) 10% (C) 6% (D) 8%

52. A shopkeeper marks his goods 20% more than the C.P. and allows a discount of 15%. Find the profit percentage of the shopkeeper.
(A) 2% (B) 4% (C) 1% (D) 3%

53. A shopkeeper marks his goods 15% above cost price, but allows 20% discount for cash. His net loss is.
(A) 3% (B) 5% (C) 10% (D) 8%

54. A shopkeeper earns a profit of 20% on selling a book at 30% discount on list price. How many percentage the list price on greater than cost price.
(A) 75% (B) 71.4%
(C) 80% (D) 70%

55. Allowing 20% and 15% successive discounts, the selling price of an article becomes Rs.3060; then the marked price will be.
(A) 4400 (B) 5000
(C) 4500 (D) 4000

56. Find the selling price of an article if a shopkeeper allows two successive discounts of 5% each on the marked price of Rs.80?

YT channel: MERCHANT NAVY CLASSES

(A) Rs.70.20 (B) Rs.70.10
(C) RS.72.00 (D) Rs.72.20

57. A shopkeeper marks his goods at such a price that after allowing a discount of 12.5% on the mark price he can earn a profit of 20%. If the article cost his Rs.1400 then find its mark price.
(A) 1920 (B) 1820
(C) 1780 (D) 1620

58. The list price (marked price) of an article is Rs.900 and is available at two successive discount of 20% and 10%; the selling price of the article, in rupees is
(A) 640 (B) 648
(C) 720 (D) 738

59. The sum of money, that will give Rs.1 as interest per day at the rate of 5% per annum simple interest is
(A) 3650 (B) 36500
(C) 730 (D) 7300

60. The simple interest of Rs.1000 is increased by Rs.20 if the time is increased by 2 years. Find the rate of interest.
(A) 2% (B) 3%
(C) 1% (D) 4%

61. A sum was put at simple interest at a certain rate for 2 years. Had it been put at 3% higher rate, it would have fetched Rs.72 more. The sum is __?
(A) 1200 (B) 1100
(C) 1400 (D) 1300

62. On a certain sum, the simple interest at the end of 6.25 years becomes 3/8 of the sum. The rate of interest is:
(A) 5% (B) 6%
(C) 7% (D) 8%

63. The difference between two simple interests on Rs.1500 for 3 years. is Rs.13.50. Find the difference between the rate of interests.
(A) 0.2% (B) 0.5%
(C) 0.3% (D) 1%

64. Ratio of the principal and the amount after 1 year is 10:12. Then the rate of interest per annum is:
(A) 12% (B) 10%
(C) 18% (D) 20%

65. If a sum of money becomes 4000 in 2 year and 5500 in 4 year 6 month at the same rate of simple interest per annum. Then the rate of simple interest is:
(A) $21\frac{3}{7}\%$ (B) $21\frac{2}{7}\%$
(C) $21\frac{1}{7}\%$ (D) $21\frac{5}{7}\%$

66. A person lent Rs.5000 partly at the rate of 4 percent pa and partly at the rate of 5 percent per annum simple interest. The total interest after 2 years is Rs.440. To find the sum of money lent at each of the above rates, Rs.5000 is to be divided in the ratio:
(A) 4:5 (B) 3:2 (C) 5:4 (D) 2:3

67. A man invests half of his capital at the rate of 10% per annum, one-third at 9% and the rest at 12% per annum. The average rate of interest per annum, which he gets, is.
(A) 12% (B) 10%

YT channel- MERCHANT NAVY CLASSES

(C) 18% (D) 20%

68. A sum of money becomes Rs.17640 in 2 years from Rs.16000, if compounded annually. Find the rate of interest.
(A) 8% (B) 6% (C) 10% (D) 5%

69. A man invested a sum of money at compound interest. It amounted to Rs.2420 in 2 years and to Rs.2662 in 3 years. Find the sum.
(A) 1000 (B) 2000
(C) 5082 (D) 3000

70. The difference between compound interest and simple interest on Rs. 2500 for 2 years at 4% per annum is.
(A) 5 (B) 4 (C) 10 (D) 8

71. If the different between compound interest and simple interest for 2 years at the rate of 5% per annum is 1.50, find the sum.
(A) 600 (B) 800
(C) 500 (D) 1000

72. If $x: y = 3: 2$ then the ratio $2x^2 + 3y^2 : 3x^2 - 2y^2$ = ?
(A) 12 : 5 (B) 6 : 5
(C) 30 : 19 (D) 5 : 3

73. If $a: b = 2: 3$ and $b: c = 4: 5$ then $a^2 : b^2 : bc$ = ?
(A) 4 : 9 : 45 (B) 16 : 36 : 45
(C) 16 : 36 : 20 (D) 4 : 36 : 20

74. If $A: B: C = 2: 3: 5$ then $\frac{B+C}{A} : \frac{C+A}{B} : \frac{A+B}{C}$ = ?
(A) 4 : 7 : 1 (B) 12 : 7 : 3
(C) 4 : 7 : 3 (D) 12 : 7 : 1

75. If m:n=3:2, then $(4m + 5n) : (4m - 5n)$ is equal to:
(A) 4:9 (B) 9:4 (C) 11:1 (D) 9:1

76. If A:B=1:2, B:C=3:4, C:D=6:9 and D:E=12:16 then A:B:C:D:E is equal to:
(A) 1:3:6:12:16 (B) 2:4:6:9:16
(C) 3:4:8:12:16 (D) 3:6:8:12:16

77. If x:y=3:4, then the value of $\frac{5x-2y}{7x+2y}$ =?
(A) $\frac{7}{25}$ (B) $\frac{7}{23}$ (C) $\frac{7}{29}$ (D) $\frac{7}{17}$

78. If A:B=3:4 and B:C=8:9, then A:B:C is
(A) 8:6:9 (B) 9:8:6
(C) 6:8:9 (D) 3:32:9

79. A capital was distributed among Rajesh, sonal and Chetan in such a way that 2 times of Rajesh's share, 3 times of sonal's share and 5 times of Chetan's share are equal. find the ratio of Rajesh, Sonal and Chetan's share.
(A) 10 : 6 : 15 (B) 15 : 10 : 6
(C) 6 : 10 : 15 (D) 15 : 6 : 10

80. The sum of three numbers is 98. If the ratio of the first to the second is 2:3 and that of the second to the third is 5:8, then the second number is:
(A) 49 (B) 48 (C) 30 (D) 20

81. The ratio of money between Ram and Gopal is 7 : 17 and ratio of Krishna and Gopal is 7 : 17. If Ram has Rs. 490 then how much Krishna have –

YT channel MERCHANT NAVY CLASSES

(A) 2890 (B) 2680
(C) 2330 (D) 1190

82. The ratio of the first and second class fares between two railway stations is 4:1 and that of the number of passengers travelling by first and second classes is 1:40. If on a day Rs.1100 are collected as total fare, the amount collected from the first class passengers is:
(A) Rs.315 (B) Rs.275
(C) Rs.137.50 (D) Rs.100

83. The income of x and y are in the ratio 4 : 3 and ratio of their expenditure is 12 : 7. If both saves Rs. 3200 at the end of the month. Then the income of x is –
(C) Rs. 4000 (D) Rs. 6000

84. The annual income of A and B are in the ratio 4:3 and the ratio of their expenditures is 3:2. If each of them saves Rs.600 in the year, the annual income of A is:
(A) 4800 (B) 1800
(C) 1200 (D) 2400

85. The ratio of income of two persons is 5:3 and that of their expenditures is 9:5. If they save Rs.2600 and Rs.1800 respectively. Their incomes are:
(A) Rs.8000, Rs.4800 (B) Rs.6000, Rs.3600
(C) Rs.10000, Rs.6000 (D) Rs.9000, RS.5400

86. If 18, x and 50 are in proportions, then find the value of x.
(A) 32 (B) 30 (C) 3 (D) 5

87. Find the average of all even numbers upto 100?
(A) 51 (B) 38.6
(C) 50.8 (D) 45.6

88. The average of certain first consecutive odd numbers is 60. Find their sum?
(A) 2551 (B) 3600
(C) 3460 (D) 2496

89. The average of four consecutive even numbers is 15. The 2nd highest number is:
(A) 12 (B) 18 (C) 14 (D) 16

90. The average of the five consecutive odd numbers is k. If the next three odd numbers are added then the new average will exceed the old average by.
(A) 2 (B) 3 (C) 4 (D) 2

91. Find the average of first 20 multiples of 3?
(A) 33 (B) 31.5 (C) 35 (D) 29

92. Average of 7 consecutive odd numbers is 31. If the pervious and next odd numbers to the 7 odd numbers are also included then what will be the new average.
(A) 7 (B) 19 (C) 82 (D) 84

93. Average of 7 consecutives even numbers is 52. If the pervious and next even number to these 7 even numbers are also included then what will be the new average.
(A) 33 (B) 52 (C) 35 (D) 29

94. Find the average of squares of consecutive even numbers from 1 to 44.
(A) 680 (B) 960
(C) 690 (D) 860

95. The average of the first and the second of three numbers is 15 more than the average of the second and the third of these numbers. What is the difference between the first and the third to these three numbers?
(A) 15 (B) 45 (C) 60 (D) 30

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96. The average of six numbers is 45.5. If a new number is added then the new average becomes 47. Find the new number?
(A) 62 (B) 56 (C) 54 (D) 52

97. The average of the marks obtained in mathematics by 5 students is 50. Later it was realized that the marks obtained by one student was read as 84 though it was originally 48. Find the correct average.
(A) 42.8 (B) 75.3
(C) 65.4 (D) 47.4

98. Average marks of 20 students are 75. One student got 30 marks instead of 80. Other 70 instead of 50. Find true average.
(A) 75 (B) 76.5 (C) 71 (D) 77

99. The average weight of five persons sitting in a boat is 38 kg. The average weight of the boat and the persons sitting in the boat is 52 kg. What is the weight of he boat?
(A) 228 (B) 122 (C) 232 (D) 242

100. Average age of 22 persons is 28 years. 2 persons left the group then average age decreases by 1.5 years. Find the average age of two persons.
(A) 43 (B) 47 (C) 49 (D) 41

101. Average age of 20 persons increases by 6 months when two persons of age 45 & 30 years replaces by new person. Find the average age of two new persons?
(A) 42.5 (B) 43.5 (C) 45 (D) 50

102. The average monthly salary of all the employees in a factory is Rs.8840. If the average salary of all the officers is Rs.15000 and that of the remaining employees is Rs.8000, then what is the percentage of the officers among the employees?
(A) 10% (B) 12% (C) 15% (D) 20%

103. The average age of a class is 15.8 years. The average age of the boys in the class is 16.4 years while that of the girls is 15.4 years. The ratio of boys to girls in the class is.
(A) 3:5 (B) 2:3 (C) 3:4 (D) 1:2

104. The average salary of the entire staff in a office is Rs.150 per month. The average salary of officers is Rs.450 and that of non-officers is Rs.80. If the number of officers is 14 then find the number of non-officers in the officer.
(A) 65 (B) 55 (C) 60 (D) 70

105. A car goes 20 metres in a second. Find its speed in km/hr.
(A) 20 (B) 18 (C) 72 (D) 36

106. The speed of a car is 72 km/hour, in how much time will it cover the distance of 720 meters.
(A) 72 sec. (B) 36 sec.
(C) 80 sec. (D) 20 sec.

107. A person crosses a 600 m long street in 5 minutes. What is his speed in km/h?
(A) $7\frac{1}{5}$ (B) $8\frac{1}{5}$ (C) $9\frac{1}{5}$ (D) 8

108. A man crosses a road 250 metre wide in 75 seconds. His speed in km/hr is:
(A) 10 (B) 12 (C) 12.5 (D) 15

109. A train is moving with the speed of 180 km/hr. its speed (in metres per second) is:
(A) 5 (B) 40 (C) 30 (D) 50

110. A train is moving at a speed of 80km/h and covers a certain distance in 4.5 hours. The speed of the train to cover the same distance in 4 hours is:
(A) 100 km/hr (B) 70 km/hr
(C) 85 km/hr (D) 90 km/hr

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111. Raj and Prem walk in opposite direction at the rate of 3km/hr and 2km per hour respectively. How far will they be from each other after 2 hours?
(A) 8 (B) 10 (C) 2 (D) 61

112. Ravindra walks at 6 km/h and Riya cycles at 11 km/h towards each other. What was the distance between them when they started if they meet after 42 minutes?
(A) 17.9 kms (B) 14.9 kms
(C) 8.9 kms (D) 11.9 kms

113. A thief steals a car at 3 pm and drives it at 60 kmph. The theft is discovered by police at 3:30 pm and seats off in another car at 80 kmph. When will the police overtake the thief?
(A) 5 pm (B) 6 pm
(C) 4 (D) 4 : 30

114. A man covers $\frac{9}{20}$ distance by bus and the remaining 10 km on foot. His total journey (in km) is.
(A) 15.6 (B) 12.8 (C) 18.18 (D) 24

115. A man travels the distance of his journey $\frac{3}{4}$ by bus, $\frac{1}{6}$ by Rickshaw and remaining 2km on foot. The total distance travelled by the man is:
(A) 12 (B) 18 (C) 20 (D) 24

116. Neha walks at a speed of 12 km/h and reaches her school 12 minutes late. If she walks at the speed of 18 km/h, she reaches her school 18 minutes early. Find the distance of her school form her house.
(A) 18 km (B) 64 km
(C) 75 km (D) 72 km

117. If I walk at 5 km/h, I miss a train by 7 minutes. However, if I walk at 6 km/h I reach the station 5 minutes before the departure of the train. The distance between my house and the station is
(A) 5 km (B) 6.5 km (C) 7 (D) 6

118. A student goes to school at the rate of 2 ½ km/h and reaches 6 minutes late. If he travels at the speed of 3 km/h he is 10 minutes early. What is the distance to the school?
(A) 1 km (B) 4
(C) $3\frac{1}{2}$ (D) $3\frac{1}{4}$

119. A thief is stopped by a policeman from a distance of 350 metres. When the policeman starts the chase, the thief also starts running. Assuming the speed of the thief as 5 km/h and that of the policeman as 7 km/h, how far the thief would have run, before he is over taken?
(A) 875 m (B) 700 m
(C) 1050 m (D) 525 m

120. Walking at 3/5 of its usual speed, a man is 20 minutes to late. Find its usual time to cover that distance.
(A) 30 m (B) 35 m
(C) 28 m (D) 40 m

121. A and B start to cover a distance at their usual speeds at the same time. But B takes 1 hour 15 minutes more than A to reach the destination if he walks 5/6th of A's speed. The time taken by B is
(A) 6 hr 45 min (B) 7 hr 15 min.
(C) 7 hr 30 min (D) 8 hr 15 min

122. A train 100 m. long of which speed is 36 km/h. How much time will it take to cross a man?
(A) 10 sec (B) 20 sec
(C) 15 sec (D) 18 sec

123. A train of length 360 m. is running at a speed of 72 km/h. In how long time will cress pole?
(A) 14 sec (B) 15 sec
(C) 16 sec (D) 18 sec

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124. A train is of length 160 m. crosses a 240 m. long platform in 20 seconds. Find the speed of the train in km/h.
(A) 72 kmph (B) 54 kmph
(C) 60 kmph (D) 80 kmph

125. The diameter of a toy wheel is 14cm, What is the distance travelled by it in 15 revolutions?
(A) 880 (B) 660
(C) 600 (D) 560

126. The perimeter (in meter) of a semicircle is numerical equal to its area (in sq. m). The length of its diameter is ($\pi=\frac{22}{7}$)
(A) $3\frac{3}{11}$ (B) $5\frac{6}{11}$ (C) $6\frac{6}{11}$ (D) $6\frac{2}{11}$

127. The difference between the circumference and diameter of a circle is 150m. the radius of that circle is ($\pi=\frac{22}{7}$)
(A) 25 (B) 30 (C) 35 (D) 40

128. The outer and inner diameter of a circular path is 728 cm and 700cm respectively. The breadth of the path is.
(A) 7 (B) 14 (C) 28 (D) 20

129. The ratio of the radii of two wheels is 3:4. The ratio of their circumference is:
(A) 4:3 (B) 3:4 (C) 2:2 (D) 3:2

130. If the circumference of a circle is reduced by 50%, its area will be reduced by
(A) 12.5% (B) 25%
(C) 50% (D) 75%

131. The area of a triangle is 216 sq. cm and its sides area in the ratio 3:4:5. The perimeter of the triangle is:
(A) 6 (B) 12 (C) 36 (D) 72

132. If the numerical value of the perimeter of an equilateral triangle is $\sqrt{3}$ times the area of it, then the length of each side of the triangle is:
(A) 2 (B) 3 (C) 4 (D) 6

133. What is the area of the triangle whose sides are 9cm, 10cm and 11cm? (in sq. cm)
(A) 30 (B) 60
(C) $30\sqrt{2}$ (D) $60\sqrt{2}$

134. From a point within an equilateral triangle, perpendiculars drawn to the three sides are 6cm, 7cm and 8cm respectively, the length of the side of the triangle is:
(A) 7 cm (B) 10.5 cm
(C) $14\sqrt{3}$ cm (D) $\frac{14\sqrt{3}}{3}$ cm

135. If the side of an equilateral triangle are increased by 20%, 30% and 50% respectively to form a new triangle the increased in the perimeter of the equilateral triangle is:
(A) 25% (B) $33\frac{1}{3}\%$
(C) 75% (D) 100%

136. The length of the diagonal of a square is 'a' cm. Which of the following represents the area of the square (in cm^2)?
(A) 2a (B) $\frac{a}{\sqrt{2}}$ (C) $\frac{a^2}{2}$ (D) $\frac{a^2}{4}$

137. If the diagonals of two squares are in the ratio of 2:5. Their area will be in the ratio of:
(A) $\sqrt{2}:\sqrt{5}$ (B) 2:5

(C) 4:25 (D) 4:5

138. The ration of length to width of a rectangular field is 3:2.If the perimeter is 80m ,then find the width.
(A) 16 (B) 18 (C) 20 (D) 25

139. The perimeter of a rectangular is 160 m, and difference of both sides is 48 m. then find the side of a square which has equal area to the rectangle.
(A) 24 (B) 32 (C) 36 (D) 48

140. Between a square of perimeter 44cm and a circle of circumference 44cm, which figure has larger area and by how much?
(A) Square, $33cm^2$
(B) Circle, 33 cm^2
(C) Both have equal area
(D) Circle, 495 cm^2

141. The diagonal of a square is $4\sqrt{2}$ cm. The diagonal of another square whose area is double that of the first square is:
(A) $8\sqrt{2}$ (B) 16 (C) $\sqrt{32}$ (D) 8

142. If the length and the perimeter of a rectangle are in the ratio 5:16. then its length and breadth will be in the ratio:
(A) 5:11 (B) 5:8 (C) 5:4 (D) 5:3

143. If the side of a square in increased by 25%, then its area is increased by:
(A) 25% (B) 55%
(C) 40.5% (D) 56.25%

144. The perimeter of a rectangle is 160 metres and the difference of two sides is 48 metres. Find the side of a square whose area is equal to the area of this rectangle?
(A) 32 (B) 8 (C) 4 (D) 16

145. The area (in m^2) of the square which has the same perimeter as a rectangle whose length is 48m and is 3 times its breadth is:
(A) 1000 (B) 1024
(C) 1600 (D) 1042

146. The perimeter of five square are 24cm, 32cm, 40cm, 76cm and 80cm respectively. The perimeter of another square equal in area to sum of the areas of these squares is:
(A) 31 (B) 961 (C) 124 (D) 62

147. A circular wire of radius 42 cm is bent in the form of a rectangle whose sides are in the ratio of 6:5. The smaller sides of the rectangle is: ($\pi=\frac{22}{7}$)
(A) 60 (B) 30 (C) 25 (D) 36

148. Area of square with diagonal $8\sqrt{2}$ cm is: (in sq. cm)
(A) 64 (B) 128 (C) 56 (D) 29

149. The perimeter and length of a rectangle are 40m and 12m respectively. It breadth will be:
(A) 10 (B) 8 (C) 6 (D) 3

150. A piece of wire when bent to form a circle will have a radius of 84 cm. If the wire is bent to form a square, the length of a side of the square.
(A) 152 (B) 168 (C) 132 (D) 225

151. The perimeter of a non-square rhombus is 20cm. On its diagonal is 8cm. the area of the rhombus is:
(A) 28 cm^2 (B) 20 cm^2
(C) 22cm^2 (D) 24 cm^2

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152. The area of a rhombus is 256 sq. cm and one of its diagonal is twice the other in length The length of its larger diagonal is:
(A) 32 (B) 48 (C) 36 (D) 24

153. The perimeter of a rhombus is 40 cm and the measure of an angle is 60°, then the area of it is: (in square cm)
(A) $110\sqrt{3}$ (B) $50\sqrt{3}$ (C) $160\sqrt{3}$ (D) 100

154. The area of a field in the shape of a trapezium measures $1440m^2$. The perpendicular distance between its parallel sides is 5:3, the length of the longer parallel side is:
(A) 75 (B) 45 (C) 120 (D) 60

155. The perimeter of the floor of a room is 18m. What is the area of the walls of the room, if the height of the room is 3m? (sq. metre)
(A) 21 (B) 108 (C) 54 (D) 42

156. If a diagonal of a cube is 4√3 cm. find its volume (in cube cm)
(A) 60 (B) 64 (C) 68 (D) 72

157. A solid cuboid of dimensions 8cm×4cm×2cm is melted and cast into identical cubes of edge 2cm. Number of such identical cubes is:
(A) 16 (B) 4 (C) 10 (D) 8

158. If the volume of two cubes are in the ratio 27:1, the ratio of thier edbe is:
(A) 3:1 (B) 27:1 (C) 1:27 (D) 1:3

159. The volume of a right circular cylinder is equal to the volume of that right circular cone whose height is 108 cm and diameter of base is 30 cm. If the height of the cylinder is 9 cm, the diameter of its base is
(A) 40 (B) 50 (C) 70 (D) 60

160. The curved surface area of a cylindrical pillar is 264 m^2 and its volume is 924 m^3 ($\pi=\frac{22}{7}$). Find the ratio of tis diameter to its height.
(A) 7:6 (B) 6:7 (C) 3:7 (D) 7:3

161. 12 sphere of the same size are made by melting a solid cylinder of 16cm diameter and 2cm height. The diameter of each sphere is:
(A) 2 (B) 4 (C) $\sqrt{3}$ (D) 3

162. If the radius of a sphere is increased by 2cm, its surface area increased by 352 cm^2. The radius of sphere before change is:
(A) 3 (B) 4 (C) 5 (D) 6

163. Each of the measure of the radius of base of a cone and that of a sphere is 8cm. Also, the volume of these two solids are equal. the slant height of the cone is:
(A) $8\sqrt{17}$ cm (B) $4\sqrt{17}$ cm
(C) $34\sqrt{2}$ cm (D) 34 cm

164. The ratio of the volume of a cube and of a solid sphere is 363:49. The ratio of an edge of the cube and the radius of the sphere is ($\pi=\frac{22}{7}$).
(A) 7:11 (B) 22:7 (C) 11:7 (D) 7:22

ANSWERS

1.	C	2.	C	3.	A	4.	C	5.	D
6.	D	7.	B	8.	A	9.	A	10.	A
11.	C	12.	C	13.	C	14.	A	15.	D

16. C	17. B	18. B	19. B	20. C
21. B	22. D	23. A	24. C	25. B
26. B	27. C	28. C	29. A	30. A
31. A	32. B	33. B	34. C	35. A
36. D	37. C	38. B	39. C	40. A
41. C	42. A	43. A	44. C	45. B
46. A	47. D	48. B	49. D	50. D
51. D	52. A	53. D	54. B	55. C
56. D	57. A	58. B	59. D	60. C
61. A	62. B	63. C	64. D	65. A
66. B	67. B	68. D	69. B	70. B
71. A	72. C	73. B	74. B	75. C
76. D	77. C	78. C	79. B	80. C
81. A	82. D	83. A	84. D	85. A
86. B	87. A	88. B	89. D	90. B
91. B	92. D	93. B	94. C	95. D
96. B	97. A	98. B	99. B	100. A
101. A	102. B	103. B	104. C	105. C
106. B	107. A	108. B	109. D	110. D
111. B	112. D	113. A	114. C	115. D
116. A	117. D	118. B	119. A	120. A
121. C	122. A	123. D	124. A	125. B
126. A	127. C	128. B	129. B	130. D
131. D	132. C	133. C	134. C	135. B
136. C	137. C	138. A	139. B	140. B
141. D	142. D	143. D	144. A	145. B
146. C	147. A	148. A	149. B	150. C
151. D	152. A	153. B	154. A	155. C
156. B	157. D	158. A	159. D	160. B
161. B	162. D	163. A	164. B	

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Tense

Tense- a grammatical concept

Time-a concept beyond grammar

Tense को हम अपनी सुविधानुसार तीन भागों में बांट सकते है।

1. Present Tense
2. Past Tense
3. Future Tense

Time और tense के बीच में one to one correspondence का होना जरूरी नहीं होता है। दूसरे शब्दों में यदि किसी sentence का tense present है तो कोई जरूरी नहीं है कि उसका Time भी Present ही हो।

Ex. He goes to school daily.

Tense-present

Time-present

Ex. A : how do you earn your livelihood?

B : I sell books.

Tense - present

Time - present/past/future

Ex. Ravi is going to Mumbai tomorrow.

Tnese - present

Time - future

Ex. You will have read this story in your ninth class.

Tense - future

Time - past

1. Simple present tense - V-I

Helping verb - do / does

1. Universal truths show करने के लिए

Ex. The sun rises in the east.

Two and two make four

The moon shines at might.

2. Idioms/proverbs में सामान्यतः इसी tense का प्रयोग करते है।

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Ex. Honesty is the best policy.

Practice makes a man perfect.

3. Scientific theories show करने के लिए

Ex. Water boils at 100 degree C.

4. Habitual actions show करने के लिए
Dogs bark at strangers.

He always helps the poor.

5. Daily routine activity show करने के लिए

Ex. I take my breakfast at 7.30am.

He studies till 11.00 pm

6. Human feelings/emotions/natural activity show करने के लिए

Ex. A baby loves its mother.

We love our country.

Leaves fall in autumn.

7. Newspaper headlines में

Ex. India win the match

India lose the match to England.

8. Games/sports commentary में

Ex. Sachin hits/strikes a six.

Pathan bowls to Afridi.

9. Stative verb का प्रयोग generally simple present tense में ही करते है।

Ex. I know Ram. (✓)

I am knowing Ram. (x)

She prefers coffee.

Alok likes sweets.

10. Future में यदि कोई fixed programme या fixed plan show करने के लिए

Ex. The P.M. leaves for America tomorrow.

The finance minister comes here next Sunday.

Our college reopens on 1 July.

11. Activity regular हो customary हो तथा almost permanent हो उन्हें show करने के लिए

Ex. He smokes.

He drinks.

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Adverbs-usually/generally/occasionally/always/often

इन Adverbs का प्रयोग सामान्यत simple present में होता है।

Ex. He always does his home work.

2. Present continuous tense- is/are/am+v-ing

1. इस tense का प्रयोग उन actions को show करने के लिए करते है जो बोलते समय (at the time of speaking) जारी हो।

Ex. The fan is moving.

They are dancing in the room - immediate present.

2. General present को show करने के लिए भी इस tense का प्रयोग होता है।

Ex. He is doing a course in English.

3. निकट भविष्य (near future) के action का बोध कराने के लिए

Ex. She is leaving for New Delhi tonight.

I am going to Mumbai tomorrow.

4. Sub (action) - Be+going+to+V-I - intention

Sub (no action)- Be+going+to+v-I - likelihood

Ex. He is going to buy this dictionary (intention)

It is going to rain today. (likelihood)

5. (now/still/now-a-days) इन adverbs का प्रयोग generally present continuous tense में करते है।

Ex. Ravi is watching a movie now.

He is still waiting for you outside.

6. Usual तथा unusual action में contract show करना हो तो usual action को simple present में तथा unusual action को present continuous में रखते है।

Ex. He generally sells only newspapes but this week he is selling magazines also.

Every month he goes to New Delhi on business but this week he is going to Mumbai.

7. Go/come/fly/leave यदि इन verb का प्रयोग आने–जाने के sense में हो तो present continuous tense का प्रयोग होता है।

Ex. He is going to Delhi tomorrow.

He is coming from Australia.

He is flying to Newyork.

3. Present perfect tense- has/have + V - 3

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1. किसी action के completion को show करने के लिए

Ex. He has done his home work.

We have taken lunch today.

2. Present perfect tense में past time marker का प्रयोग नहीं कर सकते।

Ex. I have seen the Taj Mahal. (✓)

I have seen the Taj Mahal in 2005. (x)

I saw the Taj Mahal in 2005. (✓)

3. किसी past action का present में result show करने के लिए

Ex. I have read the Ramayana.

I have seen the Taj Mahal.

4. **Stative Verbs** का प्रयोग यदि Present Perfect Tense में हो तो time show करने के लिए since/for का प्रयोग करते है।

Ex. I have been knowing him since 2005. (x)
I have known him since 2005. (✓)

He has been ill since Monday. (✓)

5. Yet/already/just/lately इन adverbs का प्रयोग generally present perfect tense में करते है।

Yet - used in negative sentence

Already - used in positive sentence

Ex. He has already done the home work.

They have not yet arrived here.

They have just come in.

Which Hindi film have you seen lately?

6. यदि किसी activity/action के होने की संख्या (number) को दर्शाना हो तो present perfect tense का प्रयोग करते है।

यदि किसी activity/action के repetition को दर्शाना हो तो present perfect continuous tense का प्रयोग करते है।

Ex. I have telephoned you 15 times today. (number)

I have been telephoning you all day. (repetition)

He has failed the exam 19 times.

He has been failing the exam since 1993.

4. **Present perfect continuous- has/have+been+v-ing**

Since - point of time

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For - period of time

Ex. Ravi has been watching television since morning.

Ravi has been watching television for 3 hours.

Rule:- Wait/knock/lie- इन तीन verbs का प्रयोग perfect tense में नहीं करते है। इनका प्रयोग continuous tense में करते है।

Ex. The duster has lain on the table for 3 days.

The duster has been lying on the table for 3 days.

I have waited for you for half an hour.

I have been waiting for you for half an hour.

Rule:- All day, all night, all morning, all evening, all afternoon- यदि sentence में ये expression हो तो (live/stay/work/sleep/rain/think) इन verbs का प्रयोग perfect continuous tense में होगा।

Ex. I have been working in the garden all day.

He has been sleeping all afternoon.

Ex. India and Pakistan have played a hundred test matches in the 80s.

India and Pakistan played a hundred test matches in the 80s.

5. Simple past tense- V-II

1. किसी past action का time बताने के लिए

Ex. I saw the Taj Mahal in 2005.

In 1526 Babar became an emperior.

Rule- कई sentences में past maker indirect/applied होता है।

Ex. I played chess at school.

He got a job in New Delhi.

Rule:- यदि simple past tense के साथ किसी adv of frequency का प्रयोग हो तो इससे past habit show होती है।

Ex. He visited us on Sundays.

The hostel inmates always complained about the bad quality of food.

Ex. It is mentioned in history that Akbar rarely slept/had slept more than 5 hours at a stretch in his life.

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Rule:- ago/yesterday/last night/just now- इन adverbs का प्रयोग simple past tense में करते है।

Ex. The train left a few minutes ago.

The plane landed just now.

He came here yesterday.

Rule:- यदि दो या दो से अधिक past actions को 'and' से जोड़े तो उन्हें sequence में रखते है तथा sentence में simple past tense का प्रयोग करते है।

Ex. He took off his shift and threw it on the floor.

He heard an explosion and he telephoned the police.

Rule:- Present perfect में पूछे question का answer देने के लिए भी simple past tense का प्रयोग करते है।

Ex. Have you taken lunch today?

Yes, I took it half an hour ago.

6. Past continuous tense-was/were+v-ing

Rule:- भूतकाल में किसी action के continuation (जारी रखना) को show करने के लिए

Ex. They were talking loudly.

She was watching television.

Rule:- point of time→ simple past tense.

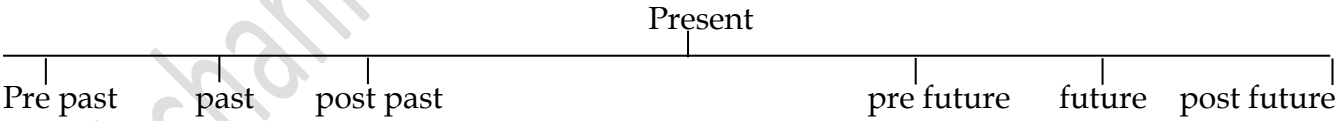
Period of time → past continuous tense.

Ex. The plane was flying vey low when I caught sight of it.

We were having supper when the lightly went off.

When I saw Ram, he was playing football.

While I was watching television, my wife was mending her dresses.



7. Past perfect tense:- Had + V-3

Rule:- यदि past में दो actions हो तो जो action पहले होता है उसे pre past कहते है तथा उसे past perfect tense में रखते है तथा जो action बाद में होता है उसे simple past में रखते है।

When/before - simple past

After - past perfect

Ex. The train had left when I reached the station.

The patient had died before the doctor came.

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The doctor came after the patient had died.

The chief guest came after the show had started.

Rule:- Contextualized Sentence- जिसमें सन्दर्भ हो।

Decontextualized sentence- जिसमें कोई सन्दर्भ हो।

यदि Sentence Decontextualized हो तो simple past tense का ही प्रयोग करते है।

Ex. I had done an M.A. in 2015. (x)

I did an M.A. in 2015. (✓)

Rule:- Single achievement या non-achievement show करने के लिए past tense was/were+able+to+V-I का प्रयोग करते है।

Ex. When the boat capsized, they could swim across the river.

When the boat capsized they were able to swim across the river.

8. Past perfect continuous tense:- Had+been+V-ing

Ex. I had done an M.A. when I joined this institute in 2015.

Pre past	past	present
Completion of M.A.		

Ex. I was/had been doing an M.A. when I joined this institute in 2015.

Pre	past	post past	present
	job	M.A. continued	

Rule:- post past के action को past perfect continuous tense में रखते है।

Ex. I have been waiting for you for an hour.

I had been waiting for you for an hour when the clock struck six.

9. Simple future tense- will/shall+V-I

1. सामान्यतः रूप से future होने वाले action को show करने के लिए

Ex. I shall meet her at 5 O' clock. (pure future)

He will go to Delhi tomorrow. (pure future)

Rule:- I/We- shall

He/she/it/they/you-will

I/we-will

He/she/it/they/you-shall

Ex. I will go to the cinema. (determination)

You shall get a new bicycle. (promise)

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He shall go there (order/command)

I will dismiss you if you don't obey my orders. (threat)

Rule:- let's से शुरू होने वाले sentence का question tag बनाने के लिए

Let's play cricket, shall we?

Let's go for a walk, shall we?

10. Future continuous tense-will/shall+be+v-ing

Ex. I'll be doing my M.A. in 2015 (action incomplete)

Rule:- Future में routine activily show करने के लिए future continuous tense का प्रयोग करते है।

Ex. I'll be seeing my doctor this evening.

11. Future perfect tense-will/shall+have+v-3

1. यदि past action को predict करने के लिए

Ex. You will have read this story at school.

2. यदि sentence by+adv. of time हो तो future perfect tense का प्रयोग करते है।

Ex. He will reach Delhi by now. (x)

He will have reached Delhi by now. (✓)

Rule:- यदि future में दो actions हो तो जो action पहले होता है उसे pre future कहते है तथा उसे future perfect में रखते है तथा जो action बाद में होता है उसे simple present में रखते है।

Ex. By the time this letter reaches you, you will have received my parcel.

Present	Pre future	Future
	Parcel received	letter

Ex. By the time you reach the theatre, the show will have started.

12. Future Perfect continuous-will/shall+have+been+v-ing

Ex. He will have completed 20 years of his prison life in 2029.

Present	Pre future	Future
		2009

He will have been completing 20 years of his prison life in 2029.

Present	future	post future
	2029	

Rule:- Post future के action का future perfect continuous tense रखते है।

English language में 3 areas और है जिनमें tense का ही प्रयोग होता है।

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1. In time clause.
2. In hypothetical expressions
3. In conditional clauses.

1. Time clauses-

Subordinator-when/before/after/as soon as/till/until/now that/since

Rule:- Time clause में future tense का प्रयोग नहीं करते है।

Ex. I will allow you to go when you will complete your work. (x)

I will allow you to go when you complete your work. (✓)

Rule- time clause में simple present या present perfect tense का प्रयोग करते है।

Ex. you can go after you will finish your homework. (x)

You can go after you finish your homework. (✓)

You can go after you have finished your homework. (✓)

Ex. All the flowers in this garden will dry up before it rams/will rain.

Ex. I will inform you as soon as I get/shall get any news in this regard.

Ex. You can't go home until you will not finish the work. (x)

You can't go home until you do not finish the work. (x)

You can't go home until you finish the work. (✓)

Rule:- 'until' के बाद वाले clause में 'not' का प्रयोग नहीं करते है।

Ex. Don't go any where until/till I come back.

Wait here untill/till I come back.

Ex. Now that you are a post graduate you must reach for a job.

2. Hypothetical expression:-

I wished (past perfect tense)

I wish

As if

As though

It is high time

Supposing → exclamatory

if only sentence

Hypothetical expressions के बाद में सामान्यतः simple past tense का प्रयोग करते है।

Ex. I wish I know. (x)

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I wish I knew. (✓)

I wished I had known. (✓)

Rule:- hypothetical expression के बाद में यदि 'be-verb 'were' आये तो हमेशा were का प्रयोग करते है।

Ex. He talks to me as if he was/were my master.

He looks pale as though he was/were dead.

Supposing 10 people came!

If only I were rich!

He looks pale as if he saw/had seen ghost.

I wish he met/had met you when you were living in England.

Rule- past की बात को hypothetical expressions में show करने के लिए past perfect tense का प्रयोग करते है।

Ex. It is high time you got ready.

It is high time the government rooted out corruption.

3. Conditional sentences- तीन प्रकार के होते है।

- ❖ Open condition
- ❖ Imaginary condition
- ❖ Destroyed condition

1. Open condition/factual condition

Ex. If you labour you will pass

(if clause) result clause

(simple present) (simple future)

If you don't labour, you will not pass.

Rule:- unless- (if+not)

Unless you don't labour, you will not pass. (x)

Unless you labour, you will not pass. (✓)

Rule:- यदि open condition में किसी fact (तथ्य) को दर्शाये गये तो result clause में भी simple present tense का प्रयोग करते है।

If you heat butter, it melts.

If you heat butter, it will melt.

If you mix blue and yellow, you get green.

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If you mix blue and yellow, you will get green.

Whenver you heat butter, it melts.

Ex. Take an umbrella in case it rains/will rain.

Rule:- यदि if-clause में should+v-I form आये तो main clause में imperative या simple future tense का प्रयोग करते है।

If anything happens, telephone the police.

If anything happens, I shall telephone the police.

If anything should happen, telephone the police.

If anything should happen, I shall telephone the police.

Ex. Should anything happen, telephone the police.

Should anything happen, I shall telephone the police.

2. Imaginary condition/hypothetical/theoretical/unreal past/remote possibility.

Ex. If you labored hard, you would pass.

(if clause) result clause

Simple past (would/should/could/mkight+V-I)

Ex. If I went to Delhi, I would buy you a new pair of shoes.

Rule:- Imaginary condition के if-clause में यदि be-verb हो तो हमेशा 'were' का ही प्रयोग करते है।

Ex. If I was/were a bird, I could fly in the air

Were I a bird, I could fly in the air.

Ex. If I were rich, I could build another Taj.

Were I rich, I could build another Taj.

Ex. If he were to come, she would be happy.

If he had some time to spare, he would spend it in a public library.

3. Destroyed condition/Rejected condition/impossible condition

Ex. If you had laboured, you would have passed.

(if clause) result clause

(had+v-3) (would/should/could/might+v-3)

Ex. If I had gone to Delhi. I would have bought you a new watch.

Had I gone to Delhi, I would have bought you a new watch.

Ex. If you had not explained, I would not have understood it.

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If she had acted on me, she might not have attacked by Malaria.

If she had acted on me, she might not have been attacked by Malaria.

If we had had John in our team, we would have won the match.

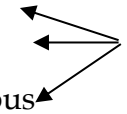
If he had a few hours to spare, he would spend them in a public library.

Rule:- Exclamatory sentences **में** continuous tense **का प्रयोग नहीं करते है।**

Ex. Here is coming the Bus!

Here comes the bus?

It (simple present)

Rule:- present perfect
present perfect continuous

since simple past

Ex. Three years have passed since I met with an accident.

It is three years since I met with an accident.

Ram has not been going to the office since he fell ill.

Ex. This is for the first time that he helps me.

This is for the first time that he helped me.

Ex. I missed the train that I usually catch and have/had to travel by the next which was a slow train.

Noun (संज्ञा)

Ram	goes	to	school	daily
↓	↓	↓	↓	↓
noun	verb	preposition	noun	adverb

We use different types of words in a language and put them in a particular order to convey our meaning. These different types of words are known as 'Parts of speech'

English **में** parts of speech **की संख्या 8 होती है।**

1. Nouns
2. Pronouns
5. Adverbs
6. Prepositions

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3. Adjectives
4. Verbs
7. Conjunctions
8. Interjections

1. Nouns (संज्ञा)

- A noun is a naming word.
- A noun is the name of a person, place, thing or idea / quality

1. John is a good player. (Name of a person)
2. Jaipur is the capital of Rajasthan. (Name of a place)
3. The book is mine. (Name of a thing)
4. Honesty is the best policy. (Name of an idea/quality)

- A noun is a word which can be used as a subject an object or a complement in a sentence.
- Nouns वह शब्द है जिसका प्रयोग हम sentence में subject, object और complement की तरह कर सकते है।

How to use Apostrophe-s ('s)

's का प्रयोग possession (अधिकार) show करने के लिए करते है।

Ram's father

Boy's house

's का प्रयोग living noun (सजीव) के साथ करते है।

जबकि non-living noun (निर्जीव) के साथ 'of' का प्रयोग करते है।

John's legs (✓)

The table's legs (x)

The legs of the table (✓)

1. यदि noun proper/common singular हो– 's

Ravi's mother

Boy's father

2. यदि noun common plural हो (जिसके अंत में s ना हो)– 's

Men's hostel

Women's beauty parlour

3. यदि noun common plural हो (जिसके अंत में s हो)– '

Boys' hostel

Girls' common room.

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Country
River
Ship
- Proper noun - 's/of
- Common noun-of

This country's population (x)
The population of this country. (✓)
India's population. (✓)
The Ganga's water (✓)
Vikrant's soldier (✓)

Time
Distance
Weight

इन शब्दों के expressions में हम 's का प्रयोग करते है।

1. Today's newspaper.
2. One month's leave.
3. A yard's distance.
4. 10 miles' distance

Ex. This is Ram's house. (✓)
This is John's and Mary's house. (x)
This is John and Mary's house. (✓)

Ex. Ravi does not want anybody's help. (✓)
Ravi does not want anybody's else help. (x)
Ravi does not want anybody else's help. (✓)

Ex. Your tone of speaking as well as your-friends is really extremely sarcastic yet we never take it seriously. (x)
Your tone of speaking as well as your friend's is really extremely sarcastic yet we never take it seriously. (✓)

Ex. America's problems are, however, not so serious and awful as India. (x)
America's problems are, however, not so serious and awful as India's. (✓)

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English में कुछ particular words है जिनके ऊपर exams में प्रश्न पूछे जाते रहे है।

Good- (U)

This is a good/goods train.

Moral - (C)

Moral-morals

He is a man of low moral/morals.

Morale- (U)

The morale of the Indian army is/are very high.

Custom- रीति रिवाज (C)

1. Dowry is a bad custom.
2. There are so many bad customs in India.
3. He is a custom/customs officer.
4. He works in the custom/customs department.

Damage - (U)

1. This has done great damage to the reputation of P.M. (✓)
2. He has claimed damage/damages in a court of law.

(Compensation क्षतिपूर्ति / भरपाई)

Manner- (C)

She does not have manner/manners. (शिष्टाचार)

Etiquette- (U)

He does not have professional etiquette/etiquettes.
(शिष्टाचार)

Premise-(C) निष्कर्ष

The meeting will be held on the premise/premises of Rajasthan University.

(building भवन / परिसर)

Colour-(C)

The president gave away colour/colours to battalion 13.

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(flag)

Fruit-(U)

1. How many fruits do you eat daily? (x)
2. How much fruit do you eat daily? (✓)
3. The fruits of labour are sweet. (✓)

Work- (U) (कार्य)

(C) Book (रचना / कृति)

1. This is a famous work of Kalidas. (✓)
2. These are some famous works of Kalidas. (✓)

Paper - (U)

1. Your papers are not in order. (✓)
   (documents)

Letter-(C)

1. Nehru was a man of letter/letters. (idiom).

Word-(C)

1. He is a man of his word/words. (idiom)
2. The officer gave the word/words to fire.
   (order)
3. The word/words of his success came abroad.
   (news)

Pain - (U)

There was pain/pains all over my body.

I have a pain in my back. (✓)

(localize करने पर pain शब्द article लेता है)

I have great pain in my back. (x)

I have great pains in my back. (✓)

[to show the intensity of 'pain']

a pain- ordinary pain

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great pains- बहुत ज्यादा pain

Take pains
Take trouble - idioms

He took pain/pains to help me.
He took trouble/troubles in help me.
He spared no pain/pains to help me.

Common Errors:

1. The water of Jaipur does not suit me. (x)
   The climate of Jaipur does not suit me. (✓)
2. There is no place/space in this compartment. (x)
   There is no room in this compartment. (✓)
3. He is my cousin brother. (x)
   He is my cousin. (✓)
4. This is my study room. (x)
   This is my study. (✓)
5. Put your sign here. (x)
   Put your signature here. (✓)
6. You should write it in your copy. (x)
   You should write it in your notebook. (✓)
7. Ravi had a bath in the sea. (x)
   Ravi had a bathe in the sea. (✓)
8. The boy is wearing a new dress. (x)
   The boy is wearing a new suit. (✓)
9. John is an usherer in the Gem cinema. (x)
   John is an usher in the Gem cinema. (✓)
10. They are my forebearers. (x)
    They are my forebears. (✓)
11. I do not know about her whereabout. (x)
    I do not know about her whereabouts. (✓)
12. I do my best but I don't understand why I do not get expected mark in the examination. (x)
    I do my best but I don't understand why I do not get expected marks in the examination.(✓)

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13. I have not read this letter so I am not aware of its <u>content.</u> (x)
    I have not read this letter so I am not aware of its <u>contents</u>. (✓)
14. This is a <u>full proof</u> system. (x)
    This is a <u>fool proof</u> system. (✓)
15. He is a <u>cheater.</u> (x)
    He is a <u>cheat.</u> (✓)
16. <u>Wait list.</u> (x)
    Waiting list. (✓)
17. <u>Passing marks.</u> (x)
    Pass marks. (✓)

➢ **<u>Noun Gender</u>**

1. Masculine – Male gender
   Boy/father/man/son/brother/husband/king/uncle/usher etc.
2. Feminine – Female gender
   Girl/mother/ sister/ daughter/ wife/ queen/ lady/ woman/ aunt/ usherette etc.
3. Common gender – Either male or female
   Child/ baby/ friend/ student/ teacher/ doctor/ dancer/ singer/ professor etc.
4. Neuter – Neither male nor female / non-living things
   Pen/ Book/ Table/ Chair/ Room/ Building/ Bike etc.

➢ **<u>Modern Grammar</u>**

1. **Countable Nouns**
2. **Uncountable Nouns**

On the basis of count ability –

Countable Noun	Uncountable Noun
1. Nouns that can be counted : - chair, boy, girl, country	1. Nouns that can't be counted :- water, tea, petrol, experience
2. Noun that can take :- a, an, one, two, three, many before them. Ex. – a boy one student, many	**2. Noun that don't take numerals before them :- a, an, one, two, three, many**
countries	
3. Nouns that can be made plural Boy – boys, city – cities, country – language, subject, disease	**3. Noun that can't be made plural** **Ex. John – Johns, Jaipur – Jaipurs** **The place on wheels – no plural – force , physics, Rabies, English, chinese** **Uncountable Noun uncountable**

- **<u>Countable Nouns:-</u>** plural from
  a/an/one – singular form
  Common Nouns and Collective nouns Countable Nouns
  Boy – Boys
  He is a boy.
  Family – families
  Ravi has a small family.
- **<u>Uncountable Nouns:-</u>** No Plural form
  No number – uncountable Nouns
  Proper Nouns/ Material Nouns/ Abstract Nouns Uncountable Nouns
  Ex. John – Johns. (x)
  A <u>John</u> is my best friend. (x)
  <u>John</u> is my best friend.
  Gold – Golds. (x)
  <u>A Gold</u> is a precious metal. (x) <u>Gold</u> is precious metal.
  Honest – Honesties (x)
  <u>An honesty</u> is the best policy. (x)

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Honesty is the best policy ( )

- **Uncountable Nouns:-**
  (equipment/ furniture / jewellery/ luggage/ baggage / poetry/ evidence/ bread/ filth/ garbage/ rubbish/ dirt/ experience/ electricity/ hair/care/ work/ crockery/ dust/ toast/ soap/ News/ work/ advice/ paper/ information/ scenery/ violence/ mischief / word etc.) – Uncountable nouns.
  1. I have an urgent work. (x)
  2. I have an urgent piece of work. ( )
  3. He gave me two advices. (x)
  4. He gave me two pieces of advice. ( )
  5. This is a good news. (x)
  6. This is good news. ( )
  7. We need to buy some furnitures. (x)
  8. We need to guy some furniture. ( )
  9. I have to write an application. Please give me two papers. (x)
  10. I have to write an application. Please give me two sheets of paper. ( )
- Note:- Uncountable Nouns singular verb
  1. The poetries of Words worth are romantic in nature.
     The poetry of Words worth is romantic in nature.
  2. The sceneries of Kashmir are wonderful.
     The scenery of Kashmir is wonderful.
- Some words are always plural in their form and sense, we can't remove –s in the ending:-
  Garments:- Breeches, Jodhpurs. Jeans, Pants, Trousers, Braces, Shoes.
  Instruments: - Scissors, Pliers, Tongs, Bellows, Pincers, gallows, Binoculars, Spectacles
- **Special words**:- Savings, Earnings, Wages, Proceeds, Means, aegis, surroundings, archives, arrears, premises, teens, auspices, thanks, alms, fireworks, embers, outskirts, rations, shambles,
  Now-a-day, Wages are increasing.
  1. The wages of sin is death. (Proverb)
  2. This is a proper means to solve this question.
  3. The terrorist entered India through illegal means.
  4. I can't afford expensive schooling because my means is/are not good.
  5. He has entered the room.
  6. He entered into the function last year.
  7. Alms were/are given to the poor.
  8. The embers of fire is/are still burning.
  9. The meeting was held on the premises of JNU
  10. The meeting ended without any premise.
  11. The police arrested some criminals on the outskirts of Jodhpur.
  12. My thanks go to the students who supported me.
- Some nouns may be singular or uncountable despite their endings in – s
- They always take a singular verb
  Ex.
  Name of subjects :- Physics, Economics, Civics, Statistics, Politics, Linguistics
  Name of Diseases :- Measles, Mumps, Rickets, Shingles
  Name of Games :- Billiards, Darts, Draughts, Tennis
- Some Special words :- Summons, Crossroads
  Ex.
  1. The court has sent him a summon/summons.
  2. The court has sent him two Summons/Summonses.
  3. Too much mathematics is/are taught at schools.
  4. His mathematics is/are week.
  5. He is good at mathematics.
  6. Rabies is/are diseases.
  7. Statistics has/have revealed multiple scams in the organization of Commonwealth Games.
  8. There is a crossroad/crossroads near the railway station.
- Some words seem singular inform but they are plural in sense :-
  4 C's :- cattle, clergy, children, cavalry
  4 P's :- Peasantry, people, poultry, police
  Electoral, Infantry, Gentry, Vermin
  Ex.
  1. Many cattle/cattles are grazing in the field.
  2. The police is looking for a fair- haired man in his twenties.
  3. Our infantry has marched forward.
  4. A police have arrested the thieves.
  5. Childrens are playing in the garden.
- Some nouns are same in singular and plurals :-
  Fish, deer, sheep, series, species, corps, aircraft, zebra, crew, team, jury, counsel, swine, pice, apparatus, stone, craft, public, offspring, committee, government, firm, innings
  Ex.
  1. Only one inning was played.
  2. Both innings have been played.
  3. In the ancient time the kings usually hunted deers.
  4. Two sheeps were grazing in the field.

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5. A team which is full of enthusiasm have a better chance of winning.
6. There are many fishes in the aquarium.

- Some Collective noun are used as both Singular and plurals forms :- Family, crew, team, jury, committee, parliament.
  Ex.
  1. The jury be unanimous on this issue.
  2. The jury was/were divided on this matter.
  3. The jury has/have given its/their majority decision.
  4. The jury be ready to give its/their decision without any dispute.
  5. The jury wants to give their unanimous decision.
  6. Parliament are unanimous on the issue of the lokpal

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Noun Numbers

- Regular Plural:-
  1. x, o, ch, sh, ss, zz, -- es
     ex. Box—boxes
     Lens lenses
     Quiz Quizzes
     Fax Faxes

     Photo—photos, Steno—stenos, Gas—Gasses, Topaz—Topazes
     Radio—Radios, dynamo—dynamos, Fuzz—Fuzzes, Flash—Flashes
     Exceptions

     mango – mangoes
     Church—churches
     Bench—benches
     Brush—brushes
     Dish—dishes
     Bush—bushes
     Boss—bosses

     Stereo—stereos
     stomach—stomachs
     Monarch—monarchs
     Buzz—buzzes
     piano—pianos
     Exceptions

  2. consonant + y – remove – y – add – ies
     Story—stories
     History—histories
     Sky—skies
     Baby Babies
     Petty Petties
     City Cities

     baby babies
     Lady ladies
     Harmony Harmonies
     Thievery Thieveries
     Army Armies

     Y (a/e/i/o/u) plural Y s
     Boy—boys
     Key—keys
     Journey Journeys
     Key Keys

     day days
     Holiday Holidays
     Guy Guys

  3. f/fe f/ fe
     Wife—wives
     Knife—knives
     Half—halves
     Leaf—leaves
     Self—selves
     Thief—thieves

     roof roofs
     chief chiefs
     hoof hoofs
     dwarf dwarfs
     café cafes
     serf serfs
     Handkerchief Handkerchiefs
     Safe Safes
     Exceptions

     Life—lives
     Sheaf—Sheaves
     cliff—cliffs
     Grief—Griefs
     Elf—Elves
     Loaf—Loaves
     Proof—Proofs
     Strife—Strifes

     <u>Irregular Plurals:-</u>
     Man—men
     Ox—oxen
     Tooth—teeth
     Mouse—mice

     woman—women
     foot—feet
     goose—geese

     Ex. This wall of this building is fifty foot high.
     This wall of this building is fifty feet high.

     Some special Plurals

- Note:- Latin Plural
  Syllabus Syllabuses

  Datum data
  Criterion Criteria
  Phenomenon Phenomena

  Syllabi

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Stratum Strata

Formula Formulas / Formulae
Medium Mediums / Media
Alumnus alumni
Basis—Bases
Oasis—Oases
Thesis—theses

Erratum errata
Sanatorium sanatoria

Radius Radii
Axis—axes
Crisis—Crises
Analysis—analyses
Synopsis—Synopses
Stadium—Stadia

1. The earth moves round the sun on its axis/axes.
2. Our country is facing so many crisis/crises these days.

1. The data is/are highly misleading.
2. What is/are your criteria for admission to this college?
3. An earthquake is a natural phenomenon/ phenomena
4. They are the alumnus/alumni of this department.
5. A language consists of several stratums/ strata.

- **Compound Nouns:-**

Sister-in-law	sisters-in-law	editor-in-chief	editors-in-chief
Father-in-law	fathers-in-law	Court-martial	Court-martials
Mother-in-law	mothers-in-law		
Brother-in-law	Brothers-in-law		

Woman teacher	women teachers	man teacher	men teachers
Man servant—men servants		woman hater—woman haters	
Man hater—	man haters	Attorney General — Attorneys General	
Woman lover	woman lovers	Looker-on	Lookers-on
Vice-admiral	vice-admirals	Hanger – on	hangers – on
Girl student	girl students	passer – by	passers – by
Boy student	boy students	runner – up	runners - up

- **Double Plural:-**

Index—indexes/indices Mathematical Operation
Cloth—Cloths—Clothes
Genius—Geniuses—Genii
Staff—Staff—Staves
Brother—Brothers—Brethren
The Brethren of this locality always remains at daggers drawn.
Hindus, Muslims and Christians are brethren.
Inventions are made by geniuses.
The rules of indices help us in Maths.
The boy was wrapped in cloths
He was wearing new clothes.

Collective nouns

A herd of cattle	a chain of mountains	a quiver of arrows	a band of musicians
a sheaf of grains	a heap of sand	a fleet of ships	a choirs of singers
A flock of birds	a shoal of fish	a lock of hair	a string of onions
A cluster of stars	a bevy of ladies	a flight of stairs	a brace of pigeons
A troupe of dancers	an army of soldiers	a board of directors	a squadron of planes
a crew of sailors	a pack of wolves	a mob of rioters	multitude of people
a queue of people	a bouquet of flowers	a congregation of pilgrims	a staff of officials

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a syndicate of merchants | an album of photos | a beam of rays | a pack of playing cards
a mass of ruins | a lump of coal | a league of nations | A jury of judge

Pronoun (सर्वनाम)

Pro - for

Noun - noun

A word which we use in place of a noun is known as 'Pronoun'

Ex. Ram is a doctor.

He is a doctor.

- हम Pronoun का प्रयोग noun के repetition को avoid करने के लिये करते है।

  Ram is a good player.

  He is my best friend.

  He lives in Delhi.

- Noun और pronoun के बीच में number और gender का agreement होता है।

  Ram- he

  Sita- she

  Boys- they

  Pen- it

  Climate- that

  Car- one

  Cars- ones

Note- Pronoun noun का substitute होता है इसलिए इसका प्रयोग sentence में subject, object और complement की तरह कर सकते है।

He beats his servant violently. (subject)

Ram beat him. (object)

It is he. (complement)

Classification of pronouns:

1. Personal pronouns-

   1st person- I/We (the person speaking or writing)

   2nd person-you (the person spoken to/written to)

   3rd person-he/she/it/they/name (the person or thing spoken about/written about)

Subjective	Objective	Possessive adjective	Possessive pronoun
↓	↓	↓	↓

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I Person	I	me	my	mine
	We	us	our	ours
II Person	You	You	Your	yours
III Person	he	him	his	his
	She	her	her	hers
	It	it	its	X
	They	them	their	theirs

Rule: Pronoun की subjective form sentence में सामान्यतः subject का कार्य करती है।

Ex. Ravi is a good student.

He is a good student.

Rule- Pronoun की objective form sentence में सामान्यतः object का काय करती है।

Ex. I saw Ravi at the party last night.

I saw him at the party last night.

Rule- Possessive adjective के बाद noun का आना अनिवार्य होता है।

Ex. This is my house.

That is your book.

They are my relatives.

Rule- Possessive pronoun के बाद noun का पयोग नहीं करते।

Structure-1- Noun/pronoun + be/of + possessive pronoun.

Ex. This book is my/mine.

He is a friend of your/yours.

Structure-2- Possessive pronoun + be + noun/adjective

Ex. My father is a doctor and your/yours is a lecturer.

Your car is blue and my/mine is red.

His efforts will bring him more success than your. (x)

His efforts will bring him more success than yours. (✓)

One's - possessive of 'one'.

Ones- plural of 'one'.

Ex. One should respect one's/ones parents.

Small cars are more economical than big one/ones.

There are 5 active workers and 3 lazy one/ones in this factory.

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Ex. It's legs are broken. (x)

Its legs are broken. (✓)

It's :- it is

It's raining outside. (✓)

It is raining outside. (✓)

2. Reciprocal pronoun:

English में reciprocal pronouns की संख्या दो है।

Each other - used for 2

One another- used for more than 2

1. Ram and Ravi looked at each other/one another.
2. The five brothers are at daggers-drawn, so they feel it below their dignity to talk to each other/one another.

3. Distributive pronouns:

Each
Either
Neither
} Verb singular pronoun singular

1. Each is present in his room.
2. Each of the boys is/are present in his room.
3. Has/have either of you seen my Rolex watch?
4. Neither of them is/are guilty of this crime.

4. Indefinite pronouns : (अनिश्चय वाचक सर्वनाम)

Indefinite pronouns की संख्या लगभग 12 होती है।

Some
No
Any
Every
} body + one + thing - verb singular + pronoun singular

1. Somebody is waiting for you outside.
2. Everyone is busy with his/their research work.
3. Nobody was/were ready to go there.
4. Everything was/were lying scattered in the room.

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5. Everybody gets frightened whenever they see a tiger. (x)

   Everybody gets frightened whenever he sees a tiger. (✓)

6. Everybody who was working in that office gave a day's pay as their contributions to the dependants of Dr. Sonu who had died on duty. (x)

   Everybody who was working in that office gave a day's pay as his contribution to the dependants of Dr. Sonu who had died on duty. (✓)

7. Nobody should lose heart when they find themselves in the teeth of difficulties. (x)

   Nobody should lose heart when he finds himself in the teeth of difficulties. (✓)

5. Demonstrative pronouns:

This/that - verb singular

These/those- verb plural

1. This is my bike.
2. These are your books.

6. Relative pronouns: (संबंध सूचक सर्वनाम)

English में relative pronoun की संख्या 3 होती है।

Who- used for persons (व्यक्ति)

Which-used for things (वस्तु)

That-used for both persons & things (दोनों)

इन्हें relative pronoun इसलिये कहते है क्योंकि ये pronoun अपने से पहले आने वाले noun/ pronoun को relate करते है।

Relative pronoun से पहले आने वाला noun/pronoun उसका antecedent कहलाता है।

The boy who is wearing a red shirt is Ram.

(antecedent)

Subjective	Objective	Possessive
who	whom	whose
which	which	of which
that	that	X

The boy who is standing there is John.
(sub)

The boy whom you met yesterday is John.
(Ob)

The book which is on the table is mine.

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(sub)

The book which you gave me is very useful.

(ob)

Rule- यदि relative pronoun के तुरन्त बाद में verb आये तो वह relative pronoun clause (उपवाक्य) में subject का कार्य करेगा।

यदि relative pronoun तथा verb के बीच में कोई दूसरा शब्द हो तो वह relative pronoun clause में object होगा।

Rule- यदि relative pronoun clause में subject का कार्य करे तो हम इसे नहीं हटा सकते।

यदि relative pronoun clause में subject का कार्य ना करे तो हम इसे हटा सकते है।

1. The boy who is standing there is John. (✓)
   The boy is standing there is John. (x)
2. The boy whom you met yesterday is John. (✓)
   The boy you met yesterday is John. (✓)
3. The book which is on the table is mine. (✓)
   The book is on the table is mine. (x)
4. The book which I bought yesterday is very useful. (✓)
   The book I bought yesterday is very useful. (✓)

Rule:- यदि **relative pronoun** से पहले कोई **preposition** हो तो **relative pronoun** को हटाकर उस **preposition** को **sentence** के अंत में लगाते है।

1. This is the house in which I live. (✓)
2. This is the house in I live. (x)
3. This is the house I live in. (✓)

I want somebody to whom I can talk. (✓)
I want somebody I can talk. (x)
I want somebody I can talk to (✓)

I need a chair on which I can sit. (✓)
I need a chair I can sit. (x)
I need a chair I can sit on. (✓)

Ram whose legs are broken is my student. (✓)

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(Person)

The table whose legs are broken s mine. (x)

(Thing)

The table the legs of which are broken is mine. (✓)

(Relative clause)

Rule:- Relative pronoun 'that' **से पहले preposition का प्रयोग नहीं कर सकते।**

This is the book about that I was talking. (x)

This is the book about which I was talking. (✓)

This is the book that I was talking about. (✓)

This is the book I was talking about. (✓)

Rule:- (only/none/all/little/any/superlative degree) यदि ये शब्द sentence में antecedent से पहले आये या antecedent का कार्य करे तो relative pronoun 'that' का ही प्रयोग करते है।

1. This is the only book which/that is in demand now-a-days.
2. There is none who/that can help you in this crucial moment.
3. All which/that glitters is not gold.
4. I can give you the little money which/that I have.
5. You can come at any time which/that you like.
6. This is the best which/that I have been able to do for you.

Ex. She hates everybody and everything who reminds her of her mistakes. (x)

She hates everybody and everything that reminds her of her mistakes. (✓)

Ex. What is there which/who is hidden from God? (x)

What is there that is hidden from God? (✓)

7. Interrogative Pronouns: **(प्रश्नवाचक सर्वनाम)**

इन Pronouns का प्रयोग प्रश्न पूछन के लिये करते है इसलिये इन्हें Interrogative Pronouns कहते है। All wh-words including-how

(who/what/which/whom/whose/when/where/why/how)-interrogative adverbs

Who-indicates indentity

Who is he?

He is Alok.

What-indicates profession.

What **is he?**

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He is a doctor.

Which-used for things.

Which is your car?

Which is your house?

Rule: Selection/choice **के लिए हमेशा which का प्रयोग करते है।**

1. Which is Ravi?

   The one in a white shirt is Ravi.

2. Who is the better student, Mohan or Sohan? (x)

   Which is the better student, Mohan or Sohan? (✓)

3. Who is the two players got selected in the team? (x)

   Which of the two players got selected in the team? (✓)

4. The owner asked who of the servants broke the jug. (x)

   The owner asked which of the servants broke the jug. (✓)

Whom-objective of who

Ex. Whom did you see yesterday?

I saw Ram yesterday.

Whose-possessive of 'who'

Ex. Whose is this house?

This house is mine. (possessive)

When-indicates time.

Ex. When did you come here?

I came here on Monday. (time)

Where-indicates place

Ex. Where do you live?

I live in Jaipur. (place)

Why- indicates reason.

Ex. Why did you not take the R.A.S. exam?

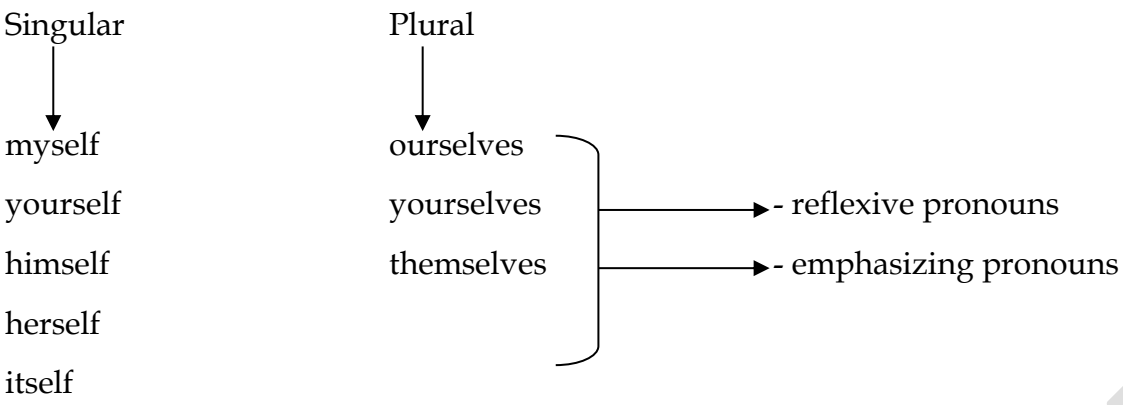
I did not take the R.A.S. exam because I met with an accident (reason)

How- indicates manner.

Ex. How does Ravi play?

Ravi plays well (manner)

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These are Reflexive pronouns and emphasizing pronouns.

8. Emphasizing/emphatic pronouns:

- **जिन pronouns के अन्त में -self/selves आये उन्हें यदि sentence से हटा दे, तो भी sentence grammatically correct रहता है वे emphasizing pronouns कहलाते है।**
- **Emphasizing pronouns का प्रयोग जिस element पर जोर देना हो उसके तुरन्त बाद अथवा sentence के अन्त में करते है।**

I talked to Lata.

I myself talked to Lata.
I talked to Lata herself. → Emphasizing Pronouns
I talked to Lata myself.

9. Relfexive Pronouns-

वे pronoun जिनके के अन्त में -self/-selves आये उन्हें यदि sentence से हटा दे तो sentence grammatically wrong हो जाता है वे reflexvie pronouns कहलाते है।

reflexive pronoun sentence में object का कार्य करते है।

He killed a tiger. (✓)
He killed himself. (✓)
He killed. (x)

She hurt herself.
She saw herself in the mirror.
He gave the largest portion to himself.
They enjoyed themselves.

He introduced as the capation of the local cricket team to the principal. (x)
He introduced himself as the captain of the local cricket team to the principal. (✓)

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He prostrated before his master who had returned from America after a year. (x)

He proostrated himself before his master who had returned from America after a year. (✓)

Relfexive Verbs-

Avail + reflexive pronouns + of + noun
Pride + reflexive pronouns + on + noun
Absent + reflexive pronouns + from + noun

1. He availed himself of the opportuntiy. (✓)
2. She prides herself on her beauty. (✓)
3. They absented themselves from the meeting. (✓)

Common Errors:

Rule: यदि sentence में व्यक्ति/वस्तु की संख्या 2 हो तो as/than के बाद में pronoun की I form का प्रयोग करते है।

I know Ram better than he. (✓)
I know Ram better than him. (✓)

दोनों sentence grammatically correct है परन्तु इन दोनों वाक्यों के अर्थ में अन्तर है।

Rule:- 'be' verb के बाद में pronoun की Ist form का प्रयोग करते है।

Be- [is, am, are, was, were]

If I were him, I should hire a taxi to go there. (x)

If I were he, I should hire a taxi to go there. (✓)

Rule- let के बाद में pronoun की II form का प्रयोग करते है।

Let you and I accomplish the task. (x)

Let you and me accomplish the task. (✓)

Rule- Preposition के बाद में pronoun की II form का प्रयोग करते है।

The secret lies between you and I. (x)

The secret lies between you and me. (✓)

Rule- Either - 2 (व्यक्ति/वस्तु) के लिए
Any - 2 से ज्यादा के लिए → [Positive Sentence]

Any of the two pictures is worth seeing. (x)

Either of the two pictures is worth seeing. (✓)

Rule:- Neither - 2 (व्यक्ति/वस्तु) के लिए
None - 2 से ज्यादा के लिए → [Neegative sentence]

Neither of the three principals is looking after his college well. (x)

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None of the three principals is looking after his college well. (✓)

Note:- 2 3 4 (used in positive sense)

2	3	4
You	he	I
	She	We
	It	
	They	

2. Second person pronoun
3. Third person pronoun
4. First person pronoun

1. He, I and you are good friends. (x)
2. You, he and I ae good friends. (✓)
3. You and he are good friends. (✓)
4. He and I are good friends. (✓)
5. You and I are good friends. (✓)

Note:- 1 2 3 (used in negative sense)

You, he and I are guilty of this crime. (x)

I, you and he are guilty of this crime. (✓)

Note- 1 2 3 (when used in the plural)

We, you and they should work for peace and prosperity

Rule:- Relative pronoun के बाद में आने वाली verb उसके पहले आने वाले noun/pronoun से fix होती है।

Antecedent ← (Who / which / That) → verb

Ex. It is I who am to blame.

It is you who are to blame.

It is he who is to blame.

Ex. This is one of the greatest events that has/have happeded this year.

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This is the only one of her novels that is/are in demand now a days.

Ex. The boy who does best he will get a prize. (x)

The boy who does best will get a prize. (✓)

Ex. The climate of Jaipur is better than Delhi. (x)

The climate of Jaipur is better than that of Delhi. (✓)

Ex. The hotels of Pune are more luxurious than Meerut.

The hotels of Pune are more luxurious than that/those of Merrut. (✓)

Rule:- यदि 'they' के बाद कोई relative pronoun आये तो they को those में बदल देते है।

Ex. They who are standing in this room should not move from their positions. (x)

Those who are standing in this room should not move from their positions. (✓)

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Prepositions

Word- Preposition → Noun
Preposition → Pronoun
Preposition → V+ing (gerund)

Ex. The book is on the table.
(noun)
The watch is for her.
(pron)
He is thinking of going to the cinema.
(v-ing)

1. In- motionless **(स्थिर)**
into- in motion **(गतिशील)**

Ex. Ravi is in the room.
Ravi is moving in the room.
Ravi came into the room.

Ex. He is swimming in the river.
He dived into the river.

2. on- motionless **(स्थिर)** Surface **से स्पर्श होनी चाहिए**
upon- in motion **(गतिशील)**

Ex. The cat is sitting on the table.
The cat jumped upon its prey.

3. Between - used for 2
Among - more than 2

Ex. Distribute these chocolates between Ram and Mohan.
Distribute these chocolates between the two sister.
Divide these apples among the children.

Ex. You can come to see me between 5.00 pm to 6.00 pm. (x)
You can come to see me between 5.00 pm and 6.00 pm. (✓)

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Rule:- यदि देशों की border की बात हो तो हमेशा between का ही प्रयोग होगा।

Ex. India is among Pakistan, Nepal, China and Bangladesh. (x)
India is between Pakistan, Nepal, China and Bangladesh. (✓)

This tournament is among India, Pakistan, and Australia. (x)
This tournament is between India, Pakistan, and Australia. (✓)

4. Over:- यदि दो points के बीच की height को vertically measure कर सके।
Under- यदि दो points के बीच की depth को vertically measure कर सके।
Above- यदि दो points के बीच की height को vertically measure ना कर सके। Simply Higher
Below- यदि दो points के बीच की depth को vertically measure ना कर सके। Simply Lower

Ex. The fan is moving over his head.
The cat is sitting under the table.
The sky is above us.
The valley is below us.

Ex. A saint should be over/above the worldly pleasures.

5. Beside- by the side of के पास में
Besides-in addition of के अतिरिक्त

Ex. Go and sit beside your parents.
Besides a car, he has a bike.

6. Since-point of time/beginning of an action.
For- period of action/duration

Ex. He has been watching television since 9 O' clock.
He has been watching television for 3 hours.

7. In spite of/despite- के बावजूद

Ex. In spite of his thorough training in the repair and maintenance, he could not get a suitable Job.
Despite his thorough training in the repair and maintenance, he could not get a suitable Job.

8. Till-the end of an action

Ex. He works in the office till 6 pm

9. Off-surface से अलग करना

Ex. He took the book off/from the table.
Wipe the dust off the table.

10. Onto- ऊपर से नीचे की ओर

Ex. When he was waiting for the bus, a ball fell onto his head.

11. Across- from one side to the other (sub-action)
- on the other side (sub-no action)

Ex. He went across the road.

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The boys swam across the river.

Ex. There is a tree across the road.

12. Through-from one end of sb/sth to the other (में से होकर)

Ex. The train passed through the tunnel.

He went through the forest to reach there in time.

You must go through these books to score good marks in the exams. (exception)

13. From-to

इन Prepositions का प्रयोग यदि time के लिए हो तो इनका एक साथ आना जरूरी होता है।

Ex. He sits in the clinic from 8.00am to 12.00pm.

यदि From-to का प्रयोग place के लिए हो तो इन preposition का एक साथ प्रयोग जरूरी नहीं होता।

Ex. He is going to New Delhi.
He is coming from Mumbai.
He is flying from Newyork to Washington.

at- clock wise time
on-day/date
in-month/year
------fixed time

Ex. I take breakfast at 7.30am.

He came here on Monday.

He was born in 1992.

Rule:- today/tomorrow/yesterday/every day- इन शब्दों से पहले on का प्रयोग नहीं करते है।

Ex. He came here on yesterday. (x)

He came here yesterday. (✓)

Rule- last/next इन शब्दों से बनने वाले dverbs से पहले 'on' का प्रयोग नहीं करते है।

He visited us on last Sunday. (x)

He visited us last Sunday. (✓)

Rule:- festivals के नाम से पहले at का प्रयोग करते है।

He came to visit us at Christmas.

He came to visit us at Diwali.

Rule:- in the morning

in the evening

in the afternoon

in the night. (at a particular time)

at night - (the whole night)

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Ex. The moon shines at night.
He came to see me in the night.
He rises early in the morning.

In the morning
In the evening
In the afternoon
In the night
→ - यदि इन expression के साथ day/date का प्रयोग किया जाये तो 'in' के स्थान पर 'on' का प्रयोग करते है।

Ex. He came here <u>on</u> Saturday morning.
Emergency was declared <u>on</u> the night of 25 June 1975.
I met him <u>on</u> the evening of 1 June.

He came in the late afternoon.
He came in the early morning of 31 June.
He came in the late afternoon of 31 June.
→ - exceptions

Rule:- at dawn
at dusk
at sunrise
at sunset/sundown
at noon
at midnight (12.00am)

Before-earlier than sb/sth
Ex. They will come here before 6 P.M.

After-later than sb/sth
Ex. We shall leave after 6.00 pm

Ex. Application forms must be submitted by 31st July.
(on 31 July or before 31 July, but not after 31 July)

In- at the end of
Within-before the end of

Ex. She will come back in an hour.
Complete the task within two days.

Rule:- at-below city (village/town)

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In-city/above city

Ex. He lives at Rampur.

He lives in Jaipur.

He lives in Rajasthan.

He lives in India.

Ex. I was born in Bikaner. (birth place को importance देने के लिए)

Rule:- 'in' preposition-village/town (proper name)

Ex. He was born in a village. (✓)

He was born in a town. (✓)

Rule:- यदि किसी जगह (place) पर किसी reason की वजह से थोड़े समय के लिए रूके और यदि action temporary nature का हो तो 'at' का प्रयोग करते है।

Ex. The P.M. stayed at London on his way to New York.

The conference will be held at Jaipur on 31 March 2014.

For

1. Abbreviations की full form से पहले

Ex. P.M. stands for prime minister.

2. 'Support' show करने के लिए

Ex. He is for the congress.

He is against the congress.

3. Purpose show करने के लिए

Ex. He has gone to London for doing a Ph.D.

4. To show 'price and exchange'

Ex. I bought this dictionary for a hundred rupees.

Ex. Thank you very much for the trouble you took in helping me.

By

1. Passive voice में agent (living) से पहले 'by' का प्रयोग करते है।

Passive voice instrument (non-living) से पहले 'with' का प्रयोग करते है।

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Ex. A snake was killed by Ram. (agent)

A snake was killed with a stick. (instrument)

Rule:- by- no article- means of transport

Any other preposition- article - means of transport 'by'

Rule:- 'Manner/method' show करने के लिए 'by' का प्रयोग करते है।

Ex. I caught him by the neck.

Bananas are sold by the dozen.

Petrol is sold by the litre.

Mangoes are sold by the kilo.

Letters are sent by post.

Ex. The payment must be made by cheque/by draft.

The payment must be made in case (exception)

Of-

Rule:- 'origin' show करने के लिए

Ex. He comes of a rich family.

Rule:- 'exact age' show करने के लिए

Ex. Alok is of 20 years.

Rule:- 'character' show करने के लिए

Ex. He is a man of his word.

He is a man of iron will.

He is a man of principle

He is a man of letters.

Rule:- 'space' में separation show करने के लिए

Ex. China is in the north of India. (x)

China is north of India. (✓)

China is to the north of India. (✓)

Bangladesh is east of India. (✓)

Pakistan is west of India. (✓)

Rule:- 'Time' में separation show भरने के लिए

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Ex. He got two promotions within a year of his service.

Rule:- 'Material' show करने के लिए

Ex. The shirt is made of cotton.

The table is made of wood.

The box is made of silver.

Ex. Wine is made of/from grapes. (from बदल जाने पर)

The cake is made with flower, sugar, ghee, butter etc. (in case of ingredients)

Ex. His name was cut in stone.

His statue was cast in bronze. - (exception)

These paintings have been done in oils.

Rule:- disease/illness के कारण death होने पर of का प्रयोग करते है।

Ex. He died of malaria.

He died of cancer.

Rule:- hunger/starvation/thirst/grief/shame के कारण death होने पर 'of' का प्रयोग करते है।

Ex. He died of hunger.

He died of grief.

Rule:- overeating/over drinking/food poisoning/wound के कारण death होने पर 'from' का प्रयोग करते है।

Ex. He died form a wound.

He died from over-eating

Rule- Medium of expression show करने के लिए 'in' का प्रयोग करते है।

Ex. He writes in English.

He writes in Hindi. (medium)

He writes in ink/pen/pencil.

Ex. He writes with a pen/a pencil.

He cut the tree with an axe. (instrument)

Exceptions:

Ex. What's the time by your watch?

It is 10 to 11. (11.00 बजने में 10 मिनट है)

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I told it to his face. (face to face)

I asked a question of him.

(to ask sth of sb)

He is good at mathematics.

He is bad at physics.

He is slow at learning a foreign language.

He is busy with his project work.

Ex. The birds are sitting in the tree.

The birds are sitting on the branches of the tree.

He was sitting under a tree.

He was sitting in the shade of a tree.

I waded through water to push the boat out.

The bus is plying on the road.

He is walking along the road. (सहारे सहारे)

I am on my way to Delhi.

Ravi is on the selection committee.

He is on the teaching staff of this college.

He is on the Indian express.

He died in debt.

Ex. I had to leave Jaipur bag and baggage.

He reached the station. (no preposition)

He made the station. (no preposition)

He got to the station.

He arrived at the station. (preposition 'at' छोटी जगह)

He arrived in Jaipur. (preposition 'in' बड़ी जगह)

He has been to Delhi.

He is leaving America. (place of departure)

He is leaving for America. (place of destination)

Ex. The train is bound for Mumbai.

He was sitting on the desk.

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He was sitting at the desk.

Don't drive at this speed.

She married beneath her. (a man of lower status)

It is beneath contempt. (hate के लायक भी नहीं)

It is beneath dignity for me to accept the offer.

The road is in bad condition.

The road is in bad shape.

The road is under repair.

The house is under construction.

He works under me.

Tansen lived under Akbar. (during the reign of)

Under these circumstance he can't help me.

She received us with a smile

His face was wet with tears.

He was groaning in pain.

The college is on the man road.

The man in a red shirt is my friend.

The boy with blue eyes is John. (physique शरीर का हिस्सा होने पर)

The girl with a bucket on her head is my sister.

The woman with a baby in her arms is my aunt.

जब कोई व्यक्ति/वस्तु हम अपने साथ carry करते हे उससे पहले with का प्रयोग करते है।

What are you listening to?

Who are you talking to?

Ex. He did not answer my question.

This is no answer to my question.

The P.M. stresses the importance of punctuality.

The P.M. put stress on the importance of punctuality.

The principal emphasized the importance of honestly.

The principal laid emphasis on the importance of honesty.

He pitied me.

He took pity on me.

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The committee consists of ten members. (✓)

The committee comprises of ten members.

The committee comprises ten members. (comprise verb के बाद में of नहीं आता) (✓)

When he was sleeping a thief entered into his house.

They entered into this agreement.

They entered into a hot exchange.

The game lasted two hours.

The game lasted for two hours.

I invited him to lunch.

He is playing the guitar.

He is playing a modern tune on the guitar.

Rule:- English में कुछ verb ऐसी है जिनके बाद preposition नहीं आता है।

Ex. They are discussing about the problem of Kashmir.

Let's decide on it right now.

The poet describes about nature in this poem.

China invaded in India in 1962.

He attacked on me with a dagger.

He approached to me for some urgent work.

She accompanied with me to the temple.

He has visited to Jaipur.

She ordered for two cups of coffee.

She advised to her child not to play with fire.

He resembles to his father.

This incident amply illustrates about the present political system of our country.

Considerations of safety override over all other concerns.

He demanded for pencils.

He married with Sita.

He got married to Sita.

Rule:- English में कुछ certain words है जिनके बाद में आने वाले preposition fixed होते है।

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For-

Affection for	desire for	eligible for
Apology for sth	fondness for	prepared for
Capacity for	match for	qualified for
Compensation for	passion for	sorry for
Canvass for sth	responsible for	hope for
Substitute for	mourn for	wait for/await

To-

access to	acceptable to	yield to
Approach to	applicable to	replay to
Object to	devoted to	damage to
Admit to	reduced to	apparent to
Attention to	entitled sb to sth	propose to sb
Indifference to	aspire to	suggest to sb
Key to	belong to	say to sb
Preface to	listen to	speak to sb
Reference to	occur to sb	prefer to
Succession to	refer to	connected to
Concession to	conducive to	

With

Acquaintance with	gifted with	tremble with
Alliance with	popular with	thronged with
Comparison with	satisfied with	acquainted with
Agree with	busy with	coincide with
Contented with	quarrel with	delighted with
Comply with	disgusted at/by/with	commensurate with

Of-

Guity of	devoid of	accused of
Ignorant of	worth of	kind of
Capable of	jealous of	afraid of
Short of	fond of	tired of
Ashamed of	envious of	deprived of

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Envious of	deprived of	sure of/about
Loss of	theft of	glimpses of

At

Astonished at	surprised at	amazed at
Startled at	quick at	slow at
Aim at	good at/bad at	look at ab/sth
Vexed at sth		

From

Prevent sb from sth

debar from

Discourage from

abstain from

refrain from

free from

Restrain from

different from

On

Compliment on	congratulate on	insist on
Succeed in	Interested in	

Angry with sb

Angry at sth

Annoyed with sb

Annoyed at sth

Ex. Ram got angry with me.

Ram was angry at his behaviour.

Part from sb

Part with sth

Ex. Lata parted with/from her mother in tears.

Blinded in (physically blind)

Blind to (not pay attention to)

Ex. He is blind in the right eye.

The government is blind to the needs of the people.

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Complain to sb

Complain against sb

Complain of/about sth

Ex. The players complained to the manager against the attendant and of the poor facilities provided in the hotel.

Damage to
Theft of
Loss of

Ex. We should prevent damage to public property.
We should prevent theft of public property.
We should prevent damage to and theft of public property.

Ex. Some animals can live both in water and on land very easily.

Warn of/against sth

Ex. I warned him of the upcoming danger.

Enquire of sb
Enquire into sth
Cash in on
Burst into
Plunged into
Affiliated to/with
Go to the cinema
Go shopping/swimming/skiing
Go the circus
Abide by
Triumph over
Anxious for sb
Anxious about sth
Overwhelmed by

Ex. The streets were thronged with people.
She burst into tears.
The whole city was plunged into darkness due to sudden power failure.

Remain in doubt
Decline of a dynasty
Decline in moral.

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Subject Verb Agreement

1. There are three types of concord in English language.
   2. Grammatical Concord
   3. Notional Concord
   4. Proximity Concord

Grammatical concord –इस concord में verb subject के अनुसार आती है।

Subject Singular – Verb singular

Subject Plural – Verb plural

Priyanshi goes to school.
S.S S.V Pre. Phrase

Anku and Ankita go to college.
Sub.P Verb+P

2. Notional Concord :- Subject important नहीं sentence के sense के according Verb का use :-

→ Dickens has criticized the philosophy in The Hindu.

→ **<u>100 Rupees</u> is not a big amount.**

(The amount of 100 rupees) → Real subject

→ **15 years is not a long period in the history of a nation.**

→ **Wuthering Heights is a great novel.**

→ **Either –or**
Neither –nor
Not only – but also
we apply proximity concord in these expressions.

→ **Either you or she is guilty of this crime.**

→ **Neither he nor I am going to the party .**

→ **Not only Kiran but also Anku and her friends are participating in this event.**

Rule => Subject + Verb + Adverb

Adverb + Verb + Subject

→ **Two miles beyond that pasture <u>were</u> seen hundreds of cattle.**

→ **Under the tree <u>was</u> sleeping the cap-seller.**

→ **Outside the hotel <u>are</u> two bars.**

Rule => **There**
The following
+ Verb + Noun

→ **There was a king**

→ **There were many kings.**

→ **The following is the problem in the execution of the plan.**

Rule=> **Relative Pronoun** के बाद आने वाली **Verb,** उसके पहले आने वाले **Antecedent** के **According** आती है।

Antecedent + **Who** / **Which** / **That** **+ Verb**

Who

→ **It is I am to blame.**

→ **It is we who are to blame.**

→ **This is one of the greatest events that have happened this year.**

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→ **Monika is one of those girls who were absent yesterday.**

→ **This is the only one of her novels that is worth reading.**

→ **Rule=>** यदि दो **gerund** अथवा दो **infinitive** को **and** से जोड़े तो **verb plural use** होगी।

Teaching and studying are her hobbies.

To walk and to swim are good exercises.

Rule => clause (subject) – verb singular

That she is an intelligent girl is certain
Sub

Rule => **A number of - verb plural**

The number of
The percentage of] **verb singular**

→ **A number of children were present in the seminar yesterday**

→ **The number of students interested in English is generally small.**

Rule => **noun(s)**
More than one ↑ + verb (S)

More than one person was killed.

Rule=> **More + (noun) plural + than one – verb plural**

More boys than one have failed in the exam.

Rule=> **The majority of – verb plural**

The majority of writers have not passed that stage.

Rule=> **The former The latter**

जिस **noun** के लिए **use** किया जाये उन्ही के अनुसार **verb**

→Of iron and gold the latter is more precious than the former.

→Keats has written many dramas and poems but the former are greater than the latter.

→The rest of the students have failed in the exam.

Rule => Optative sentence मे **singular subject** के साथ भी **plural verb** का **use** होता है।

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Long live our king!

God save the king!

Rule=> Two or more singular nouns or pronouns joined by and require a plural verb : as

Gold and silver are precious metals.

Fire and water do not agree.

Knowledge and wisdom have <u>of times</u> no connection.

Are your father and mother at home?

Rule=> if they suggest one idea →singular verb :-

Time and Tide waits for none.

The horse and carriage is at the gate.

My friend and benefactor has come.

The novelist and poet is no more.

Articles

❖ Articles दो प्रकार के होते है।

1. Indefinite articles - a/an
2. Definite articles - the

1. Indefinite articles- इनमें किसी अनिश्चित वस्तु या व्यक्ति का बोध होता है।

Ex. I saw <u>a doctor</u>.

↓

(any doctor कोई चिकित्सक)

I saw <u>the doctor</u>.

↓

(some particular doctor)

a/an-noun → countable
a/an-noun → singular
a/an-noun → non-identified

Ex.	He drinks a <u>water</u>.	(U)	(x)
	He eats a <u>mangoes</u> daily.	(Cp)	(x)
	He eats a <u>mango</u> daily.	(Cs)	(✓)

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An- यदि किसी word की पहली sound स्वर (vowel) हो।

A- यदि किसी word की पहली sound व्यंजन (consonant) हो।

Ex.	a	university	a	Member of Parliament
	a	utensil	an	Upper Divisional Clerk
	an	umbrella	a	Lower Divisional Clerk
	an	M.A.	an	honest man
	an	M.P.	an	urdu poet
	an	S.P.	an	heir
	a	D.S.P.	a	year
	an	L.D.C.		
	a	U.D.C.		

Usage of the indefinite articles a/an

1. Speed के expression में

Ex. John rides his bike at 80 kilometers the hour. (x)

John rides his bike at 80 kilometers an hour. (✓)

2. Ration के expression

Ex. Ravi takes tea 5 times the day. (x)

Ravi takes tea 5 times a day. (✓)

3. Price के expression में

Ex. Sugar sells for Rs. 50 a kilo (✓)

Rs. 10 a metre. (✓)

4. यदि किसी countable singular noun का प्रयोग पहली बार करते है तो उसके पहले a/an का प्रयोग करते है।

Ex. Once upon a time there was a king.

There is a dog at the door.

Reeta needs a job.

5. यदि कोई countable singular noun अपनी पूरी class (जाति) को represent करे तो उसके पहले a/an का प्रयोग करते है।

Ex. A house should be built well.

A dog is a faithful animal.

6. Exclamatory sentences में countable singular noun से पहले

Ex. What a beautiful lady!

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What a nice pair of trousers!

7. यदि हम किसी व्यक्ति को नहीं जानते तो उसका proper name हटाकर उसके surname से पहले a/an लगाते है।

Ex. A Mr. Sharma wants to see you.

जहाँ Indefinite articles 'a/an' का प्रयोग नहीं होता

1. Proper/material/abstract noun से पहले a/an का प्रयोग नहीं होता।

Ex. A John is my best friend. (x)

John is my best friend. (✓)

A gold is a precious metal. (x)

Gold is a precious metal. (✓)

An honesty is the best policy. (x)

Honesty is the best policy. (✓)

2. Names of meals (dinner/supper/lunch/breakfast) से पहले a/an का प्रयोग नहीं होता।

Ex. I take breakfast at 7.30 am.

He has taken lunch today.

लेकिन यदि names of meals से पहले कोई adjective हो तो इससे पहले a/an का प्रयोग अनिवार्य होता है।

Ex. I look a Heavy breakfast today.
(adjective)

I have taken a delicious lunch.
(adjective)

I invited him to dinner yesterday.
(ordinary dinner)

❖ I invited him to a dinner yesterday (exception)
(a special dinner)

Proper noun/material noun/abstract noun अपने usual meaning से हटकर कोई different meaning बताये तो ये countable हो जाते है।

Ex. India needs a Hitler
(a person like Hitler)

There is a Shakespeare on the shelf.
(a book written by Shakespeare)

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He had a smoke after lunch.

(a cigar/a cigarette)

Please bring us 6 coffees.

(6 cups of coffee)

India is a democracy.

(an example of democracy)

She is a beauty.

(an example of beauty)

The definite article "The"

Definite article 'the' से किसी निश्चित वस्तु या व्यक्ति का बोध होता है।

Ex. He is the doctor.

(some particular doctor)

1. Anaphoric use of the यदि किसी noun की दूसरी बार mention करे तो वह identified हो जाता है तथा इससे पहले 'the' का प्रयोग करते है।

Ex. I saw a woman.

The woman was carrying a baby in her arms.

The baby was crying.

2. Cataphoric use of the यदि किसी noun के बाद में post modifier का प्रयोग करें तो noun से पहले pre-modifier 'the का प्रयोग करते है।

Structure- the + noun + Post modifier

The boy there is my student.

(a word-post modifier)

The boy in the corner is any student.

(prepositional phrase + Post modifier)

The boy sitting in the corner is my student.

(present participle-Post modifier)

The boy who is sitting there is my student.

(relative-clause - Post modifier)

The boy beaten by his senior lodged a complaint with the principal

(Past Particle V-3-Post modifier)

The boy to arrive in time was Ram.

(to infinitive - Past modifier) to + V-I

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People in Darwin have accustomed to cyclone warnings. (x)

The people in Darwin have accustomed to cyclone warnings. (✓)

It is only after the war is over that people realize how bad it was. (x)

It is only after a war is over that people realize how bad it was. (✓)

It is a pity that the son of millionaire should indulge in stealing. (x)

It is a pity that the son of a millionaire should indulge in stealing. (✓)

News I have received is not good. (x)

The news I have received is not good. (✓)

He claims to be a scientist but in reality he does not know even A B C of science. (x)

He claims to be a scientist but in reality he does not know even the A B C of science. (✓)

3. Homophoric use of 'the'- जिस context में noun का प्रयोग हो यदि noun उसी context से identified हो जाये तो उससे पहले 'the' का प्रयोग करते है।

Ex. Open window. (x)

Open the window. (✓)

Close door. (x)

Close the door. (✓)

Write it on blackboard. (x)

Write it on the blackboard. (✓)

Room is stuffy. (x)

The room is stuffy. (✓)

My mother is working in a kitchen. (x)

My mother is working in the kitchen. (✓)

A doctor says that the case is hopeless. (x)

The doctor says that the case is hopeless. (✓)

4. Generie use of the- जब कोई noun अपने पूरे group का सम्बोधित करे तो इस noun से पहले the का प्रयोग कर सकते है।

Ex. The dog is a faithful animal. (✓)

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The bustard is almost extinct. (✓)

The usage of the definite article 'the'

1. Newspapers के नाम से पहले

Ex. I read the Time of India.

Alok reads the Hindu.

2. Political parties के नाम से पहले

Ex. He supports the Congress.

My brother supports the BJP.

3. Seas/River/deserts/chains of mountains के नाम से पहले

Ex. The Artic
The pacific → names of seas

The Ganga
The Brahmaputra → names of rivers

The Sahara
The Gobi → names of deserts

The Alps
The Himalayas → names of chains of mountains

4. Epic (महाकाव्य) और scriptures (धार्मिक पुस्तकें) के नाम से पहले

Ex. The Ramayana
The Mahabarata → names of epics

The Bible
The Geeta - Scriptures
The Kuran

5. Unique nouns से पहले

Ex. The earth moves round the sun.

I have seen the Taj Mahal.

6. Adjectives की superlative degree से पहले

Ex. Ravi is the best boy in t he class.

7. Organization के नाम से पहले the का प्रयोग करते है।

Ex. The U.N.O. The W.T.O.

The B.B.C. The W.H.O.

यदि organization की abbreviated forms को मिलाकर proper noun बना दिया जाए तो उन organization की abbreviated forms से पहले the का प्रयोग नहीं करते।

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NASA SAARC

8. Public Facilities के नाम से पहले

Ex. We are going to the Gem Cinema.

9. Posts (जो एक समय पर एक ही व्यक्ति Hold करे) के नाम से पहले

Ex. The P.M.

The Headmaster

The Principal

The Secretary

10. Cenema/office/theatre/museum के नाम से पहले

Ex. Alok is going to the cinema.

He is going to the office.

We are going to the museum.

11. Musical instrument (यदि purpose primary हो) से पहले

Ex. He is playing the guitar. (primary purpose)

He bought a guitar yesterday. (secondary purpose)

12. Scientific Inventions के नाम से पहले

Ex. The telephone is a useful invention.

13. Ordinals से पहले

Ex. He was the first man to arrive in time.

He was the only man to arrive in time.

14. जिन nouns से plurality वर्ग बोध होता है।

Ex. The U.S.A. The West Indies The Punjab

The U.S.S.R. The Philippines

The U.K. The Netherlands

The Hague → Exceptions

15. Morning/evening/afternoon/night/day से पहले

Ex. He came in the morning.

He came during the day.

Note- 'at night' में article 'the' नहीं आयेगा।

The moon shines at night.

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School/College/University
Court/Prison/Jain → - Primary purpose - no article
Church/temple
Hospital → - Secondary purpose 'the' article
Bad/Sea

Ex. He goes to school daily.
Alok went to the school to deposit his son's fee.

He meet with an accident and was taken to hospital.
I went to the hospital to see my ailing friend.

He goes to bed at 10.00 pm.
He looked under the bed.

Note- यदि किसी Bank के नाम में 'of' preposition आये तो उससे पहले 'the' लगाते है।

Ex. The State Bank of India.
The Reserve Bank of India.
Punjab National Bank.

Axis Bank
Indian Overseas Bank

Note- यदि किसी university का नाम किसी person के नाम पर रखा जाय तो person का नाम ही पहले आता है।

Ex. Rajasthan University (✓)
The University of Rajasthan (✓) [Name after the name of a place]
Jawaharlal Nehru University (✓)
The University of Jawaharlal Nehru (x) [named after the name of a person]

Rule- यदि dynasty/whole family की बात करनी हो तो proper name को हटाकर surname में 's' लगाते है और उससे पहले 'the' का प्रयोग करते है।

Ex. The Vermas.
The Nehrus have been ruling over this country for many years.

Rule-human body के parts के नाम से पहले 'the' का प्रयोग करते है।

Ex. I caught him by the neck.

Rule-

Following

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same

इन शब्दों से पहले 'the' → late

का प्रयोग होता है। former

latter

Ex. The is the same some book that I gave you yesterday.

The late Indira Gandhi

The president visits Rajasthan the following Month.

Rule- यदि किसी sentence/clasue की शुरूआत में comparative degree आये तो उससे पहले 'the' को प्रयोग करते है।

Ex. The higher the rate of literacy, the lower the demand for dowry.

Rule- abstract adjective से पहले या तो 'the' का प्रयोग करते है अथवा उनके बाद things शब्द का प्रयोग करते है।

The + (beautiful / wonderful / marvelous / mysterious / sublime) + things

Ex. The writer is talking about the wonderful things. (x)

The writer is talking about the wonderful. (✓)

Rule-press/radio/television इन शब्दों से पहले 'the' का प्रयोग optional होता है।

Ex. What's on television tonight? (✓)

What's on the television tonight? (✓)

I recognize you. Aren't you on television? (✓)

I recognize you. Aren't you on the television? (✓)

Rule-language के नाम से पहले article को प्रयोग नहीं करते है।

Ex. The Punjabi is the very sweet language of the Punjab. (x)

Punjabi is a very sweet language of the Punjab. (✓)

Rule-Games/sports के नाम से पहले article का प्रयोग नहीं करते है।

Ex. Hockey is the national game of India.

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Cricket is a very popular game.

यदि किसी games **या** sports **को व्यक्ति विशेष तक सीमित कर दे तो उस** games **या** sports **से पहले** the **का प्रयोग करते है।**

I am talking about the cricket that Sachin plays.

Ex. Winters are generally very cold her.

Rule- **यदि किसी वस्तु का नाम** place **(स्थान) के नाम पर रखा जाये तो उससे पहले** 'the' **का प्रयोग होता है।**

Ex. The kashmiri shawls

The Rajasthan bedsheets.

Rule-

(sort of
kind of
type of
rank of
title of
post of) + no article + noun

Ex. He was given the title of a Nawab. (x)

He was given the title of Nawab. (✓)

He is a clerk. (✓)

He was given the post of a clerk. (x)

He was given the post of clerk. (✓)

Rule-

(man
woman
life
death
science) - in general sense - no article

- in particular sense - article (a/an/the)

Ex. I saw a man crossing the road.

(particular sense)

Man is mortal

↓

(general sense)

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Ravi wept bitterly at the death of his son

(particular sense)

Death is inevitable.

↓

(general sense)

Rule-choose/elect/nominate/appoint/crown इन verbs के सबसे बाद ऐसी post हो जो एक बार में एक ही व्यक्ति hold करें तो उनसे पहले कोई article नहीं लगाते यदि इन verbs के बाद ऐसी post हो तो एक से ज्यादा व्यक्ति के पास हो तो a/an लगाते है।

Ex. We elected him the president. (x)

We elected him president. (✓)

We choose Ram monitor. (✓)

We crowned him king. (✓)

The president nominated him M.R. (✓)

Common Error:-

Ex. It is a child's play for you. (x)

It is child's play for you. (✓)

He did it in a record time. (x)

He did it in record time. (✓)

You should keep him at an arm's length. (x)

You should keep him at arm's length. (✓)

This is a part and parcel of my duty. (x)

This is part and parcel of my duty. (✓)

People pay only a lip service to Gandhi an ideas. (x)

People pay only lip service to Gandhi ideas. (✓)

English has taken a place of pride in India. (x)

English has taken place of pride in India. (x)

English has taken pride of place in India. (✓)

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I would like to take your leave now. (x)

I would like to take leave of you now. (✓)

Rule-

To take leave of sb

By mistake

In detail

In fact

Take action

Take care

Take heart

have a good/sound knowledge of sth

in a temper

in a dilemma

in a mood

in a hurry

in a shambles

make a noise

In trouble

In danger

In debt

In haste

In confusion

Before sunset/before sunrise

After sunset/after sunrise

In conclusion

In society

Give battle

Set on fire

catch fire

run a temperature

in the queue

in the affirmative

in the negative

in the singular

in the plural

in the world

on the rampage

The benefit of the doubt

The (Rich/Poor, Intelligent/Dull, Blind/lame, Deaf/dumb, Living/wounded, Dead) + People

Ex. The rich people hate poor people. (x)

The rich hate the poor. (✓)

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ADJECTIVES

→ An adjective is a word which qualifies a noun or a pronoun

Ex. Khanika is intelligent.

Noun adj.

She is intelligent.

Pronoun adj.

Adjectives are of two types :-

→ I Attributive adjectives -- इस **adjective** का **use noun** के पहले करते हैं।

Ex. Priyanshi is a wise girl.

attributive adj.

→ I Predicative adjectives -- इन **adjectives** का प्रयोग **verb** के बाद करते हैं।

Khushbu is intelligent

(predicative adj)

सामान्यत **Adjectives** का प्रयोग **attributive** तथा **predicative** दोनों तरह से कर सकते है।

Ex This is a red pen. (attributive)

This pen is red. (predicative)

He is a sick boy. (attributive)

The boy is sick. (predicative)

Rule → कुछ **adjective English** में ऐसे होते हैं। जिनका **meaning** अपने आप में **complete** होता हैं।

Ex Kiran is tall.

Rahul is handsome.

Satish is intelligent.

Rule → English में कुछ **adjective** ऐसे होते हैं। जिनका **meaning** अपने आप में **complete** नही होता है।

Ex He is fond. (ग)

He is fond of sweets. (√) → Prepositional phrase

Ramesh is afraid. (ग)

Vikash is afraid of the dogs. (√)

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→ **Rule (utter, complete, mere, sheer) --- only attributive adjectives.**

The fool is utter. (ग)

He is an utter fool. (√)

The idiot is complete. (ग)

She is a complete idiot. (√)

The nonsense is mere. (ग)

This is a sheer nonsense. (√)

→ **Rule (ill, well, asleep, fond, afraid) --- only predicative use**

He is an ill boy. (ग)

The boy is ill. (ग)

Ram is a well boy. (ग)

Ram is well. (√)

She is an asleep girl. (ग)

The girl is asleep. (√)

He is fond of music. (√)

→ **Post Posed Adjective:-**

जब **Adjectives** का प्रयोग कुछ **significant meaning** बताने के लिए **Noun** के बाद में करते हैं उसे **post posed adjective** कहते हैं।

This is a blue pen. (√)

This is a pen blue. (ग)

General knowledge. (√)

Knowledge general. (ग)

General Inspector. (ग)

Inspector General (IG). (√)

- You are a proper candidate for the post of manager. (√)
- We live in proper Jaipur. (x)
- We live in Jaipur proper. (√)
- This is an involved structure. (√)
  complex
- The involved people in the scandal will be punished. (x)
- The people involved in the scandal will be punished. (√)
  (शामिल)

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- She is a concerned mother. (√)
  (worried)
- The concerned clerk is absent today. (x)
- The clerk concerned is absent today. (√)
  (संबंधित)
- The department concerned. (√)
- The authority concerned. (√)

→ Rule => we can use 6 adjectives before a noun in English language.

Ex. all those big old round red Indian wooden chairs are mine.
1 2 3 4 5 6 noun.

The proper sequence of adjective.

1. Size (adj)
2. Age (adj)
3. Shape (adj)
4. Colour (adj)
5. Nationality (adj)
6. Material (adj)

Ex This is a Japanese small transistor. (x)

This is a small Japanese transistor. (√)

→ Rule :- English में adjectives दो प्रकार के होते हैं।

1. Subjective adj → जिनके facts fixed नहीं होते हैं।

2. Objective adj→ जिनके facts fixed होते हैं।

Rule => English में S.A. को प्रयोग पहले तथा ob. adj. को प्रयोग बाद में करतें है।

→ This is a red beautiful carpet. (ग)

→ This is a beautiful red carpet. (√)

→ This is an old fine house. (ग)

→ This is a fine old house. (√)

S.A. O.A.

Colour adjective
Nationality adjective
Material adjective
Style adjective

Noun से just पहले

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- This is a red beautiful car. (x)
- This is a beautiful car. (√)
- She is an American famous scholar. (x)
- She is a famous American scholar. (√)
- This is a silver beautiful box. (x)
- This is a beautiful silver box. (√)
- This is a leather expensive purse. (x)
- This is an expensive leather purse. (√)
- This is a Mughal costly painting. (x)
- This is a costly Mughal painting. (√)
- Open the door (command) open →

  door →
- The door is open. open
- Close the door. (adj) close → घनिष्ठ
- The door is close. verb (close) →

  Closed → adj→ बन्द
- Ravi is my close /closed friend.
- Our college stayed close /closed yesterday.

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Rule => **The+**

Rich		
Poor		
Old		
Needy		
Aged		
Dead		**+ verb (P) +P.P**
Meek		
Wicked		
Young		
Intelligent	**lame**	
Dull	**wounded**	
Deaf	**living**	
Blind	**sick**	

Ex **Rich people hate poor people.**

The rich hate the poor.

<u>The rich people</u> <u>hates</u> <u>the poors</u>.
ग ग ग

He provided the <u>blinds</u> with food and clothes.(ग)

<u>Rewrite</u>

Chinese are fond of chowmin. (ग)

<u>Rewrite</u>

Now a days the condition of the hospitals is miserable because the sick is not properly looked after. (ग)

<u>Rewrite</u>

Rule **superior**
Inferior
Junior **to**
Senior
Prefer
preferable

- Ram is junior than me. (x)

  <u>Rewrite</u>

- The death of king Edward was prior than the great war of Europe.

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DETERMINERS

→ **No Noun, no determiner.**

→ **Determiner** वह शब्द है जिसका प्रयोग **sentence** में सिर्फ **noun** के लिए ही करते है किसी दूसरे **parts of speech** के लिए नहीं करते हैं।

Example : This is a big table.(√)

This is a big. (x)

This is big.(√)

Note => We can use adjective and adverb between noun and determiner.

Example: →a/an → countable singular

There was a water. (x)

There was a kings.(x)

There was a king. (√)

→These →countable plural

These book are mine. (x)

These books are mine. (√)

But, <u>This</u> is my pen. (pronoun)

<u>This</u> book is mine (Determiner)

These <u>sort</u> of students always create problems. (x)

These <u>sorts</u> of students always create problems. (√)

Note => We use different terms for determiners in traditional grammar.

Articles	→	**A, An, The**
Cardinals	→	**One, Two, Three etc.**
Possessive adj	→	**My, our, your, his, her, its**
Distributive adj	→	**Each | every | either | neither**
Quantifiers	→	**Little | a little | Few | a few etc.**

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Interrogative adj → **What | which | whose |**

Adjective → **Enough, some**

I. => (a/an/one/each/every/either/neither/another/many a/ a certain) + Countable singular noun

Ex. There was a king.

Each boy is busy with his project work.

Many a women have achieved eminence in science. (x)

Many a woman has achieved eminence in science. (√)

I need a certain boys to accomplish the task. (x)

I need a certain boy to accomplish the task. (√)

II. => (These/those/many/both/several/few/a few/ The Few/a good many/ a great many/two+ countable plural noun.

These books are very useful.

Many women have achieved eminence in science. (√)

I have a few friends in the locality.

III. => (Much/little/a little/a good deal of + uncountable noun

Much water has flowed down the river since we last met.

There is little sugar in it.

We have done a good deal of work today.

IV. => (all/ a lot of /lots of/enough/more/most) + countable plural noun / uncountable noun

Ex. A lot of books on English grammar are available in this market (Cp)

Ex. A lot of money has already been spent on this project. (U)

Rule - both- used for → two persons

several/many → used for more than two → persons

many/certain → countable plural noun

many a/a certain → countable singular noun

<u>Difference between 'some' and 'Any'</u>

→ Do you have any money on you ?

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→ No I don't have any money on me.

→ Yes I have some money on me.

Furore → uproar →Rumpus →outcry →fuss →excitement
Ruckus → fracas → Melee

→ यदि **sentence** में **request / offer** का **sense** हो तो **interrogative** होने पर भी **some** का प्रयोग करेंगे।

→Rule → Each- pronoun/Determiner
Every- only determiner

Each is present in his class. (√) (P)

Each boy is present in his class. (√) (D)

Every is present in his room. (x)

Every boy is present in his room. (√)

Rule - each 2 के लिए **every** 2 से ज्यादा के लिए

Each of the two boys is busy with his work (√)

Everyone of the two boys is busy with his work (x)

Rule→ Each - singular noun- singular verb - s.p.

Every- s.n. - s.v.-s.p.

Each book and Magazine was in its proper place.

Every Tom, Dick and Harry drinks wine.

Each and every soldier has taken his position.

There were only two soldiers but each and every soldier was equal to five ordinary men. (x)

There were only two soldiers but each soldier was equal to five ordinary men.(√)

- (2 से ज्यादा होने पर हो each and every का use करते है।)

Difference between few and little

Rule - few-countable noun/number

little- uncountable noun/quantity

used in negative sense.

Ex. There are few intelligent students in the class.

I have little money.

Accountant - A p
Auditory - Audile
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Rule- A few-/ a little --- used in positive sense

There are a few intelligent students in the class.

I asked Ram to give me a little money.

Rule- The few - not many but all that there is

The little -not much but all that there is

Ex. I can give you the little money that I have.

The little knowledge of computer that she possessed proved a boon for her in getting a job.

Rule- little शब्द का प्रयोग यदि **size** के अर्थ में करते है।

she is a little girl.

I have a little pocket dictionary

Rule- Little - less - least

few - fewer - fewest

Ex. There were less than a hundred boys there. (X)

There were fewer than a hundred boys there. (√)

Rule - less than + Distance + time+money +noun

Less than two hours.

Less than ten miles.

Less than 500 rupees.

Rule -English में **Ordinal** का प्रयोग **Cardinal** से पहले करते है। **(ordinal - first, second, next, last)**

Ex. I have read the first two chapters of this book (√)

I have read the two first chapters of this book. (X)

Rule :- Pronoun → other + s

Determiner → other + noun + s

~~**Other**~~ **will do it. (X)**

Others will do it. (√)

Other boy will do it. (X)

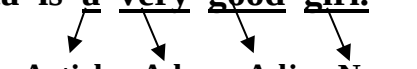
Other boys will do it.(√)

Ex. - Many beautiful cities were destroyed in communal riots. The first is Bihar and other city is unknown. (X)

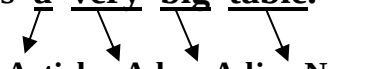
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Rule => article + adverb + adjective + noun

Monica is a very good girl.



This is a very big table.



Whole the chapter is full of errors. (x)

The whole chapter is full of errors. (√)

Rule => **so**
how
as
too
adverb + adj. + article + Noun

Ex. I have never seen ~~a so good boy~~ in my life. (x)

I have never seen so good a boy in my life. (√)

Rule - Enough + Noun

Adjective + Enough

Ex. - There is enough sugar in the jar.(√)

Ex. - Monika is brave enough to face this problem.

Rule - Pre-determiners → All/both/Half

Determiners → My/these/ the

My both hands are dirty. (x)

Both my hands are dirty.(√)

He is not as a competent man as I thought .(x)

- He is not as competent a man as I thought. (√)
- how funny a name she has. (√)
- I am too small a few. (√)

Abbreviate / Shorten

Hirsute - hairy - full of hair

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Determiners

1. There is _____milk in that jug. (some, any)
2. She wanted_____ stamps but there were't_____ in the machine.
3. I would like to buy_____ new clothes but I haven't_____money.
4. They can't have_____ more strawberries: I want _____ to make jam.
5. Where shall we sit ?
   oh,_____ where will do. (any / some)

6. The half work is over. (x)
7. She is a small girl. (x)
8. I have lots of _____(book/books) on this topic.
9. I need certain _____(boy/boys) to accomplish my work.
10. _____(Every/Each)was present in his class.
11. Three_____(thousand/thousands) students were shot dead.
12. How a funny name she has. (x)

Rewrite -

13. I am a too small she few. (x)

Rewrite

14. I have already spent_____ few rupees I had. (a, an, the)
15. How_____(many/much) eggs are there in the basket?
16. Not _____ than 10 kilos of ghee was used in the sacred fire. (fewer/much/less)
17. _____ knowledge is always dangerous. (little, a little)
18. The soldiers know_____ duty well (his, their)
19. Do you have_____plan to go abroad.(a, any, an)?
20. I have _____ Knowledge of French.(little, few).
21. each and every soldier_____(has/have) taken_____(his/their) position.
22. I have enough_____(book/books) on this topic.
23. Many a flower _____ (is/are) born to blush unseen and (waste / wastes) its sweetness on the desert.

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24. Hundred of years ago, Ashoka established tranquility in India.

25. About the result of the examination I have little patience but she has even_____(less, fewer)

26. Never before have I seen a so good boy. (x)

Rewrite-

27. He should mix his authority with_____(little/ a little) of courtesy and common sense.

28. India is better than most others film producing countries of the world.

Rewrite-

29. _____(Thousand/Thousands) of students were shot dead in china.

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Verbs

A verb is a word which indicates what sb/sth does, is, or feels

Ex. Ram plays crickets

He is watching television.

She is cooking food.

Ex. Ravi is a doctor.

Ram is a teacher.

Ex. I feel bad about it.

Classification of verbs

1. Helping verb.
2. Main verb

Rule- यदि sentence में एक verb हो तो वह finite verb/main verb होगी।

यदि sentence में verb की संख्या एक से ज्यादा हो तो वह पहली verb helping verb होगी तथा शेष सभी main verb होगी।

Ex. Ravi goes to school daily.

(main verb)

Ravi is a doctor.

(main verb)

Ravi is going to the cinema.

(H.V.) (M.V.)

1. Finite verb- जो verb tense/number/person की जानकारी दे या उनके बदलने से प्रभावित हो जाये।
2. Non finite verb जो verb tense/number/person की जानकारी ना रखे। इनके बदलने से प्रभावित ना हो।

Ex. Alok plays chess.
(finite)

Alok is a doctor.
(finite)

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It has been raining since morning.
(finite) (non-finite) (non-finite)

Rule- किसी भी verb की ये तीनों form हमेशा non-finite ही होती है।

1. To+V-1 (to infinitive)
2. V+ing (present participle)
3. V-3 (past participle)

Ex. I want to go home.

He wants to go home.

He wanted to go home.
(non-finite)

We are playing hockey.

He is playing hockey.

He was playing hockey.
(non-finite)

He has gone to Mumbai.

They have gone to Mumbai.

They had gone to Mumbai.
(non-finite)

Intensive Verb:-

Intensive Verb के बाद में compliment/adverb आ सकता है। Complement का कार्य generally noun/adjective करते है। सामान्यतः 'be' verb intensive verb होती है।

Ex. Ram is doctor.
(complement-noun)

Ram is intelligent
(complement-adjective)

Ram is from Australia.
(adverb)

Note- 'be'- is/am/are/was/were

'be' verb के अलावा लगभग 30 verb और है जिनका प्रयोग Intensive verb की तरह किया जा सकता है यदि उनके बाद में complement आयें तो।

[Get/grow/feel/become/remain/appear/seem/look/continue/taste/smell/ring/read/sound/wash/die/marry/lie/stand/sit/keep/rest/fall/run/wear/go/break/come/turn]

Ex. He felt surely of winning the game. (x)

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He felt sure of winning the game. (✓)

Mangoes taste sweetly. (x)

Mangoes taste sweet. (✓)

Roses smell sweetly. (x)

Roses smell sweet. (✓)

He became a doctor. (✓)

He turned a doctor. (x)

He turned a traitor. (x)

He turned traitor. (✓)

Rule:- 'Turn' verb का प्रयोग हमेशा negative sense करते है।

'Turn' verb के बाद में आने वाले noun से पहले article का प्रयोग नहीं करते है।

Intransitive verb- अकर्मक क्रिया

Transitive verb- सकर्मक क्रिया

1. Intransitive verb- यदि sentence में object ना हो तो verb intransitive होती है। तथा ये verb अपने बाद में adverb ले सकती है तथा ये adverb दो प्रकार का होता है।

Optional adverb- जिसे sentence से हटाया जा सके।

Obligatory adverb- जिसे sentence से हटाया ना जा सके।

Ex.- The earth moves round the sun.

The sun shines during the day.

The moon shines at night.

Birds fly in the air.

Wood floats on water.

(Optional adverbs)

He lives in Delhi.

He goes to school.

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(Obligatory adverbs)

2. Transitive verb- यदि sentence में object हो तो verb transitive होती है। तथा इन verb के बाद में object/object+obj/obj+comp./obj+adverb आ सकता है।

Object

Ex. Ram plays chess.
Sita cooks food.
He is watching a hindi film.
She is writing a letter.

Object + Object

Ex. He told me a story.
He gave me a book.
He asked me a question.
He read me the Ramayana

Object+comp.

Ex. People called him a fool.
We elected Ravi President.
We chose Ram monitor.
The police want him alive.

Object+adverb

Ex. He put the book on the table.

Ex. In all circumstances I have helped him and he knows. (x)
In all circumstances I have helped him and he knows it. (✓)

He asked for the permission to go to the cinema but his father did not give. (x)
He asked for the permission to go to the cinema but his father did not give it. (✓)

Stative verb

Dynamic verbs

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1. Dynamic verbs- वे verb जिनका प्रयोग सभी continuous tense में किया जा सके।

Ex. Ram plays cricket.

Ram is playing cricket.

1. Stative verbs- वे verb जिनका प्रयोग सभी continuous tense में न किया जा सके।

Ex. I know Ram. (✓)

I am knowing Ram. (x)

Rule- Generally verb dynamic होती है कुछ ही cases में verbs stative होती है।

Performative verb- (agree/disagree/see/understand) ये verb stative verb की category में आती है।

Ex. I am agreeing with you. (x)
I agree with you. (✓)
I disagree with you. (✓)
Oh! I see. (✓)
Ok I understand. (✓)

Verbs of perception- वे verbs, जो हमारे sense से सम्बन्धित होती है वे verbs of perception कहलाती है।

(See/hear/feel/smell/taste)

Voluntary action- ऐच्छिक – dynamic verb

Involuntary action- अनैच्छिक – stative verb

Ex. Mangoes are tasting sweet. (x)

Mangoes taste sweet. (✓)

(involuntary action)

I am tasting these mangoes. (✓)

(voluntary action)

Ex. Roses are smelling sweet. (x)

Roses smell sweet. (involuntary action) (✓)

John is smelling a rose. (voluntary action) (✓)

Are you hearing noise there? (x)

Do you hear a noise there? (✓)

(involuntary action)

Are you seeing the house on the corner? (x)

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Do you see the house on the corner? (✓)

(involuntary action)

Who is hearing your case? (✓)

Stative verbs-(like/dislike/hate/love/prefer/mind/forget/remember/recognize/recollect /hope/expect/think/suppose/believe/want/get/need/required/mean/understand/ know/matter/contain/include/depend/resemble/appear/seem/adore/belong/possess/ have/own/owe/declare)

Ex. I <u>like</u> sweets.

John <u>possesses</u> a magnificent house.

I <u>owe</u> him my success.

I <u>own</u> this car.

I <u>expect</u> that Ram will come today.

(आशा करना)

I <u>am expecting</u> a letter from her.

(प्रतीक्षा करना)

I <u>think</u> you are right.

(समझना)

I have been thinking over this matter.

(सोचना)

Ex. I <u>have</u> a racing bike. (sense of possession)

He <u>is having</u> a bath. (no sense of possession)

We <u>are having</u> a good time.

<u>To-infinitive</u>

Full infinitive - to + V-I

Bare infinitive- V-I

let

bid

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watch

See ⟶ इन verbs के बाद आने वाली verb bare infinitive होती है।

hear

feel

make

have

help

Ex. I saw him <u>to cross</u> the road. (x)

I saw him <u>cross</u> the road. (✓)

I heard you <u>to leave</u> the house. (x)

I heard you <u>leave</u> the house. (✓)

He felt the house <u>to shake.</u> (x)

He felt the house <u>shake.</u> (✓)

I watched John <u>to play</u> cricket. (x)

I watched John <u>play</u> cricket. (✓)

I bade him to <u>do</u> it. (x)

I bade him <u>do</u> it. (✓)

Rule- would rather

would sooner

had better

rather than

would rather-than

would sooner-than

→ इन सभी verb के बाद में bare infinitive का प्रयोग करते है।

Ex. I would rather <u>to wait</u> for her. (x)

I would rather <u>wait</u> for her. (✓)

You had better <u>to leave</u> Jaipur tomorrow. (x)

You had better <u>leave</u> Jaipur tomorrow. (✓)

He would rather die than <u>to surrender.</u> (x)

He would rather die than <u>surrender.</u> (✓)

He would sooner starve than <u>to beg.</u> (x)

He would sooner starve than <u>beg.</u> (✓)

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Rather than to go there by air, I will take the slowest train. (x)

Rather than go there by air, I will take the slowest train. (✓)

Rule- but
Except → - bare infinitive

Ex. I have nothing to do but wait for her. (✓)

He can do anything except lend you money. (✓)

Rule- with a view to
look forward to
take to
admit to
object to → V+ing (Gerund)
be used to
be given to
be accustomed to
be addicted to
prefer to
prefer doing sth

Ex. I am looking forward to see you in Jaipur. (x)

I am looking forward to seeing you in Jaipur. (✓)

He has gone to London with a view to do a Ph.D. (x)

He has gone to London with a view to doing a Ph.D. (✓)

Ram used to play cricket. (✓)

He is used to work in the sun. (x)

He is used to working in the sun. (✓)

He is used to sleeping in the open. (✓)

Rule- worth
mind
prefer
avoid → V+ing (gerund)
deny
contemplate

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confess

enjoy

Ex. The book is worth buying.

The movie is worth watching.

I don't mind going to the cinema.

I prefer walking to cycling.

I am contemplating going there.

I avoid smoking at public places.

Rule- It is no use
It is no good
Can't stand
Can't help
→ V+ing (gerund)

Ex. It is no use going there.

It is no good talking to Ram.

I can't stand talking to him.

I can't help laughing at Ram.

Rule- Able
Fail
→ - to + V-1

Ex. He is able of doing it. (x)

He is able to do it. (✓)

He failed in getting a success in the exam. (x)

He failed to get a success in the exam. (✓)

Rule- capable + of refrain + from

succeed + in prevent + from

insist + on discourage + from

Ex. Ram is capable to do it. (x)

Ram is capable of doing it. (✓)

He succeeded to qualify this exam. (x)

He succeeded in qualifying this exam. (✓)

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He prevented me to smoke. (x)

He prevented me from smoking. (✓)

She insists to do it. (x)

She insists on doing it. (✓)

Rule- Think- 'as' का प्रयोग नहीं कर सकते।

Consider- 'as' का प्रयोग optional है।

Regard- 'as' का प्रयोग compulsory है।

Ex. I think it as a pleasure. (x)

I think it a pleasure. (✓)

I consider him as a fool. (✓)

I consider him a fool. (✓)

I regard him as my father. (✓)

I regard him my father. (x)

Rule- 'want' verb के बाद में that clasue का प्रयोग नहीं करते जबकि to+v-I का प्रयोग कर सकते है।

Ex. I want that you should go there. (x)

I want you to go there. (✓)

Rule- 'Suggest' verb के बाद में to+v-I का प्रयोग नहीं करते जबकि that clasue अथवा gerund (v+ing) का प्रयोग कर सकते है।

Ex. I suggest to go there. (x)

I suggest going there. (✓)

I suggest that we should go there. (✓)

Rule- know
learn → - how + to + V-I

John knows to speak French. (x)

John knows how to speak French. (✓)

Rule- Awkward
Impossible → - इन शब्दों के बाद to + V-I का प्रयोग कर सकते है।
Difficult

Ex. It is awkward to look at.

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It is impossible to do.

The task is difficult to perform.

Rule- allow, Permit, recommend - Gerund (v+ing) - यदि sense general हो।

to + V-I - यदि sense particular हो।

Ex. I do not allow smoking. (general case) (✓)

I do not allow my students smoking. (x)

I do not allow my students to smoke. (✓)

(particular case)

Rule- Sub [action} - need, require - to + V-I

sub [no action]- want - v+ing (gerund)

to + be + v-3

Ex. Ravi needs to go home. (✓)
My shoes need to mend. (x)
My shoes need mending. (✓)
My watch requires to repair. (x)
My watch require repairing. (✓)
My watch requires to be repaired. (✓)

Rule-	action-1		action-2	action-2	action-1
	Forget Remember regret	+	to+v+I	forget Remember regret	 v-ing v+ing

Ex. Ravi remembers to post the letter.

Ravi remembers posting the letter.

I forgot to sign the cheque.

I forgot about signing the cheque.

I remember meeting Ram five years ago.

(a) Meet

(b) no improvement

(c) To be meeting

(d) To have been meeting.

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Rule:- do/does/did - V-1

Ex. Do you doubted about the success of this boy? (x)

Do you doubt about the success of this boy? (✓)

They did not came out victorious. (x)

They did not come out victorious. (✓)

Rule- Model + V-I

Ex. We must not held/hold in high esteem those who are dangerous to our secrety.

Rule- Modal के बाद V-I में S/es नहीं लगावें।

Ex. Alok possesses all those good qualities which every ideal student should possesses/ possess.

Rule- perfect form- modal + have + V-3

Ex. John might has done this. (x)

John might have done this. (✓)

Ex. I want you to pick up the box of eggs gently and kept it in the corner carefully. (x)

I want you to pick up the box of eggs gently and to keep it in the corner carefully. (✓)

Ex. Seminars give us an opportunity of meeting different kinds of people, listening to them and share/sharing our view with them.

Ex. Only one dispute has been settled and others left to both the parties to settle. (x)

Only one dispute has been settled and others have been left to both the parties to settle. (✓)

Ex. Much water has flown down the river since we last met. (x)

Much water has flowed down the river since we last met. (✓)

Rule-	Fly	Flew	Flown	उड़ना
	Flow	Flowed	Flowed	बहना

Ex. It must be born in mind. (x)

It must be borne in mind. (✓)

Rule-	Bear	Bore	Born	जन्म लेना
	Bear	Bore	Borne	याद रखना

Ex. The criminal was hung till death. (x)

The criminal was hanged till death. (✓)

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Rule-	hang	hanged	hanged	फांसी पे लटकाना
	Hang	hung	hung	खूंटी पे लटकाना

Ex. He shut the T.V. (x)

He turned off the T.V. (x)

He switched off the T.V. (✓)

Ex. He drives a scooter. (x)

He rides a scooter.

He drives a car. (✓)

Note-	lie	lied	lied	झूठ बोलना

Ex. It's very surprising that a seasonable man like you has lain/lied to me. (✓)

Note-	Lie	Lay	Lain	intransitive
	Lay	Laid	Laid	transitive
	Lie	Lied	Lied	

Ex. He was laying a book.

He always lies . (✓)

He had lain the book on the table. (x)

He has laid the book on the table. (✓)

Note-	fall	fell	fallen	intransitive
	fell	felled	felled	transitive

Ex. The tree felled down. (x)

The tree fell down. (✓)

He fell the tree. (x)

He felled the tree. (✓)

Note:	Rise	rose	risen	transitive
	Raise	raised	raised	transitive

Ex. The sun rises in the east. (✓)

Alok rose/raised this important issue at the meeting. (✓)

Ex. When Alok came out of the cinema hall he noticed/found that somebody had stolen his watch.

Rule-	Lift	Lifted	Lifted	उठाना
	Leave	left	left	.....

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Ex. He complained to the police that his briefcase had been stolen and he was lifted/left without any money.

Rule- hope/expect - used in positive sense.

Fear - used in negative sense.

Ex. I hope/fear the economic condition of the country may go from bad to worse.

Rule:- Marry - शादी करना।

Marry off- दूसरे से शादी कराना।

Ex. We advised him to marry/marry off his daughter because she had come to marriage able age.

Rule- Seat- किसी जगह पर बैठना।

Student- take exam

Teacher - give exam

Ex. Ravi seated in the exam twice but always failed.

Ravi took this exam twice but always failed.

Ex. One of my friends lived no longer in this world so we always remind/**remember him.**

Remind- याद दिलाना

Remember- याद करना।

Ex. He does not know the name of the person who invented/discovered America.

I saw a model applying cream to her face with practiced hands. (x)

I saw a model applying cream to her face with practised hands. (✓)

Participles:-

Participles दो प्रकार के होते है।

1. Present participle
2. Past participle

1. Present participle- जब 'v+ing' form sentence में का कार्य करे तो उसे present participle कहते है। इसे grammar में verbal adjective भी कहते है।

Ex. This is a blue chair.

This is a revolving chair.

(adj.)

This is a red car.

This is a racing car.

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ADVERBS

Adverb वह शब्द है जो **verb** को **adjective** को, और दूसरे **Adverb** को **modify** कर सकता है।

Ex. He runs fast

He ran very fast.

She is very beautiful.

Adverb वह शब्द है जो अपने अंत को **Generally - ly** में लेता है।

Ex.

Swift -	**swiftly**
Smart -	**smartly**
Quick -	**quickly**
Slow -	**slowly**
Honest-	**honestly**
Beautiful-	**beautifully**

नोट कुछ **verbs English language** में ऐसे भी है जो अपने अतं में **-ly** लेने के बावजुद **Adjectives** होते है।

Ex.

Fatherly	**ghastly**
Motherly	**costly**
Lovely	**worldly**
Friendly	**lonely**
Brotherly	
Sisterly	**deadly**
Miserly	
Cowardly	

Ex. He is friendly with me.

He never treated his brothers ~~friendly~~.

He never treated his brothers <u>in a friendly manner</u>.

Adverb वह शब्द होता है जो **time/place/manner** की जानकारी देता है, **Adverb** एक शब्द अथवा एक से ज्यादा शब्दो का हो सकता है।

Time -	**when**
Place -	**Where**
Manner -	**how**

Ex. He goes <u>to school.</u>

He arrived here <u>on Monday.</u>

Ram plays <u>well.</u>

He is <u>upstairs.</u> (adv. Single word)

He is <u>on the roof.</u> (adv. More than one word)

Note - English में ऐसे कई **words** है जिनका प्रयोग **Adjective** तथा **Adverb** दोनों की तरह से किया जा सकता है।

Ex. Hard / fast / early / late / dead / still / everyday / most / best etc.

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Ex. **He works <u>hard</u>.**
He is a <u>hard</u> worker.

He runs <u>fast</u>.
He is a <u>fast</u> runner.

He came early.
He is an early riser.

He goes to school <u>daily</u> .
This is a <u>daily</u> newspaper.

The telephone is <u>dead</u>.
I am <u>dead</u> sure about my success in the R.A.S exam.

Shahrukh is <u>still</u> sleeping.
Salman is standing <u>still</u>. (motionless)

He comes here <u>every day</u> .
This is an <u>everyday</u> occurrence.

Every day - Adverb
Everyday -Adjective

Note- Adjective की **superlative degrees** से पहले **the** का प्रयोग होता है जबकि **Adverb** की **superlative degree** से पहले **'the'** का प्रयोग नहीं होता ।

Ex. **Alok is the best boy in the class. (√)**
The boy who does the best will get a prize. (X)
The boy who does best will get a prize. (√)

Ex. **Ravi is the most competent teacher. (√)**
I like coffee the most. (X)
I like coffee most. (√)

<u>Difference between the two adverb.</u>

1. <u>Hard</u> - कठिन
   <u>Hardly</u> --------- नही ं के बराबर/मुश्किल से हो

Ex. My brother wants to stand first in the exam. So he works <u>hard</u>/ ~~hardly~~.

2.Most -to the greatest extent
Mostly -for the larger part

Ex. I like it <u>most</u>.
The audience consisted <u>mostly</u> of women.

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2. Direct- से / बिना कहीं रूके
   Directly - प्रत्यक्ष रूप सें

Ex. He went direct to the station.
The P.M. is directly responsible for this political disorder.

4 Late -विलम्ब / देरी से
Lately -हाल ही में

Ex. Have you seen any Hindi film lately.
The boys who come late will not be allowed to enter the class.

5. At present-right now \ at the time of speaking
Presently -shortly\soon
Ranbir is busy with his project at present.
I will let you know about my programme presently.
The captain came recently but he has slipped out unnoticed.

Placing of adverbs -

Note -Generally adverbs placing sentence के अतं में ही होती है।

Ex. I like coffee very much.
We elected Ram president Yesterday.
Murari runs Fast.
He works hard.
He plays well.

यदि कोई **Adverb of manner** एक शब्द का हो तो **verbs** पर जोर देने के लिए इसका प्रयोग उस **verb** से पहले भी कर सकते है।

Ex. He violently shut the door.

He slowly spelt out his name.

Rule- यदि **sentence** में ये तीनो **adverb** एक साथ हो तो
M-manner
P-place
T-time
ये इसी **sentence** में आते है।

Ex. He studied hard at the library yesterday.
She danced beautifully at the party last night.

Rule- यदि **sentence** में **come / go/ arrive/ reach** ये **verbs** आये तो
P-place
M-manner
T-time
ये **Adverbs** इसी **sequence** में आते है।

Rule- Adverb of time- form exact to less exact
Ex. He was born at 9 am on 31 March 1990.

Rule- Adverb of place-form exact to less exact.
Ex. I live at B-155,Mera Marg, Mansarover, Jaipur.

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Rule- Adverb of manner- from shorter to longer.

Ex. He pressed the knob gently with his finger.

He beat the servant violently with a stick.

3 types of Adverbs :-

1.Adverbs of frequency-

Often / always / generally / occasionally / usually / ever

2. Negative adverbs -

Seldom / never / hardly / rarely / scarcely

3. Pre-verb adverbs -
   Just / already / yet / still

Rule :- यदि **sentence** में एक **verb** हो तो इन **Adverbs** का प्रयोग उस **verb** से पहले करते हैं।

Ex. He seldom goes to the movies.

He always does his home work.

They never come late.

She rarely speaks the truth.

Rule :- यदि **sentence** में एक **verb** हो ओर वो **verb be** हो तो इन **adverbs** का प्रयोग **'be' verb** के बाद में करते हैं।

Ex. He is often late.

I am hardly interested in this is job.

She is still in the kitchen.

They are always ready for parties.

Rule :- यदि **sentence** में एक से ज्यादा **verbs** हो तो इन **adverbs** का प्रयोग **first verb** के बाद करते हैं।

Ex. He has never been to Delhi.

He is still sleeping.

I have already had lunch.

He has just come.

Rule :- यदि किसी **sentence/clause** की ठमहपददपदह में छमहंजपअम ध्तमेजतपबजपअम ंकअमतइे आये तो मदजमदबम ध्बसंनेम का जतनबजनतम पदजमततवहंजपअम ;ट्ैद्ध हो जाता है लेकिन मदजमदबम के अन्त में निसस`जवच ;ण्द्ध लगाते हैं।

Negative adverbs - Seldom / never /hardly/ rarely / scarcely / no sooner / little etc.

Restrictive adverbs - Only yesterday / only in this way / only by working hard / in no other way/ to such a degree

Ex. He seldom goes to the cinema.

Seldom does he go to the cinema.

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He <u>little</u> knows about Maths.
<u>Little</u> does he know about Maths.

It can be done <u>only</u> in this way.
<u>Only</u> in this way can it be done.

I came here <u>only yesterday</u>.
<u>Only yesterday</u> did I come here.

The problem of Kashmir can be solved <u>in no other way</u>.
<u>In no other way</u> can the problem of Kashmir be solved .

You can take a first <u>only by studying hard</u>.
<u>Only by studying hard</u> can you take a first.

As soon as the bell rang. The teacher entered the class.
<u>No sooner</u> did the bell rang <u>than</u> teacher entered the class.

Ex. I don't go to the film on weekends.
Nor do I. (√)
Nor I do. (X)

Ex. I have' t had lunch.
Neither have I. (√)
Nor have I. (√)

A: I go to the cinema on Sundays.
B: so do I. (√)
B: So I do. (X)

Rule:- hardly – when/before
Scarcely – when
No sooner – than

Ex. <u>Hardly</u> had he finished his meal ~~<u>than</u>~~ somebody knocked at the door.
<u>Hardly</u> had he finished his meal <u>when</u> somebody knocked at the door.

<u>Scarcely</u> had he finished his office work ~~<u>then</u>~~ somebody telephoned him.
<u>Scarcely</u> had he finished his office work <u>when</u> somebody telephone him.+

<u>No sooner</u> had he finished lunch ~~<u>then</u>~~ someone came to see him.
<u>No sooner</u> had he finished lunch <u>than</u> someone came to see him.

Rule:- English में दो **negative adverbs** का प्रयोग एक साथ नही करते है।
Ex. The fog was so thick that we could ~~<u>not</u>~~ hardly see anything.

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The fog was so thick that we could not see anything.

Fairly/Rather

Rule:- fairly adverb के बाद **pleasant adjectives** का प्रयोग करते है जबकि **rather' adverb** का प्रयोग **negative sense** मे करते हैं।

Fairly – positives degree

Rather – positive degree/comparative degree

Ex. The decision of the chairman was ~~fairly~~ unjustified as it might put the lives of two hundred workers in danger.

The decision of the chairman was rather unjustified as it might put the lives of two hundred workers in danger.

Rule :- almost तथा **quite** इन दोनो शब्दो का प्रयोग एक साथ नहीं कर सकते है।

He is ~~almost~~ quite competent for the post of manager.

He is quite competent for the post of manager.

Rule:- quite तथा सस इन दोनो शब्दो का प्रयोग एक साथ नहीं कर सकते है।

Ex. She is ~~quite~~ all right.

She is all right.

Rule:- 'Quite' से पहले किसी **adverb** का प्रयोग नहीं होता है।

John is ~~absolutely~~ quite alone.

John is quite alone.

Rule- too much –noun

Much too-adjective

Ex. The house is beautiful but ~~too much~~ small to accommodate all of us.

The house is beautiful but much too small to accommodate all of us.

Rule - Enough + noun

Adjective/adverb + enough

Ex. I have enough money to buy this dictionary.

He is ~~enough~~ brave to face this critical situation.

He is brave enough to face this critical situation.

Rule:- too/so/enough/very/as-adjective (positive degree)

Ex. Ramesh is ~~smarter~~/smart enough to get selected for this post.

Sachin is so ~~best~~/good a player that everybody admires him.

These problems are too ~~smallest~~/small for the government to handle efficiently.

Rule:- much-adjective (comparative degree)

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I like coffee <u>much</u>.
I am much ~~well~~ today.
I am much <u>better</u> today.

Note – very – generally with present participle (v+ing)
Much- generally with past participle (v-3)

Rule;- the movie is very <u>interesting</u>.
Ravi is much <u>interested</u> in this job.
He is much <u>worried</u> about the safety of his son.

very ← (Tired / Pleased / Delighted / contented) → exceptions

I am <u>very delighted</u> to see you again.

Rule:- 'Only' / 'almost' का प्रयोग **sentence** में **meaning** के अनुसार अलग–अलग जगह पर होता है।
Ex. Although he ~~only~~ earns 600 rupees a month, he manages to support his family.
Although he earns only 600 rupees a month, he manages to support his family.

Ex. Somebody reported to Ram that his partner had ~~only~~ died a week before.
Somebody reported to Ram that his partner had died <u>only</u> a week before.

This a big shopping mail so you can ~~almost~~ buy everything here.
This a big shopping mail so you can buy <u>almost</u> everything here.

Rule:- generally adverbs का प्रयोग **to + V-1** बीच मे नही करते है।
Ex. He asked me to ~~completely~~ forget her.
He asked me to forget her <u>completely</u>.

Rule:- out rightly – English मे ऐसा कोई **adverb** नहीं है।
The proposal was rejected ~~outrightly~~.
The proposal was rejected <u>outright</u>.

Rule:- fastly – English में ऐसा कोई **adverb** नहीं है।
Vivek runs as ~~fastly~~ as Ram.
Rahul Ram as <u>fast</u> as Ram.

Rule:- well – adjective / adverb
Good – adjective

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Ex. **She is well. (√)**
She is good. (√)
She is feeling good today. (X)
She is feeling well today. (√)

Rule:- **Just / already/ yet / lately present perfect**
Yet – used in negative sentence
Already – used in positive sentence

Ex. **He just <u>come in</u>.**
He <u>has just</u> come in.
Did you see any Hindi film <u>lately</u>? (X)
Have you seen any Hindi film <u>lately</u>? (√)

Ex. **They have not <u>yet</u> arrived here.**
He has <u>already</u> done the assignment.

Rule:- **yesterday / last night / ago / just now – simple past**

Ex. **He has come here yesterday. (X)**
He came here yesterday. (√)

Ex. **His father had died 5 days ago. (X)**
His father died 5 days ago. (√)
The plane has landed just now. (X)
The plane landed just now. (√)

Rule:- **before / after – past perfect tense**

Ex. **The train ~~left~~ a few minutes before.**
The trains <u>had left</u> a few minutes before.

Rule:- **Recently – present perfect / simple past**
Did you see any Hindi film recently?
Have you see any Hindi film recently?

Ex. **They are ~~exceptional~~ good orators so we should listen to them attentively.**
They are <u>exceptionally</u> good orators so we should listen to them attentively.

Ex. **This a ~~real~~ important issue which should be discussed at the meeting.**
This a <u>really</u> important issue which should be discussed at the meeting.

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Ex. **He never liked to treat his brothers ~~friendly~~.**
He never liked to treat his brothers in a friendly manner.

She looked at him ~~lovely~~.
She looked at him in a lovely manner.

Rule:- **No – no article – noun**
Not -article – noun

Ex. **There is disastrous drought all over India ~~no a drop~~ of water is available for thirsty bushes.**
There is disastrous drought all over India no drop \ not a drop of water is available for thirsty bushes.

Ex. **No ~~an~~ intelligent student is present in the class.**
No intelligent student is present in the class.
Not an intelligent student is present in the class.

Ex. **Where ~~else~~ did you go besides the bank?**
Where did you go besides the bank?

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VOICE

Voice-2 प्रकार के होते है।

1. Active Voice
2. Passive Voice

Ex. Ram killed a tiger. (Active)
A tiger was killed by Ram. (Passive)

Rules:-active voice to Passive voice - transform 5 types changes.

1. Active voice का object passive voice का subject बनता है।
2. Passive voice में tense के अनुसार 'be' verb का प्रयोग करते है।
3. Passive voice में main verb की v-3 का प्रयोग करते है।
4. Active voice का subject passive voice का object बनता है।
5. Passive voice में agent से पहले by का प्रयोग करते है।

Active Voice		**Passive Voice**
V-I	-	is/am//are+V-3
Is/are/am+v-ing	-	is/am/are+being+v-3
Has/have + v-3	-	his/have+been+v-3
Has/have+been+v-ing	-	X
V-II	-	was/were+v+3
Was/were+v-ing	-	was/were+being+v-3
Had+v-3	-	had+been+v-3
Had+been+v-ing	-	X
Will/shall+v-I	-	will/shall+be+v-3
Will/shall+be+v-ing	-	X
Will/shall+have+v-3	-	will/shall+have+been+v-3

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Will/shall+have+been+v-ing - X

Declarative Sentence- (Sub + V + Obj.)

Ex. John writes a novel.

A novel is written by John.

Ex. Rama is singing a song.

A song is being sung by Rama.

Ex. Alok has done the home work.

The homework has been done by Alok.

Ex. Ravi has been watching television for 3 hours.

Television has been being watched by Ravi for 3 hours.

-	'be'- finite verb	'be'- non-finite
	Is/am/are/was/were	been/being/to+be

Ravi made a mistake

A mistake was made by Ravi

Ex. Is Ram watching television?

Is television being watched by Ram?

Ex. Does he need some money?

Is some money needed by him?

Ex. Did he type this letter?

Was this letter typed by him?

Ex. Was John playing cricket?

Was cricket being played by John?

Ex. Will you help me?

Shall I be helped by you?

Ex. Does he do the assignment?

Is the assignment done by him?

Ex. Did he do it?

Was it done by him?

Wh-words

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Ex. When did you commit the murder?

When was the murder committed by you?

Ex. Where did you hide the evidence?

Where was the evidence hidden by you?

Ex. How did you do it?

How was it done by you?

Ex. Who did it?

By whom was it done?

Who was it done by?

Ex. Who typed this letter?

By whom was this letter typed?

Who was this letter typed by?

Ex. Whom did you meet yesterday?

Who was met by you yesterday?

Ex. Which colour do you like?

Which colour is liked by you?

Ex. What books do you need?

What books are needed by you?

Ex. How much fruit do you eat daily?

How much fruit is eaten by you daily?

Ex. Who taught you physics?

By whom were you taught physics? Who were you taught physics by?

By whom was physics taught to you? Who was physics taught to you by?

Ex. Whom did you give my book?

Who was given my book by you?

To whom was my book given by you?/who was my book given to by you?

Ex. Ram was playing cricket.

Cricket was being played by Ram

Ex. I shall help you.

You will be helped by me.

Ex. The C.M. will address the meeting.

The meeting will be addressed by the C.M.

Ex. He will have completed the project by tomorrow.

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The project will have been completed by him by tomorrow.

Rule:- यदि sentence में दो object हो तो एक object direct होगा तथा दूसरा object indirect होगा।

Single object-Direct object

Rule:- Object indirect- living/recipient of object direct

Object direct-non-living

Ex. He gave me a book.

OR

I was given a book by him.
A book was given to me by him

Rule:- Call/get
Buy/bring
Find/get
– यदि sentence में दो object हो ओर ये verbs आये तो Passive Od से बनेगा तथा 'for' preposition का प्रयोग करते है।

Ex. He got me a job.

OR

A job was got for me by him.

Ex. He called me a taxi.

A taxi was called for me by him.

Ex. Ravi found me a job.

A job was tound for me by Ravi.

Ex. He bought Ram a dictionary.

A dictionary was bought for Ram by him.

Ex. She brought me some sugar.

Some sugar was brought for me by her.

Ex. You appointed him clerk.

He was appointed clerk by you.

Ex. We elected Ram President.

Ram was elected president by us.

Interrogative sentence-

Helping verb-

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यदि Active voice का structure interrogative हो तो passive voice का structure भी interrogative ही रहेगा।

Ex. Do you play chess?

Is chess played by you?

Ex. What did you give Alok yesterday?

What was Alok given by you yesterday?

What was given to Alok by you yesterday?

Ex. Why did you not take the exams?

Why were the exams not taken by you?

Imperative sentence:-

Rule- V-I+ob (active)

Let+ob+be+V-3 (passive)

Ex. Shut the door

Let the door be shut.

Ex. Do it.

Let it be done

Ex. Open the window.

Let the window be opened.

Ex. Don't ask any question.

Let no question be asked.

Ex. Don't pay any money.

Let no money be paid.

Ex. Please enter by the front door.

You are requested to enter by the font door.

Please switch on the television.

You are asked to switch on the television. (request का sense होने पर)

Ex. March forward.

You are ordered/commanded to march forward

Switch on the television

You are told to switch on the television (order का sense होने पर)

Ex. Please pass me the salt.

You are requested to pass me the salt.

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Ex. Please sit down.

You are requested to sit down.

Please be seated

Ex. Post this letter, please.

You are requested to post this letter.

Ex. Do it for me kindly

You are requested to do it for me.

Ex. Don't touch the wire.

You are asked not to touch the wire.

You are forbidden not to touch the wire. (x)

You are forbidden to touch the wire. (✓)

The wire must not be touched.

Ex. Enter by the front door.

You are asked to enter by the front door.

Ex. Consult a good physician.

You are advised to consult a good physician.

Rule:- suggestion वाले passive

V-I+ob (active)

Ob+should+be+v-3 (passive)

Ex. Love the poor.

The poor should be loved.

Ex. Help the needy.

The needy should be helped.

<u>Modals</u>

1. Simple form - Modal + I - Modal + be + v-3

Ex. You can do it

It can be done by you.

2. Continuous form-Modal+be+v-ing - Modal+be+being+v-3

Ex. He must be doing it.

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It must be being done by him.

3. Perfect form-Modal+have+v-3 - Modal +have+been+v-3

Ex. John must have done it.

It must have been done by John

4. Perfect continuous form Modal+have+been+v-ing

Modal+have+been+being+v-3

Ex. He may have been beating his wife.

His wife may have been being beaten by him.

To-infinitive-

1. Simple form- To+V-I-to+be+v-3

Ex. you have to do it.

It has to be done by you.

2. Continuous form- To+be+v-ing - to+be+being+v-3

Ex. He is supposed to be doing it.

It is supposed to be being done by him.

3. Perfect form- To have+v-3 - to+have+been+v-3

Ex. Ram is believed to have typed these letters.

These letters are believed to have been typed by Ram.

4. Perfect continuous form- To+have+been+v-ing

To+have+been+being+v-3

Ex. he is supposed to have been doing it.

It is supposed to have been being done by him.

Ex. I want you to read this novel.

I want this novel to be read by you.

Ex. I want Ram to help him.

I want him to be helped by Ram.

Rule:- जब sentence का subject और to-infinitive का subject एक ही व्यक्ति हो तो passive नहीं बनता।

Ex. I have some books to study. (No passive)

Ex. There are a lot of books to study.

There are a lot of books to be studied.

Ex. There is a lot of work to do

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There is a lot of work to be done.

Ex. It is time to open the shop.

It is time for the shop to be opened.

Ex. It's impossible to do it.

It's impossible for it to be done.

It can't be done.

Rule-

	Active	Passive
Make/bid		
Watch/see	Bare infinitive	full infinitive
Hear		
Let	bare infinitive	bare infinitive

Ex. I watched Ram go to the cinema.

Ram was watched to go to the cinema by me.

I heard him sing at the concert.

He was heard to sing at the concert by me.

I saw John cross the road.

John was seen to cross the road by me.

He made me understand his problem.

I was made to understand his problem by him.

Ex. He let me go.

I was let go by him.

Participle

V+Ing-Being + V-3

Ex. I don't like people laughing at me.

I don't like being laughed at (by people)

Ex. I don't like people laughing at Ram.

I don't like Ram being laughed at (by people)

Ex. I remember father taking me to Agra to see the Taj.

I remember being taken to Agra by father to see the Taj.

Rule:- [think/suppose/believe/consider/say/estimate]

Ex. People say that he is a spy.

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It is said that he is a spy.

He is said to be a spy.

Rule- to - infinitive की simple form (to+v-1) present tense को indicate करती है।

Rule-to-infinitive की perfect form (to+have+v-3) perfect tense को indicate करती है।

Ex. People say that he was a spy.

It is said that he was a spy.

He is said to have been spy.

Ex. Alok wrote the letter and posted it.

The letter was written and (it was) posted by Alok.

Ex. The police arrested the thief and sent him to jail.

The thief was arrested and (he was) sent to jail.

Ex. The dacoits looted the house and took away many things.

The house was looted and many things were taken away by the dacoits.

Rule- यदि active voice में main verb के बाद कोई preposition हो तो passive voice में भी वह preposition रहेगा।

Ex. he laughed at me.

I was laughed at by him.

Ex. You must look into this matter.

This matter must be looked into by you.

Ex. The police enquired into the case.

The case was enquired into by the police.

Rule:- कुछ verbs ऐसे होते है जो active voice में preposition नहीं लेते, लेकिन passive voice में उनके बाद में आने वाले preposition fixed होते है।

Ex. know to

Pleased with	Surprised at	Contained in	Annoyed with sb
Satisfied with	Amazed at	Embroiled in	Annoyed at sth
Crammed with	Astonished at	Included in	
Decorated with	Startled at	Interested in	engulfed in sth
Filled with	Vexed at		
Thronged with			

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Ornamented with

Ex. I know Ram.

Ram is known to me.

Ex. His performance pleased the principal.

The principal was pleased with his performance.

Ex. The new surprised us.

We were surprised at the news.

Ex. This question vexed all the students.

All the students were vexed at this question.

Ex. Ram annoyed me.

I was annoyed with Ram.

Ex. His behaviour annoyed me.

I was annoyed at his behaviour.

Ex. The packet contains 10 cigarettes.

10 cigarettes are contained in the packet.

Ex. Music interest me.

I am interested in music.

Voice constraints-

Rule- कुछ verbs के case में passive नहीं बनाये जाते है।

Ex. I <u>have</u> a car. A car is <u>had</u> by me. (x)

She lacks confidence. He resembles his father.

It <u>suits</u> you. The shirt <u>fits</u> you.

The dress <u>becomes</u> her. It does not <u>behove</u> you.

The baby <u>weights</u> ten pounds. This hall <u>measures</u> 10 by 20 feet.

Rule- Active sentence का object यदि कोई Reciprocal pronoun हो तो no passive.

Ex. They liked each other. (No Passive)

Rule- Active sentence का object यदि कोई Reflexive Pronoun हो तो– no passive

Ex. He killed himself. (No passive)

Ex. He washed her hands.

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Her hands were washed by him.

Ex. He washed his hands.

His hands were washed by him.

The hands were washed by him.

Ex. One should keep one's promises.

Promises should be kept.

A promise should be kept.

He is a doctor (No passive)

Ex. Mangoes taste sweet.

Mangoes are sweet when they are tasted.

Quinine tastes bitter.

Quinine is bitter when it is tasted.

Ram goes to school. (No passive)

Ex. He hasn't slept in his bed. (No passive)

His bed hasn't been slept in by him.

In which class do you read?

Which class do you read in?

Someone stole my watch. (Active)

Ex. My watch was stolen by somebody. (Passive)

My watch was stolen. (Passive) (Preferable)

I was obliged to go there. (Passive)

Circumstances obliged me to go there. (Active)

Ex. He - said, "I am busy now." Narrations

Reporting verb direct speech

He said that he was busy then.
Indirect/reported speech

Rule:- जब direct speech को indirect speech में बदलते है तो 3 types के major changes होते है।

1. Change of pronouns.
2. Change of tense

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3. Change of adverbs (time/place)

1. Change of pronouns-

S O N - reporting verb

1 2 3 - Direct Speech

Rule:- direct speech में आने वाला first person pronoun (I/we) reporting verb के subject के अनुसार बदलता है।

Ex. He said to me, "I am not well today."

He told me that he was not well that day.

Rule:- Direct speech में आने वाला second person pronoun (you) reporting verb के object के अनुसार बदलता है।

Ex. Geeta said to me, "You are wrong."

Geeta told me that I was wrong.

Rule:- Direct speech में आने वाला third person pronoun (he/she/it/they) नहीं बदलता है।

Ex. He said to me, "She is a doctor."

He told me that she was a doctor.

Ex. He said, "You are a good for nothing fellow."

He told me that I was a good for nothing fellow.

Ex. He said, "We have won the match."

He said that they had won the match.

Rule:- Reporting verb में subject या object अथवा दोनों के स्थान (first/second) person pronoun हो तो direct speech में let's को we should में बदलते है।

Ex. He said to me, "let's go for a walk."

He suggested to me that we should go for a walk.

Rule:- Reporting verb में subject तथा object दोनों के स्थान पर third person pronoun/noun हो तो direct speech में let's को they should में बदलते है।

Ex. He said to ram, "let's go for a walk."

He suggested to Ram that they should go for a walk.

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2. Change of Adverbs:-

Direct		Indirect
This	-	that
These	-	those
Here	-	there
Now	-	then
Today	-	that day
Tonight	-	that night
Tomorrow	-	the next day
Next week	-	the following week
Yesterday	-	the previous day/the day before
Last night	-	the previous night
Last week	-	the previous week.
Ago	-	before
Thus	-	So
Hence	-	thence
Hereby	-	thereby
Hither	-	thither

Ex. He said, "This is my book."
He said that that was his book.
Ravi said, "I am busy now."
Ravi said that he was busy then.
He said, "Kavita lived here."
He said that Kavita had lived there.
He said to me, "John will come today".
He told me that John would come that day.

3. Change of tense:-

Rule:- यदि reporting verb present या future tense में हो तो reported speech में tense से related change नहीं होते।

Ex. He says, "I am busy now."

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He says that he is busy then.

He will say, "I am never late."

He will say that he is never late.

Rule:- यदि reporting verb past tense में हो तो reported speech में tense से related change होते है।

Ex. He said, "I am busy now."

He said that he was busy then.

Direct		Indirect
V-I		V-II
Is/am/are+v-ing	-	was/were+v-ing
Has/have+v-3	-	had+v-3
Has/have+been+v-ing	-	had+been+v-ing
V-II	-	had + V-3
Was/were+v-ing	-	had + been + v-ing
Had + V-3	-	Had + v-3
Had+been+v-ing	-	Had + been + v-ing
Will	-	would
Shall	-	should
Can	-	could
May	-	might
Must	-	must/had to/would have to

Declarative Sentence :- (S+V)

Ex. He said to me, "Are you tired"?

He asked me If I was tired.

Ex. He said to John, "Do I sing a song now?"

He asked John whether he sang a song then.

Ex. I said to her, "Did you do this?"

I asked her if she had done that.

Ex. She said to me, "Do I not accept your proposal?"

She asked me if she did not accept my proposal.

Ex. He said to me, "will you help me?"

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He asked me if I should help him.

Ex. Ravi said to me, "Have you taken lunch today?"

Ravi asked me if I had taken lunch that day?

Ex. I said to her, "Have you ever been to New Delhi.?"

I asked her if she had ever been to New Delhi.

Ex. Ram said to her, "Had you broken my pen?"

Ram asked her if she had broken his pen.

Ex. He said to me, "Shall I ever forget her?"

He asked me if he would ever forget her.

Ex. Ram said to me, "shall I open the window"

Ram asked me if he should open the window.

Ex. he said to me, "Has he invited you to lunch?"

he asked me whether he had invited me to lunch.

Ex. She said to me, "Is your father at home?"

She asked me whether my father was at home.

Ex. I said to Rohit, "Do you know how to ply hockey."

He asked Rohit if he knew how to play hockey.

Ex. He said to her, "Did you enjoy the trip to Jaipur".

He asked her whether she had enjoyed the trip to Jaipur.

Wh-Words

Ex. He said to me, "Who teaches you English?"

He asked me who taught me English.

Ex. She said to me, "Whom did you see at the party?"

She asked me whom I had seen at the party.

Ex. I said to her, "What is your name?"

I asked her what her name was.

Ex. Ravi said to her, "Whose house is this?"

Ravi asked her whose house that was.

Ex. I said to her, "Where are you going?"

I asked her where she was going.

Ex. I said to Ram, "Why did you not take the exams?"

I asked Ram why he had not taken the exams.

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Ex. He said to me, "How are you?"

He asked me how I was.

Ex. She said to me, "Which colour do you like most?"

She asked me which colour I liked most.

Ex. I said to her, "What books do you need?"

I asked her what books she needed.

Said	-	said
Said to	-	told
Connector	-	that
Structure	-	declarative

Ex. He said to me, "I am not well today".

He told me that he was not well that day.

Ex. John said to me, "you are right"

John told me that I was right

Ex. Ravi said to her, "I shall play now."

Ravi told her that he would play then.

Ex. She said to me, "This is mine."

She told me that that was her.

Ex. He said to me, "Yours is blue and mine is green."

He told me that mine was blue and his was green.

Ex. Ravi said to her. "I did not type this letter."

Ravi told her that he had not typed that letter.

Ex. Ram said, "It has been raining since morning."

Ram said that it had been raining since morning.

Ex. Alok said, "I will attend the party."

Alok said that he would attend the party.

Ex. Peter said, "I saw her".

Peter said that he had seen her.

Ex. They said, "We came home last night.

They said that they had come home the previous night.

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Interrogative sentences:-

Said	-	asked
Said to	-	asked
Helping Verb	-	if/whether (connector)
Wh-word	-	wh-word (connector)
Structure	-	Declarative
?	-	(Full stop)

Ex. John said to me, "Do you play chess?"

John asked me if I played chess.

Ex. The teacher said, "How may students are there in the class?"

The teacher asked how many students there were in the class.

Ex. I said to her, "How much fruit do you eat daily"

I asked her how much fruit she ate daily.

Ex. He said to me, "Why did you strike me?"

He asked me why I had struck him.

Ex. He said to me, "What is the matter?"

He asked me what the matter was.

Ex. She said to me, "Why were you absent yesterday?"

She asked me why I had been absent the previous day.

Imperative sentences:

Said to	-	requested me
		asked me
		ordered /commanded me
		told me
		forbade me
		advised me
		warned me
v-I	-	to + v-I
don't	-	not to + v-I
(.)	-	(.)

Ex. The principal said to the visitor, "Please sit down."

The principal requested the visitor to sit down.

He said to me, "please switch on the lights."

He asked me to switch on the lights.

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(request के sense में)

Ex. The captain said to his soldiers, "March forward"

The captain ordered his soldiers to march forward.

Ex. He said to me, "Don't touch the wire."

He told/asked me not to touch the wire.

He forbade me to touch the wire.

Ex. He said to me, "Close the door."

He told me to close the door.

Ex. John said to me, "Consult a good physician."

John advised me to consult a good physician.

Ex. The principal said to the peon, "let the boy come in."

The principal ordered the peon to let the boy come in.

The principal ordered the peon that the boy might be allowed to come in.

Ex. The teacher said to the students, "Do not waste your time."

The teacher advised the students not to waste their time.

Ex. The teacher said to the boys, "keep silence."

The teacher ordered the boys to keep silence.

Ex. The judge said, "Call the next witness."

The judge commanded them to call the next witness.

Ex. Mohan said to Ram, "Let me go home now."

Mohan requested Ram to let him go home then.

Mohan requested Ram that he might be allowed to go home then.

Let's

Said to	-	suggested to me
		Proposed to me
Connector	-	that
Structure	-	declarative
Let's	-	We should
		they should

Ex. He said to me, "let's go for a walk"

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He proposed to me that we should go for a walk.

He said to Ram, "let's go for a walk."

He proposed to Ram that they should go for a walk.

Exclamatory sentence-

Said	-	exclaimed with joy exclaimed with sorrow exclaimed with surprise/wonder exclaimed with fear/horror exclaimed with regret
connector	-	that
structure	-	declarative
!	-	(.)

Ex. He said, "Oh! I have cut my finger."

He exclaimed with sorrow that he had cut his finger.

Ex. She said, "Alas! my cat is dead."

She exclaimed with sorrow that her cat was dead.

Ex. He said, "Hurrah! We have won the match.

He exclaimed with joy that they had won the match.

How - adjective

What - (adj) noun

How - very

Ex. She said, "how beautiful the scene is!"

She exclaimed with wonder that the scene was very beautiful.

He said, "How beautiful she is!"

He exclaimed with surprise that she was very beautiful.

Rule:- What-adj+noun - very

What- noun - big/great (meaning)

Ex. She said, "what a beautiful necklace!"

She exclaimed with wonder that it was a very beautiful necklace.

Ex. He said, "what a ghastly sight it is!"

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He exclaimed with fear that it was a very ghastly sight.

Ex. He said, "what a fool Tom is!"

He exclaimed that Tom was a big fool.

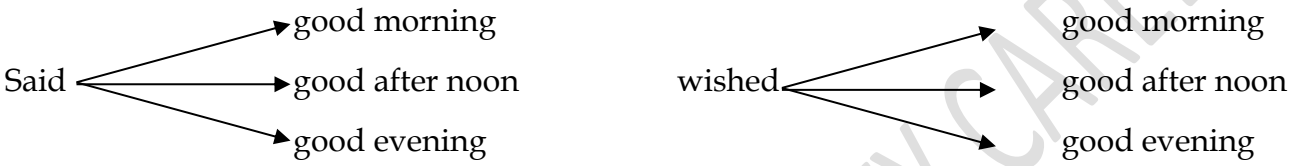
Ex. Akbar said, "fie! a soldier and afraid of fighting!"

Akbar cursed and exclaimed that a soldier should not be afraid of fighting.

Ex. The captain said, "Bravo! Well done players."

The captain applauded the players saying that they had done well.

Rule-



Ex. Ravi said to me, "Good Morning"

Ravi wished me good morning.

Ravi wished good morning to me.

Ex. We said, "Many happy return of your birthday"

We wished him many happy returns of his birthday.

Rule-



Ex. She said, "Good bye friends!"

She bade her friend good-bye.

She bade good-bye to her friends.

Ex. The countrymen said to the leader. "Farewell".

The countrymen bade the leader farewell.

Ex. He said to me, "Good morning! I have come to consult you."

He wished me good morning and told me that he had come to consult me.

Ex. He said, "Hello sir, how are you?"

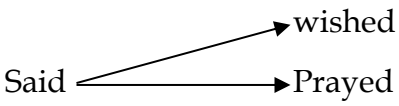
He greeted the teacher and asked him how he was.

Ex. Jaya said, "let's give a party" her husband said, "let's not"

Jaya suggested giving a party but her husband answered in the negative.

Sentences showing 'wish'

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Connector - that

May - might

Structure - declarative

! - (.)

Ex. He said, "May God grant her peace!"

He prayed that God might grant her peace.

Ex. He said to me, "May you live long!"

He prayed that I might live long.

Ex. He said to me, "May God bless you with a son!"

He wished that God might bless me with a son.

Ex. I said, "May God grant peace to the departed soul!"

I prayed that God might grant peace to the departed soul.

Ex. The teacher said, "May you succeed my boys!"

The teacher prayed that his boys might succeed.

Ex. The mother said, "May God bless you with long life and prosperity, my son!"

The mother prayed that God might bless her son with long life and prosperity.

Ex. The people said, "Long live the president!"

The people prayed that the president might live long.

Ex. He said, "Thank you."

He tanked me.

She said to us, "welcome."

She welcomed us.

He said to me, "Congratulations"

He congratulated me.

He said to Ram, "fool!"

He called Ram a fool.

Rule:- कुछ cases ऐसे होते है जिनमें reporting verb past tense में होने के बावजूद indirect speech में tense नहीं बदलते है।

Ex. My father said, "Honesty is the best policy."

My father said that honesty is the best policy. (proverb)

Ex. The teacher said, "Two and two make four."

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The teacher said that two and two make four (universal truth)

Ex. My friend said, "I go to school daily."

My friend said that he goes to school daily (daily routine activity)

Ex. The science teacher said to the students, water boils at 100 degree C."

The science teacher told the students that water boils at 100 degree C. (scientific theory)

Ex. The history teacher said, "India attained freedom in 1947."

The history teacher said that India attained freedom in 1947. (historical fact)

Ex. He said to me, "When I saw Ram, he was playing cricket in the garden.

He told me that when he saw Ram, he was playing cricket in the garden.

(past में दो actions होने पर simple past/past continuous tense)

Ex. He said, "if I were a billionaire I would build another Taj."

He said that if he were a billionaire, he would build another Taj. (hypothetical condition)

Ex. Ram said to me, "Did the news shock you as much as it shocked me?"

Ram asked me if the news shocked means much as it had shocked him.

Rule:- Would that.....!

O that ! - Said - wished

O for!

Ex. The poor man said, "would that I were rich!"

The poor man wished that he had been rich.

Ex. He said to me, "O that we could be friends forever!"

He wished that we could be friends forever.

Ex. He said, "O for a glass of water!"

He wished for a glass of water.

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Gerund

Structure :- V_1 + ing

Verbal Noun

Walking is good for health

ना / ने / नी

Noun's + Gerund

Possessive ad j+Gerund

Walking is a good exercise

(Gerund)) घुमना)

Present Participle

Stru. - V1 + ing

Verbal Adjective

This is a revolving chair.

हुआ / हुये / हुई

Noun-Present Participle

Pronoun- P.P.

— I saw Ram walking on the road

P.P. (घुमते हुए)

- I stopped him/his going out late at night.
- Radheshyam didnot approve of his son's raising stupid questions.
- I don't mind your smoking in my office.
- She left him/his dying on the road.
- She kept me waiting outside.
- I heared Tusar leaving the house.
- He objected to me/my going to the market late at night.

Rule:- Further आगे— in addition to what has just been said

Farther- दूरी में अधिक

Ex. Delhi is farther from Jaipur than Delhi

After the dispute the speaker made no further speech.

- Utmost-very important

Outer most- सबसे बाहरी

- The terrorist entered the place from the outermost gate.
- First- पहला
- Foremost- अग्रणी

Indira Gandhi was the foremost statesman of her time.

- Elder/eldest- person (blood-relation)

older/oldest- persons/things

We should respect Kaluram as he is the oldest person of our town.

- Common Errors-

This is the same book which/that I gave you.

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This is the same book as/that I gave you.

Same - as (Bre)

Same - that (Ame)

- Such - as

Such boys whom/as I knew were engineers.

- Some + noun + or + other

An intelligent man should do some work or other.

- Adjectives की तीन degrees होती है।

1. Positive - so - as (N) - as - as (both)

2. Comparative -

3. Superlative

Positive	**Comparative**	**Superlative**
Small	Smaller	the smallest
long	longer	the longest
beautiful	more beautiful	the most beautiful

Salman is better than Vivek.

Rule:- जब एक ही व्यक्ति की दो Qualities की तुलना की जाये तो comparative degree में more का use करेंगे।

- Salman is more brave than wise.

Rule:- English में एक साथ Double Comparative Degree का प्रयोग नहीं करते।

- She is the most prettiest girl in our college.

Rule:- यदि comparative degree में than की जगह पर of का use हो तो comparative degree से पहले the का use होगा।

Rahim is better than Ravi.

Ram is the better of the two boys.

of the two routes, this is the longer.

Rule:- than-any other-noun (singular)

This fort is older than any other fort in our city.

- Keats is greater than any other poet in the world.

Rule:- एक sentence में एक ही व्यक्ति/वस्तु के लिए प्रयोग किये जाने वाले Adjective समान degree के होने चाहिए।

Alock is the best and talented boy of our class. (x)

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Alok is the best and most talented boy of our class. (✓)

- She is more beautiful but not as wise as her sister. (x)
- She is more beautiful than but not as wise as her sister. (✓)

Rules:-यदि sentence में व्यक्तियों की संख्या 2 हो तो पहले वाले को The former तथा बाद वाले को the latter कहंगे।

Shakespeare wrote many dramas and many poems but the former are superior to the latter.

Rule:- यदि sentence में व्यक्तियों/वस्तुओं की संख्या दो से ज्यादा हो तो पहले वाले को the first तथा अन्तिम वाले को the last कहेंगे।

of these 3 questions, the last was most difficult.

Rule:- Unique

Ideal
Extreme
Major
Excellent
} No. degree

She is a unique singer of this college.

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CONJUNCTIONS

Conjunctions दो प्रकार के होते है।

1. Coordinators
2. Subordinators

1. Compound Sentence-Compound sentence के दोनों clause का meaning complete होता है इसलिए दोनों Clause Independent होते है तथा दोनों clause को जोड़ने वाला शब्द coordinator कहलाता है।

Ex. Ram is my brother (Main clause) — and (coordinator) — Sohan is my cousin (main clause)

Independent clause

Principal clause

2. Complex sentence- Complex sentence में एक clause का meaning complete होता है जिसे main clause कहते है तथा दूसरे clause का meaning incomplete होता है, जिसे dependent clause कहते ह। तथा दोनों clause को जोड़ने वाला शब्द subordinator कहलाता है।

Ex. The train had left (Main clause) — when (subordinator) — I reached the station (dependent clause)

Subordinate clause

Coordinators-And/or/nor/yet/but/still/so/therefore

And

1. दो Information को जोड़ने के लिए

Ex. John wears jeans and he grows long hair.

2. To show 'sequence'

Ex. She washed the utensils and she dried them.

He heard the explosion and he telephoned the police.

3. To show 'positive condition'

Ex. Give me a hundred rupees and I will do the work.

(If you give me a hundred rupees, I will do the work)

Ex. Work had and you will pass.

(If you work hard, you will pass)

Or

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1. Choice/alternative show करने के लिए

Ex. you can stay here or you can go to a hotel.

You can use my car or you can go there by bus.

2. 'Negative condition' show करने के लिए

Ex. Do you homework or your teacher will punish you.

(If you do not do your homework, your teacher will punish you.)

Ex. Keep quiet or I will throw you out

(If you don't keep quiet. I will throw you out)

Nor

1. दो negative information को जोड़ने के लिए

Ex. He does not study.

He does not play.

He does not study nor does he play

Yet
But
Still — Contrast (विरोधाभास) Show करने के लिए

Ex. She is fat but she runs fast.

He is intelligent yet he failed in the exam.

I am not interested in this job still I will try my level best to get it.

So
Therefore — 'result' show करने के लिए

Ex. I am running a temperature so I can't see you today.

He is sick therefore he cannot go to the office.

Either- or
Neither- nor
Not only-but also
Both-and

इन Correlatives का प्रयोग same part of speech से पहले करते है दूसरे शब्दों में यदि एक शब्द का प्रयोग Adjective से पहले होगा तो दूसरा शब्द भी Adjective से पहले ही आयेगा।

Ex. He is both a poet and a solider.

Either the action was just or unjust. (x)

The action was either just or unjust. (✓)

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Ex. She neither drinks tea nor coffee. (x)

She drinks neither tea nor coffee. (✓)

Ex. He neither drinks nor smokes. (✓)

Ex. He is not only anxious to acquire knowledge but also to display it. (x)

He is anxious not only to acquire knowledge but also to display it. (✓)

Ex. He is anxious not only to acquire knowledge but also eager to display it. (x)

He is not only anxious to acquire knowledge but also eager to display it. (✓)

Ex. He not only visited Jaipur but also Jodhpur. (x)

He visited not only Jaipur but also Jodhpur. (✓)

Ex. It is neither good for the centre nor for the states. (x)

It is good neither for the centre nor for the states. (✓)

Ex. Sachin is a dangerous batsman. He is not only famous in India but also in Abroad. (x)

Sachin is a dangerous batsman. He is famous not only in India but also in Abroad. (✓)

Subordinate clause-

Subordinate clause तीन प्रकार के होते है।

1. Noun clause
2. Relative clause
3. Adverb clause

1. <u>Noun clause</u>

Subordinators-that/if/whether/all wh-words including 'how'

Ex. That John will come today is certain.

I know that Ravi is a doctor.

Rule- that → used for certainty

If/whether → used for uncertainly

Ex. I know that/if Ram is a doctor.

I doubt that/if he will quality this entrance exam

I don't know if/that he will come today.

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I am not certain that/if he is an engineer.

Rule:- if → one sided condition

Whether → both-sided condition

Ex. I don't know whether/if he is conspiring against his friends.

I don't know if/whether he is conspiring against his friends or not

Ex. What is your name? (direct question)

I know what is your name. (x) (indirect question)

I know what your name is. (✓)

Rule:- indirect question का structure हमेशा declarative (s+v) होता है।

Ex. I know where do you live. (x)

I know where you live. (✓)

Ex. He knows who are you. (x)

He knows who you are. (✓)

Ex. I know where is he going? (x)

I know where he is going. (✓)

2. Relative clause-

Relative clause relative pronoun से बनते है।

Relative pronoun तीन प्रकार के होते है।

Who- persons

Which- for things

That- persons and things

Subjective	Objective	Possessive
Who	whom	whose
Which	which	of which
That	that	X

Ex. The boy who is wearing a red shirt is Ram

The boy whom you met in the party is John.

The boy you met in the party is John.

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This is the house in which I live.

This is the house I live in.

This is the man of whom I spoke to you.

This is the man I spoke to you of.

This is the only book that is in demand now a days.

3. Adverb clause

Adverb clause mainly **8 प्रकार के होते है।**

1. Adverb clause of time-

Subordinators-when/while/before/after/till/until/as/since/now that/as soon as/as long as/no sooner-than

Rule- When-Period of time + Point of time

While-Period of time

Ex. When/while I reached the station, the train had left.

When/while I was in Delhi, I visited him every Sunday.

Ex. The train had left when I reached the station.

The patient had died before the doctor came.

The chief guest came after the show had started.

Ex. You can't go until you will not finish the work. (x)

You can't go until you finish the work. (✓)

Ex. I saw him as he was getting off the bus.

Six months have passed since his father died.

It is six months since his father died.

It is long since I last saw you.

It is ages since I last saw you.

Ex. As soon as the bell rang, the teacher entered the class.

No sooner did the bell ring than the teacher entered the class.

2. Adverb clause of place-

Subordinators-where/wherever

Ex. Stand where you are.

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She picks quarrels wherever she goes.

3. Adverb clause of reason/cause-

Subordinators-as/since/because/for

Ex. As I am ill, I cannot see you today.

Ex. Since she is busy, she cannot spare anytime for you.

Ex. he cannot go to the office because he is ill.

Rule:- for-used in formal English

For-it comes only at the end of the sentence.

For-always preceded by a comma.

Ex. The president has invited the P.M. over a cup of tea, for she has something very urgent to talk about.

Ex. I cannot see you today because of my illness.

I cannot see you today owing to my illness.

I cannot see you to day on account of my illness.

Rule:- 'Due to' से पहले noun का प्रयोग होना चाहिए।

Ex. I cannot see you today due to my illness. (x)

His absence is due to his illness. (✓)

4. Adverb clause of result-

Subordinators-so that

So - that [so (dj)-that]

Such a-that [such a (noun)- that]

Ex. He worked hard so that he got excellent marks in the exam.

He is so good that everybody likes him.

He is such a good boy that everybody praises him.

He is such a/an honest man that everybody respects him.

They are such a dishonest people that everybody hates them.

5. Adverb clause of purpose-

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Subordination- So that

In order that

Lest

Rule- 'Purpose show करने के लिए present tense में 'may' तथा past tense में 'might' का प्रयोग करते है।

Ex. He studies hard so that he can/may pass the exam (purpose)

He worked hard so that he might support his family

He did his homework hurriedly in order that he might/should go for a picnic.

Rule- Lest-Should

Lest के बाद वाले clause में 'not' का प्रयोग नहीं करतें है।

Ex. He works hard lest he may not fail the exam (x)

He works hard lest he should fail the exam. (✓)

He wore goggles so that the sun did not harm his eyes.

He wore goggles so that the sun might not harm his eyes.

6. Adverb clause of contrast/concession-

Subordinators- Though (yet)

Although- (yet)

Even if/no matter/whatever/however

Ex. Though he is poor but/yet he is honest.

Though/although he is poor, he is honest.

Ex. Whatever you may say, I will not support him.

However much you may give him, he will ask for more.

No matter how busy you are you can spare sometime for me.

I will go to the cinema even if I have to pawn my Rolex watch.

Rule- Come what may

Try as you will

Say what you will holophrases (fixed expressions)

Be that as it may

Ex. Come what may, I will stand by him

Say what you will I will not do it.

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Be that as it may, the removal of the P.M. can't be justified

Try as you will you will not make it.

7. Adverb clause of comparison-

1. Manner

Subordinators-as/as if/as though

Ex. Do as directed.

(you) do as (you have been) directed.

He did as I advised him.

Ex. He looks pale, as if he were dead. (hypothetical)

It seems as if he has no friends. (exception)

2. Degree

Positive degree- as-as

so-as (sentence negative)

Comparative degree- er-than

more-than

Superlative degree-est

most

Ex. He is as brave as me/I. (✓)

He is so wise as Ram. (x)

He is as wise as Ram. (✓)

He is not so wise as Ram. (✓)

He is not as wise as Ram. (✓)

Ex. This is older than any/any other fort in our city.

Ex. Shakespeare is greater than any other poet/poets.

Ex. Ravi is the best boy in the class.

This is the longer of the two routes.

8. Adverb clause of condition

Subordinators- if/unless/on condition that/provided that/as long as/in case

Rule- Conditional clause में future tense का प्रयोग नहीं करते।

Ex. If you will labour, you will pass.

Ex. I can give you the book provided that you will promise to return it in a week.

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Ex. Take an umbrella in case it will rains.

Rule-unless-(if+not)

Ex. Unless you do not labour, you will not pass (x)

Unless you labour, you will not pass. (✓)

9. Adverb clause of proportions:-

Ex. The more he studies, the more he understands.

The higher the rate of literacy, the lower the demand for dowry.

10. Adverb clause of preference

Ex. Rather than ask him to do it, I will do it myself.

Rather than go there by air, I shall take the slowest train.

Rule:- एक coordinator meaning के अनुसार दूसरे coordinator को modify कर सकता है।

Ex. He is intelligent yet he failed in the exam.

He is intelligent and yet he failed in the exam.

Rule:- एक subordinator दूसरे subordinator को modify नहीं कर सकता।

As/since he is busy, he cannot see you today. (x)

As he is busy, he cannot see you today. (✓)

Rule:- एक Sentence में एक coordinator तथा एक subordinator का प्रयोग नहीं करते।

Ex. As he is ill, so he cannot go the office. (x)

As he is ill, he cannot go to the office. (✓)

He is ill so he cannot go to the office. (✓)

Rule:- 'as' और 'so' का प्रयोग proportion के sense में एक साथ कर सकते है।

Ex. As you sow so shall you reap.

As time went on, so my interest in teaching waned.

Rule:- consider if

supposing if- इन expressions में ये दोनों subordinators एक साथ नहीं आ सकते।

Ex. Consider/if he gives you a cheque how will you encash it?

Rule:- until-जब तक की नहीं (time)

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Unless- यदि नहीं (Condition)

Ex. You cannot enter the interview hall until/unless the first candidate comes out.

You can't qualify this exam unless/until you labour hard.

Rule:- same-as/that no other-than hardly-when/before

such-as lest-should scarcely-when

Nothing-but no sooner-than

Ex. It is nothing than/but hypocrisy that all politician take advantage on the name of religion.

option

Rule- no (other) choice - but + to + v-1

alternative

As there was only one taxi, I had no other option but to pay the fare. (✓)

Ex. He has no other Job but/than back biting of others.

Rule- hyponymy का relation दिखाने के लिए like/such as का प्रयोग करते है।

Broad term ex

ex

ex

ex

The captain asked the players to go to market and buy some fruits such as/like apples, oranges, mangoes etc.

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PRACTICE QUESTIONS

1. She did it ___ her own.
   (a) by
   (b) on
2. It is not ____ mine.
   (a) same as
   (b) same like
   (c) the same as
   (d) the same like
3. I didn't see ____ people there.
   (a) many
   (b) many of
4. What ___ you like for dinner tonight?
   (a) do
   (b) would
5. She did it ____ Tuesday.
   (a) last
   (b) on last
6. If I ___ you, I'd tell her immediately.
   (a) was
   (b) were
7. The radio was ___ from my car last week.
   (a) robbed
   (b) stolen
8. The person ____ interviewed me was very friendly.
   (a) who
   (b) which
9. I was certain ___ she would be there.
   (a) of
   (b) that
   (c) whether
10. If I go, I ____ speak to her.
    (a) will
    (b) would
11. If I went, I ____ speak to her.
    (a) will
    (b) would
12. I'll be ____.
    (a) their
    (b) there
13. I won't be ____.
    (a) their
    (b) there
14. Every ____
    (a) day
    (b) days

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15. She lives ___ Anne.
    (a) at
    (b) with
16. I haven't seen her ____ ages.
    (a) for
    (b) since
17. That was rude- you ____ have done it.
    (a) mustn't
    (b) shouldn't
18. She found it easy ____ the test.
    (a) finish
    (b) to finish
    (c) finishing
19. She lives ____.
    (a) alone
    (b) own
    (c) her own
    (d) lonely
20. What is the correct spelling?
    (a) necesary
    (b) neccesary
    (c) necessary
    (d) neccessary
21. Which is the American spelling?
    (a) gray
    (b) grey
22. It looks ____ good.
    (a) pretty
    (b) prettily
23. She's good ____ maths.
    (a) in
    (b) at
    (c) on
24. Vegetables are good ____ you.
    (a) to
    (b) by
    (c) at
    (d) in
    (e) for
25. She spoke ____ him yesterday.
    (a) to
    (b) with
    (c) Either could be used here.
26. If I'd been there, I _____ that.
    (a) wouldn't do
    (b) wouldn't have done
    (c) Either could be used here.
27. She kissed him ___ the lips.
    (a) in

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(b) on
(c) at
(d) to
(e) by

28. It was ___ hot that we couldn't concentrate.
(a) so
(b) such

29. It was ____ hot to drink.
(a) so
(b) such
(c) too
(d) very

30. ____ I saw her yesterday, she was on her way home.
(a) When
(b) Whenever

31. She dropped her kids ____ at school on her way to work.
(a) of
(b) off
(c) up
(d) to

32. It ____ a nation of millions to hold us back.
(a) makes
(b) takes

33. There are only fifteen questions ____ in this test.
(a) go
(b) going
(c) to go

34. I'll look it ___ in the dictionary.
(a) in
(b) on
(c) at
(d) to
(e) up

35. I have ____ finished.
(a) near
(b) nearly

36. It's ____ trouble.
(a) too much
(b) too many

37. She climbed _____ Everest.
(a) Mount
(b) the Mount
(c) The Mount

38. The shop is in ____ Street.
(a) Oxford
(b) the Oxford

39. She tried ____.
(a) hard
(b) hardly

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40. She ___ tried.
    (a) hard
    (b) hardly
    (c) Either could be used here.
41. If I move to the country, it will mean ____ a long way every day.
    (a) travel
    (b) travelling
    (c) to travel
42. Has she ____ to him about it?
    (a) said
    (b) spoken
    (c) told
43. She handed the application form ___ at reception.
    (a) in
    (b) out
44. They ____ up a while ago and got divorced last month.
    (a) split
    (b) quit
45. I would be grateful if you ____ send me further information.
    (a) could
    (b) would
    (c) Either could be used here.
46. There are 50 questions ___ this test.
    (a) in
    (b) on
47. It ____ good.
    (a) is seeming
    (b) seem
    (c) seems

1. _____ you finish, you'll be in trouble.
    (a) Otherwise
    (b) Unless
2. The company _____ accountant was arrested last week has gone bankrupt.
    (a) which
    (b) whose
3. I saw her just _____ day.
    (a) another
    (b) other
    (c) the other
4. This is the town _____ I was born in.
    (a) where
    (b) when
    (c) which
5. This is the town _____ I was born.
    (a) where

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(b) when
(c) which

6. I'm not used to ____ that.
(a) do
(b) doing

7. ____ your help, I would never have managed.
(a) But for
(b) Not
(c) Unless
(d) Despite

8. The two brothers do look ____.
(a) like
(b) alike

9. No sooner ____ got there than they started annoying me.
(a) I had
(b) had I

10. Only when ____ left, did we notice what she had done.
(a) did she
(b) she did
(c) she had
(d) had she

11. I'd rather ____ go if you don't mind.
(a) not
(b) not to

12. It's high time you ____ that.
(a) stop
(b) to stop
(c) stopped
(d) stopping

13. The company ____ chairman was sacked last week has called in the receivers.
(a) which
(b) whose
(c) whom

14. I wish I ___ go there on Friday.
(a) could
(b) should
(c) would

15. If he ___, I won't stay.
(a) come
(b) came
(c) comes
(d) will come

16. He denied ____ it.
(a) say
(b) saying
(c) to say

17. You ____ better hurry.

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(a) should
(b) had
(c) would

18. I wish they ____ be quiet; they know how it annoys me.
(a) could
(b) should
(c) would

19. She took ___ painting after she retired.
(a) down
(b) from
(c) up

20. She didn't let them ___ it.
(a) do
(b) doing
(c) to do

21. They weren't allowed ____ it.
(a) do
(b) doing
(c) to do

22. By the time I reached ____, everyone had left.
(a) the party
(b) to the party

23. Hurry up- the taxi's ____ to arrive.
(a) about
(b) on the verge

24. You ____ be better off taking the train.
(a) had
(b) should
(c) would

25. This is the first time I ____ to her.
(a) spoke
(b) have spoken

26. ____ anyone came.
(a) Almost
(b) Hardly

27. Do you _____ his story?
(a) believe
(b) believe in

28. The negative of 'noble is ___-
(a) innoble
(b) unnoble
(c) ignoble
(d) disnoble

29. The adverb from 'public' is____.
(a) Publicly
(b) Publically

30. I doubt___ she'll be there.

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(a) if
(b) that

31. They made us ____ welcome
(a) feel
(b) to feel
(c) feeling

32. Let's go tonight, ___?
(a) will we
(b) shall we

33. She fell ____.
(a) asleep
(b) sleepy
(c) sleeping

34. ___ a matter of fact, I did finish it on time.
(a) As
(b) Like
(c) Alike

35. How old is he?
(a) **I'm not sure, but he is getting on a bit now**.
(b) I'm not sure, but he is getting in a bit now.

36. Thanks for your____.
(a) encourage
(b) encouragement
(c) encouraging

37. I don't know why she ____ it.
(a) did
(b) has done
(c) Either could be used here.

38. The first time I ___ her was at my brother's wedding.
(a) met
(b) have met
(c) Either could be used here.

39. It's time we ____.
(a) leave
(b) left
(c) Either could be used here.

40. She had her dog put ___ sleep because it had a malignant tumour.
(a) through
(b) down
(c) to

41. They were in ____ trouble.
(a) such
(b) so much
(c) Either could be used here.

42. The government was anxious about the results of the official ____.
(a) enquiry
(b) inquiry

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(c) Either could be used here.

43. I don't know how I'd have got ____ it without your support.
    (a) through
    (b) over
    (c) Either could be used here.
44. It ____ ten minutes to do the test.
    (a) only took him
    (b) took him only
    (c) Either could be used here.
45. It was ____ better than I was expecting it to be.
    (a) quite
    (b) rather
46. I ____ enjoyed meeting them
    (a) quite
    (b) rather
    (c) Either could be used here.
47. She went ___ to become a minister.
    (a) in
    (b) on
48. I tired ___ it.
    (a) of doing
    (b) to do
    (c) doing
49. They have yet to ____ the details of the plan.
    (a) end
    (b) finalise
50. I spent twenty minutes ___ the test.
    (a) do
    (b) to do
    (c) doing

1. I spoke to ____.
   (a) she
   (b) her
2. Where ____ you come from?
   (a) do
   (b) are
3. What time does she ___ up?
   (a) get
   (b) gets
4. Where ___ he live?
   (a) do
   (b) does
5. 'I am not ____ this film.'
   (a) liking

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(b) enjoying

6. I am seeing her ____ three o'clock.
   (a) in
   (b) at
   (c) on
7. Easter is ___ March this year.
   (a) on
   (b) at
   (c) in
8. She's ____ work all day today.
   (a) at
   (b) on
9. I go ___ by bus.
   (a) home
   (b) to home
10. 'Do you like it?'
    (a) 'Yes, I like.'
    (b) 'Yes, I do.'
11. It's the second road ___ the left.
    (a) in
    (b) at
    (c) on
12. He's arriving ___ the station at six.
    (a) at
    (b) on
    (c) to
13. I ___ what she's saying.
    (a) can't understand
    (b) am not understanding
14. She lives ___ London.
    (a) on
    (b) at
    (c) in
15. I'm going ___ the bank to get some cash.
    (a) at
    (b) to
16. I went there ____ foot.
    (a) in
    (b) by
    (c) at
    (d) on
    (e) to
17. ____ is a cinema in the shopping centre.
    (a) There
    (b) It
18. I went ____ with my sister.
    (a) their

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(b) there

19. She ____ as a doctor.
    (a) is
    (b) works
20. She ___ a doctor.
    (a) is
    (b) works
21. How ___ did your journey take?
    (a) long
    (b) long time
22. He comes ____ the north of the country.
    (a) to
    (b) from
23. She ____ goodbye.
    (a) said
    (b) told
    (c) spoke
24. They are ___ love.
    (a) in
    (b) on
    (c) at
    (d) to
    (e) by
25. You _____ drink and drive.
    (a) may
    (b) must
    (c) may not
    (d) must not
26. She's studying law ____ university.
    (a) in
    (b) at
27. The bus stopped ___ the traffic lights.
    (a) in
    (b) on
    (c) at
28. She ____ English twice a week.
    (a) study
    (b) studys
    (c) studeis
    (d) studies
29. ____ is it?
    (a) Who's
    (b) Whose
30. ___ you afford it?
    (a) Do
    (b) Can
31. They aren't selling ____ tickets.

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(a) much
(b) many

32. There isn't _____ for all of us to get in.
(a) room enough
(b) enough room

33. The police ___ called.
(a) was
(b) were

34. In a shop, you are a ____.
(a) client
(b) customer

35. It's bigger ____ the old one.
(a) as
(b) than
(c) then

36. '___ did she do it?' 'Yesterday.'
(a) Why
(b) Where
(c) When
(d) How

37. How ___ you spell it?
(a) do
(b) can

38. How much did you ___ in the shop?
(a) cost
(b) spend

39. 'How ___ did you buy?' 'Three kilos.'
(a) much
(b) many

40. I don't work ____ the morning.
(a) on
(b) in
(c) at

41. I saw her ____.
(a) last week
(b) the last week

42. It costs ____ dollars.
(a) forty
(b) fourty

43. I heard it ___ the radio.
(a) in
(b) at
(c) on

44. There ____ a lot of complaints.
(a) was
(b) were

45. There ____ a lot of sugar in the coffee.

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(a) was
(b) were

46. He didn't work ____.
(a) enough hard
(b) hard enough

47. I ____ her.
(a) am not knowing
(b) don't know
(c) doesn't know

48. ___ girlfriend left him.
(a) He
(b) Him
(c) His

49. There are too ____ questions in this test.
(a) much
(b) many

50. I hope you do ___ in this test.
(a) good
(b) well

1. There's a bus ____ quarter of an hour.
(a) every
(b) all
(c) all of
2. "All" can be used as a pronoun.
(a) **True** (b)False
3. "Every" can be used as a pronoun.
(a) True (b)**False**
4. Yesterday, it rained ____ day.
(a) all
(b) every
5. We had ____ sorts of problems doing it.
(a) all
(b) every
6. I tried ____ single thing I could think of, but it still didn't work.
(a) all
(b) every
7. I've read ____ her books.
(a) all
(b) every
8. I've seen ____ film he's made.
(a) all
(b) every
9. It took me ____ day to finish.
(a) all

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(b) every

10. I was tired, ___ I managed to finish it.
    (a) already
    (b) yet
11. Have you ____ finished?
    (a) already
    (b) yet
12. As ___, I haven't has time to call her.
    (a) already
    (b) yet
13. He has ____ to start the project.
    (a) already
    (b) yet
14. Have they arrived ___?
    (a) already
    (b) yet
15. Have they ____ finished?
    (a) already
    (b) yet
16. I was shocked that they had ____ done it.
    (a) already
    (b) yet
17. I was surprised that they had ___ to decide what to do.
    (a) already
    (b) yet
18. She has ___ to arrive.
    (a) already
    (b) yet
19. He has ___ arrived.
    (a) already
    (b) yet
20. I had ____ had enough.
    (a) already
    (b) yet
21. I hate getting up ____.
    (a) early
    (b) soon
22. Ciao- see you ____.
    (a) early
    (b) soon
23. Come on, let's wait a bit longer- I'm sure he'll be here ____.
    (a) early
    (b) soon
24. I arrived a bit ____ so I had a coffee while I waited.
    (a) early
    (b) soon
25. Could you do it as ____ as possible- I'm in a real hurry.

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(a) early
(b) soon

26. She's in her ____ sixties.
(a) early
(b) soon
27. It happened in the ____ 1990s.
(a) early
(b) soon
28. I'm not going out tonight because I want an ____ night.
(a) early
(b) soon
29. It's cloudy- I think it's going to rain ____.
(a) early
(b) soon
30. Tell her I'm sorry she's ill and to get well ____.
(a) early
(b) soon
31. It was _____ so we didn't get it.
(a) expensive enough
(b) too expensive
(c) enough expensive
32. It's ___ to read; I don't understand it at all.
(a) enough difficult
(b) too difficult
(c) difficult enough
33. They didn't sell _____ to make it worthwhile.
(a) tickets enough
(b) enough tickets
(c) too tickets
34. There were _____ people there.
(a) too
(b) too many
35. It's not ____ to sort things out.
(a) enough late
(b) late enough
(c) too late
36. It's ____ to pass.
(a) enough difficult
(b) too difficult
37. I left because I'd had _____ their arguing.
(a) enough
(b) enough of
(c) too
38. He ate ____ and felt ill.
(a) enough much
(b) much enough
(c) too much

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39. I'm shattered; I didn't ____ last night.
    (a) enough sleep
    (b) sleep enough
    (c) too sleep
    (d) too much sleep
40. It's ____ to walk- I'll take the bus.
    (a) enough far
    (b) far enough
    (c) too far
    (d) too much far
41. I didn't get ____ last night and I'm exhausted.
    (a) enough sleep
    (b) sleep enough
    (c) too sleep
42. It was _____ so we didn't get it.
    (a) expensive enough
    (b) too expensive
    (c) enough expensive
43. It's ___ to read; I don't understand it at all.
    (a) enough difficult
    (b) too difficult
    (c) difficult enough
44. They didn't sell _____ to make it worthwhile.
    (a) tickets enough
    (b) enough tickets
    (c) too tickets
45. There were _____ people there.
    (a) too
    (b) too many
46. It's not ____ to sort things out.
    (a) enough late
    (b) late enough
    (c) too late
47. It's ____ to pass.
    (a) enough difficult
    (b) too difficult
48. I left because I'd had _____ their arguing.
    (a) enough
    (b) enough of
    (c) too
49. He ate ____ and felt ill.
    (a) enough much
    (b) much enough
    (c) too much
50. I'm shattered; I didn't ____ last night.
    (a) enough sleep
    (b) sleep enough

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(c) too sleep
(d) too much sleep

51. It's ____ to walk- I'll take the bus.
(a) enough far
(b) far enough
(c) too far
(d) too much far

52. I didn't get ____ last night and I'm exhausted.
(a) enough sleep
(b) sleep enough
(c) too sleep

53. I don't have ____ free time.
(a) much
(b) many

54. I haven't met her ___ times.
(a) much
(b) many

55. Was there ___ trouble yesterday?
(a) much
(b) many

56. She doesn't have _____ problems with her clients.
(a) much
(b) many

57. How ____ did they cost?
(a) much
(b) many

58. How ____ police were at the scene?
(a) much
(b) many

59. How ____ police officers were at the scene?
(a) much
(b) many

60. I didn't have ____ time to think.
(a) much
(b) many

61. Not ____ people are that brave.
(a) much
(b) many

62. There wasn't ____ we could do to help.
(a) much
(b) many

63. I bought a pair of _____ shoes.
(a) black leather
(b) leather black

64. It was a ____ car.
(a) red fast
(b) fast red

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65. It's a ____ building.
    (a) big round
    (b) round big
66. I bought ____ knife.
    (a) a Swiss army
    (b) an army Swiss
67. It's ____ film.
    (a) a beautiful old
    (b) an old beautiful
68. He's ____ man.
    (a) an unfriendly rich
    (b) a rich unfriendly
69. It's ____ phone.
    (a) a mobile expensive
    (b) an expensive mobile
70. It's ____ village.
    (a) an old lovely
    (b) a lovely old
71. The ____ visitors were Japanese.
    (a) two last
    (b) last two
72. He's got ____ eyes.
    (a) blue big
    (b) big blue
73. It's a ____ house.
    (a) nice new
    (b) new nice
74. It's ____ airline.
    (a) a popular American
    (b) an American popular
75. It's ____ company
    (a) a family old
    (b) an old family
76. It's a ____ restaurant.
    (a) cheap good
    (b) good cheap
77. There were ..... many questions to answer so I only did three
    (a) too
    (b) enough
78. The coffee was .... to drink so I left it for a minute to cool
    (a) hot enough
    (b) too hot
79. The room wasn't ..... so I turned the heating on
    (a) hot enough
    (b) too hot
80. I didn't buy the jacket because it was .....
    (a) big enough

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(b) too big

81. The shoes were ..... so I didn't buy them
    (a) small enough
    (b) too big
82. It's far ..... to fit inside the bag
    (a) too big
    (b) small enough
83. It was far ..... to pick up
    (a) too heavy
    (b) heavy enough
84. I didn't get it because it was .....
    (a) too expensive
    (b) expensive enough
85. I haven't got ..... to take a holiday this year
    (a) too much time
    (b) enough time
86. It was ..... to carry
    (a) heavy enough
    (b) too heavy
87. There wasn't ..... for everybody to get in
    (a) enough room
    (b) too much room
88. He didn't do ..... to pass the exam
    (a) too much work
    (b) enough work
89. There wasn't ..... to finish so I didn't manage to
    (a) enough time
    (b) too much time
90. She can't sleep because she drinks ..... coffee
    (a) enough
    (b) too much
91. I didn't know him ....., but I was still upset when he died
    (a) too well
    (b) good enough
92. I left the job because I was making .....
    (a) too little money
    (b) too few money
93. I put a scarf on because it was .....
    (a) cold enough
    (b) too cold
94. This town isn't ..... for both of us
    (a) big enough
    (b) too big
95. He isn't ..... to go back to work yet
    (a) well enough
    (b) too well
96. I don't think the exam is ..... for me to have a good chance of passing

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(a) easy enough
(b) too difficult

97. This room isn't ..... for all the guests; we'll have to get a bigger one
(a) big enough
(b) too big

98. She isn't ..... to start driving until next year
(a) too old
(b) old enough

99. I haven't got ..... to pay the bill until I get my salary
(a) enough money
(b) too much money

100. Everything was much ..... for us to get anything
(a) too expensive
(b) cheap enough

101. He speaks far ..... for us to understand him unless we ask him to say it again
(a) quickly enough
(b) too quickly

102. They haven't got ..... to buy it
(a) enough money
(b) too much money

103. They took ..... on the first part and didn't finish the rest
(a) too much time
(b) enough time

104. There's never ..... to do everything I would like to do
(a) too much time
(b) enough time

105. The photograph didn't come out because it was .....
(a) dark enough
(b) too dark

106. I did it ____.
(a) yesterday
(b) tomorrow

107. I ____ there today.
(a) go
(b) am going

108. My plane leaves ____.
(a) yesterday
(b) tomorrow

109. I haven't seen her ____.
(a) yesterday
(b) today

110. I haven't seen her since ____.
(a) yesterday
(b) today

111. I saw them ____ morning.
(a) yesterday
(b) in yesterday

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112. I'll do it first thing ____.
(a) yesterday
(b) today
(c) tomorrow
113. I'm going there ____ night.
(a) tomorrow
(b) at tomorrow
114. I could do it ____ if you like.
(a) yesterday
(b) tomorrow
115. ____ is her birthday; she'll be fifteen.
(a) Today
(b) Tomorrow
116. I might go there ____.
(a) yesterday
(b) tomorrow
117. She's been married ____ her husband for twenty years.
(a) to
(b) with
118. He's married ____ two kids.
(a) to
(b) with
119. It is similar ____ mine.
(a) from
(b) to
(c) than
120. He's interested ____ music.
(a) at
(b) in
(c) on
(d) of
121. She was responsible ____ the mess.
(a) for
(b) of
(c) in
(d) on
122. That's typical ____ them.
(a) in
(b) of
(c) off
(d) upon
123. I am sick ____ their behaviour.
(a) of
(b) off
(c) with
124. They're capable ____ passing.
(a) in

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(b) of
(c) off
(d) through

125. I'm a bit short ____ cash; could you lend me some?
(a) at
(b) of
(c) off
(d) to

126. I wasn't aware ____ what had happened.
(a) at
(b) of
(c) off
(d) tup
(e) upon

127. They spent _____ time arguing.
(a) all
(b) whole
(c) the
(d) the whole

128. They argue _____ time.
(a) all
(b) whole
(c) all the

129. ____ people at the meeting were bored.
(a) All
(b) Whole
(c) All the
(d) Whole the

130. The book is ____ about the royal family.
(a) all
(b) whole
(c) the all
(d) the whole

131. The ____ family was there.
(a) all
(b) whole

132. He is ____ responsible.
(a) whole
(b) wholly

133. That's ____ point.
(a) all
(b) whole
(c) the all
(d) the whole

134. ____ in ____, it was a success.
(a) All ... all
(b) Whole ... whole

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135. I'd travel ____ world to find her.
(a) all
(b) the whole
136. It's ____ in the mind.
(a) all
(b) whole
(c) the all
(d) the whole
(e) all the

VOCAB

Synonyms And Antonyms Meaning

Word	Synonyms	Antonyms
abandon	**desert, forsake**	**keep**
ability	**skill, aptitude incompetence,**	**inability**
abundant	**ample, sufficient scanty,**	**insufficient**

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achieve	**accomplish, attain**	**fail**
adjourn	**postpone, recess**	**recommence, continue**
amateur	**beginner, novice**	**professional**
artificial	**fake, synthetic**	**real, authentic**
barren	**unproductive, infertile fertile,**	**productive**
bashful	**shy, timid outgoing, assured**	
beginning	**start, initiate**	**finish, end**
blend	**combine, mix**	**separate**
broad	**wide, expansive**	**narrow**
calm	**quiet, tranquil, still**	**excited, turbulent**
chubby	**plump, pudgy**	**thin, skinny**
coarse	**bumpy, rough**	**fine, smooth**

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competent	**capable, qualified**	**incompetent, inept**
complex	**complicated, intricate**	**simple**
condemn	**censure, denounce**	**approve**
conflict	**fight, battle, struggle**	**peace, harmony**
congested	**overcrowded, stuffed**	**empty, unfilled**
conscientious	**scrupulous, virtuous**	**neglectful, careless**
consecutive	**successive, continuous**	**interrupted**
contaminate	**pollute, defile, infect**	**purify**
conventional	**customary, traditional**	**unusual**
courteous	**polite, civil**	**rude**
cranky	**cross, irritable**	**good-humored**
daring	**bold, audacious**	**cautious**

YT channel- MERCHANT NAVY CAREER

decay	**rot, spoil bloom, flourish**	
delicate	**fragile, dainty**	**sturdy**
desolate	**barren, forsaken**	**dense, verdant**
despise	**hate, detest, loathe**	**love**
detach	**separate, unfasten, remove**	**attach**
difficult	**hard, challenging**	**easy**
diminish	**curtail, lessen, decrease**	**increase, amplify**
dormant	**sleeping, inactive**	**awake, active**
drastic	**severe, extreme, tough**	**mild,moderate**
dumb	**stupid, dense**	**smart**
eccentric	**peculiar, unusual**	**normal**
ecstasy	**joy, rapture, elation**	**sadness, depression**

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enormous	**vast, immense, colossal**	**tiny, microscopic**
eternal	**always, perpetual, everlasting**	**temporary, passing**
exhilarated	**overjoyed, ecstatic, elated**	**depressed, dejected, sad**
extravagant	**extreme, excessive, luxurious**	**meagre**
fabulous	**marvelous, amazing**	**unexciting**
face	**confront, meet**	**avoid**
fantastic	**incredible, extra-ordinary,**	**ordinary, normal**
feasible	**possible, attainable, practical**	**impossible**
ferocious	**fierce, savage, brutal,**	**tame, gentle**
flaw	**defect, fault, blemish**	**perfection**
flimsy	**frail, fragile, delicate**	**sturdy, strong**
flippant	**impudent, sassy**	**polite, respectful**

YT channel- MERCHANT NAVY CAREER

fluid	**liquid**	**solid**
foe	**enemy, adversary, opponent**	**friend**
follow	**succeed, trail lead,**	**precede**
forbid	**prohibit, ban, bar**	**permit**
frivolous	**trivial, unimportant, silly**	**important, serious**
furious	**angry, enraged, infuriated**	**calm, placid**
gaudy	**showy, garish, vulgar**	**tasteful, refined**
gaunt	**scrawny, skinny, thin**	**overweight, plump**
generous	**giving, selfless, big-hearted**	**selfish, stingy**
gentle	**tender, mild**	**rough, harsh**
genuine	**real, authentic, sincere**	**fake, phony**
gigantic	**immense, colossal, enormous**	**tiny, minute**

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gloomy	**dark, dismal, depressing**	**cheery, bright**
glorious	**splendid, magnificent, superb**	**terrible, awful**
good	**nice, fine, well-behaved**	**bad, awful**
gorgeous	**ravishing, dazzling, stunning**	**hideous, unattractive**
gratitude	**thankfulness, appreciation**	**ungratefulness**
hold	**grasp, grip, retain**	**release, discharge**
honest	**truthful, sincere, frank**	**untruthful, dishonest**
hospitable	**welcoming, cordial, gracious**	**rude, unfriendly**
hostile	**antagonistic, aggressive, militant**	**friendly, cordial**
huge	**vast, immense, great**	**small, tiny**
humble	**modest, unpretentious**	**vain, showy**
humiliate	**embarrass, disgrace, dishonour**	**honour, dignify**

YT channel- MERCHANT NAVY CAREER

immaculate	**spotless, pure**	**dirty, filthy**
immature	**childish, inexperienced**	**mature, adult**
immune	**resistant, exempt**	**susceptible**
impartial	**neutral, unbiased, fair**	**prejudiced, partial**
impatient	**eager, anxious, intolerant**	**patient**
imperfect	**marred, defective, faulty**	**perfect, flawless**
impetuous	**impulsive, rash, reckless**	**restrained, careful**
join	**connect, unite, link**	**separate, disconnect, detach**
jolly	**merry, jovial, joyful**	**sad, grim, glum**
jubilant	**overjoyed, delighted, elated**	**dejected, depressed**
keep	**save, protect, guard**	**discard, lose**
kind	**considerate, tender, thoughtful**	**mean, cruel, inconsiderate**

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lenient	**kind, unrestrained, easy**	**harsh, strict**
listless	**lethargic, tired**	**active, energetic**
logical	**sensible, sane, rational**	**illogical, unreasonable**
migrant	**drifting, traveling, transient**	**stationary, immovable**
militant	**combative, aggressive,warlike**	**peaceful**
minor	**lesser, inferior,**	**secondary major**
mirth	**merriment, fun, laughter**	**gloom, sadness**
mischievous	**naughty, impish**	**well-behaved, angelic**
misfortune	**hardship, catastrophe, mishap**	**good luck, fortune**
mobile	**moveable, changeable**	**immobile, stationary**
moderate	**temperate, lenient, medium**	**extreme, harsh**
momentous	**important, powerful, outstanding**	**unimportant, insignificant**

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monotonous	**boring, tedious,**	**humdrum, interesting**

One Word Substitutions

A. Pertaining to Government:

1.	**To give up a throne or another office of dignity**	**Abdicate**
2.	**A diplomatic minister of the highest order sent by one country to another**	**Ambassador**
3.	**Absence of government**	**Anarchy**
4.	**A person who is out to destroy all government and order**	**Anarchist**
5.	**A person liable to be called to account for his actions**	**Answerable**
6.	**Government by one Autocracy**	**Despotism**
7.	**Government by the officials/ departments of state**	**Bureaucracy**
8.	**Government by the nobles**	**Aristocracy**
9.	**The right of self-government**	**Autonomy**
10.	**Government of the people, for the people, by the people**	**Democracy**

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B. Pertaining to Sciences and Arts:

1.	**The study of all heavenly bodies and the earth in relation to them**	**Astronomy**
2.	**One who studies the sky and stars**	**Astronomer**
3.	**The study of mankind/ the science which treats of man**	**Anthropology**
4.	**The study of physical life or living matter**	**Biology**
5.	**The study of plants**	**Botany**
6.	**The art of beautiful hand-writing**	**Calligraphy**
7.	**The art of practicing by statesman and ambassadors**	**Diplomacy**
8.	**One who thinks of the welfare of women**	**Feminist**
9.	**The study of coins**	**Numismatics**
10.	**The study of birds/ the science which treats of the birds**	**Ornithology**

C. Pertaining to the Literary Arts:

1.	**A work whose writer is unknown**	**synonymous**

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2.	**A record of one's life written by himself**	**Autobiography**
3.	**The history of the life of a person**	**Biography**
4.	**A humorous play, having a happy ending**	**Comedy**
5.	**A list of books in a library**	**Catalog**
6.	**A book in which the events of each day are recorded**	**Diary**
7.	**A book containing the words of a language with their definitions, in alphabetical order**	**Dictionary**
8.	**A book of names and address**	**Directory**
9.	**A book containing information on all branches of knowledge**	**Encyclopedia**
10.	**A speech delivered without earlier preparation**	**Extempore**

D. Pertaining to Religion:

1.	**One who is not sure of the existence of God**	**Agnostic**
2.	**One who renounces his religious vows or forsakes his religious principles**	**Apostate**

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3.	**One who does not believe in the existence of God**	**Atheist**
4.	**One who believes in the existence of God**	**Theist**
5.	**One who has narrow and prejudiced religious views**	**Bigot**
6.	**One who is present everywhere**	**Omnipresent**
7.	**One who knows everything**	**Omniscient**
8.	**One who is all-powerful**	**Omnipotent**
9.	**One who serves the public interest and feels very sympathetic towards human beings**	**Humanitarian**

E. Pertaining to Death:

1.	**To destroy completely**	**Annihilate**
2.	**Dead and decaying flesh especially of animals**	**Carrion**
3.	**A monument set up for persons who are buried elsewhere**	**Cenotaph**
4.	**To preserve a dead body from decaying**	**Embalm**

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5.	**Words inscribed on a tomb**	**Epitaph**
6.	**An examination of a dead body**	**Autopsy/Postmortem**
7.	**An article in the newspaper about the life of the deceased**	**of one Obituary**
8.	**The property left to someone by a will**	**Legacy**
9.	**Something occurring after death**	**Posthumous**
10.	**The act of killing a human being**	**Homicide**

F. Pertaining to Marriage/ Family/ Children:

1.	**One who marries a second wife/ husband while the legal spouse is alive**	**Bigamist**
2.	**One vowed to a single or unmarried life**	**Celibate**
3.	**One engaged to be married**	**Fiancee**
4.	**Children whose parents are dead**	**Orphan**
5.	**A hater of marriage**	**Misogamist**

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6.	**One has more than one wife or husband at a time**	**Polygamist**
7.	**State of growth between boyhood and youth**	**Adolescence**
8.	**Marrying more than one husband/ wife at a time**	**Polygamy**
9.	**Marrying more than one husband at a time**	**Polyandry**
10.	**Hater of mankind**	**Misanthrope**

G. Pertaining to Profession:

1.	**One who writer books**	**Author**
2.	**One who makes or sells candles**	**Chandler**
3.	**One who drives a motor car**	**Chauffeur**
4.	**One skilled in the care of hands and feet**	**Chiropodist**
5.	**One who sells sweets and pastries**	**Confectioner**
6.	**One who works in a coal mine**	**Collier**

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7.	**One who attends to the teeth**	**Dentist**
8.	**One who shoes horses**	**Ferrier**
9.	**One who studies rocks and soils**	**Geologist**
10.	**One who travels from place to place selling articles**	**Hawker**

H. Denoting Numbers

1.	**A collection of poems**	**Anthology**
2.	**A number of merchant ships protected by warships**	**Convoy**
3.	**A number of stars grouped together**	**Constellation**
4.	**A number of hired applauders, i.e., persons paid to clap**	**Claque**
5.	**A number of people at church**	**Congregation**
6.	**A number of people gathered together for some common purpose**	**Gathering, assembly**
7.	**People who get together to work for some cause of common interest**	**Coterie**

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8.	**A number of workmen, prisoners, thieves**	**Gang**
9.	**A number of sheep**	**Flock**
10.	**A number of ships**	**Fleet**

I. Miscellaneous

1.	**Loud enough to be heard**	**Audible**
2.	**Not distinct enough to be heard**	**Inaudible**
3.	**Fit to be eaten**	**Edible**
4.	**Unfit for human consumption**	**Inedible**
5.	**Fit to be chosen or selected**	**Eligible**
6.	**Not having the qualities of being chosen**	**Ineligible**
7.	**Writing that is easy to read**	**Legible**
8.	**Writing that is easy to decipher**	**Illegible**

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9.	**Able to read**	**Literate**
10.	**Unable to read**	**Illiterate**
11.	**Born of married parents**	**Legitimate**
12.	**Born of unmarried parents**	**Illegitimate**
13.	**To send back a person to his own country**	**Repatriate**
14.	**To banish from one's country**	**Expatriate**
15.	**To move from one country to another**	**Migrate**
16.	**One who leaves his country to settle in another**	**Emigrant**
17.	**One who comes into a foreign country to settle there**	**Immigrant**
18.	**Incapable of being redeemed from evil, i.e., beyond correction**	**Incorrigible**
19.	**That which cannot be rubbed out or blotted out**	**Ineffaceable**
20.	**That which cannot be conquered**	**Invincible**

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Idioms and Phrases

Cry for the moon	**Ask for the impossible**
To in a tight corner	**In a very difficult situation**
Cock and bull story	**Made up story that one should not believe**
Cat and dog life	**Life full of quarrels**
Like a cat on hot bricks	**Very nervous**
Have a card up one's sleeve	**Have a secret plan in reserve**
Tell against	**To prove adverse to; to go against**
Run through	**to squander or waste**
Run over	**to drive over**
Alive and kicking	**Active and healthy**
To burn one's fingers	**To lose on account of foolish behaviour**

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At sea	**Confused or lost**
Behind one's back	**In the absence of**
To call a spade a spade	**To be frank and truthful**
To break the ice	**To break the silence**
To bell the cat	**To face the risk**
A cold fish	**A person devoid of emotions**
To cross the Rubicon	**To take a decisive step**
Double	**dealing – Deceiving**
To die hard	**Change with great difficulty**
To eat humble pie	**To apologize in a humble manner**
To fight a losing battle	**Struggle without hope of success**
A deadlock	**A position where no progress can be made**

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A daredevil	**A person who does not care for any consequences**
Once in a blue moon	**Very rarely**
At a premium	**Difficult to get or obtain**
To play the game	**To act honestly**
To die in harness	**To work till the last day of one's life**
To plough a lonely furrow	**To work without help or support**
Lock, stock and barrel	**Completely**
The long and the short	**The main point**
Bring to light	**Disclose**
The bottom line	**The final criteria**
To kick one's heels	**To wait with nothing to do**
Rain cats and dogs	**Rain heavily**

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A hard nut to crack	**A difficult problem**
In high spirits	**Very happy**
Like a fish out of water	**In a strange situation**
Smell a rat	**Suspect something foul**
Read between the lines	**Understand the hidden meaning**
Tooth and nail	**With all one's power**
Spread like wild fire	**Spread quickly**
Hit the nail on the head	**Do or say the exact thing**
Burn the midnight oil	**Work or study hard**
Under his thumb	**Under his control**
With a high hand	**Oppressively**
Gain ground	**Become popular**

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By leaps and bounds	**Rapidly**
Make off with	**to run away with**
Pass away	**to die**
To make a pig of oneself	**To overeat**
Donkey's years	**A very long time**
To pull one's leg	**to joke; tease someone**
To keep one's fingers crossed	**To hope fervently**
To call in question	**To doubt**
To talk through one's hat	**To talk ignorantly**
To make up one's mind	**To decide**
Hard and fast rules	**Rigid rules**
To make one's mouth water	**To stimulate one's appetite**

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By word of mouth	**By spoken message**
To get the better of someone	**To take advantage of someone**
To rub shoulders	**To come into close contact**
A snake in the grass	**A hidden enemy**
Between the devil and the deep blue sea	**Between two dangers**
Chew the cud	**To discuss matters in a ponderous manner, especially reminiscing**
To kill two birds with one stone	**To achieve two results with one effort**
A close shave	**A narrow escape**
Chip off the old block	**someone as good as one's father**
To pull a long face	**To look sad**
Above board	**Open and honest**
To turn a deaf ear	**To ignore**

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To turn the tables	**To reverse the situation**
To call someone names	**To abuse or insult someone**
Above board	**Honest and open**
At arm's length	**To keep at a distance**
At the eleventh hour	**At the last moment**
An apple of discord	**A cause of quarrel**
To burn one's boats	**Make a final decision, destroy all possible ways of going back a situation**
To build castles in the air	**Make imaginary schemes**
A burning question	**An important topic**
A brown study	**Daydreaming**
Sitting on the fence	**Hesitate between two decisions**

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